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(1973); S. Weinberg, *ibid.* **7**, 2887 (1973); R. Jackiw, Ref. 4.

¹²The analysis of the physical interpretation of $\Gamma(\psi, G)_{\text{bare}}$ is an adaptation to the present context of the corresponding argument for $\Gamma(\psi)_{\text{bare}}$. This discussion is due to K. Symanzik, *Commun. Math. Phys.* **18**, 48 (1970). We

learned it from S. Coleman, in proceedings of the Lectures given at the International Summer School of Physics "Ettore Majorana," 1973 (unpublished).

¹³R. Dahlen, B. Hasslacher, and A. Neveu, *Phys. Rev. D* (to be published).

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Confinement of quarks^a

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A mechanism for total confinement of quarks, similar to that of Schwinger, is defined which requires the existence of Abelian or non-Abelian gauge fields. It is shown how to quantize a gauge field theory on a discrete lattice in Euclidean space-time, preserving exact gauge invariance and treating the gauge fields as angular variables (which makes a gauge-fixing term unnecessary). The lattice gauge theory has a computable strong-coupling limit; in this limit the binding mechanism applies and there are no free quarks. There is uniformly no Lorentz (or Euclidean) invariance in the strong-coupling limit. The strong-coupling expansion involves sums over all quark paths and sums over all surfaces (on the lattice) joining quark paths. This structure is reminiscent of relativistic string models of hadrons.

1. INTRODUCTION

The success of the quark-constituent picture both for resonances and for deep-inelastic electron and neutrino processes makes it difficult to believe quarks do not exist. The problem is that quarks have not been seen. This suggests that quarks, for some reason, cannot appear as separate particles in a final state. A number of speculations have been offered as to how this might happen.¹

Independently of the quark problem, Schwinger observed many years ago² that the vector mesons of a gauge theory can have a nonzero mass if vacuum polarization totally screens the charges in a gauge theory. Schwinger illustrated this result with the exact solution of quantum electrodynamics in one space and one time dimension, where the photon acquires a mass $\propto e^2$ for any nonzero charge e [e has dimensions of (mass)^{1/2} in this theory]. Schwinger suggested that the same effect could occur in four dimensions for sufficiently large couplings.

Further study of the Schwinger model by Lowenstein and Swieca³ and Casher, Kogut, and Susskind⁴ has shown that the asymptotic states of the model contain only massive photons, not electrons. Nevertheless, as Casher *et al.* have shown in detail, the electrons are present in deep-inelastic processes and behave like free pointlike

particles over short times and short distances. The polarization effects which prevent the appearance of electrons in the final state take place on a longer time scale (longer than $1/m_\gamma$, where m_γ is the photon mass).

A new mechanism which keeps quarks bound will be proposed in this paper. The mechanism applies to gauge theories only. The mechanism will be illustrated using the strong-coupling limit of a gauge theory in four-dimensional space-time. However, the model discussed here has a built-in ultraviolet cutoff, and in the strong-coupling limit all particle masses (including the gauge field masses) are much larger than the cutoff; in consequence the theory is far from covariant.

The confinement mechanism proposed here is soft (long-time scale). However, in the model discussed here the cutoff spoils the possibility of free pointlike behavior for the quarks.

The model discussed in this paper is a gauge theory set up on a four-dimensional Euclidean lattice. The inverse of the lattice spacing a serves as an ultraviolet cutoff. The use of a Euclidean space (i.e., imaginary instead of real times) instead of a Lorentz space is not a serious restriction; the energy eigenstates (including scattering states) of the lattice theory can be determined from the "transfer-matrix" formalism as has been discussed by surf⁵ and reviewed by Wilson and Kogut.⁶ A brief discussion of the

transfer-matrix method is given in Sec. III.

In Schwinger's speculations about four dimensions, the photon mass would be zero for any charge e less than a critical coupling e_c ; for $e > e_c$ the photon mass would be nonzero and vary with e . Figure 1 shows how a plot of m_γ vs e might look. The point e_c is a point of nonanalyticity. Similar nonanalytic points, called critical points, occur in solid-state physics at certain types of phase transitions. Consider, for instance, a ferromagnet in the absence of an external field. For any temperature above the Curie temperature T_C , the spontaneous magnetization M is 0. Below T_C , M is nonzero and a function of temperature. At T_C there may be either a first-order phase transition (in which case M is discontinuous at T_C) or a second-order phase transition (critical point) for which $M \rightarrow 0$ as $T \rightarrow T_C$ from either side.

By analogy with the solid-state situation one can think of the transition from zero to nonzero photon mass as a change of phase: this analogy is best understood by imagining the particles of quantum electrodynamics to be the excitations of a medium (the ether). In this case it is the ether which undergoes a change of phase at e_c . There is again a question whether this change of phase is first-order (cf. Fig. 2) or second-order (Fig. 1). (Coleman and Weinberg¹⁰ have found a nontrivial example of a first-order transition in another context.)

The model discussed in this paper is a single Abelian gauge field coupled (with strength g) to massive quarks. In weak coupling the gauge field behaves like a normal free zero-mass field (despite modifications introduced in the lattice quantization) and the quarks are unbound. In strong coupling the gauge field is massive and the quarks are bound, showing the existence of the second phase. Thus there should be a phase transition at some intermediate value of g . Nothing is known about this transition at the present time.

The quantization procedure and strong-coupling approximation described in this paper can be applied to non-Abelian gauge theories also. This will be explained briefly in Sec. III.

An extraordinary feature of the strong-coupling expansion of the lattice theory (see Sec. IV) is that it has the same general structure as the relativistic string models of hadrons.¹¹ The vacuum expectation values of the gauge theory involve (in the strong-coupling expansion) sums over all quark paths and sums over all surfaces connecting these paths; the surfaces are generated by the gauge field treated in strong coupling. The paths and surfaces are defined on a discrete lattice. There are geometrical difficulties in relat-

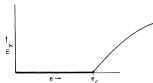


FIG. 1. Speculative plot of photon mass vs renormalized charge e , in unknown units. The transition at e_c is second-order (see text).

ing the surfaces on the lattice to the continuum surfaces of the string models; it is not known at present whether these difficulties can be overcome.

In Sec. II the nature of the quark binding mechanism will be discussed, qualitatively. In Sec. III the gauge theory will be formulated on a discrete lattice, both classically and quantum mechanically. In Sec. IV the strong-coupling expansion for the lattice gauge theory is explained. In Sec. V a cursory discussion of weak coupling is given. In Sec. VI there is a brief discussion of the problem of Lorentz invariance and the relation to string models.

II. QUARK BINDING MECHANISM

The binding mechanism will be explained in this section using the Feynman path-integral picture. The path-integral framework will be used in an intuitive rather than a formal way. Consider the current-current propagator

$$D_{\mu\nu}(x) = \langle \Omega | T J_\mu(x) J_\nu(0) | \Omega \rangle, \quad (2.1)$$

whose Fourier transform determines the e^+e^- annihilation cross section into hadrons. Assume that the currents $J_\mu(x)$ are built from quark fields as in the quark-parton model. Assume that the quarks interact through a single gauge field. (The restriction to one field is only for simplicity.) In the Feynman path-integral picture the propagator

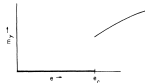


FIG. 2. Speculative plot of photon mass vs renormalized charge e if there is a first-order transition at e_c .

$D_{\mu\nu}(x)$ is given by a weighted average over all possible classical quark paths and all possible classical values of the gauge field. The currents $J_q(x)$ and $J_{\bar{q}}(0)$ are thought of as producing a quark pair at the origin which later annihilate at the point x . One has to sum over all paths joining the points 0 and x for each of the pair of quarks.

An example of paths for the quark and antiquark are shown in Fig. 3. The vacuum can also emit closed loops as illustrated in Fig. 4. All possible loops must be summed over too. It is also possible to have independent loops for the points 0 and x (Fig. 5), but this possibility will not be important here.

The weight associated with a given quark path or set of paths includes a factor of $\exp[i\int A_\mu(x) \times ds^\mu]$, where $A_\mu(x)$ is the gauge field. Here $\int \cdots ds^\mu$ is a line integral or a sum over line integrals for each of the quark-antiquark loops, including the loop joining the points 0 and x . The constant g is the coupling constant of the gauge theory. There are further weight factors independent of A . Finally, independently of the quark paths there is another weight factor, namely, the exponential of the free action for the gauge field. The combined weight factor is then averaged over all quark paths and all gauge fields $A_\mu(x)$ to give the current-current propagator.

In order that quarks exist as separate final-state particles it must be possible to have quark-antiquark loops with well-separated quark and antiquark lines, at least when x and 0 are far apart. This is illustrated in Fig. 6(a). If the quark and antiquark paths are unlikely to separate beyond a fixed size, say 10^{-13} cm [see Fig. 6(b)], then clearly no detector will see a quark or antiquark in isolation.

It is assumed in this discussion that vacuum loops are not important. If vacuum loops are important enough then space will be filled with a high density of vacuum-produced quark-antiquark pairs. In particular, there will be many quark-antiquark pairs inside a detector of macroscopic size. The question then is whether there can be an excess of quarks over antiquarks, or vice versa, in a region of macroscopic size. This is



FIG. 3. An example of quark (q) and antiquark ($\bar{q}$) paths connecting the points 0 and x .



FIG. 4. Example of current loop (as in Fig. 3) with an extra vacuum loop.

a more difficult question to answer and will not be discussed in this paper.

Note that x must be large: If x is small there is little likelihood of finding a large size loop. This may seem a bit peculiar: One expects quarks to appear in the final state of e^+e^- annihilation only at large virtual-photon momentum q if they appear at all, and large q means small x , not large x . The answer to this paradox has two parts. First, the important paths in the Feynman path integral bear no detailed relation to possible physical final states (the paths are paths of bare particles, not physical particles). Secondly, large x does not necessarily mean small q . In fact the study of whether well-separated quark-antiquark paths exist for large x is really a search for a quark-antiquark threshold in e^+e^- annihilation, which would contribute a term $\sim \exp(2m(\sqrt{s})^2)$ to the current-current propagator for large x , where m is the quark mass. (Here $\sim$ means up to a power of x^2 .) Such a term corresponds in momentum space to the singularity at the threshold $q^2 = (2m)^2$.

Suppose the gauge-field averaging is performed before the quark-paths averaging. Then one determines the average over all gauge fields of the weight factor $\exp[i\int g A_\mu(x) \times ds^\mu]$ weighted further with the exponential of the free gauge-field action. For an Abelian gauge theory this average can be computed explicitly: It is



FIG. 5. Example of separate quark loops for the points 0 and x . (Integration over the gauge field produces gauge propagators which connect these loops.)

$$\exp \left[-g^2 \oint ds^\mu \oint ds'^\mu D_{\mu\mu}(y-y') \right],$$

where $D_{\mu\mu}(y-y')$ is the free gauge-field propagator.

The quark binding mechanism can be seen by comparing the above expression for one space dimension and three space dimensions. In three space dimensions this calculation gives no binding (the binding occurs only with a modified gauge-field action: see Sec. IV), while there is binding in one space dimension. In three space dimensions $D_{\mu\mu}(y-y')$ behaves as $(y-y')^{-2}$. In consequence large values of $(y'-y)$ are negligible in the double line integral. Hence, the double integral is proportional to P , where P is the length of the loop. Unfortunately the integral is divergent at $y'=y$; a cutoff is needed for the integral to make sense. Since a cutoff will be introduced anyway in this paper, this is not a major concern. For simple loops, the perimeter P is roughly of order $(x^2)^{1/2}$ (ignoring the case that x is close to the light cone). Thus one has an exponential of the type one expects when free quarks are present.

In one space dimension, $D_{\mu\mu}(y-y')$ behaves as $\ln[(y-y')^2]$ for $y'-y$ large, and y' and y can freely range separately over the loop. In this case the double integral is proportional to P^2 . Now the gauge-field average behaves as $e^{-cA P^2}$, where c is a constant. In this case the contribution of large loops is heavily suppressed and there are no free quarks. [The case of nearby quark-antiquark pairs as in Fig. 6(b) is special—in this case large $y-y'$ is unimportant due to cancellation between the quark path and the nearby and oppositely directed antiquark path. In this case the double integral behaves as P , not P^2 , but in this case there are no isolated quarks.]

In the strongly coupled lattice gauge theory described in later sections, the gauge-field average of $\exp[i g^2 A_\mu(x) ds^\mu]$ behaves as $\exp(ic'A)$, where A is the *enclosed area* of the loop. This heavily suppresses large loops, such as in Fig. 6(a), where A is of order P^2 . One can think of one factor P as being roughly $(x^2)^{1/2}$, the other P as being analogous to a mass multiplying $(x^2)^{1/2}$. Since $P \rightarrow \infty$ as $x \rightarrow \infty$, the quark-antiquark threshold is at infinite mass.

In all these calculations one can have a large loop if there is a nearby vacuum loop (Fig. 7). In this case one always gets $e^{i c' P}$ behavior. For example, in the strong-coupling case the relevant enclosed area is the area between the two loops which is proportional to the perimeter P provided the separation of the two loops is fixed independently of P . This is in accord with Schwinger's picture. While an isolated well-separated loop

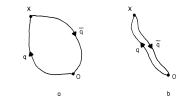


FIG. 6. (a) Loop with well-separated quark and antiquark. (b) Loop with small separation between quark and antiquark.

may be suppressed (due to P^2 or A dependence in the exponential) a loop closely shielded by a vacuum-polarization loop is always unsuppressed.

The binding mechanism proposed here is soft: The exponential damping is associated with large size loops having large areas. The behavior of small loops is irrelevant to the binding mechanism. Also for small loops both their area and perimeter are small and neither is of great importance in an exponential.

The mechanism discussed here works equally well for Dirac quarks or scalar quarks. This is in contrast to the Higgs mechanism which can wipe out the charge of scalar particles only.

III. LATTICE QUANTIZATION OF GAUGE FIELDS

A. Classical action on a lattice

In this section the gauge-field action (space-time integral of the Lagrangian) will be defined on a discrete lattice with spacing a in Euclidean space-time. The simplest way to proceed is to consider a continuum action, substitute finite-difference approximations for derivatives, and replace the space-time integral by a sum over the lattice sites. However, the result of this is an action which is not gauge-invariant for nonzero a . Because of the vagaries of renormalization this is likely to mean that the quantized theory still lacks

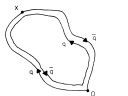


FIG. 7. Quark-antiquark loop with nearby vacuum loop.

gauge invariance in the limit $a \rightarrow 0$ (if such a limit exists). The alternative is to formulate gauge invariance for a lattice theory, and tinker with the action so that it is gauge invariant for any a . This alternative will be pursued here.

For convenience a charged Dirac field ψ coupled to a single gauge field A_μ will be discussed in detail. Generalizations to non-Abelian gauge groups will be noted later.

On the lattice the fields are $\bar{\psi}_n$, $\bar{\psi}_{n+1}$, and $A_{n,\mu}$, where n is a four-vector with integer components referring to points on a simple hypercubic lattice. A simple action on the lattice for the Dirac field (ignoring the gauge field for now) has the form

$$A_D = -a^4 \sum_n m_0 \bar{\psi}_n \psi_n + \frac{1}{2} a \sum_n \sum_\mu \bar{\psi}_n \gamma_\mu (\psi_{n+\hat{\mu}} - \psi_{n-\hat{\mu}}), \quad (3.1)$$

where m_0 is the bare mass; $\hat{\mu}$ is a unit vector along the axis μ ; $a \sum_n$ replaces the space-time integration of the continuum theory, and $(\psi_{n+\hat{\mu}} - \psi_{n-\hat{\mu}})/2a$ replaces ∇_μ . There is no over-all factor of i due to the Euclidean metric. A gauge transformation on the lattice can be defined as follows:

$$\bar{\psi}_n \rightarrow e^{i\gamma_4 \epsilon} \bar{\psi}_n, \quad (3.2)$$

$$\bar{\psi}_n \rightarrow e^{-i\gamma_4 \epsilon} \bar{\psi}_n, \quad (3.3)$$

$$A_{n,\mu} \rightarrow A_{n,\mu} + (\gamma_{n,\mu} - \gamma_{n-\hat{\mu},\mu})/a, \quad (3.4)$$

where g is the coupling constant and γ_μ is arbitrary. The terms $\bar{\psi}_n \psi_{n+\hat{\mu}}$ and $\bar{\psi}_{n+\hat{\mu}} \psi_n$ are not invariant to this transformation; the corresponding gauge-invariant expressions are $\bar{\psi}_n \psi_{n+\hat{\mu}} \exp(iagA_{n,\mu})$ and $\bar{\psi}_{n+\hat{\mu}} \psi_n \exp(-iagA_{n-\hat{\mu},\mu})$. Thus, a gauge-invariant form for A_D is

$$A_D = \frac{1}{2} a^4 \sum_n \sum_\mu (\bar{\psi}_n \gamma_\mu \psi_{n+\hat{\mu}} e^{iagA_{n,\mu}} - \bar{\psi}_{n+\hat{\mu}} \gamma_\mu \psi_n e^{-iagA_{n-\hat{\mu},\mu}}) - a^4 \sum_n m_0 \bar{\psi}_n \psi_n. \quad (3.5)$$

It is convenient to define

$$A = -c \sum_n \bar{\psi}_n \psi_n + K \sum_n \sum_\mu (\bar{\psi}_n \gamma_\mu \psi_{n+\hat{\mu}} e^{i\epsilon A_{n,\mu}} - \bar{\psi}_{n+\hat{\mu}} \gamma_\mu \psi_n e^{-i\epsilon A_{n-\hat{\mu},\mu}}) + \frac{1}{2K\epsilon} \sum_n \sum_\mu e^{i\epsilon A_{n,\mu}}, \quad (3.12)$$

with $c = m_0/a^4$, $K = a^3/2$. This action reduces to the usual continuum action for $a \rightarrow 0$; for finite a it is gauge invariant and periodic in the gauge field. Note, however, that the continuum limit is a classical limit in which the lattice variables $\bar{\psi}_n$, $\bar{\psi}_n$, and $A_{n,\mu}$ approach continuum functions $\bar{\psi}(x)$, $\bar{\psi}(x)$, and $A_\mu(x)$ with $x = na$. The continuum limit of the quantized theory is much harder to discuss owing to renormalization problems.

$$B_{n,\mu} = agA_{n,\mu}. \quad (3.6)$$

In the action A_D , the field $B_{n,\mu}$ acts like an angular variable: A_μ is periodic in $B_{n,\mu}$ with period 2π . The free gauge-field action will be defined to preserve this property. This does not mean that $A_{n,\mu}$ is an angular variable in the continuum limit. Owing to the relation (3.6), $A_{n,\mu}$ becomes an infinitesimal angular variable $A_\mu(x)$ for $a \rightarrow 0$; such a variable has the range $-\infty < A_\mu(x) < \infty$ without any periodicity.

A gauge-invariant lattice approximation for

$$\nabla_\mu A_\nu - \nabla_\nu A_\mu$$

$$F_{n,\mu\nu} = (A_{n,\mu} \hat{\nu}_\nu - A_{n,\nu} \hat{\mu}_\mu - A_{n+\hat{\mu},\nu} + A_{n,\mu+\hat{\nu}})/a. \quad (3.7)$$

It is convenient to define a rescaled form of $F_{n,\mu\nu}$:

$$f_{n,\mu\nu} = a^2 g F_{n,\mu\nu} = B_{n,\mu} \hat{\nu}_\nu - B_{n,\nu} \hat{\mu}_\mu - B_{n+\hat{\mu},\nu} + B_{n,\mu+\hat{\nu}}. \quad (3.8)$$

A simple lattice action for the gauge field which preserves periodicity is

$$A_g = \frac{1}{2a^2} \sum_n \sum_{\mu < \nu} e^{i f_{n,\mu\nu}}. \quad (3.9)$$

In the continuum limit, $f_{n,\mu\nu} \rightarrow 0$ due to the factor a^2 in the definition of $f_{n,\mu\nu}$. Thus for small a , one can write

$$A_g = \frac{1}{2a^2} \sum_n \sum_{\mu < \nu} (1 + i f_{n,\mu\nu} - \frac{1}{2} f_{n,\mu\nu}^2 + \dots). \quad (3.10)$$

The constant term is irrelevant. The linear term in $f_{n,\mu\nu}$ is 0 because $f_{n,\mu\nu}$ is odd in the indices μ and ν . The quadratic term gives

$$A_g = -\frac{1}{4} a^4 \sum_n \sum_{\mu < \nu} F_{n,\mu\nu}^2, \quad (3.11)$$

which is the conventional gauge-field action in a lattice approximation. The terms involving $f_{n,\mu\nu}^3$, $f_{n,\mu\nu}^4$, etc., all vanish for $a \rightarrow 0$ even after removing a factor a^4 to convert $\sum_n$ into an integral.

The full action may now be written

8. Quantization

The problem of principal interest here is the quantization of the gauge field. Therefore, the gauge field will be quantized by itself to start with. Later the quantization of ψ will be discussed. At the end of this section the generalizations to non-Abelian gauge theories will also be described. The quantization of the lattice gauge theory will

be carried out in two steps. The first step will be to define a lattice version of Euclidean vacuum expectation values, starting from a lattice version of the Feynman path integral. The second step will be to define a quantum theory on the lattice, which will allow the introduction of a real time variable and the definition of particle states and scattering amplitudes. In both cases the lattice provides an ultraviolet cutoff and there is no Lorentz invariance. Lorentz invariance can only be achieved in the limit a (lattice spacing) $\rightarrow 0$, if at all, and in practice this is a difficult limit to evaluate.

As discussed in Sec. I, one would like to calculate the gauge-field average of $\exp[ig\bar{\psi}\psi(x)]$ weighted with the gauge-field action. On a lattice the line integral becomes a sum over a closed path P on the lattice (see, e.g., Fig. 8). The sum has the form: $i\sum_p (x)B_{x,p}$, where a particular $B_{x,p}$ is present in the sum if the path connects the sites x and $x+\hat{\mu}$ ($-\hat{\mu}$) appears if the path goes from x to $x+\hat{\mu}$ to x .

On the lattice, an average over all gauge fields involves integrating over all values of the $B_{x\mu}$ for all x and μ . Normally one would have integrals over an infinite range: $-\infty < B_{x\mu} < \infty$, but because of the periodicity in $B_{x\mu}$ there is no point to integrating over more than a single period. Thus the lattice version of the gauge-field average is

$$\langle RP \rangle = Z^{-1} \left(\prod_x \prod_{\mu} \int_{-\pi}^{\pi} dB_{x\mu} \right) \times \exp \left(i \sum_p (x) B_{x,p} + \frac{1}{2g^2} \sum_{x,\mu} e^{i f_{x\mu}} \right), \quad (3.13)$$

with

$$Z = \left(\prod_x \prod_{\mu} \int_{-\pi}^{\pi} dB_{x\mu} \right) \exp \left(\frac{1}{2g^2} \sum_{x,\mu} e^{i f_{x\mu}} \right). \quad (3.14)$$

Note that no gauge-fixing term has been added to the action. The finite range of $B_{x\mu}$ makes a gauge-fixing term unnecessary. In continuum gauge theories where $A_\mu(x)$ has an infinite range $[-\infty < A_\mu(x) < \infty]$ a gauge-fixing term is essential to have a convergent functional integral. The reason for this is that the volume in path-integral space generated by all possible gauge transformations is infinite; the gauge-fixing term provides a convergence factor in this volume.¹⁰ In the lattice theory the total volume of integration is finite if the lattice itself is of finite extent; no convergence factor is required. For a lattice of infinite extent there are divergences due to the infinite number of integrations (in other words, the in-

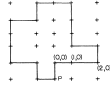


FIG. 8. Example of a lattice path P .

finite lattice volume), but these divergences are normally removed by the division by Z (this division is equivalent to removing all vacuum loops in perturbation theory).

One can define more conventional vacuum expectation values in a similar manner. For instance, one can define a propagator. In the absence of a gauge-fixing term it is awkward to define a propagator for the gauge field $B_{x\mu}$ itself; instead one can define a gauge-invariant propagator as

$$D_{x\mu, y, \nu} = Z^{-1} \left(\prod_x \prod_{\mu} \int_{-\pi}^{\pi} dB_{x\mu} \right) \times \exp \left(i f_{x\mu} - i f_{y, \nu} + \frac{1}{2g^2} \sum_{x,\mu} e^{i f_{x\mu}} \right). \quad (3.15)$$

This is a propagator for the operator $e^{i f_{x\mu}}$; it is defined only for the lattice points x of a Euclidean space-time lattice. If the lattice spacing is a , this means the propagator is defined only for imaginary times of the form $i n_0 a$, where n_0 is an integer.

A theory defined only for discrete imaginary values of the time leaves much to be desired. Fortunately, one can generalize the theory to define a Hamiltonian for a quantized theory. The particle eigenstates and scattering amplitudes of the theory can then be obtained, in principle, by diagonalizing the Hamiltonian. The Hamiltonian will be defined using the transfer-matrix formalism. Only a brief discussion of the transfer-matrix approach will be given here. For a review of the ideas see Wilson and Kogut.⁸ A detailed discussion including approximate calculation of single-particle energies and scattering amplitudes in a simple scalar field theory is given by Suri.⁷

Consider the expression for Z . Introduce finite bounds on the lattice coordinate n_μ , say

$$-N < n_\mu < N. \quad (3.16)$$

Introduce periodic boundary conditions (see below). Then one can write Z as the trace of a matrix V , more precisely

$$Z = \text{Tr } V^{2N+1}. \quad (3.17)$$

This formula is made possible by the fact that each term in the action A involves no more than two adjacent values of n_i .

To set up the matrix V , one must first understand the space on which it acts. The space used here is the space of all functions $\psi(B_{\tau i})$ (periodic in each $B_{\tau i}$ with period 2π), where the index i runs from 1 to 3 only, and the lattice variable $\tilde{n}$ has only three components (n_1, n_2, n_3). The matrix V will be defined as a function of two sets of arguments, say $B_{\tau i}^+$ and $B_{\tau i}^-$, then two sets of arguments referring to the space-time fields $B_{\alpha i}$ for two adjacent values of n_i . Matrix multiplication of two V 's involves integrations over a set of variables $\{B_{\tau i}^{\pm}\}$. Define V as

$$V = \prod_{\tau} \int_{-\pi}^{\pi} dB_{\tau i} e^{iA}. \quad (3.18)$$

The quantity U as written out below looks complicated, but all it is is that part of the action A referring to a given nearest-neighbor pair of values for n_i ; terms in A referring to a single value of n_i are divided equally between the matrices connecting n_i to $n_i + 1$ and n_i to $n_i - 1$. The result is

$$U = \frac{1}{4g^2} \sum_{\tilde{n}} \sum_i \sum_j (e^{i/2 B_{1j}} + e^{i/2 B_{2j}}) + \frac{1}{4g^2} \sum_i \sum_j (e^{i/2 B_{3i}} + e^{i/2 B_{3j}}), \quad (3.19)$$

where

$$f_{\tau i j}^{\pm} = B_{\tau i}^{\pm} + B_{\tau+1, i}^{\pm} - B_{\tau+1, j}^{\pm} - B_{\tau j}^{\pm} \quad (3.20)$$

and

$$f_{\tau i j}^{\pm} = B_{\tau i}^{\pm} + B_{\tau+1, i}^{\pm} - B_{\tau+1, j}^{\pm} - B_{\tau j}^{\pm}. \quad (3.21)$$

With the definition of V given here, the trace $\text{Tr } V^{2N+1}$ is easily seen to reproduce all the integrations involved in the equation for Z , and the sum of the $2N+1$ exponents U reproduces the action A except for some additional terms coupling the boundary $n_i = N$ to the boundary $n_i = -N$ to achieve a periodic structure.

Note that the 0 components of $B_{\alpha i}$ have received special treatment. This is because there are no terms in A involving $B_{\alpha i}$ for more than one value of n_i ; this makes it possible to include integrations over $B_{\alpha i}$ in the definition of V rather than in the definition of matrix multiplication.

The matrix V is used to define the quantized theory. Briefly, this is accomplished as follows. V is a Hermitian matrix, i.e.,

$$V(B, B')^* = V(B', B). \quad (3.22)$$

[This result can be verified by close examination

of Eqs. (3.18)–(3.21). One must remember that the variables $B_{\tau i}^{\pm}$ are integrated out in the definition of V . This means that in forming the complex conjugate of V one can also make the change of variable $B_{\tau i}^{\pm} \rightarrow -B_{\tau i}^{\pm}$. Hence V has a complete orthogonal set of eigenstates ψ and eigenvalues λ . The Hamiltonian H is now defined as follows: The eigenstates of H are the eigenstates of V ; the eigenvalues of H are given by

$$E = -a^{-1} \ln \lambda, \quad (3.23)$$

where λ is the corresponding eigenvalue of V . The reason for the factor a^{-1} will be evident shortly. The reason for using the logarithm is so that the energies of multiparticle scattering states will be the sum of single-particle energies (see *surf*).³

A problem arises with this definition if V has any negative eigenvalues λ . If this were to happen, H would have complex eigenvalues. This did not happen in the case studied by *surf*³; whether it happens here the author does not know; this question must be studied further. Even if V has negative eigenvalues, they may be irrelevant in the limit $a \rightarrow 0$ if such a limit exists.

The definition (3.23) means that

$$V = e^{-aH}. \quad (3.24)$$

This means V is the operator which propagates a state through an imaginary time ia . It is a consequence of this that the propagator $D_{\alpha i, \tau, \sigma}$ is a vacuum expectation value for imaginary time ia in the theory with Hamiltonian H . For proof of this statement see Refs. 5 and 6.

The lattice quantization procedure can be extended to non-Abelian gauge theories. This is done as follows. In place of the single variable $B_{\alpha i}$, one has a set of variables $B_{\alpha i}^{\mu}$, where α is an internal index. For each α and μ , $B_{\alpha i}^{\mu}$ is to parametrize an element $b_{\alpha i}$ of the gauge group. In place of $\exp(iB_{\alpha i})$ one substitutes $U(b_{\alpha i})$, where U is the unitary matrix representing $b_{\alpha i}$ in the quark representation. The product $\tilde{\psi}_i \psi_i \exp(iB_{\alpha i})$ is replaced by $\tilde{\psi}_i \gamma_{\alpha} U(b_{\alpha i}) \psi_{i+\hat{\alpha}}$. A gauge transformation is defined by a set of group transformations y_i ; under these transformations

$$\psi_i \rightarrow U(y_i) \psi_i, \quad (3.25)$$

$$\tilde{\psi}_i \rightarrow \tilde{\psi}_i U^{\dagger}(y_i), \quad (3.26)$$

$$b_{\alpha i} \rightarrow y_i b_{\alpha i} y_{i+\hat{\alpha}}^{-1}, \quad (3.27)$$

$$U(b_{\alpha i}) \rightarrow U(y_i) U(b_{\alpha i}) U^{\dagger}(y_{i+\hat{\alpha}}), \quad (3.28)$$

where $y_i b_{\alpha i} y_{i+\hat{\alpha}}^{-1}$ is computed according to the multiplication law of the group. A simple gauge-invariant action for the gauge field is

$$A_g = \frac{1}{2g^2} \sum_{a \in \mathfrak{g}} \text{Tr } U(g) b_{a+1,1} b_{a+1,2} b_{a+1,3} \cdots b_{a+1,n} b_{a+1,n}^{-1} \quad (3.29)$$

where the unitary transformation U_A is taken from the adjoint representation of the group. (Any representation will do as well.) In the classical continuum limit, b_a becomes an infinitesimal group transformation $[B_a^\alpha = g \alpha A_a^\alpha(na)]$ with A_a^α fixed as $g \rightarrow 0$; in this limit one can show, with some effort, that A_g reduces to the standard continuum Yang-Mills action. When the non-Abelian theory is quantized, the integrations over b_{a+1} are group integrations over all elements of the group. These are compact integrations (for compact groups) and no gauge-fixing term is required.

Since quark binding can be illustrated using Abelian theory, the non-Abelian theory will not be studied further.

Finally, the quantization of the Dirac field will be discussed. It is convenient initially to quantize the Dirac field in an analogous manner to the gauge field. The only problem that occurs is to define "integration" for a Fermi field. This can be done.¹¹

The property of the integral that is crucial for quantization is translational invariance in the integration variable. For example, when quantizing a scalar field ϕ on a lattice the field averaging involves the integral $\int_{-\infty}^{\infty} d\phi_a$ which has the translational invariance

$$\int_{-\infty}^{\infty} d\phi_a f(\phi_a + J_a) = \int_{-\infty}^{\infty} f(\phi_a) d\phi_a \quad (3.30)$$

for any integrable function f and any constant J_a . It is this translational invariance that makes the Feynman path integral provide a realization of Schwinger's action principle (see, e.g., Ref. 12 for further discussion). Analogously one needs to define an integration over Fermi fields with the same translational invariance. Stated abstractly, one wants to define a bracket operation $\langle \cdots \rangle$ defined on functions of purely anticommuting Fermi fields ψ_a with the property

$$\langle f(\psi_a + \eta_a, \bar{\psi}_a + \bar{\eta}_a) \rangle = \langle f(\psi_a, \bar{\psi}_a) \rangle \quad (3.31)$$

where η_a and $\bar{\eta}_a$ are anticommuting c-numbers (these have been introduced by Schwinger). The bracket operation should produce a number for every function f ; it should also be a linear operation. Thus for a finite lattice it is sufficient to specify the bracket $\langle \cdots \rangle$ for all monomials in the ψ_a and $\bar{\psi}_a$. Because of the anticommutation rules, ψ_a^2 and $\bar{\psi}_a^2$ vanish (more correctly, ψ_{a+1}^2 and $\bar{\psi}_{a+1}^2$ vanish where ψ_{a+1} and $\bar{\psi}_{a+1}$ are any component of ψ_a and $\bar{\psi}_a$), therefore, there are only a finite number of possible monomials. It is now easy to see that

the bracket $\langle \cdots \rangle$ must vanish for all products of ψ 's and $\bar{\psi}$'s, except the product containing all possible ψ 's and $\bar{\psi}$'s. For example, suppose there are two lattice sites 0 and 1, and ψ_a and $\bar{\psi}_a$ are single component fields. Then the brackets must be

$$\begin{aligned} \langle 1 \rangle &= 0, \\ \langle \psi_0 \rangle &= \langle \bar{\psi}_0 \rangle = \langle \bar{\psi}_1 \rangle = 0, \\ \langle \psi_0 \bar{\psi}_0 \rangle &= \langle \bar{\psi}_0 \psi_1 \rangle = \cdots = 0, \\ \langle \bar{\psi}_0 \psi_0 \bar{\psi}_0 \rangle &= \cdots = 0, \\ \langle \bar{\psi}_0 \bar{\psi}_0 \bar{\psi}_0 \psi_0 \rangle &= 1, \end{aligned} \quad (3.32)$$

where 1 is a constant which was chosen arbitrarily. This definition of the bracket operation satisfies translational invariance: for example,

$$\langle (\bar{\psi}_0 + \bar{\eta}_0)(\psi_0 + \eta_0)(\bar{\psi}_1 + \bar{\eta}_1)(\psi_1 + \eta_1) \rangle = \langle \bar{\psi}_0 \bar{\psi}_1 \psi_1 \psi_0 \rangle = 1$$

because the terms multiplying the η 's are all 0. Note also that the anticommutation rules mean that, for example,

$$\langle \bar{\psi}_0 \bar{\psi}_0 \bar{\psi}_1 \psi_1 \rangle = -1. \quad (3.33)$$

(In analogy to the scalar case, one requires $\bar{\psi}_0 \bar{\psi}_0 = -\bar{\psi}_0 \psi_0 \psi_0$, not $\bar{\psi}_0 \bar{\psi}_0 = 1 = \bar{\psi}_0 \psi_0$.)

One can now define the Feynman path integral on a lattice for the complete gauge theory including the Dirac fields. For example, the current-current propagator on the lattice is

$$D_{ab}(x) = Z_{\text{tot}}^{-1} \left(\prod_a \prod_b \int_{-\infty}^{\infty} dB_{ab} \right) \langle \bar{\psi}_a \gamma_\mu \psi_a \bar{\psi}_b \gamma_\nu \psi_b e^{iA} \rangle \quad (3.34)$$

where A is the full action of Eq. (3.12) and

$$Z_{\text{tot}} = \left(\prod_a \prod_b \int_{-\infty}^{\infty} dB_{ab} \right) \langle e^{iA} \rangle. \quad (3.35)$$

This formulation of the path integral is different from the formulation discussed in Sec. II. However, one can easily derive a lattice form of the path integrals of Sec. II from the present expression. The procedure is to expand Eq. (3.34) in powers of K , where K is the coefficient of the nearest-neighbor coupling terms $\bar{\psi}_a \gamma_\mu \psi_{a+\hat{\mu}} e^{iA_{a+\hat{\mu}}}$ etc. This nearest-neighbor coupling term can be represented diagrammatically by a line from the site a to the site $a+\hat{\mu}$. The expansion is best described by studying an example of a term from the expansion of the numerator of Eq. (3.34), which will now be discussed. An example of a term in the expansion is represented diagrammatically in Fig. 9. The expression for this term is

$$\left(\prod_{\alpha} \prod_{\gamma} \int_{-\pi}^{\pi} dB_{\alpha\gamma} \right) K^4 (\bar{\psi}_1 \gamma_{\mu} \psi_1 \bar{\psi}_2 \gamma_{\nu} \psi_2 \bar{\psi}_3 \gamma_{\rho} \psi_3 \bar{\psi}_4 \gamma_{\sigma} \psi_4 e^{i R_{\alpha\sigma} \gamma_{\sigma} \bar{\psi}_1 \psi_1} e^{i R_{\beta\gamma} \gamma_{\gamma} \bar{\psi}_2 \psi_2} e^{-i R_{\alpha\gamma} \gamma_{\gamma} \bar{\psi}_3 \psi_3} e^{-i R_{\beta\sigma} \gamma_{\sigma} \bar{\psi}_4 \psi_4}) = D \quad (3.36)$$

(D for diagram), where the four lattice sites involved are $(n_0, n_1) = (0, 0)$, $(1, 0)$, $(0, 1)$, and $(1, 1)$; the values of n_2 and n_3 are constant and have been suppressed in the notation of Eq. (3.36). The action A_2 omits the K term, and is

$$A_2 = -c \sum_{\alpha} \bar{\psi}_{\alpha} \psi_{\alpha} + \frac{1}{2g^2} \sum_{\alpha} \sum_{\mu} e^{i f_{\alpha\mu}}. \quad (3.37)$$

The calculation of D has two parts, one being the integration over all gauge fields $B_{\alpha\mu}$, the other being the calculation of the $\bar{\psi}, \psi$ bracket. These are independent calculations, i.e., D factors into $D_{\psi} D_B$. The quantity D_B is

$$D_B = \left(\prod_{\alpha} \prod_{\mu} \int_{-\pi}^{\pi} dB_{\alpha\mu} \right) \exp \left[i (B_{21,1} + B_{12,1} - B_{21,2} - B_{12,2}) + \frac{1}{2g^2} \sum_{\alpha} e^{i f_{\alpha\mu}} \right]. \quad (3.38)$$

This is an example of a gauge-field average of the exponential of a line integral over a closed loop, the loop being the loop of Fig. 9. The ψ bracket calculation can be factorized further into separate bracket calculations for each lattice site, since A_2 contains no terms involving ψ or $\bar{\psi}$ and coupling different lattice sites. Consider only the four lattice sites on the loop, for simplicity. By moving the ψ 's around some (using the anticommuting rule) the bracket becomes

$$(\bar{\psi}_{20} \gamma_{\sigma} \psi_{12} e^{-i R_{12} \gamma_{\sigma} \bar{\psi}_1 \psi_1} e^{-i R_{11} \gamma_{\sigma} \bar{\psi}_1 \psi_1} e^{-i R_{21} \gamma_{\sigma} \bar{\psi}_2 \psi_2} e^{-i R_{22} \gamma_{\sigma} \bar{\psi}_2 \psi_2} e^{-i R_{12} \gamma_{\sigma} \bar{\psi}_3 \psi_3} e^{-i R_{11} \gamma_{\sigma} \bar{\psi}_3 \psi_3} e^{-i R_{21} \gamma_{\sigma} \bar{\psi}_4 \psi_4} e^{-i R_{22} \gamma_{\sigma} \bar{\psi}_4 \psi_4}).$$

To make a product of all possible ψ 's and $\bar{\psi}$'s means one must have products of all possible $\psi_{\alpha\mu}$'s and $\bar{\psi}_{\alpha\mu}$'s, all possible $\psi_{\alpha\mu}$'s and $\bar{\psi}_{\alpha\mu}$'s, etc. In summary the complete bracket may be written as a product of four separate brackets. Define

$$D_1^{\pm} = (\bar{\psi}_{10} e^{-i R_{10} \gamma_{\sigma} \bar{\psi}_1 \psi_1}), \quad (3.39)$$

$$D_{12}^{\pm} = (\bar{\psi}_{11} e^{-i R_{11} \gamma_{\sigma} \bar{\psi}_1 \psi_1} \bar{\psi}_{12} \gamma_{\sigma} \psi_{12}). \quad (3.40)$$

Both $D_1^{\pm}$ and $D_{12}^{\pm}$ are matrices in spin space due to the spinor indices implied for ψ_{10} , $\bar{\psi}_{10}$, ψ_{12} , and $\bar{\psi}_{11}$. The full bracket is simply

$$D_B = -K^4 \text{Tr} (D_1^{\pm} \gamma_1 D_{12}^{\pm} \gamma_2 D_1^{\pm} \gamma_1 D_{12}^{\pm} \gamma_2). \quad (3.41)$$

The matrices $D_1^{\pm}$ and $D_{12}^{\pm}$ are easily determined. For example, $D_1^{\pm}$ explicitly is

$$D_{10,1}^{\pm} = (\bar{\psi}_{10\alpha} \exp(-c \sum_{\beta} \bar{\psi}_{10\beta} \psi_{10\beta}) \bar{\psi}_{10\beta}). \quad (3.42)$$

The exponential can be expanded in powers of c ; assuming the spinors have four components only the c^2 term can produce a product of all four $\bar{\psi}_{10}$'s times all four $\bar{\psi}_{10}$'s; the result is

$$D_{10,1}^{\pm} = -\delta_{\alpha\beta} c^2 = -c^2 \delta^2 \quad (3.43)$$

(the minus sign comes from the convention that the bracket is positive when $\bar{\psi}_{10}$ appears to the left of ψ_{10}). A similar calculation gives

$$D_{12}^{\pm} = c^2 \gamma_2. \quad (3.44)$$

The results of this example are easily generalized. A term of general order K^4 is nonzero only

if the nearest-neighbor couplings combine to form closed loops (the lattice site at the endpoint of an open line would have an extra ψ or $\bar{\psi}$, so the bracket at it would give 0). The bracket calculation for a closed loop gives a trace involving K times a γ matrix for each line in the loop and D 's for each lattice site in the loop (except the points n and 0 where there are currents). The average over gauge fields involves an exponential of a sum of $B_{\alpha\mu}$'s around each loop. There can be any number of loops.

IV. STRONG-COUPLING APPROXIMATION

The gauge field average $\langle P \rangle$ which determines whether quarks are bound was defined on a lattice in Sec. III (Eq. 3.13). There are two limits in which this average can be calculated. The most interesting limit is the strong-coupling limit $g \rightarrow \infty$. This is the limit which exhibits quark binding. A strong-coupling expansion will be derived in this section.



FIG. 9. Elementary square on the lattice.

The strong-coupling expansion will be the basis for a reformulation of the gauge-field theory as a string model. This will also be explained in this section.

Consider specifically the numerator of Eq. (3.13), to be denoted $I_g(P)$:

$$I_g(P) = \left(\prod_{\alpha} \prod_{\beta} \int_{-\pi}^{\pi} dB_{\alpha\beta} \right) \times \exp \left[i \sum_{\alpha} \langle \alpha | B_{\alpha\beta} + \frac{1}{2g^2} \sum_{\alpha\beta\gamma} e^{i f_{\alpha\beta\gamma}} \right]. \quad (4.1)$$

$$I_g^{(1)}(P) = \frac{1}{2g^2} \left(\prod_{\alpha} \prod_{\beta} \int_{-\pi}^{\pi} dB_{\alpha\beta} \right) \sum_{\alpha\beta\gamma} \exp \left[i \sum_{\alpha} \langle \alpha | B_{\alpha\beta} + \theta_{1+\alpha} \right]. \quad (4.3)$$

The quantity $f_{\alpha\beta\gamma}$ is itself a line integral of the gauge field; it is the line integral around a square originating at the lattice site I of size a (unit square). The integral for $I_g^{(1)}(P)$ will vanish unless it is possible to find a unit square such that $f_{\alpha\beta\gamma}$ cancels completely the line integral $\sum_{\alpha} \langle \alpha | B_{\alpha\beta}$. This is possible only if the path P is itself a unit square. Otherwise the first-order term vanishes and one must study the terms of order g^{-4} or higher.

The term of order g^{-4} has the form

$$I_g^{(4)}(P) = \frac{1}{k!} \left(\frac{1}{2g^2} \right)^4 \left(\prod_{\alpha} \prod_{\beta} \int_{-\pi}^{\pi} dB_{\alpha\beta} \right) \sum_{i_1 i_2 i_3 i_4} \cdots \sum_{i_k i_{k+1} i_k} \exp \left[i \sum_{\alpha} \langle \alpha | B_{\alpha\beta} + \theta_{1+i_1 i_2 i_3} + \cdots + \theta_{i_k i_{k+1} i_k} \right]. \quad (4.4)$$

The only nonzero terms in this sum are those for which

$$\sum_{\alpha} \langle \alpha | B_{\alpha\beta} + f_{i_1 i_2 i_3} + \cdots + f_{i_k i_{k+1} i_k} = 0. \quad (4.5)$$

[See Eq. (3.8) for the definition of $f_{i_1 i_2 i_3}$ in terms of $B_{\alpha\beta}$.] This equation can be understood geometrically. Each $f_{i_1 i_2 i_3}$ corresponds to a square of size a on the lattice. For this sum to vanish the set of squares defined by $f_{i_1 i_2 i_3}, \dots, f_{i_k i_{k+1} i_k}$ must combine to make a surface with boundary P . (To be precise, each $f_{i_1 i_2 i_3}$ corresponds to a line integral around a square, and when these squares are joined to make a surface the line integrals must cancel along all internal lines of the surface. The line integrals along the boundary P of the surface must run in the opposite direction to the original path P .) See Fig. 10.

For a given path P the lowest nonzero order in $I_g(P)$ is determined by the minimal area A enclosed by P , the area A being the area of any surface built of unit squares on the lattice with boundary P . Then $I_g(P) \sim (g^{-2})^{-A/a^2}$, apart from a numerical factor.

This is the result promised in Sec. II: The gauge-field average $I_g(P)$ behaves as $\exp[-A(\ln g^2)/a^2]$, i.e., exponentially in the area enclosed by P . Hence, according to the arguments of Sec. II, quark paths will not separate

Expanding in powers of $1/g^2$, the zeroth-order term is

$$I_g^{(0)}(P) = \left(\prod_{\alpha} \prod_{\beta} \int_{-\pi}^{\pi} dB_{\alpha\beta} \right) \exp \left[i \sum_{\alpha} \langle \alpha | B_{\alpha\beta} \right]. \quad (4.2)$$

This term vanishes, since for any $B_{\alpha\beta}$ which appears in $\sum_{\alpha} \langle \alpha | B_{\alpha\beta}$, there is an integral $\int_{-\pi}^{\pi} dB_{\alpha\beta} \times \exp(i B_{\alpha\beta})$ which is zero. Thus one must seek higher-order terms in g^{-2} which cancel the $B_{\alpha\beta}$ in the line integral. The first-order term is

macroscopically, and there will be no quarks among the final-state particles.

Consider higher-order terms in the expansion of $I_g(P)$ for given P . There are many such terms because there are many surfaces with boundary P . In particular, there are many ways to combine subsets of f 's to add to zero so such subsets can be added to any minimal sum of f 's which forms a surface with boundary P . The simplest example of a set of f 's which add to zero are the set of f 's corresponding to the six faces of a unit cube. Written out, this gives

$$0 = f_{\alpha\beta\gamma} - f_{\alpha\gamma\beta} + f_{\beta\alpha\gamma} - f_{\beta\gamma\alpha} + f_{\gamma\alpha\beta} - f_{\gamma\beta\alpha}, \quad (4.6)$$

which is easily checked using Eq. (3.8). [Equation

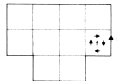


FIG. 10. Filling of enclosed area of path P by elementary squares.

(4.6) is the lattice analog of the equation $\epsilon^{2+1} \epsilon \Phi_{\mu\nu}(x) = 0$.

Let $A(P)$ be the minimal area as defined above enclosed by P . Since one can place a unit cube anywhere on this minimal area, it means that there are roughly $A(P)/a^3$ more terms of order $g^{-2} (g^{-2})^{-A(P)/a^3}$ in the expansion of $I_g(P)$ than there are terms of order $(g^{-2})^{-A(P)/a^3}$. [One can place unit cubes anywhere in space, not just on the minimal surface; but when one divides $I_g(P)$ by Z all disconnected terms cancel, as usual.] This suggests that the $1/g^2$ expansion is not very useful in the limit $A(P) \rightarrow \infty$, which is the limit of interest for quark binding. However, experience with related problems suggests that $I_g(P)$ is not the appropriate quantity to expand; instead one should try writing

$$I(P) \sim Z^{-1} I_g(P) \\ \sim (g^{-2})^{-A(P)} e^{-c(P, a^2)}, \quad (4.7)$$

and expand $c(P, g^2)$ in powers of g^{-2} instead. One would expect $c(P, g^2)$ to be dominated by a term proportional to $A(P)$, say

$$c(P, g^2) \sim A(P) f(g^{-2}) + O(P) \quad (4.8)$$

(where P is the length of the path P). The crucial question is the nature of the series for $f(g^{-2})$. Past experience with similar types of expansions (namely, the high-temperature expansions of statistical mechanics: see, e.g., Ref. 13) suggests that $f(g^{-2})$ will have a convergent expansion at least for g^{-2} less than a critical value g_c^{-2} . However, no calculations have been done in the gauge-field theory for $f(g^{-2})$ as yet.

Consider the complete expansion of $I(P)$. Each nonzero term in the expansion corresponds to a surface with perimeter P . The complete expansion corresponds to a sum over all possible surfaces with given perimeter P . "All possible" surfaces include surfaces which intersect themselves (to take into account terms where a given $f_{\mu\nu}$ appears several times in the sum $f_{1\mu\nu} + \dots + f_{1\mu\nu} + \dots$). There is a weight factor for each surface, aside from the power of g^{-2} determined by the area of the surface. For a simple surface, the weight factor is 1; the weight is more complicated for self-intersecting surfaces.

Thus, the strong-coupling expansion for the current-current propagator has the same general structure as in string models of hadrons. One is actually dealing here with a double expansion. An expansion in the coefficient K (appearing in the Dirac field action) was needed to define quark loops on the lattice; the sum of the K expansion is a sum over all possible quark loops. The g^{-2} expansion is needed to define surfaces filling in

the quark loops. The sum of the g^{-2} expansion is a sum over all such surfaces. This is precisely the structure appearing in string models: combined sums over quark loops and interpolating surfaces. However, the loops and surfaces of the gauge-field theory are defined on a lattice whereas the loops and surfaces of the string models are defined on a continuum. It may not be easy to derive quantitative relations between the two types of surfaces.

V. WEAK-COUPLED APPROXIMATION

The weak-coupling approximation will be discussed briefly, leaving many questions open. Only the pure gauge field will be discussed. Consider again the expression

$$I_g(P) = \left(\prod_{\mu} \prod_{\nu} \int_{-\pi}^{\pi} dB_{\mu\nu} \right) \\ \times \exp \left[\sum_{\mu} \sum_{\nu} \left(\frac{1}{2g^2} \sum_{\mu\nu} B_{\mu\nu} + \frac{1}{2g^2} \sum_{\mu\nu} B_{\mu\nu}^2 \right) \right]. \quad (5.1)$$

Suppose the integration variables were $f_{\mu\nu}$ rather than $B_{\mu\nu}$. For small g , only small values of $f_{\mu\nu}$ would be important in the integral, in order that $\text{Re} e^{if_{\mu\nu}}$ be near its maximum value 1. One would then expand:

$$\frac{1}{2g^2} \sum_{\mu\nu} e^{if_{\mu\nu}} \approx \frac{1}{2g^2} \sum_{\mu\nu} (1 - \frac{1}{2} f_{\mu\nu}^2). \quad (5.2)$$

With this approximation one could extend the limits of integration on $f_{\mu\nu}$ from $-\pi$ to $+\infty$, with negligible error; one would then have a set of Gaussian integrals to evaluate.

In practice the integration variables are the $B_{\mu\nu}$, not the $f_{\mu\nu}$. However, one can make a change of variable from the $B_{\mu\nu}$ to the $f_{\mu\nu}$. It is not possible to eliminate all the $B_{\mu\nu}$ by this transformation, and not all the variables $f_{\mu\nu}$ are independent. Nevertheless, the transformation is sufficient to make $I_g(P)$ calculable for small g^2 .

To make the change of variables precise, consider a system of finite size ($1 \leq \mu \leq N$) with periodic boundary conditions. Then one can change variables from the $B_{\mu\nu}$ to a subset of the $f_{\mu\nu}$, plus some gauge transformation variables ϕ_μ , plus four extra variables ξ_μ , as follows:

(i) For $\mu_0 \neq N, \mu_1, \mu_2, \mu_3$ arbitrary, $B_{\mu_0 \mu_1}$ is replaced by $f_{\mu_0 \mu_1}$. For $B_{\mu_0 \mu_2}$ one writes

$$B_{\mu_0 \mu_2} = \phi_{\mu_1 \mu_2} - \phi_{\mu_0} \quad (5.3)$$

and replaces $B_{\mu_0 \mu_3}$ by ϕ_{μ_1} . This is the essence of the transformation from $B_{\mu\nu}$ to $f_{\mu\nu}$ for $\mu \neq 0$ and from $B_{\mu\nu}$ to ϕ_μ . To complete the transformation one must discuss the surface $\mu_0 = N$.

(ii) For $n_0 = N$, $n_1 \neq N$, and n_2 and n_3 arbitrary, B_{n_2} ($\mu + 1$) is replaced by f_{n_2} ; B_{n_1} is replaced by ϕ_n , with

$$B_n = \phi_{n+1} \dot{\zeta} - \phi_n \dot{\zeta}. \quad (5.4)$$

(iii) For $n_0 = n_1 = N$, $n_2 \neq N$, and n_3 arbitrary, B_{n_2} ($\mu + 2$) is replaced by f_{n_2} ; B_{n_3} is replaced by ϕ_n , with $B_{n_3} = \phi_{n+1} \dot{\zeta} - \phi_n \dot{\zeta}$.

(iv) For $n_0 = n_1 = n_2 = N$, and $n_3 \neq N$, B_{n_3} ($\mu + 3$) is replaced by f_{n_3} ; B_{n_3} by ϕ_n , where $B_{n_3} = \phi_{n+1} \dot{\zeta} - \phi_n \dot{\zeta}$.

(v) For $n_0 = n_1 = n_2 = n_3 = N$ one writes $B_{n_0} = \dot{\zeta}_n$ with $\dot{\zeta}_n$ being the new variables. One also sets $\phi_n = 0$ for this value of n .

The variables $f_{n_{\mu\nu}}$ which are not integration variables (for example, $f_{n_{\mu\nu}}$ with $\mu \neq 0$ and $\nu \neq 0$, for $n_0 \neq N$) can be expressed in terms of the independent $f_{n_{\mu\nu}}$ variables using Eq. (4.6); neither the ϕ_n nor $\dot{\zeta}_n$ appear in these expressions.

When an arbitrary $B_{n_{\mu\nu}}$ is expressed in terms of the new independent variables, one finds ($\dot{\zeta}_n$ is present only if $n_0 = N$):

$$I_g^{\mu\nu}(P) = \left(\prod_{n_{\mu\nu}} \int_{-\infty}^{\infty} dB_{n_{\mu\nu}} \right) \exp \left(i \sum_{\mu\nu} (\dot{\zeta}_n B_{n_{\mu\nu}} - \frac{\alpha}{2} \sum_{\mu\nu} [B_{n_{\mu\nu}} - B_{n_{\mu\nu}+\mu, \nu}]^2 - \frac{1}{4g^2} \sum_{n_{\mu\nu}} f_{n_{\mu\nu}}^2) \right), \quad (5.6)$$

where the α term is a lattice version of a $(\nabla_\mu A_\nu)^2$ gauge-fixing term. This integral can be computed by explicit Gaussian integration methods rather more easily than the $f_{n_{\mu\nu}}$ integrations for $I_g^{\mu\nu}(P)$. In addition, $I_g^{\mu\nu}(P)$ can also be reduced to an integral over a subset of the $f_{n_{\mu\nu}}$, using the same change of variables as for $I_g^{\mu\nu}(P)$. The result is different in this case due to the α term which couples the ϕ 's to the f 's; also there is no convergence factor for the $\dot{\zeta}$ integration. To make $I_g^{\mu\nu}(P)$ well defined and equal to $I_g^{\mu\nu}(P)$, one must (a) put in a convergence factor for the $\dot{\zeta}$ integral, i.e., a term $-i\beta[\sum_{n_{\mu\nu}} B_{n_{\mu\nu}}]^2$ and (b) add a quadratic form in the f 's to compensate for the result of the ϕ integration of the gauge-fixing term. The author has not carried through this calculation, but since the net result is still that $I_g^{\mu\nu}(P) \sim I_g^{\mu\nu}(P)$ is a Gaussian integration in the B 's, the result will presumably be similar to the conventional free-field calculation reported in Sec. II.

VI. PHASE TRANSITIONS

In the strong-coupling limit ($g \rightarrow \infty$, $K \rightarrow 0$) the gauge theory is far from being Lorentz-invariant. More precisely, since the action was defined on a Euclidean metric, it is Euclidean invariance that is missing. In the strong-coupling limit, vacuum expectation values decrease rapidly at separations of only a few lattice sites (there is a factor g^{-2} or

$$B_{n_{\mu\nu}} = \dot{\zeta}_n + \phi_{n+\mu} \dot{\zeta} - \phi_n \dot{\zeta} + (\text{linear combination of } f_{n_{\mu\nu}}), \quad (5.5)$$

i.e., the ϕ variables define a gauge transformation and $\dot{\zeta}_n$ represents a translation of some of the B 's. It is easily verified that the integrand of $I_g^{\mu\nu}(P)$ involves only the f 's: It is independent of both the ϕ 's and $\dot{\zeta}_n$ (the latter does not appear because any closed path P has as many $-B_{\mu\nu}$ terms as $+B_{\mu\nu}$ terms on the sublattice $n_0 = N$). Hence the ϕ_n and $\dot{\zeta}_n$ integrations can be computed trivially. The f integrations are nontrivial because of the constraint (4.6).

What one wants to accomplish is to reduce the lattice theory for small g to something like a conventional free gauge-field theory. This means restoring the $B_{\mu\nu}$ as the integration variables, but with infinite limits of integration, and with a gauge-fixing term included. Suppose, for example, one starts with

K or both for each unit lattice spacing of separation). This corresponds to the existence of masses much larger than the cutoff. [The usual rule is that if a propagator falls as $e^{-m|x|}$ for x large then the lowest mass intermediate state contributing to the propagator has mass $1/\xi$. If the propagator behaves as x^{-2K} for distances $x = na$, then the corresponding mass is $2(\ln g)/a$. This is larger than the cutoff momentum π/a if g is large.]

Thus, one is interested in practice in values of g and K such that the correlation length ξ is much larger than the lattice spacing a , in order that the corresponding mass is much less than the cutoff. One knows from statistical mechanics that large correlation lengths are associated with second-order phase transitions (critical points). Thus one seeks special values g_c and K_c for g and K at which there is a phase transition.¹⁴

It has already been argued that there are two distinct phases for the gauge field, a strong-coupling phase for large g which binds quarks, and a weak-coupling phase for small g which does not bind quarks. The arguments given neglected quark vacuum loops, which is reasonable if K is small. There should be a transition between these two phases which would occur at a critical value g_c for any g and any small value of K . This is one possible phase transition; it is this transition which was discussed in Sec. II. But, as will be argued below, this is probably a first-order tran-

sition rather than second order.

Suppose one wishes to construct a model of strong interactions using the lattice theory of this paper with the gauge group separate from ordinary $SU(3) \times SU(3)$ symmetry. Then the gauge fields would all be $SU(3) \times SU(3)$ singlets, while the quark fields would carry $SU(3) \times SU(3)$ quantum numbers as well as gauge-group quantum numbers. A little thought shows that in the strong-coupling limit ($g \rightarrow \infty$, $K=0$), $SU(3) \times SU(3)$ is an exact symmetry rather than a spontaneously broken symmetry. Varying g does not change this situation; so one must hope that by increasing K one can change the exact $SU(3) \times SU(3)$ into broken $SU(3) \times SU(3)$. If this does not work one is free to introduce additional terms into the quark field action in hopes of forcing a spontaneous breaking of $SU(3) \times SU(3)$. Suppose, for simplicity, that $SU(3) \times SU(3)$ can be broken by increasing K . Then there will be a phase transition at a critical value K_c for K where one changes from exact $SU(3) \times SU(3)$ to spontaneously broken $SU(3) \times SU(3)$. If this transition is a second-order transition then there will be a large correlation length for K near K_c ; in this case the theory might be a realistic model of broken $SU(3) \times SU(3)$ for K slightly greater than K_c (with g large enough to maintain quark binding).

In summary, the transition of real interest is a transition in K (or some other parameter introduced into the quark action) rather than g .

Apart from special limits ($g \rightarrow \infty$ and $K=0$, or g small) it is very difficult to solve the lattice theory. It is especially difficult to solve the lattice theory near a critical point with a large correlation length. Various methods have been developed by statistical mechanicians to deal with this problem. In the remainder of this section these methods will be discussed briefly. There are essentially three approaches to consider:

(1) mean-field techniques, (2) series expansions, and (3) the renormalization-group approach.

Mean-field techniques¹⁸ are the simplest and crudest methods for studying a critical point; invariably they are the methods one uses first in studying a new situation. They are used to determine if there is a phase transition, whether it is first or second order, and to give rough estimates of the behavior near the transition. None of the results of a mean-field calculation are entirely trustworthy. Examples of mean-field calculations will be given later.

An example of a series expansion would be the expansion of the current-current propagator for small momentum (momentum $\ll 1/a$) in powers of g^{-2} and K , to high order in g^{-2} and K . One then uses Padé-approximant techniques to look for singularities in either g or K that would be as-

sociated with a mass approaching 0. In simple statistical-mechanical problems one can generate 12 terms or so in analogous expansions. The expansion for the lattice theory of this paper is more complicated, but one could hope to generate maybe 6 or 7 orders with some practice. Series expansions require considerably more effort than mean-field calculations; they apply mainly to propagators, being very awkward to perform on three- and four-point functions, and one must have a clear idea of what one is trying to learn before attempting such calculations. See Ref. 16 for one of the best series-expansion formalisms; see Ref. 13 for a general review.

The renormalization-group approach is potentially the most powerful and accurate method for studying lattice theories near a critical point, but at present the renormalization-group techniques are too limited in scope to be applicable to the present problem. See Refs. 6 and 17.

Return to mean-field ideas.¹⁸ The prototype mean-field calculation is a calculation of the magnetization as a function of the external field for an Ising ferromagnet. Let s_n be the spin at site n with values ± 1 only; let the interaction be

$$\frac{-H}{kT} = K \sum_{\langle n, n' \rangle} s_n s_{n'} + h \sum_n s_n, \quad (6.1)$$

where K is related to the spin-spin coupling and h is proportional to the external field. Then

$$M = \langle \dots \rangle \left(\left\langle \exp \left(K \sum_{\langle n, n' \rangle} s_n s_{n'} + h \sum_n s_n \right) \right\rangle \right), \quad (6.2)$$

where $\langle \dots \rangle$ means a sum over all configurations of all spins, and

$$Z = \left\langle \exp \left(K \sum_{\langle n, n' \rangle} s_n s_{n'} + h \sum_n s_n \right) \right\rangle. \quad (6.3)$$

In the mean-field approximation, one assumes that the spins s_i coupled to s_n can be replaced by their average value M . As a result, the formula for M simplifies to a sum over s_n only, namely

$$M = Z^{-1} \sum_{s_n = \pm 1} s_n e^{(2dK + h)s_n}, \quad (6.4)$$

with d being the dimensionality (3 usually) and

$$Z_0 = \sum_{s_n = \pm 1} e^{(2dK + h)s_n}. \quad (6.5)$$

The result is

$$M = \tanh(2dKM + h). \quad (6.6)$$

If $2dK < 1$ this equation has a unique solution for M as a function of h ; in particular, $M=0$ for $h=0$. For $2dK > 1$ the solution is multiple-valued; stability considerations show that one must choose a solution with $M=0$ when $h=0$.

In this approximation one has actually replaced $\sum_{\nu} s_{\nu}$ by $2dM$, which is a good approximation if d is large. This is generally true of mean-field theories.

An analogous mean-field calculation can be performed for the lattice gauge theory. In this case a simple external field term has the form

$$k \sum_{\nu} (e^{iA_{\nu}} + e^{-iA_{\nu}})$$

(this is to be added to the gauge-field action), and M can be defined to be the expectation value of $e^{iA_{\nu}}$. The question is whether M is zero in the limit $k \rightarrow 0$. In the gauge-field theory $e^{iA_{\nu}}$ couples to a product of three other exponentials; as a mean-field approximation one replaces this product by M^3 . The result of this is that

$$M = Z_0^{-1} \int_{\mathcal{F}} dB_{\nu} e^{iB_{\nu}} \times \exp \left[\left(\frac{2(d-1)}{2g^2} M^3 + k \right) (e^{iA_{\nu}} + e^{-iA_{\nu}}) \right], \quad (6.7)$$

where d is the space-time dimensionality, and

$$Z_0 = \int_{\mathcal{F}} \exp \left[\left(\frac{2(d-1)}{g^2} M^3 + k \right) (e^{iA_{\nu}} + e^{-iA_{\nu}}) \right]. \quad (6.8)$$

The result of this is that

$$M = f \left(\frac{(d-1)}{g^2} M^3 + k \right), \quad (6.9)$$

where f is a ratio of Bessel's functions. If g is large the solution to this equation is unique and

$M=0$ for $k=0$. If g is small then there are solutions with $M \neq 0$ for $k=0$, and stability considerations show again that the $M \neq 0$ solutions are preferred.

In the magnetic case, one finds that the spontaneous magnetization M goes to zero for $2d-1$ [from Eq. (6.6)]. However, the gauge-field case never has a solution for $k=0$ with M small but nonzero. Thus there is a first-order transition at the value of g for which M changes from zero to being nonzero.

A nonzero value of M in the limit $k \rightarrow 0$ means one has spontaneous breaking of the gauge-field symmetry. So for small g the theory shows spontaneous breaking.

A much more thorough discussion of the mean-field approximation has been given by Balian, Drouffe, and Itzykson.¹⁸ A Hamiltonian formulation of the lattice gauge theory has been given by Kogut and Susskind.¹⁹ A clear review of quark confinement in the lattice theory is given in Ref. 20. Another formulation of the connection between strongly coupled gauge theories and string models is given in Ref. 21.

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PHYSICAL REVIEW D

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Elementary and composite particles in asymptotically free gauge theories

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We demonstrate that in asymptotically free gauge theories the wave-function renormalization constants for spin-1/2 fields do not vanish. (The scalar fields, if incorporated, also have this property.) However, there exists a subclass of such theories where the Z 's for the gauge fields themselves tend to zero in the limit of infinite cutoff. These features are shown to be gauge-independent. This suggests the possibility of constructing asymptotically free strong-interaction theories in which the only elementary fields are "quarks" and all other hadrons are bound states.

1. INTRODUCTION

Gauge-invariant quantum field theories based on non-Abelian groups are being actively investigated. A number of rather unique properties have been discovered about such theories. These features, when interpreted optimistically, indicate that non-Abelian gauge theories may provide the framework within which theories of strong interactions (or even unified theories of all elementary-particle forces) may be constructed.

Gauge theories are renormalizable.¹ It has been shown, by way of the renormalization-group equation,² that the origin of the coupling-constant space is an ultraviolet-stable fixed point³ only in non-Abelian gauge theories.⁴ This asymptotically free nature of the theory provides us with a field-theoretical explanation of Bjorken scaling—rather the explanation of how Bjorken scaling is approached in the deep Euclidean limit.⁵ This same property indicates that the effective couplings can be large in the infrared limit—it just may provide the desired quark-confinement mechanism.⁶

There are also a number of works suggesting intriguing connections of gauge theories to dual models and relativistic string models of hadrons.⁷

This confluence of field-theoretical and S-matrix approaches to strong-interaction physics is also indicated by the works of Grisaru, Schnitzer, and Tsao.⁸ These authors have demonstrated that vector mesons and spin- $\frac{1}{2}$ fermions in such field theories satisfy the usual criteria of Reggeization: factorization of Born amplitudes and Mandelstam countings. However, as possible candidates for strong-interaction theories, the class of gauge theories investigated in their works may have some drawbacks: These gauge theories are not asymptotically free and while spin- $\frac{1}{2}$ particles (quarks?) lie on Regge trajectory, scalar fields do not. (It would seem an unattractive picture of having spin- $\frac{1}{2}$ fermions composite, but not all other particles.) In this paper we shall use another criterion for the compositeness of fields appearing in a Lagrangian field theory, i.e., the vanishing of the wave-function renormalization constants. Our results suggest the possibility of constructing strong-interaction field theories which are asymptotically free and in which the only elementary fields are quarks and all other hadrons are composite.

In Sec. II we shall demonstrate, through a straightforward exercise of solving the renormal-

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页码范围：P02-p001 – P02-p008

Analytic Smearing of $SU(3)$ Link Variables in Lattice QCD

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An analytic method of smearing link variables in lattice QCD is proposed and tested. The differentiability of the smearing scheme with respect to the link variables permits the use of modern Monte Carlo updating methods based on molecular dynamics evolution for gauge-field actions constructed using such smeared links. In examining the smeared meson plaquette and the static quark-antiquark potential, no degradation in effectiveness is observed as compared to link smearing methods currently in use, although an increased sensitivity to the smearing parameter is found.

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1. INTRODUCTION

The extraction of hadron masses and matrix elements from Monte Carlo estimates of Euclidean-space correlation functions in lattice QCD can be done more reliably and accurately when operators which couple more strongly to the states of interest and less strongly to the higher-lying contaminating states are used. For states containing gluons, a crucial ingredient in constructing such operators in lattice QCD is link variable smearing. Operators constructed out of smeared or fuzzed links have dramatically reduced mixings with the high frequency modes of the theory. The use of such operators has been shown to especially benefit determinations of the glueball spectrum [1], hybrid meson masses [2, 3], the tolon spectrum [4], and excitations of the static quark-antiquark potential [5].

Link variable smoothing is also playing an increasingly important role in the construction of improved lattice actions. In Ref. [6], the use of so-called fat links in a staggered quark action was shown to significantly decrease flavor symmetry breaking. Smeared links were subsequently used [7] to construct hypercubic fermion actions having improved rotational invariance. A staggered fermion action using a link smearing transformation [8] known as hypercubic (HYP) fat links was shown to improve flavor symmetry by an order of magnitude relative to the standard action. The so-called Asqad improved staggered quark action [9, 10, 11] also makes use of link fattening to reduce flavor symmetry breaking. Another variant of fermion actions which exploit smeared link variables is the fat link irrelevant clover (FLIC) action [12]. FLIC fermions are Wilson-like and described by an action which includes an irrelevant clover improvement term constructed using smeared links. Fat links have also been used to construct a gauge action [13] with reduced discretization errors using approximate renormalization group transformations.

The link fuzzing algorithm most often used in gluonic operator construction is that described in Ref. [14] in which every spatial link $U_i(x)$ on the lattice is replaced

by itself plus a real weight ρ times the sum of its four neighboring (spatial) staples, projected back into $SU(3)$. Such a fuzzing step is iterated n_ρ times to obtain the final fuzzed link variables. Empirically, one finds that the projection into $SU(3)$ is a crucial ingredient of the smearing. Link smearings which do not apply such a projection are found to be much less effective. The projection into $SU(3)$ is not unique and must be carefully defined so that all symmetry properties of the link variables are preserved. Various ways of implementing this projection have appeared in the literature. Given a 3×3 matrix V , its projection U into $SU(3)$ is often taken to be the matrix $U \in SU(3)$ which maximizes $\text{ReTr}(UV^\dagger)$. An iterative procedure is required to perform such a maximization. Alternatively [15], the projected matrix U can be defined by $U = V (V^\dagger V)^{-1/2} \det(V^{-1} V^\dagger)^{1/6}$.

Although the above smearing procedure works well in practice, the projection is somewhat unpalatable since it may be viewed as an arbitrary and abrupt way to remain within the group. More importantly, the branch cuts and lack of differentiability inherent in the projection can hinder or even make impossible the application of modern Monte Carlo updating techniques, such as hybrid Monte Carlo (HMC) [16], which require knowing the response of the action to a small change in one of the link variables.

A link smearing method which circumvents these problems is proposed and tested in this paper. The link smearing method is analytic everywhere in the finite complex plane and utilizes the exponential function in such a manner to remain within $SU(3)$, eliminating the need for any projection back into the group. Because of this construction, the algorithm is useful for any Lie group. The method is described in Sec. II and its practical implementations when constructing operators and when computing the response of the action to a change in a link variable are detailed in Secs. III and IV, respectively. Numerical tests of the smearing are presented in Sec. V. Using the smeared plaquette and the effective mass $aE_{\text{eff}}(t)$ associated with the static quark-antiquark potential, no degradation in effectiveness is observed as compared to the standard link smearing method, al-

though an increased sensitivity to the smearing parameter is found. The results also suggest that lattice actions and operators constructed out of smeared link variables may be much less afflicted by radiative corrections since the usually-dominant large tadpole contributions are drastically reduced.

II. ANALYTIC LINK SMEARING

A method of smearing link variables which is analytic, and hence differentiable, everywhere in the finite complex plane can be defined as follows. Let $C_\mu(x)$ denote the weighted sum of the perpendicular staples which begin at lattice site x and terminate at neighboring site $x+\hat{\mu}$:

$$C_\mu(x) = \sum_{\nu \neq \mu} \rho_{\mu\nu} \left(U_\nu(x) U_\mu(x+\hat{\nu}) U_\mu^\dagger(x+\hat{\mu}) + U_\mu^\dagger(x-\hat{\nu}) U_\mu(x-\hat{\nu}) U_\nu(x-\hat{\mu}) \right), \quad (1)$$

where $\hat{\mu}, \hat{\nu}$ are vectors in directions μ, ν , respectively, having the length of one lattice spacing in that direction (fundamental lattice translation vectors). The weights $\rho_{\mu\nu}$ are tunable real parameters. Then the matrix $Q_\mu(x)$, defined in $SU(N)$ by

$$Q_\mu(x) = \frac{i}{2} (\Omega_\mu^\dagger(x) - \Omega_\mu(x)) - \frac{i}{2N} \text{Tr}(\Omega_\mu^\dagger(x) - \Omega_\mu(x)), \\ \Omega_\mu(x) = C_\mu(x) U_\mu^\dagger(x), \quad (\text{no summation over } \mu) \quad (2)$$

is Hermitian and traceless, and hence, $e^{iQ_\mu(x)}$ is an element of $SU(N)$. We use this fact to define an iterative, analytic link smearing algorithm in which the links $U_\mu^{(n)}(x)$ at step n are mapped into links $U_\mu^{(n+1)}(x)$ using

$$U_\mu^{(n+1)}(x) = \exp(iQ_\mu^{(n)}(x)) U_\mu^{(n)}(x). \quad (3)$$

The fact that $e^{iQ_\mu(x)}$ is an element of $SU(N)$ guarantees that $U_\mu^{(n+1)}(x)$ is also an element of $SU(N)$, eliminating the need for a projection back onto the gauge group. This fuzzing step can be iterated n_μ times to finally produce link variables which we call *stout links*[17], denoted by $\tilde{U}_\mu(x)$:

$$U \rightarrow U^{(1)} \rightarrow U^{(2)} \rightarrow \dots \rightarrow U^{(n_\mu)} = \tilde{U}. \quad (4)$$

One common choice of the staple weights is

$$\rho_{jk} = \rho, \quad \rho_{4\mu} = \rho_{\mu 4} = 0, \quad (5)$$

which yields a three-dimensional scheme in which only the spatial links are smeared. Another common choice is an isotropic four-dimensional scheme in which all weights are chosen to be the same $\rho_{\mu\nu} = \rho$. Note that such a scheme was used in Ref. [18] as a field transformation to eliminate a certain interaction term from the most general $O(a^2)$ on-shell improved gauge action.

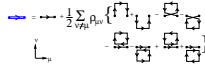


FIG. 1: The expansion up to first order in the $\rho_{\mu\nu}$ of the new link variable $U_\mu^{(1)}$ in terms of paths of the original links. Each closed loop includes a trace with a factor $1/N$ in $SU(N)$.

It is not difficult to show that, given an appropriate choice of weights $\rho_{\mu\nu}$, the stout links have symmetry transformation properties identical to those of the original link variables. Under any local gauge transformation $G(x)$, the link variables transform according to $U_\mu(x) \rightarrow G(x) U_\mu(x) G^\dagger(x+\hat{\mu})$, where the $G(x)$ are $SU(N)$ matrices. Thus, under such a gauge transformation,

$$Q_\mu(x) \rightarrow G(x) Q_\mu(x) G^\dagger(x), \quad (6)$$

$$e^{iQ_\mu(x)} \rightarrow G(x) e^{iQ_\mu(x)} G^\dagger(x), \quad (7)$$

$$U_\mu^{(1)}(x) \rightarrow G(x) e^{iQ_\mu(x)} G^\dagger(x) G(x) U_\mu(x) G^\dagger(x+\hat{\mu}) \\ = G(x) U_\mu^{(1)}(x) G^\dagger(x+\hat{\mu}), \quad (8)$$

as required. One can also easily show that if the weights $\rho_{\mu\nu}$ respect the rotation and reflection symmetries of the lattice, the stout links obey the required transformation properties under all rotations and all reflections in a plane containing the link. Lastly, consider a reflection in a plane normal to a link $U_\mu(x)$ and passing through its midpoint $x+\frac{1}{2}\hat{\mu}$. Under such an operation, the link $U_\mu(x)$ transforms according to $U_\mu(x) \rightarrow U_{-\mu}(x+\hat{\mu}) = U_\mu^\dagger(x)$ and for $\nu \neq \mu$, $U_\nu(x) \leftrightarrow U_\nu(x+\hat{\mu})$ and $U_\mu^\dagger(x+\hat{\mu}) \leftrightarrow U_\mu^\dagger(x)$, yielding $C_\mu(x) \rightarrow C_\mu^\dagger(x)$ assuming the $\rho_{\mu\nu}$ are real, so that

$$\Omega_\mu(x) \rightarrow C_\mu^\dagger(x) U_\mu(x) = U_\mu^\dagger(x) \Omega_\mu^\dagger(x) U_\mu(x), \quad (9)$$

$$Q_\mu(x) \rightarrow -U_\mu^\dagger(x) Q_\mu(x) U_\mu(x), \quad (10)$$

$$U_\mu^{(1)}(x) \rightarrow U_\mu^\dagger(x) e^{-iQ_\mu(x)} = U_\mu^{(1)\dagger}(x), \quad (11)$$

as required. Given that $U_\mu^{(1)}(x)$ transforms under all symmetry operations in the same manner as $U_\mu(x)$, it follows that the stout links $\tilde{U}_\mu(x)$ have symmetry transformation properties identical to those of the original link variables.

Since the exponential function has a power series expansion with an infinite radius of convergence, each stout link may be viewed as an incredibly large and complicated sum of paths. For small $\rho_{\mu\nu}$, the paths which make up the link variable $U_\mu^{(1)}$ to first order in the $\rho_{\mu\nu}$ are shown in Fig. 1. Note that the standard smearing method, defined by

$$U_\mu^{(1)}(x) = \mathcal{P}_{SU(3)} \left\{ U_\mu(x) + C_\mu(x) \right\}, \quad (12)$$

where $\mathcal{P}_{SU(3)}$ denotes the projection into $SU(3)$, yields the same sum of paths at first order in $\rho_{\mu\nu}$.

A few further remarks are worthy of note. First, an alternative smearing scheme in which the weights $\rho_{\mu\nu}$ are chosen to be imaginary does not reduce the couplings to the high-frequency modes of the theory. Secondly, it is not possible to remove the leading $O(a^2)$ discretization errors in the Wilson gauge action by expressing the action in terms of stout links; diminishing the $O(a, a^2)$ errors by reducing their tadpole contributions is the best which can be achieved. Thirdly, the use of stout links in Symanzik-improved actions may eliminate the need to use tadpole-improvement in determining the couplings in the action. Lastly, the definition of $C_\mu(x)$ given in Eq. (1) is not unique. Analytic smearing schemes can be constructed as outlined above using sums of other paths from site x to $x + \hat{\mu}$, such as those used in fat link[6] and HYP[8] smearings.

III. IMPLEMENTATION OF THE SMEARING

To implement this analytic link smearing scheme, the efficient evaluation of $\exp(iQ)$, where Q is a traceless Hermitian 3×3 matrix, is required. The Cayley-Hamilton theorem states that every matrix is a zero of its characteristic polynomial, so that

$$Q^3 - c_1 Q - c_0 I = 0, \quad (13)$$

where

$$c_0 = \det Q = \frac{1}{3} \text{Tr}(Q^3), \quad (14)$$

$$c_1 = -\frac{1}{2} \text{Tr}(Q^2) \geq 0. \quad (15)$$

The Hermiticity of Q requires $27c_0^2 - 4c_1^3 \leq 0$ and the definition of Q given in Eq. (2) restricts the possible values of c_1 . Thus, the coefficients c_0 and c_1 satisfy

$$-c_0^{\text{max}} \leq c_0 \leq c_0^{\text{max}}, \quad 0 \leq c_1 \leq c_1^{\text{max}}, \quad (16)$$

where

$$c_0^{\text{max}} = 2 \left(\frac{c_1}{3} \right)^{3/2}, \quad (17)$$

$$c_1^{\text{max}} = \frac{1}{32} (69 + 11\sqrt{33}) \left(\sum_{\nu \neq \mu} \rho_{\mu\nu} \right)^2, \quad (18)$$

for each μ . Eq. (13) implies that Q^n for integer $n \geq 3$ can be expressed in terms of Q^2 , Q , and the identity matrix I . Hence, we can write

$$e^{iQ} = f_0 I + f_1 Q + f_2 Q^2, \quad (19)$$

where the three scalar coefficients $f_j = f_j(c_0, c_1)$ are basis independent, depending only on c_0 and c_1 . Eq. (19) is valid for any 3×3 Hermitian, traceless matrix Q .

Let q_1, q_2, q_3 denote the three eigenvalues of Q . Since Q is Hermitian and traceless, we know that these are real

numbers satisfying $q_1 + q_2 + q_3 = 0$. These eigenvalues are the three roots of a cubic polynomial which can be easily determined:

$$q_1 = 2u, \quad (20)$$

$$q_2 = -u + w, \quad (21)$$

$$q_3 = -q_1 - q_2 = -u - w, \quad (22)$$

where

$$u = \sqrt{\frac{2}{3}} c_1 \cos\left(\frac{1}{3}\theta\right), \quad (23)$$

$$w = \sqrt{c_1} \sin\left(\frac{1}{3}\theta\right), \quad (24)$$

$$\theta = \arccos\left(\frac{c_0}{c_1^{3/2}}\right). \quad (25)$$

Given these eigenvalues, the matrix Q can be written

$$Q = M \Lambda_Q M^{-1}, \quad \Lambda_Q = \begin{bmatrix} q_1 & 0 & 0 \\ 0 & q_2 & 0 \\ 0 & 0 & q_3 \end{bmatrix}, \quad (26)$$

where M is a unitary matrix. Then it easily follows that

$$e^{i\Lambda_Q} = f_0 I + f_1 \Lambda_Q + f_2 \Lambda_Q^2. \quad (27)$$

Explicitly, we have the following linear system of equations to solve:

$$\begin{bmatrix} 1 & q_1 & q_1^2 \\ 1 & q_2 & q_2^2 \\ 1 & q_3 & q_3^2 \end{bmatrix} \begin{bmatrix} f_0 \\ f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} e^{iq_1} \\ e^{iq_2} \\ e^{iq_3} \end{bmatrix}. \quad (28)$$

If all three eigenvalues are distinct, this system of equations has a unique solution, but when two of the eigenvalues are exactly the same, the solution to Eq. (28) is not unique and one of the f_j 's can be freely set to any value. The case of degenerate eigenvalues occurs when $c_0 \rightarrow \pm c_0^{\text{max}}$.

In practice, it is extremely unlikely that an exact degeneracy will be encountered during a numerical simulation. Much more likely is the possibility that two eigenvalues are nearly, but not quite, equal. Encounters with both near and exact degeneracies can be handled by expressing the f_j in terms of u and w in order to isolate and tame the numerically sensitive part. One finds that the f_j factors can be written

$$f_j = \frac{h_j}{(9u^2 - w^2)}, \quad (29)$$

where the h_j are well-behaved functions given by

$$h_0 = (u^2 - w^2)e^{2iu} + e^{-iu} \{8u^2 \cos(w) + 2iu(3u^2 + w^2)\xi_0(w)\}, \quad (30)$$

$$h_1 = 2ue^{2iu} - e^{-iu} \{2u \cos(w) - i(3u^2 - w^2)\xi_0(w)\}, \quad (31)$$

$$h_2 = e^{2iu} - e^{-iu} \{\cos(w) + 3iu\xi_0(w)\}, \quad (32)$$

defining

$$\xi_0(w) = \frac{\sin w}{w}. \quad (33)$$

The $w \rightarrow 0$ problem as $c_0 \rightarrow c_0^{\text{max}}$ has been completely contained in ξ_0 . To numerically evaluate this factor, one uses, for example,

$$\xi_0(w) = \begin{cases} 1 - \frac{1}{2}w^2 \left(1 - \frac{1}{20}w^2 \left(1 - \frac{1}{42}w^2\right)\right), & |w| \leq 0.05, \\ \sin(w)/w, & |w| > 0.05. \end{cases}$$

However, the $w \rightarrow 3u \rightarrow \sqrt{3}/2$ limit as $c_0 \rightarrow -c_0^{\text{max}}$ has not been tamed. Fortunately, this problem can be circumvented using the following symmetry relations under $c_0 \rightarrow -c_0$:

$$f_j(-c_0, c_1) = (-)^j f_j'(c_0, c_1). \quad (34)$$

Thus, the determination of the f_j coefficients for $c_0 < 0$ can be achieved by computing the coefficients for $|c_0|$ and utilizing Eq. (34). This means that a situation in which the denominator in Eq. (29) becomes nearly zero will never be encountered. It should be emphasized that the f_j coefficients are smooth, non-singular functions of c_0 and c_1 . There are no actual singularities as $c_0 \rightarrow \pm c_0^{\text{max}}$. The numerical evaluation of the f_j coefficients to machine precision in these limits simply requires additional care.

As an aside, if one writes $Q = \frac{1}{2} \sum_{a=1}^8 Q_a \lambda^a$ in terms of the eight Gell-Mann matrices λ^a , then

$$e^{iQ} = u_0 + \frac{1}{2} \sum_{a=1}^8 u_a \lambda^a, \quad (35)$$

$$u_0 = f_0 + \frac{1}{2} c_1 f_2, \quad (36)$$

$$u_a = f_1 Q_a + \frac{1}{2} f_2 d^{abc} Q_b Q_c, \quad (37)$$

$$c_0 = \frac{1}{12} d^{abc} Q_a Q_b Q_c, \quad (38)$$

$$c_1 = \frac{1}{2} Q_a Q_a, \quad (39)$$

where d^{abc} and f^{abc} are the real symmetric and antisymmetric structure constants, respectively.

IV. MOLECULAR DYNAMICS EVOLUTION

The analyticity of the stout-link scheme permits the use of modern Monte Carlo updating methods for gauge-field actions constructed using stout links. Molecular dynamics evolution forms the core of many of the Markov processes used to generate ensembles of field configurations needed for unquenched QCD simulations [16, 19, 20, 21]. Since a small change to the underlying gauge fields always leads to a small change in the stout links, then provided the small change to the stout links causes small changes to the action of the theory, the force term on the underlying links is always well-defined. The importance of defining a link smearing scheme which permits the computation of the molecular dynamics force term has been emphasized recently in Ref. [22].

The $\Sigma_\mu(x)$ force field describes how the action S changes when the gauge-field link variables $U_\mu(x)$ change, holding the momenta $P_\mu(x)$ and the pseudofermion field $\phi(x)$ fixed. For this reason, we shall write the action $S = S[U]$ in this section, ignoring its dependence on the pseudofermion field. The transpose of the force field is defined by the derivative of the action with respect to the link variables:

$$\Sigma_\mu(x) = \left(\frac{\partial S[U]}{\partial U_\mu(x)} \right)^T. \quad (40)$$

Assume that the action can be written as a sum of a term $S_{\text{th}}[U]$ constructed entirely of the original thin link variables and a term $S_{\text{st}}[\tilde{U}]$ constructed completely out of stout links:

$$S[U] = S_{\text{th}}[U] + S_{\text{st}}[\tilde{U}]. \quad (41)$$

Then the force field may be written

$$\Sigma_\mu(x) = \Sigma_\mu^{(\text{th})}(x) + \Sigma_\mu^{(0)}(x), \quad (42)$$

where

$$\Sigma_\mu^{(\text{th})}(x) = \left(\frac{\partial S_{\text{th}}[U]}{\partial U_\mu(x)} \right)^T, \quad \Sigma_\mu^{(0)}(x) = \left(\frac{\partial S_{\text{st}}[\tilde{U}]}{\partial U_\mu(x)} \right)^T. \quad (43)$$

The computation of $\Sigma_\mu^{(\text{th})}(x)$ is usually straightforward, and nothing more about its determination needs discussion here. We now focus on the evaluation of $\Sigma_\mu^{(0)}(x)$.

Recall that the stout links are constructed iteratively starting with the original links $U \rightarrow U^{(1)} \rightarrow U^{(2)} \rightarrow \dots \rightarrow U^{(n_\ell)} = \tilde{U}$. The computation of the force field proceeds similarly in an iterative fashion, except that the order is reversed $\tilde{\Sigma} = \Sigma^{(n_\ell)} \rightarrow \Sigma^{(n_\ell-1)} \rightarrow \dots \rightarrow \Sigma^{(1)} \rightarrow \Sigma^{(0)}$. The sequence starts by computing

$$\tilde{\Sigma}_\mu(x) = \left(\frac{\partial S_{\text{st}}[\tilde{U}]}{\partial U_\mu(x)} \right)^T. \quad (44)$$

This step depends on the form of the gauge and fermion action in terms of the stout links and is usually just as straightforward as the computation of $\Sigma_\mu^{(\text{th})}(x)$. In subsequent steps, $\Sigma^{(k)}$ and $U^{(k-1)}$ are used to compute $\Sigma^{(k-1)}$, where the effective force at level k is defined by

$$\Sigma_\mu^{(k)}(x) = \left(\frac{\partial S_{\text{st}}[\tilde{U}]}{\partial U_\mu^{(k)}(x)} \right)^T. \quad (45)$$

The recursive mapping

$$\left\{ \Sigma^{(k)}, U^{(k-1)} \right\} \longrightarrow \Sigma^{(k-1)} \quad (46)$$

is repeated until $\Sigma^{(0)}$ is finally evaluated.

In Monte Carlo methods based on molecular dynamics, such as HMC [16] and the R -algorithm [21], the flow of

lattice configurations through phase space via Hamilton's equations is parametrized by a fictitious simulation time coordinate τ . To determine the mapping in Eq. (46), it is convenient to express the force field in terms of the rate of change of the action with respect to this time coordinate τ :

$$\frac{d}{d\tau} S_{st}[\tilde{U}] = 2 \sum_{x,\mu} \text{ReTr} \left\{ \Sigma_\mu^{(k)}(x) \frac{d}{d\tau} U_\mu^{(k)}(x) \right\}, \quad (47)$$

for $k = 0, 1, \dots, n_p$. One then proceeds using the chain rule of differentiation:

$$\begin{aligned} \frac{dU_\mu^{(k)}(x)}{d\tau} &= e^{iQ_\mu^{(k-1)}(x)} \frac{dU_\mu^{(k-1)}(x)}{d\tau} \\ &\quad + \frac{d(e^{iQ_\mu^{(k-1)}(x)})}{d\tau} U_\mu^{(k-1)}(x). \end{aligned} \quad (48)$$

To simplify matters, $Q_\mu^{(k-1)}(x)$, $U_\mu^{(k-1)}(x)$, $\Sigma_\mu^{(k-1)}(x)$ shall simply be written as Q , U , Σ , respectively, in the calculations which follow, and $\Sigma_\mu^{(k)}(x)$ and $U_\mu^{(k)}(x)$ shall be written as Σ' and U' , respectively. The Cayley-Hamilton theorem gives us

$$\begin{aligned} \frac{d(e^{iQ})}{d\tau} &= \frac{d}{d\tau} (f_0 + f_1 Q + f_2 Q^2), \\ &= \frac{df_0}{d\tau} + \frac{df_1}{d\tau} Q + \frac{df_2}{d\tau} Q^2 \\ &\quad + f_1 \frac{dQ}{d\tau} + f_2 \frac{dQ}{d\tau} Q + f_2 Q \frac{dQ}{d\tau}. \end{aligned} \quad (49)$$

Since the f_j coefficients are functions $f_j = f_j(u, w)$ of u and w only, one has

$$\frac{df_j}{d\tau} = \left(\frac{\partial f_j}{\partial u} \right) \frac{du}{d\tau} + \left(\frac{\partial f_j}{\partial w} \right) \frac{dw}{d\tau}, \quad (50)$$

and since u and w are functions of c_0 and c_1 only, then

$$\frac{du}{d\tau} = \left(\frac{\partial u}{\partial c_0} \right) \frac{dc_0}{d\tau} + \left(\frac{\partial u}{\partial c_1} \right) \frac{dc_1}{d\tau}, \quad (51)$$

$$\frac{dw}{d\tau} = \left(\frac{\partial w}{\partial c_0} \right) \frac{dc_0}{d\tau} + \left(\frac{\partial w}{\partial c_1} \right) \frac{dc_1}{d\tau}. \quad (52)$$

Next, one finds that

$$\frac{dc_0}{d\tau} = \frac{1}{2} \frac{d}{d\tau} \text{Tr}(Q^2) = \text{Tr} \left(Q^2 \frac{dQ}{d\tau} \right), \quad (53)$$

$$\frac{dc_1}{d\tau} = \frac{1}{2} \frac{d}{d\tau} \text{Tr}(Q^2) = \text{Tr} \left(Q \frac{dQ}{d\tau} \right), \quad (54)$$

and

$$\begin{aligned} \frac{\partial u}{\partial c_0} &= \frac{1}{2(9u^2 - w^2)}, & \frac{\partial u}{\partial c_1} &= \frac{u}{(9u^2 - w^2)}, \\ \frac{\partial w}{\partial c_0} &= \frac{-3u}{2w(9u^2 - w^2)}, & \frac{\partial w}{\partial c_1} &= \frac{3u^2 - w^2}{2w(9u^2 - w^2)}. \end{aligned} \quad (55)$$

Now define

$$r_j^{(1)} = \frac{\partial \tilde{h}_j}{\partial u}, \quad r_j^{(2)} = \frac{1}{w} \frac{\partial \tilde{h}_j}{\partial w}, \quad (56)$$

and

$$b_{1j} = \frac{2ur_j^{(1)} + (3u^2 - w^2)r_j^{(2)} - 2(15u^2 + w^2)f_j}{2(9u^2 - w^2)^2}, \quad (57)$$

$$b_{2j} = \frac{r_j^{(1)} - 3ur_j^{(2)} - 24uf_j}{2(9u^2 - w^2)^2}, \quad (58)$$

then

$$\frac{df_j}{d\tau} = b_{1j} \text{Tr} \left(Q \frac{dQ}{d\tau} \right) + b_{2j} \text{Tr} \left(Q^2 \frac{dQ}{d\tau} \right), \quad (59)$$

where

$$\begin{aligned} r_0^{(1)} &= 2 \left(u + i(u^2 - w^2) \right) e^{2iu} \\ &\quad + 2e^{-iu} \left\{ 4u(2 - iu) \cos w \right. \\ &\quad \left. + i(9u^2 + w^2 - iu(3u^2 + w^2)) \zeta_0(w) \right\}, \end{aligned} \quad (60)$$

$$\begin{aligned} r_1^{(1)} &= 2(1 + 2iu)e^{2iu} + e^{-iu} \left\{ -2(1 - iu) \cos w \right. \\ &\quad \left. + i(6u + i(w^2 - 3u^2)) \zeta_0(w) \right\}, \end{aligned} \quad (61)$$

$$r_2^{(1)} = 2ie^{2iu} + ie^{-iu} \left\{ \cos w - 3(1 - iu) \zeta_0(w) \right\}, \quad (62)$$

$$\begin{aligned} r_0^{(2)} &= -2e^{2iu} + 2iue^{-iu} \left\{ \cos w \right. \\ &\quad \left. + (1 + 4iu) \zeta_0(w) + 3u^2 \zeta_1(w) \right\}, \end{aligned} \quad (63)$$

$$\begin{aligned} r_1^{(2)} &= -ie^{-iu} \left\{ \cos w + (1 + 2iu) \zeta_0(w) \right. \\ &\quad \left. - 3u^2 \zeta_1(w) \right\}, \end{aligned} \quad (64)$$

$$r_2^{(2)} = e^{-iu} \left\{ \zeta_0(w) - 3iu \zeta_1(w) \right\}, \quad (65)$$

with

$$\zeta_0(w) = \frac{\sin w}{w}, \quad (66)$$

$$\zeta_1(w) = \frac{\cos w}{w^2} - \frac{\sin w}{w^3}. \quad (67)$$

Given the above results,

$$\begin{aligned} \frac{d(e^{iQ})}{d\tau} &= \text{Tr} \left(Q \frac{dQ}{d\tau} \right) B_1 + \text{Tr} \left(Q^2 \frac{dQ}{d\tau} \right) B_2 \\ &\quad + f_1 \frac{dQ}{d\tau} + f_0 \frac{dQ}{d\tau} Q + f_2 Q \frac{dQ}{d\tau}, \end{aligned} \quad (68)$$

where

$$B_i = b_{0i} + b_{1i} Q + b_{2i} Q^2. \quad (69)$$

The numerically sensitive $w \rightarrow 0$ limit has been totally absorbed into ζ_0 and ζ_1 , and numerical problems as $c_0 \rightarrow$

$-c_0^{\text{max}}$ can be circumvented by exploiting the following symmetry relation:

$$b_{ij}(-c_0, c_1) = (-)^{i+j+1} b_{ij}^*(c_0, c_1). \quad (70)$$

Once again, we emphasize that these coefficients are well-behaved, non-singular functions of c_0 and c_1 .

The rate of change of the stout links with respect to the simulation time is then given by

$$\begin{aligned} \frac{dU'}{d\tau} = & e^{iQ} \frac{dU}{d\tau} + \left\{ \text{Tr} \left(Q \frac{dQ}{d\tau} \right) B_1 + \text{Tr} \left(Q^2 \frac{dQ}{d\tau} \right) B_2 \right. \\ & \left. + f_1 \frac{dQ}{d\tau} + f_2 \frac{dQ}{d\tau} Q + f_2 Q \frac{dQ}{d\tau} \right\} U, \end{aligned} \quad (71)$$

and hence,

$$\text{ReTr} \left(\Sigma' \frac{dU'}{d\tau} \right) = \text{ReTr} \left(\Sigma' e^{iQ} \frac{dU}{d\tau} \right) - \text{ReTr} \left(i \Lambda \frac{d\Omega}{d\tau} \right), \quad (72)$$

defining

$$\Lambda = \frac{1}{2} (\Gamma + \Gamma^\dagger) - \frac{1}{2N} \text{Tr} (\Gamma + \Gamma^\dagger), \quad (73)$$

$$\begin{aligned} \Gamma = & \text{Tr} (\Sigma' B_1 U) Q + \text{Tr} (\Sigma' B_2 U) Q^2 \\ & + f_1 U \Sigma' + f_2 Q U \Sigma' + f_2 U \Sigma' Q. \end{aligned} \quad (74)$$

One sees that $\Gamma_\mu^{(k-1)}(x)$, and hence, $\Lambda_\mu^{(k-1)}(x)$, are defined on each lattice link in terms of the link variables $U_\mu^{(k-1)}(x)$ and $\Sigma_\mu^{(k)}(x)$. At this point, the details of the staple construction in $C_\mu(x)$ must be included. Given Eq. (1) and utilizing trace cyclicity and translational invariance, one eventually obtains the following recursion relation:

$$\begin{aligned} \Sigma_\mu(x) = & \Sigma'_\mu(x) \left\{ f_0 I + f_1 Q_\mu(x) + f_2 Q_\mu^2(x) \right\} \\ & + i C_\mu^{(1)}(x) \Lambda_\mu(x) - i \sum_{\nu \neq \mu} \left\{ \right. \\ & \rho_{\nu\mu} U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x) \Lambda_\nu(x) \\ & + \rho_{\mu\nu} U_\mu^\dagger(x - \hat{\nu} + \hat{\mu}) U_\mu^\dagger(x - \hat{\nu}) \Lambda_\nu(x - \hat{\nu}) U_\nu(x - \hat{\nu}) \\ & + \rho_{\nu\mu} U_\mu^\dagger(x - \hat{\nu} + \hat{\mu}) \Lambda_\nu(x - \hat{\nu} + \hat{\mu}) U_\mu^\dagger(x - \hat{\nu}) U_\nu(x - \hat{\nu}) \\ & - \rho_{\nu\mu} U_\mu^\dagger(x - \hat{\nu} + \hat{\mu}) U_\mu^\dagger(x - \hat{\nu}) \Lambda_\nu(x - \hat{\nu}) U_\nu(x - \hat{\nu}) \\ & - \rho_{\nu\mu} \Lambda_\nu(x + \hat{\mu}) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\mu^\dagger(x) \\ & \left. + \rho_{\mu\nu} U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) \Lambda_\mu(x + \hat{\nu}) U_\nu^\dagger(x) \right\}, \end{aligned} \quad (75)$$

in which unprimed quantities refer to step $(k-1)$ and primed quantities refer to step k .

Although the force field is related to the underlying link variables in a complicated manner, each step in the recursive computation of the force field outlined above is straightforward, facilitating a natural and efficient implementation in software. Analyticity in the entire (finite) complex plane is an important property here; if $\Sigma_\mu^{(k)}(x)$ is well behaved, then $\Sigma_\mu^{(0)}(x)$ must be also. Note that the above formalism can be easily adapted for other definitions of $C_\mu(x)$.

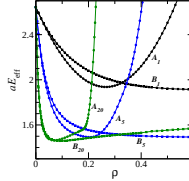


FIG. 2: The effective energy $aE_{\text{eff}}(r)$ defined in Eq. (76) for a static quark-antiquark pair separated by a distance $r = 5a$ for several levels of smearing $n_\rho = 1, 5$, and 20 against the smearing parameter ρ . These results were obtained on a 12^4 lattice using the Wilson gauge action with coupling $\beta = 5.7$. Curves labeled A_{n_ρ} indicate results using the spatially-isotropic three-dimensional version of the analytic stout link smearing scheme with $n_\rho = 1, 5$, and 20 levels, while the curves labeled B_{n_ρ} show the results for links smeared using Eq. (12) with the projection method of Ref. [15].

V. NUMERICAL TESTS

The efficacy of stout links both as a gluonic-operator smearing algorithm and for reducing the effects of ultraviolet gluon modes in short-distance quantities was tested in several Monte Carlo simulations.

The energy of gluons in the presence of a static quark-antiquark pair separated by a distance r can be extracted from $r \times t$ Wilson loops $W(r, t)$ as the temporal extent t becomes large. The Wilson loop can be viewed as the correlation function of a gauge-invariant operator consisting of a static quark-antiquark pair connected by a product of link variables following a straight-line path between the quark and the antiquark. Extraction of the lowest energy can be done much more reliably if the couplings of the quark-antiquark-gluon operator with higher-lying states are small. This can be achieved by utilizing a product of smeared links connecting the quark and the antiquark, instead of the original link variables. In other words, the ground state energy of gluons in the presence of a static quark-antiquark pair can be more reliably determined from Wilson loops $W(r, t)$ constructed using smeared spatial links in the r -link paths on the initial and final time slices. A measure of how well the couplings to the higher levels are reduced is the effective

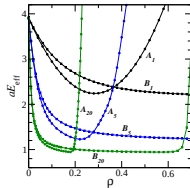


FIG. 3: The effective energy $aE_{\text{eff}}(r)$ defined in Eq. (76) for a static quark-antiquark pair separated by a distance $r = 10a$ for several levels of smearing $n_\rho = 1, 5$, and 20 against the smearing parameter ρ . Results were obtained on a 24^4 lattice using the Wilson gauge action with coupling $\beta = 6.2$. Curves labeled A_{n_ρ} indicate results using the spatially-isotropic three-dimensional version of the analytic stout link smearing scheme with $n_\rho = 1, 5$, and 20 levels, while the curves labeled B_{n_ρ} show the results for links smeared using Eq. (12) with the projection method of Ref. [15].

energy for a single time step, defined by

$$aE_{\text{eff}}(r) = -\ln \left(\frac{\tilde{W}(r, a)}{W(r, 0)} \right). \quad (76)$$

Under unexceptional circumstances for a lattice gauge action with a positive-definite transfer matrix, such as the Wilson action, reducing the excited-state contamination tends to lower the effective energy given above.

The effective energy defined in Eq. (76) was used to test the ability of the stout link smearing scheme to reduce excited-state contamination in gluonic operators. Results for $r = 5a$ on a 12^4 lattice using the Wilson gauge action with coupling $\beta = 5.7$ are shown in Fig. 2, and results for $r = 10a$ on a larger 24^4 lattice with $\beta = 6.2$ are shown in Fig. 3. The results are compared to those obtained with links smeared using Eq. (12) and the projection method of Ref. [15].

Both figures show that link smearing can dramatically reduce contamination from the high-lying modes of the theory. Consider first the results obtained using the spatially-isotropic three-dimensional version of the stout link smearing scheme ($\rho_{1k} = \rho$, $\rho_{4k} = \rho_{\text{rot}} = 0$). For a given number of smearing iterations n_ρ , the effective energy decreases initially as the smearing parameter ρ is increased from zero. Eventually an optimal value for the smearing parameter is reached at which the effective energy is minimized. This optimal value decreases as the

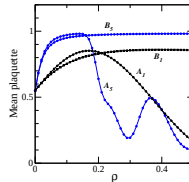


FIG. 4: The mean smeared plaquette against the smearing parameter ρ . These results were obtained using the Wilson gauge action on a 12^4 lattice with coupling $\beta = 5.7$. Curves labeled A_{n_ρ} indicate results obtained using the isotropic four-dimensional version of the analytic stout link smearing scheme with $n_\rho = 1$ and 5 levels, while the curves labeled B_{n_ρ} show the results with links smeared using Eq. (12) and the projection method of Ref. [15].

number of smearing levels n_ρ is increased, and the reduction in the effective energy at this optimal ρ value is substantial. Further increasing ρ beyond its optimal value then results in a sharply increasing effective energy. The rapidity of both the fall and rise of the effective energy about its minimum is more pronounced for larger n_ρ . Also, the minimum value decreases as n_ρ increases until eventually a saturation point is reached beyond which no additional reduction occurs.

The results obtained with links smeared using Eq. (12) and the projection method of Ref. [15] display essentially the same trends, except that each minimum in the effective energy is much broader. The increased sensitivity of the stout link smearing scheme to the parameter ρ is not surprising since ρ occurs inside an exponential function. However, it is important to note that both smearing methods produce nearly the same minimum values of the effective energy. The stout link smearing scheme is just as effective at reducing excited-state contamination in gluonic operators as the standard link fuzzing scheme in current use, although more careful tuning is necessary due to the increased sensitivity to the smearing parameter ρ . Note that this analytic link smearing method has already been successfully applied in computing the spectrum of torelon excitations[4].

The main purpose of using smeared links in lattice gauge and fermion actions is to reduce discretization effects caused by the ultraviolet modes in the lattice theory. Often such effects are dominated by large contributions from tadpole diagrams. A simple measure of how well a

link smearing scheme can reduce artifacts from tadpole diagrams is the mean smeared plaquette. A value of the mean smeared plaquette near unity indicates a substantial reduction of the tadpole contributions.

Results for the mean smeared plaquette on a 12^4 lattice using the Wilson gauge action with coupling $\beta = 5.7$ are shown in Fig. 4. For isotropic four-dimensional versions ($p_{\text{iso}} = p$) of both smearing methods, the mean smeared plaquette initially increases towards unity as ρ increases from zero until a maximum is reached. The two smearing methods produce nearly the same maximum value. As ρ is increased further, the mean smeared plaquette in the stout link scheme quickly begins to fall, whereas the standard smearing scheme changes little. As n_ρ initially increases from zero, the maximum value of the mean plaquette also increases, but eventually a saturation point is reached. Note that the maximum values for $n_\rho = 5$ are very near to unity, suggesting a dramatic reduction of tadpole contributions. In summary, the analytic stout link smearing scheme is observed to be just as efficacious as the standard smearing scheme in reducing discretization effects from the ultraviolet gluon modes, but with an increased sensitivity to the smearing parameter ρ .

VI. CONCLUSION

Link-variable smoothing is a crucial ingredient in constructing gluonic operators which have dramatically reduced mixings with the high frequency modes of the theory. Link smearing is also playing an increasingly important role in the construction of improved lattice actions. The lack of differentiability with respect to the underly-

ing link variables of standard smearing schemes prevents the use of efficient Monte Carlo updating methods based on molecular dynamics evolution.

A link smearing method which circumvents these problems was proposed and tested in this paper. The link smearing method is analytic everywhere in the finite complex plane and utilizes the exponential function in such a manner to remain within $SU(3)$, eliminating the need for any projection back into the group. Because of this construction, the algorithm is also useful for any Lie group. An efficient implementation of this smearing scheme, as well as the recursive computation of the force term describing the change of the action in response to a variation of the link variables, was described. Although the force field is related to the underlying link variables in a very complicated manner, each step in the recursive computation of the force field is actually straightforward, allowing a natural and efficient design in software. The smeared mean plaquette and the effective energy associated with the static quark-antiquark potential were used to show that no degradation in effectiveness is observed as compared to link smearing methods currently in use, although an increased sensitivity to the smearing parameter was found.

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A novel quark-field creation operator construction for hadronic physics in lattice QCD

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A new quark-field smearing algorithm is defined which enables efficient calculations of a broad range of hadron correlation functions. The technique applies a low-rank operator to define smooth fields that are to be used in hadron creation operators. The resulting space of smooth fields is small enough that all elements of the reduced quark propagator can be computed exactly at reasonable computational cost. Correlations between arbitrary sources, including multi-hadron operators can be computed *a posteriori* without requiring new lattice Dirac operator inversions. The method is tested on realistic lattice sizes with light dynamical quarks.

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I. INTRODUCTION

One of the goals of lattice calculations is to predict the low-energy hadron spectrum of confined quarks and gluons, starting solely from the QCD lagrangian. This approach to spectroscopy necessitates methods for measuring the two-point correlation functions of field operators with the selected quantum numbers under investigation. A complete understanding of the QCD spectrum must also include excitations of the mesons and baryons as well as exotic states. Describing the resonances seen in scattering experiments as states in QCD has long presented a challenge to lattice calculations as direct access to the matrix elements related to decay widths is usually missing in Euclidean formulations of field theory. The way to circumvent this is well known in principle: decay properties can be inferred from a detailed study of the dependence of the spectrum in a finite box on the volume of the box [1, 2]. The best means of making solid determinations of energy levels is to employ a large basis of operators and then use the variational method to find good approximations to energy eigenstates. As soon as states above threshold are to be explored, the basis should include operators that resemble multi-hadron systems where the constituents have well-defined momenta. Access to all elements of the quark propagator [3] relevant to long-range correlation enables these measurements and also enables more detailed investigations into physical observables whose accurate measurement has often eluded lattice practitioners. One example is the isoscalar meson sector, where mass determinations require the evaluation of disconnected diagrams as part of the two-point correlation function. As well as this ambitious list of requirements, precision data will be crucial for these calculations [4].

In a Monte Carlo calculation on the lattice, the physically relevant signal in a correlation function falls exponentially and is rapidly dominated by statistical fluctuations. Operators that create low-lying energy eigenstates quickly are invaluable and improve the quality of data extracted exponentially. The most useful tool at the lattice practitioner's disposal in building good creation operators is smearing. The best means of understanding the low-energy degrees of freedom of confinement must thus come from a smearing method that can address all these features; it must be numerically accessible, do a good job of projecting onto the lowest energy states and it must facilitate easy evaluation of correlation functions involving a broad set of creation operators, including those exciting multi-hadron states.

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A new generation of experiments devoted to hadron spectroscopy, including GlueX at Jefferson Lab, PANDA at GSI/FAIR and BES III intend to make measurements with unprecedented precision and in previously unexplored mass ranges and quantum-number sectors. The main aim of the *Hadron Spectrum Collaboration* is to extract from realistic lattice simulations predictions for masses, decay properties and relevant matrix elements in these domains.

Spectroscopy investigations are almost as old as non-perturbative Monte Carlo calculations on the lattice [5, 6]. As the first studies progressed, it was quickly realised that operators creating hadrons which are constructed directly from the fields in the lattice lagrangian have significant overlap with a large tower of states and that extracting the lightest elements of the spectrum was difficult. The solution was to build operators from “smeared” fields, where some linear operator was applied first to the quark degrees of freedom on the appropriate time-slice before the creation operator was formed. The aim of constructing a smeared field was to reduce the component of the fields close to the cut-off substantially, since these modes do not contribute significantly to long-range correlation functions. Initially, the techniques considered smeared fields that were not gauge covariant. Gauge-covariant schemes were introduced soon after [7, 8]. In particular, Ref. [8] introduced the Jacobi smearing algorithm, in which a lattice approximation to the three-dimensional gauge-covariant laplacian is constructed and applied iteratively to the field. This acts naturally as a long-wavelength filter on the modes constituting the quark field.

In this paper, a new method for smearing quark fields is described and tested in large-scale realistic simulations. In Sec. II, the formulation of the new algorithm is presented, including detail on its application to measurements of hadron two-point correlation functions and three-point functions where an arbitrary current is inserted. Sec. III presents the results of initial tests of the effectiveness of the method, and Sec. IV discusses practical issues that arise with the method and outlines some future research directions.

II. OPERATOR CONSTRUCTION

The energy of an eigenstate of the hamiltonian of a quantum field theory can be determined by computing the correlation function between creation and annihilation operators χ at Euclidean times t and t' :

$$C(t', t) = \langle \chi(t') \chi^\dagger(t) \rangle. \quad (1)$$

Inserting a complete set of eigenstates of the hamiltonian, such that $\hat{H}|k\rangle = E_k|k\rangle$, this correlation function decomposes into a sum of contributions from all states in the spectrum with the same quantum numbers as the source operators,

$$C(t', t) = \sum_k |\langle \chi | k \rangle|^2 e^{-E_k(t'-t)}. \quad (2)$$

In order to measure energies of low-lying states, it is crucial to construct operators that overlap predominantly with these light modes. This exposes the asymptotic behaviour of the correlator at earlier time-separations and enables more statistically accurate determinations of energies. There is a good deal of freedom in the construction of appropriate operators. Once the constraints of symmetries and temporal localisation are imposed, any function of the fields in the path integral can be used in principle.

Smearing is a well-established means of defining an initial step in the construction of creation operators. Rather than applying a creation operator to the fields directly, a smoothing function is applied first. The function should preserve as many symmetries as possible while effectively removing the presence of short-range modes, which make an insignificant contribution to the low-energy correlation function. In contemporary simulations, a popular form of this operation in theories where fermion fields couple to gauge bosons is the Jacobi smearing method [8].

Gauge-covariant quark smearings based on the lattice Laplacian start with the simplest representation of the second-order three-dimensional differential operator

$$-\nabla_{xy}^2(t) = 6\delta_{xy} - \sum_{j=1}^3 \left(\tilde{U}_j(x, t)\delta_{x+j, y} + \tilde{U}_j^\dagger(x-j, t)\delta_{x-j, y} \right), \quad (3)$$

where the gauge fields, $\tilde{U}$ may be constructed from an appropriate covariant gauge-field-smearing algorithm [9]. After defining the Laplace operator, a simple smearing can be written

$$J_{\sigma, n_\sigma}(t) = \left(1 + \frac{\sigma \nabla^2(t)}{n_\sigma} \right)^{n_\sigma}, \quad (4)$$

where σ and n_σ are tunable parameters that can be used to optimise projection onto the states under investigation. For large n_σ , this approximates the exponential of $\sigma \nabla^2$, *i.e.*

$$\lim_{n_\sigma \rightarrow \infty} J_{\sigma, n_\sigma}(t) = \exp(\sigma \nabla^2(t)). \quad (5)$$

The resulting exponential suppression of higher eigenmodes of the lattice Laplace operator means only a small number of the lowest modes contribute substantially to J .

This observation suggests the smearing operator can be approximated by forming an eigenvector representation, truncated to the lowest modes. Since the smearing operators used in lattice calculations are approximated well by low-rank constructions, a definition of smearing can be chosen to enforce this more absolutely. Let V_M be the vector space of scalar fields charged under the fundamental representation of the gauge group on a particular time-slice. V_M has rank $M = N_c \times N_c \times N_c \times N_c$ where N_c is the number of colours, and N_t, N_x and N_y are the extents of the lattice in the three spatial directions. Now define smearing to be a well-chosen operator of rank $N \ll M$. This class of operators will be called “distillation” operators. There is a substantial benefit to doing this: if the rank of the operator is sufficiently small, all elements of the propagation matrix from this space can be constructed at an affordable computational cost (which is proportional to the rank of the space, N). Consequently, correlation functions involving arbitrarily intricate hadron creation operators can be measured with a fixed inversion overhead.

Define the distillation operator on time-slice t as a product of an $M \times N$ matrix and its hermitian conjugate:

$$\square(t) = V(t)V^\dagger(t) \implies \square_{xy}(t) = \sum_{k=1}^N v_x^{(k)}(t)v_y^{(k)\dagger}(t). \quad (6)$$

The k^{th} column of $V(t)$ contains the k^{th} eigenvector of ∇^2 evaluated on the background of the spatial gauge fields of time-slice t , once the eigenvectors have been sorted by eigenvalue. This is the projection operator into V_N , the subspace spanned by these eigenmodes, so $\square^2 = \square$. When the number of eigenvectors included is the same as the dimension of V_M , *i.e.* $N = M$, the distillation operator becomes the identity, and fields acted upon are unsmeared.

The Laplace operator inherits many symmetries of the vacuum. It transforms like a scalar under rotations, is covariant under gauge transformation and is parity and charge conjugation invariant. If the action of one of these symmetries on the Laplace operator maps ∇^2 onto $\tilde{\nabla}^2$, then there is a unitary transformation, R on V_M such that

$$R\tilde{\nabla}^2 R^\dagger = \nabla^2. \quad (7)$$

This implies that if v is an eigenvector of ∇^2 then the eigenvectors of $\tilde{\nabla}^2$ are Rv . Considering the definition of the distillation operator given in Eq. 6, the transformed operator must then obey

$$R \square R^\dagger = \square, \quad (8)$$

and so correlation functions constructed using distilled fields have the same symmetry properties on the lattice as those constructed using Laplacian smearing methods.

A. Meson two-point correlation functions

Consider the momentum-projected creation and annihilation operators of an isovector meson, $\bar{u}\Gamma^A d$ and $\bar{d}\Gamma^B u$, where Γ acts in spin and color as well as coordinate space. Applying the distillation operator $\square$ onto each quark field, the creation operator at three-momentum $\vec{p}$ is written as

$$\chi_M^A(\vec{p}, t) = \bar{u}_\alpha(t)\square_{xy}(t) \cdot e^{-i\vec{p}\cdot\vec{y}} \Gamma_{\beta\delta}^A(t) \cdot \square_{z\omega}(t)d_\omega(t), \quad (9)$$

where there is an implied summation over repeated spatial indices. In a shorthand notation the correlation function can be written as

$$C_M^{(2)}(t', t) = \left\langle \bar{d}(t')\square(t')\Gamma^B(t')\square(t')u(t') \cdot \bar{u}(t)\square(t)\Gamma^A(t)\square(t)d(t) \right\rangle. \quad (10)$$

After evaluating the quark field path-integral and inserting the outer-product definition of the distillation operator $\square$ from Eq. 6, the correlator can be written

$$C_M^{(2)}(t', t) = \text{Tr} \left[\Phi^B(t') \tau(t', t) \Phi^A(t) \tau(t, t') \right], \quad (11)$$

where

$$\Phi_{\alpha\beta}^A(t) = V^\dagger(t) \left[\Gamma^A(t) \right]_{\alpha\beta} V(t) = V^\dagger(t) \mathcal{D}^A(t) V(t) S_{\alpha\beta}^A, \quad (12)$$

and

$$\tau_{\alpha\beta}(t', t) = V^\dagger(t') M_{\alpha\beta}^{-1}(t', t) V(t), \quad (13)$$

with M the lattice representation of the Dirac operator and where the quark spin indices, α, β of Φ and τ have been explicitly written. Φ has a well-defined momentum, while there is no explicit momentum projection in the definition of τ . Often, Φ can be decomposed into terms that act only within coordinate and color space $\mathcal{D}^A$ and only within spin space S^A . Note that Φ and τ are square matrices of dimension $N \times N_\sigma$ where N_σ is the number of components in a lattice Dirac spinor. Therefore it requires just $N \times N_\sigma$ operations of the inverse of the fermion matrix on a vector in order to compute all elements of τ , the “perambulator”. Notice also that the choice of source and sink operators is entirely independent of the computation of τ ; any source and sink operators can be correlated *a posteriori* once all elements of the τ matrix have been computed and stored. The method straightforwardly extends to the determination of correlation functions for mesons composed of different, non-degenerate quark flavors.

In the determination of an isoscalar meson correlation function, evaluation of disconnected terms is required. The disconnected diagram can be similarly represented once distilled fields are used in creation and annihilation operators. Such a term would comprise two separate traces over the distillation space:

$$C_M^{(2, \text{disc})}(t', t) = \text{Tr} \left[\Phi^A(t) \tau(t, t) \right] \text{Tr} \left[\Phi^B(t') \tau^\dagger(t', t') \right]. \quad (14)$$

An exact determination of this expression requires computing $\tau(t, t)$ for all time-slices t .

Since distilled single-particle operators can be projected onto definite momentum at both source and sink, multi-meson correlators can also be constructed using creation and annihilation operators of the form

$$\chi_{MM}(\vec{q} = \vec{p}_1 - \vec{p}_2; t) = \chi_M(\vec{p}_1, t) \chi_M(-\vec{p}_2, t), \quad (15)$$

where p_1 and p_2 are the three-momenta of the single particle operators in Eq. 9. After integration over the quark fields, the resulting diagrams can again be computed by taking traces over products of construction operators in the distillation space with the perambulators. The precise structure of these functions depends on the quark flavor content of the source and sink operators, so details are not presented here. As before, these matrices are small compared to the dimension of the space of quark fields, and once the quark propagation is encoded in the perambulator, these multi-hadron correlation function can be computed.

B. Baryon two-point correlation functions

The factorization technique can also be applied to baryons. To illustrate the concepts involved, consider just the isospin-1/2 sector although the technique generalizes naturally to other baryon multiplets. An annihilation operator involving displacements as well as coefficients in spin space is written:

$$\chi_B(t) = e^{abc} S_{\alpha_1 \alpha_2 \alpha_3} (\mathcal{D}_1 \square d)_{\alpha_1}^a (\mathcal{D}_2 \square u)_{\alpha_2}^b (\mathcal{D}_3 \square u)_{\alpha_3}^c(t), \quad (16)$$

where the color indices of the quark fields acted upon by the displacement operators $\mathcal{D}_i$ are contracted with the antisymmetric tensor, and repeated spin indices are summed. After integration over quark fields, the correlation function

$$C_B^{(2)}(t', t) = - \left\langle \chi_B(t') \bar{\chi}_B(t) \right\rangle \quad (17)$$

can be factored into perambulator terms, Eq. 13 as well as creation and annihilation operators, where the latter can be written as

$$\Phi_{\alpha_1 \alpha_2 \alpha_3}^{(i, j, k)}(t) = e^{abc} \left(\mathcal{D}_1 v^{(i)} \right)^a \left(\mathcal{D}_2 v^{(j)} \right)^b \left(\mathcal{D}_3 v^{(k)} \right)^c(t) S_{\alpha_1 \alpha_2 \alpha_3}. \quad (18)$$

The spin terms factor from the antisymmetric contraction of the vectors v .

There are two terms in the resulting correlation function that involve tensor contractions of the creation and annihilation operators with the perambulators

$$C_B^{(2)}[\tau_d, \tau_u, \tau_u](t', t) = \Phi^{(i,j,k)}(t') \tau_d^{(i,\bar{i})}(t', t) \tau_u^{(j,\bar{j})}(t', t) \tau_u^{(k,\bar{k})}(t', t) \Phi^{(l,\bar{l},\bar{k})^*}(t) \\ - \Phi^{(i,j,k)}(t') \tau_d^{(i,\bar{i})}(t', t) \tau_u^{(j,\bar{j})}(t', t) \tau_u^{(k,\bar{k})}(t', t) \Phi^{(l,\bar{l},\bar{k})^*}(t), \quad (19)$$

where there is an implicit tensor contraction over the internal spin indices, and where the quark labels d and u are used to denote the corresponding flavors of the quark perambulator terms in Eq. 13.

Similar to the meson case, the choice of source and sink operators is independent of the computation of the perambulators τ_d and τ_u . The computation of these matrices can also be shared with computation of meson correlators. A baryon correlation function can be evaluated *a posteriori* using the contractions of the vectors with displacements in Eq. 18 which can also be shared among the source and sink operators. These contractions do not involve spin components, thus making the storage of Φ manageable.

C. Meson three-point correlation functions

A generic meson three-point function is written

$$C^{(3)}(t_f, t, t_i) = \left\langle \left(\bar{d} \square \Gamma^B \square u \right) (t_f) \cdot \left(\bar{u} \Gamma u \right) (t) \cdot \left(\bar{u} \square \Gamma^A \square d \right) (t_i) \right\rangle, \quad (20)$$

where there is no smearing in the operator on timeslice t . The completely connected Wick contraction can be expressed as

$$C_{\text{conn}}^{(3)}(t_f, t, t_i) = \text{Tr} \left[\Phi^B(t_f) J(t_f, t, t_i) \Phi^A(t_i) \gamma_5 \tau^t(t_f, t_i) \gamma_5 \right] \quad (21)$$

where the “generalized perambulator” is defined by

$$J_{\alpha\beta}(t_f, t, t_i) = V^\dagger(t_f) \left(M^{-1}(t_f, t) \Gamma(t) M^{-1}(t, t_i) \right)_{\alpha\beta} V(t_i). \quad (22)$$

In general, a disconnected term may also appear; this can be factorised as the product of a two-point function (evaluated according to the algorithm of Sec. II A) and a disconnected insertion. Since the disconnected trace does not involve distilled fields, the evaluation of this insertion is not discussed further at this point. To compute the connected three-point correlation function, there are two distinct sets of inversions needed. The first is from the distillation vectors at the source timeslice t_i and the second from the sink timeslice t_f . The generalized perambulator of Eq. 22 is then constructed by contracting the two solutions from these two sets of inversions. The current insertion is encoded in the choice of operator Γ . Any momentum insertion at the current operator involves a Fourier transform on each time-slice t .

D. Baryon three-point correlation functions

Baryon three-point correlation functions can also be expressed in terms of the generalized perambulators of Eq. 22. Consider a three-point correlation function where a baryon is created on time-slice t_i , then is acted on by a current operator on time-slice t and is subsequently annihilated at t_f :

$$C_B^{(3)}(t_f, t, t_i) = - \left\langle B(t_f) \cdot \left(\bar{u} \Gamma u \right) (t) \cdot \bar{B}(t_i) \right\rangle. \quad (23)$$

After quark-field integration, this can be written as a sum of both a connected three-point contribution and the product of a disconnected insertion and a two-point correlator. As before, the discussion here is restricted to isospin-1/2 light baryons although the technique is quite general. Note that the disconnected insertion involves unmeasured fields and must be evaluated through some other technique. The two-point contribution follows from Sec. II B. For an up-quark insertion in the correlation function

$$C_{\text{conn}}^{(3)}(t_f, t, t_i) = - \epsilon^{abc} S_{\alpha_1 \alpha_2 \alpha_3} \bar{S}_{\beta_1 \beta_2 \beta_3} \\ \left\langle \left((\bar{D}_1 \square d)_{\alpha_3}^a (\bar{D}_2 \square u)_{\alpha_2}^b (\bar{D}_3 \square u)_{\alpha_1}^c \right) (t_f) \right. \\ \left. \cdot \left(\bar{u} \Gamma u \right) (t) \cdot \left((\bar{d} \square \bar{D}_1)_{\beta_3}^a (\bar{u} \square \bar{D}_2)_{\beta_2}^b (\bar{u} \square \bar{D}_3)_{\beta_1}^c \right) (t_i) \right\rangle, \quad (24)$$

there are four Wick contractions after quark-field integration. The up-quark generalized perambulator from Eq. 22 is denoted as

$$\tilde{\tau}_U(t_f, t, t_i) = V^\dagger(t_f) M_U^{-1}(t_f, t) \Gamma(t) M_U^{-1}(t, t_i) V(t_i) \quad (25)$$

and similarly for the down quark. Since the quarks in the insertion come in pairs, the connected three-point correlator can be written as the sum of two two-point correlation functions with each up-quark perambulator substituted with the corresponding generalized perambulator in turn;

$$C_u^{(3)}(t_f, t, t_i) = C^{(2)}[\tau_D, \tau_U; \tilde{\tau}_U](t) + C^{(2)}[\tau_D, \tilde{\tau}_U, \tau_U](t). \quad (26)$$

The down-quark contribution can be written as

$$C_d^{(3)}(t_f, t, t_i) = C^{(2)}[\tilde{\tau}_D, \tau_U, \tau_U](t). \quad (27)$$

This kind of construction is a generic feature of the three-point functions.

As in the previous two- and three-point correlator examples, the choice of source and sink operators is independent of the perambulator and generalized perambulator computations. These latter generalized perambulators can be shared among meson three-point correlator calculations. In addition, for degenerate quarks, only one generalized perambulator need be computed, independent of the flavor of the quark line where the current insertion is to occur. Thus, the overall computation of distinct three-point correlation functions has a manageable computational cost.

III. TEST RESULTS

To test the efficacy of the method, a first study of meson and baryon two-point correlators was performed. In addition, the method is demonstrated for a two-meson system. The background gauge fields were generated by the *Hadron Spectrum Collaboration* [10, 11]. They are $16^3 \times 128$ and $20^3 \times 128$ anisotropic lattices, with the ratio of spatial to temporal lattice spacings $a_s/a_t = 3.5$ and 2+1 flavors of dynamical quarks. The bare light-quark mass is $a_t m_q = -0.0840$, and when the Ω baryon mass is used to set the scale, the corresponding pion mass is 383 MeV. The valence-quark action is the same as the light-dynamical-quark action.

The numerical techniques for the iterative solution of the sparse linear system for the Sheikholslami-Wohlerl improved discretisation of the Dirac equation are described in Ref. [10]. In particular, the EigCG inverter [12] was used for the solution of these linear systems. This method uses eigenvector deflation to accelerate convergence. The first few inversion calls are needed to construct a sufficient basis, and subsequent inversions are accelerated by a factor of about three or more compared to the undeflated solver. For the lattices considered here, this asymptotic rate is found for $N \geq 4$.

A. The profile of the distillation operator

The distillation operator is written as a low-rank projection operator into a vector space of smooth fields. As such, there is little physical intuition to suggest it resembles more traditional smearing algorithms in generating a field that is roughly localized in some confining region while still remaining smooth. To examine the properties of the operator, its spatial distribution was computed. Defining

$$\Psi(r) = \sum_{x,t} \sqrt{\text{Tr}(\Box_{x,x+r}(t) \Box_{x+r,x}(t))}, \quad (28)$$

means Ψ is a gauge-invariant measure of the degree to which a field is smeared by the application of the distillation operator. This observable was measured for various values of N and for a range of different off-set vectors r on a small ensemble of the 20^3 spatial lattices and the results are presented in Figure 1. The data clearly shows that for N between 8 and 64 the distillation operator is very well described by a gaussian wavefunction. As N increases, so the radius of this distribution reduces. This behaviour is to be anticipated since when $N = M$ the distillation operator is the identity and has no spatial extent at all. The measurements presented in Fig. 1 also clearly illustrate the distillation operator has good rotational symmetry, indicating it is only very weakly influenced by dynamics at the lattice cut-off. The artifacts seen at larger values of r in the very broad $N = 8$ distribution arise solely from the finite lattice volume used in these measurements.

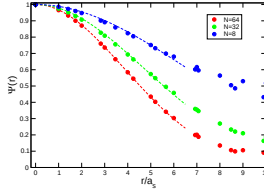


FIG. 1: The spatial distribution of the distillation operator constructed from the lowest eigenvalues of the Laplace operator on the lattice. Data is from a 20^3 spatial volume and statistical fluctuations are much smaller than plot symbols. The dashed lines are fits to a gaussian distribution.

Name	Construction
γ_i	$\bar{\psi}\gamma_i\psi$
$a_0 \times \nabla_{T_1}$	$\bar{\psi}\nabla_i\psi$
$\rho \times \mathbb{D}_{T_1}$	$\epsilon_{ijk} \epsilon_{lmn} \bar{\psi}\gamma_l\nabla_m\nabla_n\psi$
$\pi \times \mathbb{B}_{T_1}$	$\epsilon_{ijk}\bar{\psi}\gamma_n\nabla_j\nabla_k\psi$

TABLE I: Derivative-based meson operators used in these tests. $\bar{\nabla}$ is the one-link discretised covariant derivative

B. Meson correlation functions

In this section, connected meson correlation functions formed using the derivative-based operators presented in Refs. [13, 14] are shown, with particular focus on the T_1^- channel. Fig 2 shows plots of the effective mass, defined by

$$m_{\text{eff}}(t) = -\frac{1}{\delta t} \ln \left(\frac{C(t + \delta t)}{C(t)} \right) \quad (29)$$

with $\delta t = 5a_0t$, for four diagonal correlators while varying the number of distillation vectors. Also shown are the corresponding effective masses computed using the traditional point-to-all propagator construction with both unsmeared and Jacobi-smeared sources. The four operators used are defined in Table I. In all cases, gauge links in the operators are stout-smeared [9]. 90 configurations of the $16^3 \times 128$ lattice were used for this comparison.

A variational analysis [15, 16] of matrices of two-point correlation functions has proven to be a powerful method in extracting the spectrum and matrix elements of excited states [17]. Consequently, cost estimates in terms of the computing time required to form such a matrix of correlators and not just any single correlator should be considered. As a measure of the relative computational demands of distillation versus the traditional point-to-all approach, note that in the point-to-all approach, ten forward propagators, generated by applying $1, \nabla_{i=2,3,x}, \nabla_i \nabla_j|_{i \neq j}$ to a point source at the origin are required to produce the set of T_1^- correlation functions defined above. Each of these sources then requires twelve inversions of the Dirac matrix ($N_c \times N_p$). To compare with the distillation methods, note that approximately the same total inversion cost arises from using 32 distillation vectors since

$$(N_{\text{vec}} = 10) \times (N_c = 3) \times (N_p = 4) = 120 \approx (N = 32) \times (N_p = 4) = 128. \quad (30)$$

In the correlation function for the $\bar{\psi}\gamma_i\psi$ operator, it is clearly seen that increasing the number of vectors causes the effective mass to approach that of the unsmeared point-to-all correlation function, since $\sum_{k=1}^M v^{(k)} v^{(k)} = 1$. The

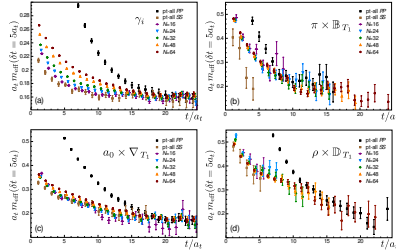


FIG. 2: Effective masses (five timeslice shift) for diagonal correlators in the T_1^{--} channel computed on $16^3 \times 128$ lattices. The operator construction is shown in table I. Correlators are constructed using distilled quark fields with $N \in \{16, 24, 32, 48, 64\}$ vectors and using point-to-all methods with standard gauge-invariant quark smearing (SS, brown) and without (PP, black).

statistical variance is observed to decrease rapidly as the number of vectors is increased. For this range of N it is always smaller than the point-to-all correlators. A similar but less rapid convergence of the signal is observed in the $a_0 \times \nabla T_1$ channel.

On the other hand, the $\pi \times \mathbb{B} T_1$ signal appears to be essentially unchanged over this range of vector number but the noise reduces considerably with increasing vector number. The $\rho \times \mathbb{D} T_1$ channel shows similar behaviour. Note that these operators sample non-trivial spatial dependence, and this may require a larger space of eigenvectors. The rapid scaling of noise with number of vectors can be quantified as follows: consider the correlator noise-to-signal ratio r on a timeslice as a function of the number of vectors used, N and model this behaviour with

$$r = a + \frac{b}{N^p}. \quad (31)$$

Fitting the parameters to best reproduce the Monte Carlo data from 450 configurations and averaging over many time-slices gives the following estimates for the scaling exponent

$$p(\gamma_i) = 0.8(1); \quad p(a_0 \times \nabla T_1) = 1.1(2); \quad p(\pi \times \mathbb{B} T_1) = 1.3(3); \quad p(\rho \times \mathbb{D} T_1) = 1.8(2).$$

In all cases, the noise reduction is better than the behaviour expected from simple statistical scaling, $p = 0.5$.

As a realistic test of the efficacy of distillation in the extraction of an excited-state spectrum, a matrix of correlation functions using twelve operators with T_1^{--} quantum numbers was constructed. In Fig. 3 the low-lying spectrum extracted as a function of number of vectors used is shown. Clearly, even with a relatively small number of vectors this method can be used to construct two-point correlation matrices from which the spectrum of low-lying states can be extracted with some precision. The masses, up through the fourth excited state, become more consistent with increasing number of vectors.

Now consider the issue of the scaling of this method with increasing lattice volume. Fig. 4 shows the effective masses of correlators computed on both 16^3 and 20^3 lattices, the spatial-volume ratio here being very close to 2 ($\frac{20^3}{16^3} = 1.95$). It is evident that the correlation function with N vectors on the 16^3 volume is essentially the same as the corresponding case with $2N$ vectors on the 20^3 lattice. Note that although the number of inversions for the same

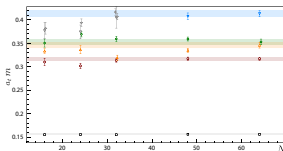


FIG. 3: T_1^{--} variational mass spectrum as a function of number of eigenvectors used. The error bars are purely statistical and do not take account of possible systematic errors due to fitting.

signal is doubled when the lattice volume increases from 16^3 to 20^3 , there is a reduction in the noise of approximately a factor of two.

A strikingly simple explanation for this effect comes through consideration of the eigenvalues belonging to these eigenvectors of the Laplacian operator, $\nabla^2 v^{(k)} = -\lambda_k v^{(k)}$. Fig. 5(a) shows the eigenvalues measured on 16^3 and 20^3 lattices. One could consider choosing the rank of the distillation operator by using all eigenvectors with eigenvalue below a certain cut-off. It is clear from the plot that on a larger volume this will require more eigenvectors. This is displayed more directly in Fig. 5(b) where the number of eigenvectors with eigenvalue less than a cut-off is plotted. Approximately twice as many eigenvectors in the 20^3 case are required to have the same cut-off as in the 16^3 case. It appears from this analysis that the meson correlator signal is effectively determined by the cut-off in eigenvalue used to construct the distillation space. This is equivalent to using a Heaviside approximation to smearing: $J_\lambda(t) = \Theta(\lambda + \nabla^2(t)) = \sum_{k=1}^M \Theta(\lambda - \lambda_k) v^{(k)}(t) v^{(k)\dagger}(t) = \sum_{k=1}^{N(\lambda)} v^{(k)}(t) v^{(k)\dagger}(t)$.

C. Baryon correlation functions

Baryon correlation functions were evaluated using the displaced-quark operators described in Refs. [18, 19] and employed in spectrum studies Refs. [20, 21]. The distillation method is demonstrated on four nucleon G_{1g} operators, two of which are local to a single site with the remaining two having a singly displaced quark field. The corresponding 4×4 matrix of correlators was computed on 310 configurations of the $16^3 \times 128$ lattice. Fig. 6 shows the effective-mass plots for the diagonal correlators. A clear trend is observed, similar to that seen for the mesons, of the correlator having larger excited-state contribution but smaller statistical fluctuations as the number of eigenvectors is increased. Modelling the correlator noise-to-signal ratio with Eq. 31 gives best-fit exponents $p \sim 1.1(2)$, showing again that increasing the number of vectors decreases the noise considerably faster than simple statistical scaling. The variational method [15, 16] was used to extract the masses of the lowest two states in the G_{1g} spectrum. Fig. 7 shows the dependence of the effective masses of these states on the number of eigenvectors used to form the distillation operator. Consistent masses are found, with an increase in statistical precision as the number of vectors is increased. Fig. 8 shows the diagonal effective masses computed on 16^3 and 20^3 lattices. The similarity between the signals found before and after doubling the number of vectors is not as obvious here as in the meson case.

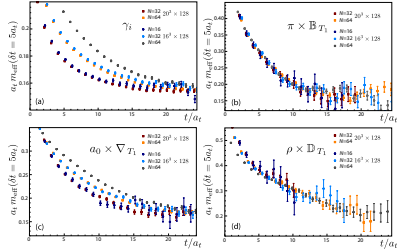


FIG. 4: Effective masses (five timeslice shift) for diagonal correlators in the T_1^- channel on 16^3 and 20^3 spatial volumes. Operator construction shown in table I.

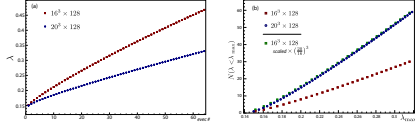


FIG. 5: (a) Eigenvalues of the laplacian on 16^3 and 20^3 lattices. (b) Number of eigenvectors with eigenvalue below $\lambda_{\max}$ on 16^3 and 20^3 . Also shown is the 16^3 data scaled by a factor of $(\frac{16}{20})^3$.

D. Multi-meson correlation functions

One of the main motivations for defining distillation methods is to enable efficient evaluation of correlation functions of creation operators which resemble multi-hadron states where the component hadrons have definite momentum. Reliable and efficient means of including these states in the operator basis are important for addressing scattering and resonance properties. The simplest case was tested, which comprises the isospin-two two-pion state formed using Eq. 15 with the single particle operators $M(\vec{p}_{rel})$ constructed using the local operator $\bar{u}\gamma_5 d$. For this example, consider the two-pion operator

$$\chi_{MM}(|\vec{p}_{rel}|^2, t) = \sum_{k \in \mathcal{R}(\vec{p}_{rel})} \chi_M(k, t) \chi_M(-k, t) \quad (32)$$

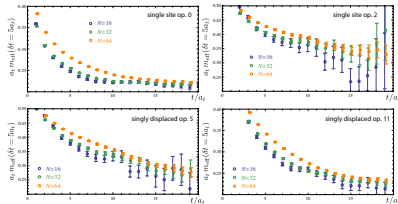


FIG. 6: Effective-mass plot for two single-site and two singly displaced operator correlators in the nucleon G_{10} channel.

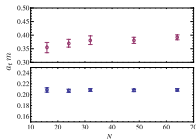


FIG. 7: The fitted energies for the first-excited (top) and ground state (bottom) as functions of N .

with total momentum zero, and the single-particle operators to have relative momentum $\vec{p}_{rel}$. The sum is over the set of momentum vectors $\mathcal{R}(\vec{p}_{rel})$ related to $\vec{p}_{rel}$ by lattice rotations, to create a state that transforms trivially under rotations and so the operator forms an A_1 irreducible representation of the cubic group. In this test, all values of $|\vec{p}_{rel}|$ between 0 and $2 \times \frac{\pi}{L}$ were considered, corresponding to $\vec{p}_{rel} = \frac{2\pi}{L}\vec{n}$ with $\vec{n} \in \{(0, 0, 0), (1, 0, 0), (1, 1, 0), (1, 1, 1), (2, 0, 0)\}$.

A 5×5 correlation matrix is formed between all of these two-pion operators which is diagonalized by solving the generalized eigenvalue problem. The effective mass of the lowest three principal correlation functions measured on 100 configuration is shown in Fig. 9. Clear indication of a signal for the second excited state is seen, even with this relatively low level of statistics. Note that when using the conventional point-to-all method, the source operator cannot be projected onto definite momentum so the correlation matrix described above cannot be constructed in practice.

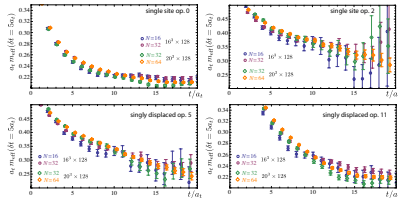


FIG. 8: Effective-mass plot for two single-site and two singly displaced operator correlators from two volumes: $16^3 \times 128$ and $20^3 \times 128$

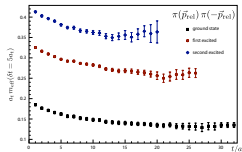


FIG. 9: Effective masses (five-timeslice shift) for the first three principal correlators in the isospin-two $\pi(\vec{p}_{\text{rel}})\pi(-\vec{p}_{\text{rel}})$ channel (A_1^{++}) computed on $16^3 \times 128$ lattices. The 5×5 correlation matrix is constructed from $|\vec{p}_{\text{rel}}|^2 \in [0, 4]$. Correlators constructed through distillation with $N = 64$ vectors on 100 configurations.

IV. DISCUSSION AND FUTURE DIRECTIONS

The results of Sec. III demonstrate clearly that the light hadrons can be studied using operators that excite only a very small number of extended degrees of freedom on each time-slice. The restricted set of modes needed to create a hadron means all elements of quark propagation needed to construct the correlation function can be measured directly. This does not preclude using stochastic methods [3] to reduce computational costs.

In the straightforward recipe described in this paper, the distillation space was constructed from the lowest eigenvectors of the lattice three-dimensional Laplace operator on a time-slice. This definition is not unique and more experimentation with different choices of distillation space might prove useful. Also, the operator used in this work is diagonal in the quark spin index. Alternative choices of the distillation space might include non-trivial spin structure, using for example the vector space spanned by the lowest modes of the three-dimensional Dirac operator.

Recall that in the simplest implementation tested here when $N = M$, the distillation operator becomes the identity and no smearing is performed. This is clearly an undesirable feature that is circumvented by the obvious variation on the method to include different weights for the eigenmodes. This more general distillation operator is then

$$\mathcal{J}(t) = \sum_{k=1}^N v^{(k)}(t) f(\lambda_k(t)) v^{(k)\dagger}(t). \quad (33)$$

Including weight functions on the eigenvectors can be used to suppress more rapidly fluctuating contributions when many eigenvectors are employed. Using exponential weights leads to increasingly accurate approximations to the conventional smearing algorithm of Eq. 4.

No increase in the number of eigenvectors would be expected as the continuum limit is taken at a fixed physical volume since, as is seen in the data of Sec. III A, the smooth distilled modes are almost completely decoupled from the cut-off dynamics. This expectation is yet to be tested numerically. One substantial possible drawback with the method is the need to increase the rank of the distillation operator as the three-dimensional lattice volume increases. The corresponding linear increase in the cost of solving the Dine equation leads to an algorithm for spectroscopy that scales like $\mathcal{O}(V^2)$. It appears from the results we have presented that with the required increase in rank with increasing volume one also obtains a decrease in statistical fluctuations. The possible solution to the issue of volume scaling may be to combine the method with a suitably designed stochastic estimation scheme. This is under investigation.

Note that the method does not give direct access to all elements of the quark propagator, only those relevant to low-energy spectroscopy. This would be a limitation in calculations involving isoscalar operator insertions, where disconnected loops with all momentum components of the quark field are needed. A well-known example of such a computation would be the evaluation of the strangeness content of the nucleon.

V. SUMMARY

In this paper, a simple new quark-field smearing has been proposed and tested in numerical investigations using QCD gauge fields with $N_f = 2 + 1$ dynamical flavors. There is substantial freedom in the construction of creation operators for hadrons. This has been exploited in this work to define an algorithm that applies a low-rank projection operator to the quark fields of the path integral on each time-slice in order to define smooth modes for subsequent operator construction. This makes affordable the evaluation of all information about quark propagation required to measure distilled hadronic correlation functions. Once the propagation matrix has been evaluated, correlations between arbitrary choices of creation and annihilation operators can be determined without requiring any further inversions. The theoretical framework needed to extend the scope of measurements using this technology to include matrix elements was described briefly.

Some initial tests of the method demonstrate that restricting quark fields to lie in a very low-dimensional space of smooth fields does not substantially degrade the quality of Monte Carlo evaluations of correlation functions and in many cases, statistical accuracy is enhanced. The most substantial drawback, still to be overcome, is the growth in cost of the algorithm as the lattice volume in physical units increases. Since the method is concerned with long-distance physics, no extra difficulties in approaching the continuum limit are anticipated, but this would need to be confirmed by practical testing.

The method is currently in its infancy and a number of research directions are outlined in this paper. The primary focus of the development efforts of this collaboration is to include stochastic evaluation of correlation functions in an optimal way and results will appear soon. The collaboration will use the techniques outlined in this paper in a range of spectroscopy determinations in the near future.

Acknowledgements

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P04: 高动量强子新夸克涂抹

主题归属	规范理论、格点与算符基础
底稿	arXiv:1602.05525
arXiv / DOI	1602.05525
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阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P04-p001 – P04-p017

Novel quark smearing for hadrons with high momenta in lattice QCD

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Hadrons in lattice QCD are usually created employing smeared interpolators. We introduce a new quark smearing that allows us to maintain small statistical errors and good overlaps of hadronic wave functions with the respective ground states, also at high spatial momenta. The method is successfully tested for the pion and the nucleon at a pion mass $m_\pi \approx 295$ MeV and momenta as high as 2.8 GeV. We compare the results obtained to dispersion relations and suggest further optimizations.

PACS numbers: 12.38.Gc, 14.40.-n, 14.20.-c, 13.60.-r

I. INTRODUCTION

Lattice QCD simulations predict an ever increasing number of observables that are relevant to particle and hadron physics phenomenology. These results are usually extracted from expectation values of n -point functions at large Euclidean time separations. Due to the decay of these functions with time, statistical noise over signal ratios increase exponentially as time separations are taken large (with the notable exception of pseudoscalar mesons at zero momentum). Fortunately, there exists some freedom in the construction of interpolators for the creation of mesonic and baryonic states. Employing interpolators that resemble the spatial structure of the ground state wave function enables asymptotic results to be extracted at time separations where the signal over noise ratio is still large.

Many applications nowadays demand hadrons that carry momentum. For instance, pushing the calculation of semileptonic decay form factors for $B \rightarrow \pi \ell \bar{\nu}_\ell$ or $\Lambda_b \rightarrow p \ell \bar{\nu}_\ell$ [1] towards small virtualities requires spatial momenta of the size of the mass difference between the B meson and the pion or between the Λ_b baryon and the proton, respectively. Another important type of application are parton distribution functions (PDFs) and their generalizations, in particular transverse momentum dependent parton distribution functions (TMDs), or Wigner distributions as a whole. Also these quantities are extracted from matrix elements of the type $\langle H(\mathbf{p}') | \mathcal{O} | H(\mathbf{p}) \rangle$, where $H(\mathbf{p})$ is a hadron state with momentum $\mathbf{p}$. The operator $\mathcal{O}$ cannot have an extent in Minkowski-time on the Euclidean lattice. Therefore, in order to extrapolate to light front kinematics for an inherently non-local operator, one has to work in a frame of reference where hadrons carry high spatial momenta $\mathbf{p}, \mathbf{p}'$. This fact has been known for quite some time in

the context of lattice calculations of TMDs [2–4] and is illustrated very clearly in recent work on these distributions in the pion [5]. For the same reason fast hadrons on the lattice are highly desirable in a new scheme proposed to relate quasi parton distributions to light front distributions in a controlled manner [6, 7]. First lattice computations in this direction have started [8, 9]. Earlier suggestions to compute quasi distribution amplitudes in position space [10–12] equally require pions or nucleons at high momenta. Unfortunately, up to now no satisfactory techniques for hadrons carrying high momenta existed to suppress excited state contributions.

To be more specific, a two-point function is given as

$$C_H(t) = \langle H(t) H^\dagger(0) \rangle = \sum_{j>0} \frac{|\langle j | \hat{H}^\dagger | 0 \rangle|^2}{2E_{H,j}} e^{-E_{H,j}t}, \quad (1)$$

where $E_{H,j}$ denotes the j th energy level within the tower of states created by the interpolator $H^\dagger$. Obviously, the contribution of the j th excited state is suppressed relative to the ground state not only by $\exp[-(E_{H,j} - E_{H,0})t]$ but also by the ratio $|\langle j | \hat{H}^\dagger | 0 \rangle|^2 / |\langle 1 | \hat{H}^\dagger | 0 \rangle|^2$; increasing the ground state overlap factor $|\langle 1 | \hat{H}^\dagger | 0 \rangle|$, relative to $|\langle j | \hat{H}^\dagger | 0 \rangle|$ for $j > 1$, results in an additional suppression of excitations.

Reducing excited state overlaps by employing extended interpolators was first pursued in computations of the glueball spectrum. In this case the gauge links within the corresponding interpolators can be iteratively “APE smeared” [13], “fuzzed” [14] or “HYPER smeared” [15], to better approximate the (smooth) ground state wave function, see also Ref. [16]. This gauge link smearing was subsequently generalized to iterative smearing of quark fields within interpolators that create mesonic and baryonic states, in particular gauge covariant Wuppertal (i.e. Gauß) smearing [17–19], hydrogen-like smearing [19] as well as Jacobi smearing [20, 21]. Additionally, in Refs. [18, 19, 22] APE smearing was employed for the spatial gauge transporters within the quark smearing while in Ref. [22] linear combinations of different levels of Wuppertal smearing were utilized.

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Since Gaussian smearing functions may not be optimal for creating, e.g., p -waves, even when adding derivatives to the interpolator, iterative smearing was later-on combined with displaced quark sources (fuzzing) in Ref. [23], a generalization of which was suggested in Ref. [24]. Finally, in Ref. [25] “free form smearing”, folding Gaussian smearing with an arbitrary function in a gauge covariant way, was invented. Preceding and in parallel to gauge covariant iterative smearing functions, gauge fixed sources have been utilized: wall sources for zero [26] and non-zero momentum [27], box [28] sources, Gaussian “shell sources” [29] and sources with nodes [30]. These gauge fixed methods and free form smearing share the disadvantage that smearing the sink requires all quark positions to be summed over individually, turning this prohibitively expensive. Having identical source and sink interpolators, however, is very desirable as only this guarantees the positivity of the coefficients of the spectral decomposition Eq. (1) and thus the convexity of two-point functions. For completeness, we also mention the “distillation” (or Laplacian-Heaviside) method of Ref. [31] since this is closely related to gauge covariant smearing.

Large momenta increase the energy of the state and result in faster decaying two- and three-point functions and, therefore, in inferior noise to signal ratios. Moreover, as we shall see, excited state suppression becomes far less effective when using conventional quark smearing methods. Some attempts have been made [32, 33] to introduce an anisotropy into Wuppertal smearing [17, 18], aiming at Lorentz contracting the interpolating wave function according to the boost factor $1/\gamma = m/E(\mathbf{p})$, along the direction of the spatial momentum $\mathbf{p}$. However, this did not result in the ground state enhancement that one would have hoped for. Here we will argue and demonstrate that to achieve satisfactory results at high momenta, additional phase factors need to be incorporated into quark smearing functions.

This article is organized as follows. First, in Sec. II, we discuss the basic idea behind the new class of smearing functions that we introduce. Then, in Sec. III we are more specific, modifying Wuppertal smearing as a generic example and suggest further improvements. In Sec. IV we discuss our simulation parameters and expectations for the nucleon and pion energies. After the stage is set, in Sec. V we investigate the feasibility of the method in a realistic numerical study, optimize the smearing parameters and pursue a comparison between the new and the conventional method. Finally, we study the pion and nucleon dispersion relations, before we conclude.

II. MOMENTUM SMEARING: THE BASIC IDEA

As discussed above, quark smearing within hadronic sources or sinks is essential in lattice simulations to increase the overlap with the desired physical state, reflecting the fact that hadrons are extended objects, rather

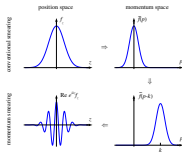


FIG. 1. Conventional smearing versus momentum smearing for the example of a Gaussian wave function in $d = 1$ spatial dimensions. The momentum k shifts the centre of the distribution in momentum space, resulting in an oscillatory behaviour in position space.

than pointlike. A smearing operator F is diagonal in time, trivial in spin and acts on the position and colour indices of quark fields:

$$(Fq)_x = \sum_{y \in (xZ)^d} f_{x-y} G_{xy} q_y, \quad (2)$$

where f is a scalar function, G is a gauge covariant transporter, which in the free case will be a unit matrix in colour and position space, and d is the number of spatial dimensions, usually $d = 3$. Note that the field q_x is usually periodic in $x = y$ on the lattice, whereas f_{x-y} need not be periodic in $x = y$. In the free case, the convolution Eq. (2) becomes a product in Fourier space

$$\sum_{x \in (xZ)^d} e^{ip \cdot x} (Fq)_x = \hat{f}(\mathbf{p}) \hat{q}_{\mathbf{p}}. \quad (3)$$

For the special case of a Gaussian,

$$f_{x-y} = f_0 \exp\left(-\frac{|x-y|^2}{2\sigma^2}\right), \quad (4)$$

the Fourier transformed smearing kernel again is a Gaussian:

$$\hat{f}(\mathbf{p}) = \sum_{x \in (xZ)^d} e^{ip \cdot x} f_x = \hat{f}(\mathbf{0}) \exp\left(-\frac{\sigma^2 \mathbf{p}^2}{2}\right). \quad (5)$$

Thus, the smeared quark operator has maximal overlap with a quark at rest, $\mathbf{p} = \mathbf{0}$. Non-zero velocities are suppressed in accordance with the above Gaussian momentum distribution. Clearly, for hadrons carrying significant spatial momenta, such a smearing may be counterproductive.

Having identified the problem, it is easy to modify the smearing to perform well for moving hadrons. We aim at a momentum distribution of quarks centred around a finite momentum $\mathbf{k}$, so we need to shift the smearing kernel in momentum space:

$$\tilde{f}(\mathbf{p}) \mapsto \tilde{f}(\mathbf{p} - \mathbf{k}), \quad (6)$$

as illustrated in Fig. 1. This translates to the replacement in position space:

$$f_{\mathbf{x}} \mapsto e^{i\mathbf{k} \cdot \mathbf{x}} f_{\mathbf{x}}. \quad (7)$$

Our modified smearing operator $F_{(\mathbf{k})}$, where $F_{(\mathbf{0})} = F$, can thus be formally expressed as

$$\begin{aligned} (F_{(\mathbf{k})}q)_{\mathbf{x}} &= \sum_{\mathbf{y} \in \langle \mathbf{x} \rangle^d} F_{(\mathbf{k})}(\mathbf{x}\mathbf{y}) q_{\mathbf{y}} \\ &= \sum_{\mathbf{y} \in \langle \mathbf{x} \rangle^d} e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} f_{\mathbf{x} - \mathbf{y}} G_{\mathbf{x}\mathbf{y}} q_{\mathbf{y}}. \end{aligned} \quad (8)$$

The only new ingredient is the additional phase factor $\exp[-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})]$. Note that the quark momentum shift $\mathbf{k}$ need not be a lattice momentum, i.e. it is not restricted to discrete values $\mathbf{k} \in (2\pi/L)\mathbb{Z}^d$. The smearing kernel f and the gauge dependent factor G can be taken over from any existing smearing method. For an iterative smearing method, the extra phase factor can easily be integrated into the elementary smearing step. Below we demonstrate this for the example of Wuppertal smearing. In principle, there could be additional effects like a Lorentz contraction of the wave function. This was studied, e.g., in Refs. [32, 33], and we will also address this possibility.

The new, modified smearing operator $F_{(\mathbf{k})}$ of Eq. (8) inherits important properties from the smearing operator $F_{(\mathbf{0})}$ it is based on. If the unmodified smearing operator is self-adjoint, then so is the modified smearing operator $F_{(\mathbf{k})}$, because from $f_{\mathbf{x} - \mathbf{y}} = f_{\mathbf{y} - \mathbf{x}}$ and $G_{\mathbf{x}\mathbf{y}} = G_{\mathbf{y}\mathbf{x}}$ it follows that $F_{(\mathbf{k})}\mathbf{a}\mathbf{b}\mathbf{y} = F_{(\mathbf{k})}^\dagger\mathbf{a}\mathbf{b}\mathbf{x}$. The underlying smearing operator $F_{(\mathbf{0})}$ should transform according to an irreducible representation (irrep) of the cubic group O_h . Usually, this will be the trivial A_1 representation but the smearing operator can also be used to inject angular momentum or to add non-trivial gluonic excitations [23]. Obviously, $F_{(\mathbf{k})}$ with a momentum shift $\mathbf{k} \neq \mathbf{0}$ breaks the cubic symmetry. However, when used in conjunction with a momentum projection that selects hadrons with a momentum $\mathbf{p}$, then as long as $\mathbf{k} \parallel \mathbf{p}$, the smearing operator remains within the A_1 representation of the O_h little group corresponding to the momentum direction $\mathbf{p}$.

III. MOMENTUM SMEARING AND HADRONIC TWO-POINT FUNCTIONS

We recapitulate Wuppertal smearing as a generic example of an iterative smearing algorithm. Then to clarify notations we discuss the standard construction

of hadronic two-point functions, before generalizing the smearing by introducing a momentum shift as described above. The discussion can easily be extended to incorporate generic hadrons and n -point functions with $n \geq 2$. We conclude with a suggestion how to further improve the method. This is relevant for fine lattices where the iteration count becomes large. In the Appendix we discuss how an additional Lorentz boost factor can be implemented.

A. Wuppertal smearing

The most prominent gauge covariant realization of a smearing function is Wuppertal smearing [17, 18], where $F = \Phi^n$, with Φ being defined as

$$(\Phi q)_{\mathbf{x}} = \frac{1}{1 + 2dc} \left[q_{\mathbf{x}} + c \sum_{j=1}^{d/2} U_{\mathbf{x},j} q_{\mathbf{x}+\mathbf{j}} \right]. \quad (9)$$

Again $d = 3$ denotes the dimension of space. The gauge transporters

$$U_{\mathbf{x},-j} = U_{-\mathbf{j},\mathbf{x}}^\dagger, \quad U_{\mathbf{x},j} = U_{\mathbf{x},j} \quad \text{with } \mathbf{x}_4 \text{ fixed}, \quad (10)$$

can also be spatially APE smeared links [13, 19, 22]. $\mathbf{j}$ denotes a vector of length of one lattice unit $\mathbf{a}$, pointing into the direction $\mathbf{j}$. In Eq. (9) we suppressed the index x_4 since the smearing is diagonal in time. c is a positive constant and the arbitrary normalization factor $1/(1 + 2dc)$ is introduced to avoid a numerical overflow for large iteration counts n .

Φ (and by implication Φ^n) is self-adjoint, a unit matrix in spinor space and transforms according to the A_1 representation of O_h or its little groups.¹ The replacement $q \mapsto \Phi^n q$ will therefore not interfere with the symmetry properties of any interpolator.

The smearing operator Φ is related to a discretized covariant Laplacian Δ :

$$\Phi q = q + \frac{c}{1 + 2dc} a^2 \Delta q, \quad (11)$$

$$a^2 (\Delta q)_{\mathbf{x}} = -2d q_{\mathbf{x}} + \sum_{j=1}^{d/2} U_{\mathbf{x},j} q_{\mathbf{x}+\mathbf{j}}. \quad (12)$$

Introducing a fictitious time $\tau = n\Delta\tau$ and defining $q(\tau) = \Phi^n q(0)$, i.e. $q(\tau + \Delta\tau) = \Phi q(\tau)$, results in the diffusion (or heat) equation

$$\frac{\partial q(\tau)}{\partial \tau} \approx \frac{q(\tau + \Delta\tau) - q(\tau)}{\Delta\tau} = \alpha \Delta q(\tau), \quad (13)$$

¹ O_h symmetry will be reduced to C_{4v} for momenta along a lattice axis (C_{4v}), a spatial diagonal (C_{2v}) or a planar diagonal (C_{2v}), see, e.g., Ref. [34].

where

$$\alpha = \frac{a^2}{\Delta\tau} \frac{\varepsilon}{1 + 2d\varepsilon}. \quad (14)$$

This is solved by $q(\tau) \approx \exp(\alpha\tau\Delta)\bar{q}(0)$. Starting from δ -sources in position and colour space $d_a^b(0) = \delta_{ab}\delta^{ab}$, assuming the free case $U_{x,j} = \mathbf{1}_3$ and large distances $r = |x| \gg a$, Eq. (13) is obviously solved by a Gaussian with

$$\sigma(\tau) = \sqrt{2\alpha\tau} = \sqrt{2na^2} \sqrt{\frac{\varepsilon}{1 + 2d\varepsilon}} \quad (15)$$

being the square root of the variance, i.e. the smearing corresponds to the smearing kernel Eq. (4). Employing a parallel transporter within the covariant Laplacian Eqs. (10) and (12) that is close to unity, like d dimensional APE smeared gauge links, means that this Gaussian shape is a good approximation, see, e.g., Ref. [35].

The diffusivity α obviously is maximal for large values of the parameter ε ($\alpha \rightarrow \frac{1}{2}a^2/\Delta\tau$ for $\varepsilon \rightarrow \infty$). For small values ($\alpha \approx \varepsilon a^2/\Delta\tau$), the iteration count n to achieve a given smearing radius² $\sqrt{3}\sigma$ is larger but the resulting wave function will be smoother. Equation (15) highlights that, to keep the smearing radius fixed in physical units, the iteration count needs to be increased in proportion to the square of the inverse lattice spacing. Obviously, this becomes computer time intensive towards the continuum limit. We remark that at small lattice spacings the diffusion equation (13) should be solved using a smarter method than iteratively applying the smearing operator Φ defined in Eq. (10). We will return to this in Sec. III E.

B. Construction of two-point functions

As an example we discuss the construction of momentum projected pion and nucleon two-point functions:

$$C_\pi(\mathbf{p}, t) = \langle \pi_{\mathbf{p}}(t) \pi_{\mathbf{p}}^\dagger(0) \rangle, \quad (16)$$

$$C_N(\mathbf{p}, t) = \langle N_{\mathbf{p}}^\alpha(t) \bar{N}_{\mathbf{p}}^\alpha(0) \rangle \mathbb{P}^{\alpha\beta},$$

where $\mathbb{P} = \frac{1}{2}(\mathbf{1} + \gamma_4)$ denotes a projector onto positive parity.³ Without smearing the pion and nucleon interpolators are local quark bilinears and trilinears:

$$\pi_{\mathbf{p}} = a^3 \sum_{\mathbf{x}} e^{i\mathbf{p}\cdot\mathbf{x}} \bar{u}_{\mathbf{x}} \gamma_5 d_{\mathbf{x}}, \quad (17)$$

$$N_{\mathbf{p}}^\alpha = a^3 \sum_{\mathbf{x}} e^{i\mathbf{p}\cdot\mathbf{x}} [u_{\mathbf{x}}^t C \gamma_5 d_{\mathbf{x}}] u_{\mathbf{x}}^\alpha, \quad (18)$$

² Note that the root mean squared radius in d dimensions reads $\sqrt{d}\sigma$. Also note that the width of the gauge invariant combination $u_{\mathbf{x}}^t d_{\mathbf{x}}$ Eq. (46) is smaller by a factor $\sqrt{2}$.

³ Note that for $\mathbf{p} \neq \mathbf{0}$ this projector will not completely remove the negative parity contribution. To improve on this, instead one can employ the oblique projector [30] $\mathbb{P}_{\mathbf{p}} = \frac{1}{4}(\mathbf{1} + i\mathbf{p}_{N_1}/E_{N_1}(\mathbf{p})\gamma_4)$, where m_{N_1} and E_{N_1} denote mass and energy of the nucleon's negative parity partner. Since this state is higher in mass than the nucleon, for simplicity, here we abstain from attempting this.

where C is the charge conjugation operator and $u_{\mathbf{x}}$ and $d_{\mathbf{x}}$ annihilate up and down quarks, respectively, at the spatial position $\mathbf{x}$.

For the pion the Wick contraction then gives

$$C_\pi(\mathbf{p}, t) = -L^3 a^3 \sum_{\mathbf{x}} e^{i\mathbf{p}\cdot\mathbf{x}} \left\langle \bar{u}_{\mathbf{x}\gamma_5} d_{\mathbf{x}} d_0^\dagger \gamma_5 \bar{u}_0^\dagger \right\rangle \quad (19)$$

$$= -L^3 a^3 \sum_{\mathbf{x}} \langle \text{tr}[(dd)_{\mathbf{x}0} \gamma_5 (u\bar{u})_{0\mathbf{x}} \gamma_5] \rangle e^{i\mathbf{p}\cdot\mathbf{x}}$$

$$= -L^3 a^3 \left\langle \sum_{\mathbf{x}} \text{tr} \left(M_{\mathbf{x}0}^{-11} M_{\mathbf{x}0}^{-1} \right) e^{i\mathbf{p}\cdot\mathbf{x}} \right\rangle,$$

where $\mathbf{x} = (\mathbf{x}, t)$ and the trace is over spin and colour. Momentum projection at the source is not necessary, due to the translational symmetry of expectation values and $L^3 = N^3 a^3$ denotes the three-volume that corresponds to this omitted sum. M is a Wilson-like lattice Dirac matrix with the quark mass $m_{\mathbf{x}} = m_d$ and we have used⁴ $d^\dagger \gamma_5 u^\dagger = d^\dagger \gamma_5 \gamma_4 u = -d^\dagger \gamma_5 u$, the Grassmann nature of the quark field creation and annihilation operators and replaced $(d\bar{d})_{\mathbf{x}} = (u\bar{u})_{\mathbf{x}} = M^{-1}$, where the subscripts d, u denote integrating out the respective Grassmann field on a given gauge background. In the last step we also exploited γ_5 -Hermiticity: $\gamma_5 M^{-1} \gamma_5 = M^{-1\dagger}$. For the nucleon one can easily work out an analogous expression that contains $e^{i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{p}'\cdot\mathbf{x}} (M^{-1})_{\mathbf{x}0}^{\beta\alpha} (M^{-1})_{\mathbf{x}0}^{\beta\beta} (M^{-1})_{\mathbf{x}0}^{\alpha\alpha}$, where we have suppressed the spinor indices and $a, b, c, e, e', e'' \in \{1, 2, 3\}$ run over fundamental colour. Equation (19) can now be evaluated, solving the linear systems

$$\sum_{\mathbf{x}, \alpha, \beta} M_{\mathbf{x}}^{\beta' \alpha' \alpha} G^{\alpha \alpha' \beta \beta'} = \delta_{\mathbf{x}' \mathbf{x}} \delta^{\beta' \beta'} \delta^{\alpha' \alpha} \quad (20)$$

for twelve δ -sources (for each source spin β and colour b) to obtain the point-to-all propagator S with sink position, spin and colour indices $\mathbf{x}, \alpha$ and a , respectively. Then,

$$C_\pi(\mathbf{p}, t) \propto \left\langle \sum_{\mathbf{x}} \text{tr} [S_{\mathbf{x}}^\dagger S_{\mathbf{x}}] e^{i\mathbf{p}\cdot\mathbf{x}} \right\rangle, \quad (21)$$

where again $\mathbf{x}_t = t$. The construction of the analogous nucleon two-point function from three point-to-all propagators is straightforward. Note that in that case no Hermitian adjoint will appear.

Smearing is diagonal in Euclidean time (hence we suppressed the t dependence) and trivial in Dirac spin. So, obviously, the smearing operator $F = \Phi^a$ commutes with any Γ structure, however, it does not

⁴ Note that the assignment $d = d^\dagger \gamma_5$ is convenient but not consistent in Euclidean spacetime, where the Dirac operator is γ_5 -instead of $(\gamma_4 \equiv \gamma_5)$ -Hermitian. This is the reason for the negative sign in Eq. (19). In the simulation, we correct for this phase and obtain a positive two-point function.

commute with covariant derivatives that may appear within the hadronic interpolator. Smearing can easily be implemented at the source, replacing $S = M^{-1}\delta$ by $S^F = M^{-1}F\delta$, see Eq. (20). Note that, due to the fact that F is a unit matrix in spinor space, the same smeared source can be employed for all four spinor components. To obtain a so-called smeared-smeared two-point function, the argument of the sum in Eq. (21) will usually be replaced by $\text{tr}[(FS^F)^\dagger(FS^F)]e^{ip \cdot x}$; every source smearing requires new inversions while sink smearing needs to be carried out on all time slices of interest.

We remark that momentum sources have been used for quite some time, see, e.g., Refs. [37–39]. Injecting momentum into quark sources is necessary (and has been done) in the context of the one-end-trick of Refs. [40, 41], where one usually employs colour diagonal (complex) $\mathbb{Z}_2$ or $U(1)$ random sources. Generalizing this to $SU(3)$ noise is in fact equivalent to using a non-gauge fixed wall source [42], which does not change expectation values. Momentum was however also injected into gauge fixed wall sources in Refs. [27, 43] (“color wave propagator”), favourably affecting not only statistical errors but also ground state overlaps. While these latter references share some of our motivation, the method presented here is quite different. For instance, in our case the hadron’s total momentum still needs to be injected at the source and $\mathbf{k}$ is not quantized.

Finally, we remark that the asymmetry of only carrying out the position sum at the sink often is exploited to reduce the statistical errors of heavy-light meson correlation functions, by only smearing the heavy quark with $F^2 = F^\dagger F$ at the source, instead of smearing each quark with F . Clearly, it would be advantageous if for each momentum smearing parameter of interest only the heavy quark propagator had to be recomputed. Unfortunately, momentum smearing (as well as traditional smearing) as an operation will not commute with momentum projection, unless $\mathbf{p} = \mathbf{0}$. Therefore, a different distribution of

the smearing between the quarks will result in a different and not necessarily optimal ground state overlap.

C. Momentum (Wuppertal) smearing

As explained in Sec. II, if we inject a momentum $\mathbf{p}$ into the hadron, it may be a good idea to distribute at least part of it among the quarks. Their smearing functions should therefore ideally be centred around a value $\mathbf{k} \neq \mathbf{0}$, where $\mathbf{k}$ is some fraction of $\mathbf{p}$, to best resemble the wave function of the physical state we wish to create.

The Fourier transform of a Gaussian, centred about $\mathbf{k}$ in momentum space, reads

$$f_{(\mathbf{k})\mathbf{x}=\mathbf{y}}^G \propto \exp \left[-\frac{(\mathbf{x}-\mathbf{y})^2}{2\sigma^2} - i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y}) \right]. \quad (22)$$

Similarly, momentum can be injected also into differently shaped smearing functions, a possibility that we shall not explore here. $q(\tau) = F_{(\mathbf{k})}^{(\tau)} q(0)$ solves the heat equation with a constant drift term

$$\frac{\partial q(\tau)}{\partial \tau} = \alpha (\nabla + i\mathbf{k})^2 q(\tau). \quad (23)$$

In analogy to the discussion of Sec. III A, in the free case the above smearing function can iteratively be constructed from “momentum” Gauß smearing steps, $F_{(\mathbf{k})} = \Phi_{(\mathbf{k})}^\dagger$, introducing a phase into Eq. (9):

$$(\Phi_{(\mathbf{k})} q)_{\mathbf{x}} = \frac{1}{1 + 2d\epsilon} \left[q_{\mathbf{x}} + \epsilon \sum_{j=1}^{d/2} U_{\mathbf{x},j} e^{i\mathbf{k} \cdot \mathbf{j}} q_{\mathbf{x}+\mathbf{j}} \right], \quad (24)$$

where $\Phi_{(0)} = \Phi$. One can easily show that the variance $\sigma^2(\tau)$ is still given as in Eq. (15). Moreover, as expected $\Phi_{(\mathbf{k})}$ remains self-adjoint:

$$\begin{aligned} \Phi_{(\mathbf{k})\mathbf{x}\mathbf{y}} &= \frac{1}{1 + 2d\epsilon} \left\{ \delta_{\mathbf{x}\mathbf{y}} + \epsilon \sum_{j=1}^3 \left[\delta_{\mathbf{x}+\mathbf{j},\mathbf{y}} U_{\mathbf{x},j} e^{i\mathbf{k} \cdot \mathbf{j}} + \delta_{\mathbf{y}+\mathbf{j},\mathbf{x}} U_{\mathbf{y},j}^\dagger e^{-i\mathbf{k} \cdot \mathbf{j}} \right] \right\}, \\ \Phi_{(\mathbf{k})\mathbf{y}\mathbf{x}}^\dagger &= \frac{1}{1 + 2d\epsilon} \left\{ \delta_{\mathbf{y}\mathbf{x}} + \epsilon \sum_{j=1}^3 \left[\delta_{\mathbf{y}+\mathbf{j},\mathbf{x}} U_{\mathbf{y},j}^\dagger e^{-i\mathbf{k} \cdot \mathbf{j}} + \delta_{\mathbf{x}+\mathbf{j},\mathbf{y}} U_{\mathbf{x},j} e^{i\mathbf{k} \cdot \mathbf{j}} \right] \right\} = \Phi_{(\mathbf{k})\mathbf{x}\mathbf{y}}. \end{aligned} \quad (25)$$

We will use the same smearing for quarks and antiquarks. For the π^- meson this amounts to the replacements $d \mapsto \Phi_{(\mathbf{k})}^\dagger d$, $u \mapsto \Phi_{(-\mathbf{k})}^\dagger u$, $\bar{d} \mapsto d\Phi_{(\mathbf{k})}$ and $\bar{u} \mapsto u\Phi_{(-\mathbf{k})}$ within Eq. (19). Note that the $\Phi_{(-\mathbf{k})}$ smear-

ing is needed as in the contraction with the momentum projector, $(\delta_{\mathbf{x}\mathbf{x}'} e^{i\mathbf{p} \cdot \mathbf{x}}) \Phi_{(\mathbf{k})\mathbf{x}\mathbf{y}}^\dagger \Phi_{(\mathbf{k})\mathbf{x}'\mathbf{y}'}$, transposing the ordering of the indices of the first smearing operator gives $\Phi_{(\mathbf{k})\mathbf{x}\mathbf{y}}^\dagger = \Phi_{(-\mathbf{k})\mathbf{y}\mathbf{x}}$ in the free case, see Eq. (29). Explicit-

ing the Hermiticity of $\Phi_{\mathbf{k}}^{\dagger}$, we then obtain

$$C_{\pi}(t) \propto \sum_{\mathbf{x}} \left\langle \text{tr} \left\{ \left[\left(\Phi_{(-\mathbf{k})}^{\dagger} M^{-1} \Phi_{(-\mathbf{k})}^{\dagger} \right)_{x0} \right]^{\dagger} \times \left(\Phi_{(\mathbf{k})}^{\dagger} M^{-1} \Phi_{(\mathbf{k})}^{\dagger} \right)_{x0} \right\} \right\rangle e^{i\mathbf{p} \cdot \mathbf{x}}. \quad (26)$$

In contrast, for baryonic two-point functions, where all quarks propagate in the forward direction, only $\Phi_{(+\mathbf{k})}^{\dagger}$ will appear. The pion two-point function can now be constructed in analogy to Eq. (21) from

$$C_{\pi}(\mathbf{p}, t) \propto \left\langle \text{tr} \left[\left(\Phi_{(-\mathbf{k})}^{\dagger} S_{(-\mathbf{k})} \right)_a^{\dagger} \left(\Phi_{(\mathbf{k})}^{\dagger} S_{(\mathbf{k})} \right)_a \right] e^{i\mathbf{p} \cdot \mathbf{x}} \right\rangle, \quad (27)$$

where the source-smearing point-to-all propagator is defined as [see Eq. (20)]

$$S_{(\mathbf{k})_x}^{aa'b} = \sum_{x', a', b', c, d} (M^{-1})_{x, x'}^{aa'c'} \delta^{a'c'} \delta^{b'c'} \Phi_{(\mathbf{k})_{x'}}^{a'c'} \delta_{x'0} \delta_{a'b} \delta^{cb}, \quad (28)$$

or, in shorthand notation: $S_{(\mathbf{k})} = M^{-1} \Phi_{(\mathbf{k})}^{\dagger} \delta$. For mesons the above twelve linear systems need to be solved, both

for δ -sources smeared with $\Phi_{(\mathbf{k})}^{\dagger}$ and with $\Phi_{(-\mathbf{k})}^{\dagger}$, while for baryons smearing with $\Phi_{(\mathbf{k})}^{\dagger}$ is sufficient.

We remark that the momentum smearing Eq. (24) can very easily be implemented by substituting the (APE smeared) transporters $U_{x,j} \mapsto U_{x,j} e^{i\mathbf{k} \cdot \mathbf{j}}$ and then iterating the usual Wuppertal smearing Eq. (9) on these modified $U(3)$ links.

D. Free field investigation

Having defined the smearing and how the contractions are to be carried out, we are now in the position to address the question what value of $\mathbf{k}$ should be chosen. Naively, one may expect $\mathbf{k} \approx \mathbf{p}/2$ and $\mathbf{k} \approx \mathbf{p}/3$ to be optimal for mesons and baryons, respectively, that are composed of degenerate quarks and carry a total momentum $\mathbf{p}$. This is indeed what we will find here for the non-interacting case. The interacting case somewhat deviates from this as we will see in Sec. V below.

Setting $U_{x,j} = 1$ within Eq. (24) gives the free case smearing function Eq. (22) for a large volume $L^3 \gg \sigma^3$ and number of smearing iterations so that $\sigma \gg a$. Smearing the quark and antiquark annihilation operators at the sink with $F_{(\mathbf{k})}$ and $F_{(-\mathbf{k})}$, respectively, we obtain the momentum projected smeared pion interpolator,

$$\begin{aligned} \pi_{\mathbf{p}} &\propto \sum_{\mathbf{x}, \mathbf{y}, \mathbf{y}'} \bar{u}_{\mathbf{y}'} f_{(-\mathbf{k})_{\mathbf{y}'-x}} e^{i\mathbf{p} \cdot \mathbf{x}} f_{(\mathbf{k})_x - \mathbf{y}} \gamma_5 d_{\mathbf{y}} \\ &\propto \sum_{\mathbf{x}, \mathbf{y}, \mathbf{y}'} \exp \left[-\frac{(\mathbf{y}-x)^2 + (\mathbf{y}'-x)^2}{2\sigma^2} \right] e^{i\mathbf{p} \cdot \mathbf{x}} e^{-2i\mathbf{k} \cdot \mathbf{x}} e^{i\mathbf{k} \cdot (\mathbf{y}+\mathbf{y}')} [\bar{u}_{\mathbf{y}'} \gamma_5 d_{\mathbf{y}}] \\ &\propto \exp \left[-\frac{\sigma^2}{2} (\mathbf{p}-2\mathbf{k})^2 \right] \sum_{\mathbf{Z}, \mathbf{\Delta}} \exp \left[-\frac{\mathbf{\Delta}^2}{\sigma^2} \right] e^{i\mathbf{p} \cdot \mathbf{Z}} [\bar{u}_{\mathbf{Z}+\mathbf{\Delta}} \gamma_5 d_{\mathbf{Z}-\mathbf{\Delta}}], \end{aligned} \quad (29)$$

where we have defined centre and relative coordinates

$$\mathbf{Z} = \frac{\mathbf{y} + \mathbf{y}'}{2}, \quad \mathbf{\Delta} = \frac{\mathbf{y}' - \mathbf{y}}{2}. \quad (30)$$

Note that the components of $\mathbf{Z}/a$ and $\mathbf{\Delta}/a$ can be integer or half-integer valued, subject to the constraints $(Z_j + \Delta_j)/a \in \mathbb{Z}$. The result of Eq. (29) is indeed maximized for $\mathbf{k} = \mathbf{p}/2$: in the free case $\mathbf{k} = \mathbf{p}/2$ is the optimal smearing parameter for mesons with mass-degenerate valence quarks while we encounter an exponential suppression in $\mathbf{p}^2$ for the conventional $\mathbf{k} = \mathbf{0}$ smearing. The broader the wave function in coordinate space, the more important becomes the correct choice of $\mathbf{k}$ as one can see from the first exponent in Eq. (29) (as well as from Eq. (31)). Unsurprisingly, for baryonic interpolators $\mathbf{k} = \mathbf{p}/3$ would be the optimal choice.

We have demonstrated that in the free case $\mathbf{k}$ can be interpreted as the momentum carried by the smeared

quark. If the quarks differ in mass or, like for the example of the nucleon, the interpolator is not symmetric with respect to the quark flavour, injecting different $\mathbf{k}$ -values into different quark fields may be advisable. In the interacting case the interpretation is not as straightforward: neither is the interpolator directly related to any of the usual definitions of a wave function nor will all momentum be carried by the valence quarks.

E. Non-iterative (momentum) smearing

We propose replacing iterative smearing, where the iteration count diverges with a^{-7} towards the continuum limit, by a more refined method. In this context we show how to introduce phase factors (and shape functions) into the “distillation” (or Laplacian-Heaviside) method

of Ref. [31]. Although we already present the basics of the method here, systematic tests are yet to be completed. Another natural extension, which we will investigate numerically in Sec. V, is to Lorentz boost the smearing function.

We define eigenvectors $|v_l\rangle$ and eigenvalues $-\omega_l^2$ of a covariant Laplacian with (smeared) gauge transporters, at a fixed Euclidean time:

$$(\Delta + \omega_l^2) |v_l\rangle = 0. \quad (31)$$

Since the Laplacian is self-adjoint, the bra-ket notation is convenient in the present context. For a lattice of N^d points per time slice ($L = Na$), $3N^d$ linearly independent eigenvectors $|v_l\rangle$ with components $v_{\mathbf{k}l}$ exist. Such eigenvectors were for instance used in Ref. [31]. The heat equation (13) is solved by

$$\begin{aligned} |q(\tau)\rangle &= \exp(\alpha\tau\Delta) |q(0)\rangle \\ &= \sum_l |v_l\rangle \exp(-\alpha\tau\omega_l^2) \langle v_l|q(0)\rangle. \end{aligned} \quad (32)$$

If we start from a δ -function at $\tau = 0$, this results in a Gaussian of variance $\sigma^2 = 2\alpha\tau$.

To implement momentum smearing one can easily replace $U_{x,j} \mapsto U_{x,j} e^{i\mathbf{k}\cdot\mathbf{j}}$ within the covariant Laplacian $\Delta \mapsto \Delta_{\mathbf{k}} \sim (\nabla + i\mathbf{k})^2$ and then recompute the eigenvectors and -values:

$$(\Delta_{\mathbf{k}} + \omega_{\mathbf{k}l}^2) |v_{\mathbf{k}l}\rangle = 0. \quad (33)$$

The natural generalization of Eq. (32) for moving particles then reads:

$$|q_{\mathbf{k}l}(\tau)\rangle = \sum_l |v_{\mathbf{k}l}\rangle \exp(-\alpha\tau\omega_{\mathbf{k}l}^2) \langle v_{\mathbf{k}l}|q(0)\rangle. \quad (34)$$

The motivation for computing eigenvectors of the covariant Laplacian is to truncate Eqs. (32) or (34) at a value ℓ_{max} where $\omega_{\ell_{\text{max}}+1}^2 > \omega_{\ell_{\text{max}}}^2$. As $\alpha\tau = \sigma^2/2$, achieving a suppression by a factor e^{-2} requires $\omega_{\ell_{\text{max}}} \approx 2/\sigma$. In general, the wider the smearing function in coordinate space the less eigenvectors will be needed. The extreme opposite limit $\alpha\tau = 0 \Rightarrow \sigma = 0$ corresponds to the Laplacian-Heaviside method proposed in Ref. [31], where summing over all eigenvectors will ultimately result in a δ -function in position space while truncating at some finite ℓ_{max} value gives a bell shape [31]. (In the free case the modulus would be a sum of sines.) The same holds for the sources suggested in Ref. [38] that correspond to sums of eigenvectors of the non-interacting Laplacian.

It is trivial to work out more details in the free case, where one basically encounters a one dimensional problem. For instance, setting $\omega = j(2\pi/L) \gtrsim 2/\sigma$, where $j \in \mathbb{Z} \setminus \{0\}$ means $|j| \gtrsim L/(\pi\sigma)$. Therefore, in d dimensions $\ell_{\text{max}} \gtrsim 3 \times [\pi^{d/2}/\Gamma(d/2+1)] \times [L/(\pi\sigma)]^d$, where the factor 3 is due to colour and the next factor is the volume of a d dimensional unit-sphere ($4\pi/3$ for $d=3$): the number of required eigenvectors increases in proportion

to the spatial volume in physical units but is independent of the lattice spacing a . In contrast, in the case of iterative smearing, reducing the lattice spacing at a fixed value of σ increases the iteration count in proportion to a^{-3} , independent of the volume. For our smearing size $\sigma \approx 0.45 \text{ fm}$ and a spatial volume $L^3 = (6 \text{ fm})^3$, which would ensure $Lm_\pi > 4$ even at the physical mass point, we obtain $\ell_{\text{max}} \approx 960$ while for a $(3 \text{ fm})^3$ box about 120 eigenvectors should suffice. Therefore, satisfactory results in terms of ground state overlaps appear to be within reach, employing moderately large numbers of eigenvectors. This is at present under investigation.

Non-Gaussian shapes can easily be modelled too, e.g., by multiplying in a “free form” weight function, $\langle v_l | v_l \rangle_{2\mathbf{p}} \mapsto \sum_{\mathbf{y}} h_{2\mathbf{p}} \exp[\alpha\tau(\mathbf{x} - \mathbf{y})^2] \langle v_l | v_l \rangle_{2\mathbf{p}}$, in analogy to Ref. [25]. However, the numerical complexity of this operation is $\mathcal{O}(N^{2d})$ for each time slice and eigenvector. At the source this can be reduced to $\mathcal{O}(N^d)$ if the solution is only required for a fixed δ -source position $q(0) = |\delta_{\mathbf{y}_s}\rangle \langle (\mathbf{y}_s, \delta_{\mathbf{y}_s}^a) = (\delta_{\mathbf{y}_s}^a)_{\mathbf{y}_s} = \delta_{\mathbf{y}_s}^a \delta^{\mathbf{a}} \rangle$. Other, numerically less expensive options include placing non-linear functions of ω_l^2 in the exponents of Eqs. (32) and (34), employing linear combinations of Gaussians or substituting the covariant Laplacian by a different operator within Eqs. (31) and (33), e.g., introducing an anisotropy.

Once the eigenvectors Eq. (31) or Eq. (33) of the APE or HYP smeared covariant Laplacian have been computed, any smearing radius can be realized, replacing $\alpha\tau$ by the targeted $\sigma^2/2$ value within the exponent of Eq. (32) or Eq. (34). Another potential advantage of the non-iterative smearing is that, instead of solving for smeared sources $|q_{\mathbf{k}l}(\tau)\rangle$ that have evolved from a δ -source $|q(0)\rangle = |\delta_{\mathbf{y}_s}\rangle$, one can also directly apply the inverse lattice Dirac operator to the eigenvectors, thereby constructing so-called perambulators [31]:

$$\tau_{\mathbf{k}|m\ell|0}^{\alpha\tau} = \langle v_{\mathbf{k}|m\ell} | (M^{-1})_{00}^{\alpha\tau} | v_{\mathbf{k}|0} \rangle. \quad (35)$$

The inner product above is over spatial position and colour, replacing these indices by the eigenvector labels ℓ at the sink and m at the source. The respective Euclidean times are explicitly shown as additional eigenvector (and perambulator) subscripts. These perambulators can then be folded with the appropriate eigenvalue Gaussians during the construction of hadronic n -point functions.

We abstain from repeating here the steps outlined in Ref. [31] how to construct hadronic two-point functions from these perambulators as there is only one difference: In the contractions over ℓ and m weight factors $\exp(-\sigma^2\omega_{\mathbf{k}|m\ell}^2/2)$ and $\exp(-\sigma^2\omega_{\mathbf{k}|m\ell}^2/2)$ should be introduced, where $\omega_{\mathbf{k}|m\ell}^2$ refers to the m th eigenvalue of the negative covariant Laplacian defined on time slice t .

On one hand using perambulators will require a larger number of inversions and computationally more expensive contractions than starting from δ -sources. On the other hand, due to volume averaging, the statistical errors of the perambulator method will be smaller and this method allows for more flexibility in the subsequent con-

struction of hadronic n -point functions. Whether the use of perambulators or of traditional (smeared) point-to-all propagators is preferable will therefore depend on the problem at hand and in particular on the number of different hadrons we are interested in. We have shown that the same quark smearing can be employed in both cases.

Finally, we remark that in certain situations recomputing the eigenvectors for different values of $\mathbf{k}$ may be avoidable. In the interacting case, gauge covariant derivatives ∇_i^2 will depend on all spatial coordinates, including x_i with $i \neq j$, and will therefore not commute with each other. However, our intuition was based on the free case and also the spatially APE [13] or HYP [15] smeared gauge covariant transporters are close to unity. Assuming translational invariance, the coordinates can be separated and the components of an eigenvector read $v_{\mathbf{k}}^a \propto \sin(\omega_1^a x_1) \cdots \sin(\omega_d^a x_d)$, where $\omega_i^a = 2\pi n_i^a/L$ and n_i^a are integer valued. The corresponding eigenvalue of the negative Laplacian is given as $\omega^2 = \omega^2$ where the frequencies for different colour components are constrained: $\omega^2 := \omega^{i^2} = \omega^{j^2} = \omega^{k^2}$. Defining

$$|v_{(\mathbf{k})\mu}\rangle = e^{-i\mathbf{k}\cdot\mathbf{x}} |v_{\mathbf{k}}\rangle \quad (36)$$

then gives $[(\nabla + i\mathbf{k})^2 + \omega^2] |v_{(\mathbf{k})\mu}\rangle = 0$.

In the interacting case this will not exactly solve the heat equation with drift (23) but the approximation $\omega_{(\mathbf{k})\mu} \approx \omega_{\mathbf{k}}$, $|v_{(\mathbf{k})\mu}\rangle \approx |v_{(\mathbf{k})\mu}\rangle$ should be sufficient to construct a gauge covariant Gaussian shape with the intended phase factors. Note that the phases Eq. (36) appear both in the bra- and in the ket-vector of Eq. (34), such that only relative phases between two spatial positions matter and the choice of the zero point becomes irrelevant. The clear disadvantage of the explicit multiplication by phase factors Eq. (36) is that these have to obey the lattice periodicity, i.e. $\mathbf{k}_j \in (2\pi/L)\mathbb{Z}$. However, such a restriction may be tolerable on large lattices. Also introducing twisted fermionic boundary conditions [44, 45] may provide a way to increase the flexibility of the choice of $\mathbf{k}$.

IV. LATTICE ENSEMBLE AND DISPERSION RELATIONS

We study the new momentum smearing method on 200 effectively decoupled configurations of $N_f = 2$ non-perturbatively improved Wilson fermions with the Wilson gluon action, generated by QCDSF and RQCD. This constitutes a subset of ensemble IV of Ref. [46]. Note that in hadron structure studies we typically employ several sources on about 2000 configurations [46], i.e. the present statistics are very moderate. Nevertheless, we will report meaningful signals for momenta as high as 2.8 GeV. The parameter values of this $32^3 \times 64$ ensemble read $\beta = 5.29$, $\kappa = 0.13632$, corresponding to the inverse lattice spacing $a^{-1} \approx 2.76$ GeV and the (finite volume) pion mass $m_\pi \approx 295$ MeV. This gives a spatial

extent $L \approx 2.29$ fm $\approx 3.42/m_\pi$. Note that realizing the physical pion mass at this lattice spacing, while keeping $L \gtrsim 3.5/m_\pi$, requires $L > 70a$. As we are aiming at momenta exceeding the hadron masses in any case, we do not expect qualitative changes of results towards smaller pion masses and have chosen the present parameters as a sufficiently realistic but still affordable compromise.

The momentum on a finite cubic lattice with even numbers of points $N = L/a$ in each spatial direction can take the discrete values

$$\mathbf{p} = \frac{2\pi}{L} \mathbf{P}, \quad P_i \in \left\{ -\frac{L}{2a} + 1, -\frac{L}{2a} + 2, \dots, \frac{L}{2a} \right\}. \quad (37)$$

This means the smallest non-trivial $|\mathbf{P}| = |(1, 0, 0)|$ gives a momentum $|\mathbf{p}| \approx 0.54$ GeV while $\mathbf{P} = (3, 3, 3)$ corresponds to $|\mathbf{p}| \approx 2.82$ GeV.

The pion and nucleon masses have already been determined with high statistics in Ref. [47] and read

$$am_\pi = 0.10675(32), \quad am_N = 0.3855(45). \quad (38)$$

We will compare our pion and nucleon ground state energies to expectations from continuum and lattice dispersion relations, using these reference values. The continuum dispersion relation reads

$$aE = a\sqrt{m^2 + \mathbf{p}^2}. \quad (39)$$

In addition, we will compare the pion energies to the lattice dispersion relation for a free naively discretized scalar particle,

$$\cosh(aE_\pi) = \cosh(am_\pi) + \frac{a^2}{2}\mathbf{p}^2, \quad (40)$$

and the nucleon energies to the dependence expected for a free Wilson fermion with the Wilson parameter $r = 1$, see, e.g., Ref. [48],

$$\cosh(aE_N) = 1 + \frac{(e^{am_N} - 1 + a^2\mathbf{p}^2/2)^2 + a^2\mathbf{p}^2}{2(e^{am_N} + a^2\mathbf{p}^2/2)}. \quad (41)$$

Above, we used the standard abbreviations

$$\hat{p}_\mu = \frac{2}{a} \sin\left(\frac{ap_\mu}{2}\right), \quad \tilde{p}_\mu = \frac{1}{a} \sin(ap_\mu), \quad (42)$$

and $\hat{\mathbf{p}}^2 = \sum_j \hat{p}_j^2$, $\tilde{\mathbf{p}}^2 = \sum_j \tilde{p}_j^2$. We remark that the mass parameters m_0 within the naive propagators (e.g., for the scalar case: $1/(m_0^2 + \hat{\mathbf{p}}_\mu \hat{p}_\mu)$) differ from the rest frame energies by lattice artefacts. The above masses are defined to satisfy $m = E(0)$, as it should be. Their conversions to m_0 are given as $m_\pi = 2a^{-1} \sinh(am_0/2)$ and $m_N = a^{-1} \ln(1 + am_0)$, respectively.

Obviously, the continuum dispersion relation should become violated towards large momenta. In this case, we would not expect the lattice dispersion relations, that apply to free pointlike particles, to accurately describe the data either. However, the difference between the continuum and the lattice formulae can serve as a naive estimate of the expected size of lattice artefacts.

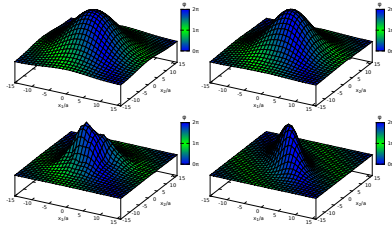


FIG. 2. Cross sections of the smearing density profile $\rho(\mathbf{x})$ Eq. (45) in the x_1 - x_2 plane. The colour encodes the phase Eq. (46) and the momentum smearing $\mathbf{k}$ parameter has the direction $(1, 1, 0)$. Top left: free field case. Top right: APE smeared gauge links. Bottom left: original gauge links. Bottom right: APE smeared links with an additional boost factor $\gamma = 5.3$.

V. RESULTS

We first describe and check our implementation of momentum Wuppertal smearing. Then, in Sec. VB we optimize the smearing parameters and test the effectiveness of the method. In Sec. VC we investigate whether introducing an additional Lorentz contraction is advantageous, before determining pion and nucleon dispersion relations up to momenta of 1.94 GeV and 2.82 GeV, respectively, in Sec. VD.

A. Implementation of the smearing

We iterate the momentum Wuppertal smearing on our $d + 1 = 3 + 1$ dimensional gauge ensemble described in Sec. IV, employing spatially APE smeared [13] gauge transporters within Eq. (24). These are iteratively constructed as follows.

$$U_{a,i}^{(n+1)} = P_{\text{SU}(3)} \left(U_{a,i}^{(n)} + \delta \sum_{[j] \neq i} U_{a,j}^{(n)} U_{a+3,j}^{(n)} U_{a+3,i}^{(n)} \right), \quad (43)$$

where $i \in \{1, 2, 3\}$, $j \in \{\pm 1, \pm 2, \pm 3\}$: the sum is over the four spatial “staples” surrounding $U_{a,i}$ where, again, we suppressed the time index as the smearing is local in time. $P_{\text{SU}(3)}$ is a gauge covariant projector onto the

SU(3) group, defined by maximizing $\text{Re} \text{tr}[A^\dagger P_{\text{SU}(3)}(A)]$. We iterate over the three diagonal SU(2) subgroups to achieve this. Other projection possibilities can, e.g., be found in Refs. [22, 49]. We iterate Eq. (43) 15 times, using the weight factor $\delta = 2.5$.

Momentum Wuppertal smearing Eq. (24) is implemented, multiplying the APE smeared links for a given $\mathbf{k}$ value by phases, $U_{a,j} \mapsto U_{a,j} e^{i\mathbf{k} \cdot \mathbf{j}}$, and then iterating the usual Wuppertal smearing Eq. (9) on these links. Within this article we set $\varepsilon = 0.25$ to obtain smooth smearing functions at tolerable iteration counts. Starting from three δ -functions (one for each source colour b) at the spatial position $\mathbf{0}$, we can define “wave functions” for three different source colours,

$$\psi_{\mathbf{k}|\mathbf{a}}^{a'b} = \sum_{\mathbf{x}' \neq \mathbf{0}} \left(\Phi_{\mathbf{k}}^a \right)_{\mathbf{a}\mathbf{x}'}^{a'} \psi_{\mathbf{x}'\mathbf{0}}^{b'}, \quad (44)$$

and the associated gauge invariant density:

$$\rho(\mathbf{x}) = \frac{\sum_{a,b} |\psi_{\mathbf{k}|\mathbf{a}}^{a'b}|^2}{a^3 \sum_{a,b} |\psi_{\mathbf{k}|\mathbf{a}}^{a'b}|^2}. \quad (45)$$

$\rho(\mathbf{x})$ does not carry any information relating to the U(1) phases. We therefore define

$$\varphi(\mathbf{x}) = \arg \left(\frac{\psi_{\mathbf{k}|\mathbf{a}}^{11}}{\psi_{\mathbf{k}|\mathbf{a}}^{11}} \right) \in [0, 2\pi), \quad (46)$$

as the phase of momentum smearing, relative to the standard Wuppertal smearing. Above, we have singled out one particular colour component but any diagonal component will give the same phase function $\varphi(\mathbf{x})$. In the case of a free configuration, i.e. employing trivial links $U_{\mu,i} = 1$, Eqs. (45)-(46) simplify since $\psi_{(\mathbf{k})\mu}^{11} = \psi_{(\mathbf{k})\mu}^{22} = \psi_{(\mathbf{k})\mu}^{33}$ and colour off-diagonal elements vanish. Moreover, $\varphi(\mathbf{x}) = \arg(\psi_{(\mathbf{k})\mu}^{11})$ in this case.

In Fig. 2 we display $\rho(\mathbf{x})$ in lattice units for the two dimensional cross section $x_3 = 0$ and different settings. The colour encodes the phase $\varphi(\mathbf{x})$. We employ $n = 200$ momentum Wuppertal smearing iterations with $\varepsilon = 0.25$ and set $\mathbf{K} = (1, 1, 0) = [L/(2\pi)]\mathbf{k}$, i.e. $\mathbf{k}$ lies within the depicted plane and its modulus is given as $|\mathbf{k}| \approx 0.77 \text{ GeV}$. Up to discretization and finite volume effects, with this smearing we expect the free case variance $\sigma^2 \approx (6.3a)^2$ from Eq. (15), i.e. $\rho(\mathbf{x}) \approx 6.3a \approx \varepsilon^{-1}\rho(0)$. In the top left panel we show the free case result from our iterative smearing, which is consistent with this expectation. In the top right panel of Fig. 2 we repeat this on one time slice of one of our gauge configurations, after having APE smeared the gauge links. The resulting shape is slightly narrower but otherwise indistinguishable from the free case and almost invariant with respect to continuous rotations. However, rotations take place in colour space: plotting individual components of ρ_a^b (not shown) gives a less smooth behaviour. In particular, the off-diagonal components do not vanish anymore. The smoothness of the gauge invariant density $\rho(\mathbf{x})$ means that the differences relative to the free case can be removed almost completely by a suitable gauge transformation.

In the bottom left panel we apply momentum Wuppertal smearing to a time slice of the original, not APE smeared gauge links. In this case the resulting density is less symmetric and less smooth. As one may expect from mean field arguments [56], the average smearing radius is somewhat reduced in a way consistent with multiplying ε by the fourth root of the average plaquette. Also in this case, the U(1) phase information is intact. A comparison with the top right panel of Fig. 2 clearly demonstrates the advantage of additional gauge link smearing.

Finally, in the lower right panel we apply boosted momentum Wuppertal smearing Eqs. (54)-(56), using the APE smeared gauge links. We set $\gamma = 5.3$, which corresponds to the ratio of the pion energy for a momentum $\mathbf{p} \approx 2\mathbf{k}$ over the pion mass $m_\pi \approx 295 \text{ MeV}$. The smearing parameter is converted according to Eq. (58). Indeed, the perpendicular shape in the central region is basically unaltered relative to the top panels while in the direction parallel to the boost the density is contracted by the γ factor. Due to the large numerical value of this factor there are slight deviations from the theoretical expectation but these discretization related effects can be removed by reducing the smearing parameter ε' and increasing the iteration count n , keeping σ constant, see Eqs. (15) and (58).

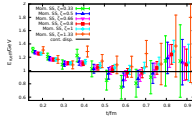


FIG. 3. Effective pion energies for $\mathbf{P} = (1, 1, 0)$, corresponding to $|\mathbf{p}| \approx 0.94 \text{ GeV}$ and different ratios ζ , see Eq. (47). The line is the expectation from the continuum dispersion relation. Symbols are shifted horizontally to enhance the legibility.

B. Optimization and test of momentum smearing

We now compute smeared-point and smeared-smeared pion and nucleon two-point functions at different lattice momenta Eq. (37), where we denote momentum and smearing vectors in physical units as $\mathbf{p}$ and $\mathbf{k}$, respectively, and integer component lattice momenta as $\mathbf{P} = [L/(2\pi)]\mathbf{p}$. Note that $\mathbf{K} = [L/(2\pi)]\mathbf{k}$ does not need to have integer valued components. We (mostly) restrict ourselves to

$$\mathbf{K} = \zeta \mathbf{P}, \quad (47)$$

where the naive expectation would be $\zeta = 1/2$ for the pion and $\zeta = 1/3$ for the nucleon, see Sec. IIID. From the resulting two-point functions, we define effective energies

$$E_{H,aH}(\mathbf{p}, t + a/2) = \frac{1}{a} \ln \left[\frac{C_H(\mathbf{p}, t)}{C_H(\mathbf{p}, t + a)} \right], \quad (48)$$

where $H \in [\pi, N]$. For the non-perturbatively improved action that we use, which contains a clover term that couples adjacent time slices, the meaningful range of t values is $t \geq 2a$, i.e. we plot effective energies, starting at $2.5a \approx 0.18 \text{ fm}$.

We realized different numbers of iteration counts n both for momentum and for conventional Wuppertal smearing. In both cases the best results in terms of the ground state overlaps for pions and nucleons at different momenta were obtained within the range $200 \leq n \leq 400$. For the results we present here we set $n = 200$, $\varepsilon = 0.25$, corresponding to $\sigma \approx 6.3a \approx 0.45 \text{ fm}$, see Fig. 2. We average the pion two-point function propagating in the forward and backward time directions (folding). For the nucleon we can only make use of the forward correlation function, since we only employed the projector $\mathbf{P} = \frac{1}{4}(1 + \gamma_4)$, see Eq. (16).

In Fig. 3 we show effective energies from smeared-smeared two-point functions for different values of ζ . In this case $\zeta = 0.8$ (red squares) appears to be the best

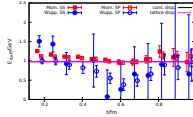


FIG. 4. Effective pion energies Eq. (48) for the lattice momentum $\mathbf{P} = (1, 1, 1)$, corresponding to $|\mathbf{p}| \approx 0.94$ GeV. In the case of momentum smearing (squares), we set $\mathbf{K} = \zeta \mathbf{P}$ with $\zeta = 0.8$. Solid symbols correspond to smeared-smeared, open symbols to smeared-point two-point functions. Some data points are shifted horizontally to enhance the legibility. The expectations from the continuum and lattice dispersion relations can be found in Eqs. (39) and (40), respectively. Symbols are shifted horizontally for better legibility.

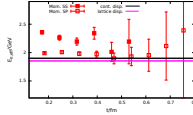


FIG. 5. The same as Fig. 4 for $\mathbf{P} = (1, 1, 1)$, corresponding to $|\mathbf{p}| \approx 1.88$ GeV. The effective energies without momentum smearing cannot be determined, due to prohibitively large errors and non-monotonic behaviour of the central values of the respective two-point functions.

choice, however the results are relatively robust against increasing or decreasing this by 20%. In general, at our pion mass $m_\pi \approx 295$ MeV and momenta up to ~ 3 GeV, the optimal ζ values came out to be $\zeta \approx 0.8 > 1/2$ for the pion and $\zeta \approx 0.45 > 1/3$ for the nucleon. As discussed in Sec. III D, we can only interpret k as the momentum carried by a single quark in the non-interacting case. Nevertheless, finding values that are larger than the naive expectation, rather than smaller, was somewhat unexpected. Since the deviation from the free field case is not uniform but bigger for the pion than for the nucleon, it would be interesting to extend our study to non-Gaussian smearing functions.

In Figs. 4 and 5 we compare effective pion energies from smeared-point (SP) and smeared-smeared (SS) correla-

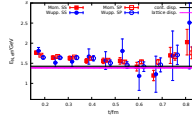


FIG. 6. The same as Fig. 4 for the nucleon and $\zeta = 0.5$. The horizontal lines correspond to the expectations from the continuum and lattice dispersion relations Eqs. (39) and (41), respectively.

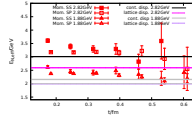


FIG. 7. The same as Fig. 6 for $\mathbf{P} = (2, 2, 2)$ ($\zeta = 0.4$) and $\mathbf{P} = (3, 3, 3)$ ($\zeta = 0.44$), corresponding to $|\mathbf{p}| \approx 1.88$ GeV and $|\mathbf{p}| \approx 2.82$ GeV, respectively. Again, the conventional Wuppertal smearing data are too noisy to allow for the extraction of effective energies.

tion functions with the expectations from the continuum and lattice dispersion relations Eqs. (39) and (40), using the pion mass Eq. (38) (horizontal lines). Note that the SS effective energies should be monotonic functions of t while this need not be the case for SP energies. It is well known that statistical errors of SP correlators are smaller than in the SS case and we also confirm this. For the momentum smeared two-point functions the data are consistent with plateaus for $t \gtrsim 0.5$ fm and we find agreement with the expectations. For $|\mathbf{p}| \approx 0.94$ GeV (Fig. 4) also the effective energies from conventionally smeared two-point functions are consistent with the expectation, however, the errors are too large to allow for quantitatively meaningful statements. For $|\mathbf{p}| \approx 1.88$ GeV (Fig. 5), within our statistics of one source position on 200 gauge configurations, it turned out to be impossible to obtain effective energies without momentum smearing at all.

The same comparison is shown for the nucleon in Figs. 6 and 7, where in Fig. 7 we also include a mo-

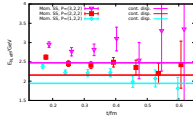


FIG. 8. The nucleon smeared-smeared effective energy with one and the same momentum smearing $\mathbf{K} = (0.5, 0.8, 0.8)$, optimized for $\mathbf{P} = (2.2, 2.2)$, but for momentum pointing into different directions. Symbols are shifted horizontally for better legibility.

momentum as high as $|\mathbf{p}| \approx 2.82 \text{ GeV}$. In this case we show the lattice dispersion relation Eq. (41) of a free Wilson fermion with the nucleon masses given in Eq. (38). The statistical errors of the momentum smeared data, again are much smaller than for the non-momentum smeared cases. Like for the pion, it was impossible within our statistics to extract effective energies for momenta larger than 1.5 GeV . At $|\mathbf{p}| \approx 0.94 \text{ GeV}$ the data agree with the expectation for $t \gtrsim 0.94 \text{ fm}$ while for the higher momenta, where the statistical errors are larger, the data are consistent with plateaus starting at $t \gtrsim 0.45 \text{ fm}$. For the high momenta shown in Fig. 7 the data seem to prefer the continuum dispersion relation over the lattice dispersion relation Eq. (41).

It is clear from the results shown above that the gain of using momentum smearing is tremendous. The only drawback, in addition to the computational overhead from the smearing itself, is that each value of the parameter $\mathbf{k}$ employed at the source requires us to recompute the respective quark propagator. A natural question therefore is whether it is possible to efficiently realize several momenta $\mathbf{p}$ using one and the same smearing vector $\mathbf{k}$. We already know from Fig. 3 that for $\mathbf{p} \parallel \mathbf{k}$ the proportionality constant ζ defined in Eq. (47) can be varied by about 20% without a significant deterioration and by much more if one is willing to accept a compromise: comparing Fig. 3 with Fig. 4 reveals that even a much less than perfect momentum smearing is a tremendous improvement over the conventional $\zeta = 0$ case. We may also ask whether it is possible to (slightly) vary the direction of $\mathbf{p}$ relative to $\mathbf{k}$. This of course is potentially dangerous since the interpolator used will not transform anymore according to an irrep of the little group of O_h (or its double cover), associated to the momentum direction. However, if for instance we are only interested in the ground state mass of a spin-0 or spin-1/2 hadron this should not be a major problem. Figure 8 demonstrates that to a certain extent varying the momentum direction

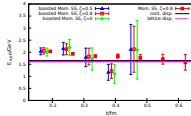


FIG. 9. Effective smeared-smeared pion energies for $\mathbf{P} = (3, 0, 0)$, corresponding to $|\mathbf{p}| \approx 1.63 \text{ GeV}$: momentum smearing with the optimized parameter $\zeta = 0.8$ and boosted momentum smearing Eq. (54) for $\gamma \approx 5.6$ with $\zeta = 0.5, 0.8, 0.9$. Symbols are shifted horizontally for better legibility.

for a fixed $\mathbf{k}$ appears feasible too. In all the cases shown it is impossible to extract any meaningful effective energies without momentum smearing.

C. Test of boosted (momentum) smearing

It has been suggested by two groups [32, 33] that introducing an anisotropy and thereby Lorentz boosting the smearing function may improve the overlap of high momentum interpolators with the respective hadronic ground states. We generalise these ideas to off-axis momentum directions, also incorporating phase factors, in the Appendix, see Eqs. (54)–(58). Length contractions depend on the choice of coordinates and in particular on time differences in the moving frame relative to the rest frame. Since in Euclidean spacetime all distances are spacelike and any real time information is lost, it is not clear to us why spatial distances should be subjected to a Lorentz boost. Our numerical observations are negative.

In Fig. 9, which is representative for our experiences, we show effective pion energies for $\mathbf{P} = (3, 0, 0)$, corresponding to $|\mathbf{p}| \approx 1.63 \text{ GeV}$ and a boost factor $\gamma = E_x(\mathbf{p})/m_\pi \approx 5.6$, close to the one that corresponds to the bottom right panel of Fig. 2 ($\gamma = 5.3$). In this case it was again not possible to extract effective energies using the conventional Wuppertal smearing. All Lorentz contracted (boosted) interpolators give results much inferior to the one obtained using the unboosted momentum smearing. This also holds for different contraction factors $1/\gamma$ (not shown). At the same time the boost enhances the signal relative to unboosted Wuppertal smearing, without the momentum phase factor. In the figure we show boosted smearing results for different momentum shifts: $\mathbf{k} = 0$, the naive expectation $\mathbf{k} = 0.5\mathbf{p}$ and the case that is close to optimal without the boost applied, $\mathbf{k} = 0.8\mathbf{p}$. We find the effective energies and their uncertainties are quite insensitive to the $\mathbf{k}$ value. This is

not surprising since the support of the smearing function in the direction of the momentum is quite small. At the same time this small support may explain why the boost outperforms conventional Wuppertal smearing as broad wave functions are disfavoured at high momenta, unless the $\mathbf{k}$ vector is introduced, see Eq. (29).

In summary, substantially contracting the smearing function in the direction of the momentum ameliorates the phase mismatch discussed in this article. Therefore, some improvement over the conventional isotropic smearing case can be achieved. However, only momentum smearing correctly accounts for this effect and we see no indication that injecting a momentum alters the optimal shape of the modulus of the smearing function Eq. (45).

D. Comparison with dispersion relations

Our main aim here was to demonstrate the effectiveness of momentum smearing. For this purpose it was sufficient to consider only one source position on 200 individual gauge configurations. The present state-of-the-art, however, is to realize multiple sources on ten times as many configurations. In the near future we will compute a multitude of physically interesting observables with enhanced statistics. The masses shown in Eq. (38) were already obtained with high statistics and in Figs. 3-9 we have compared effective energies against the continuum and lattice dispersion relations Eqs. (39)-(41), using these values.

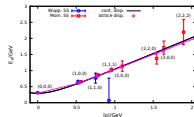


FIG. 10. Pion energies for different lattice momenta, in comparison to the continuum (solid curve) and lattice (points connected by dotted lines) dispersion relations Eqs. (39) and (40).

In all cases the smeared-smeared effective energies from optimized momentum smearing were in agreement with plateaus from $t \geq t_{\text{min}} = 8.5a \approx 0.61$ fm onwards, where $t = 8.5a$ corresponds to the effective energy obtained from the correlation function at 8a and 9a, see Eq. (48). In many cases t_{min} could be chosen smaller. For the moment being, we conservatively approximate the energies by $E_H(\mathbf{p}) \approx E_{H,\text{eff}}(\mathbf{p}, t_{\text{min}})$. The results as a function of $\mathbf{p}$ are shown in Figs. 10 and 11 and compared to the

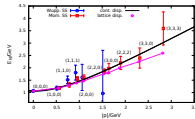


FIG. 11. Nucleon energies for different lattice momenta, in comparison to the continuum (solid curve) and lattice (points connected by dotted lines) dispersion relations Eqs. (39) and (41).

dispersion relation expectations. We also display results obtained with conventional smearing for small momenta where this is possible. For the two-point functions studied here the precision of the conventional results can be improved at little computational overhead by averaging over (for the absolute momentum values shown) six, eight or twelve equivalent directions. We have not done this, to allow for a “fair” comparison of the efficiency of the smearing methods. It is clear from the figures, however, that the maximally possible error reduction, assuming different momentum direction results to be statistically uncorrelated, would not affect any of our conclusions.

We do not expect either parametrization shown in Figs. 10 and 11 to perfectly describe the data as the lattice dispersion relations are for point particles, assuming a particular form of the effective Lagrangian. However, differences between the two functions are indicative for the size of possible lattice effects. While in the pion case differences between the parametrizations are on the present level of statistics insignificant, the nucleon data appear to be better described by the continuum dispersion relation. In the near future we will further investigate this, increasing our statistics and also employing a different smearing as described in Sec. III E.

VI. CONCLUSION

In many lattice gauge theory applications hadrons carrying high momenta are required. Due to the exponential increase of relative errors of n -point functions with Euclidean time distances and diminishing ground state sampling, high momenta previously were very difficult or impossible to achieve. In Sec. II we have introduced a new class of quark smearing methods for the construction of hadronic interpolators that address and substantially mitigate these problems. One particular realization of these methods, that is trivial to implement and comes with very little computational overhead, is momentum

Wuppertal smearing, defined in Eq. (24). We tested this very successfully in Sec. VB, enabling us to determine pion and nucleon energies for momenta as high as almost 2 GeV and 3 GeV, respectively, with just 200 measurements, see Figs. 10 and 11. These figures also include a comparison with the conventional method.

In Sec. VC we investigated the possibility of introducing an (additional) Lorentz boost [32, 33]. With and without a momentum phase factor included, this gave some improvement over unboosted Wuppertal smearing, possibly due to a dampening of the phase mismatch by the more rapid fall-off of the interpolating wave function in the direction of the momentum. However, the results obtained were inferior to those of momentum Wuppertal smearing, without any boost applied. We conclude that the intuition of contracting the wave function may be unjustified since the hadron is moving in real time but not in imaginary (Euclidean) time.

Iterative methods (momentum smearing or not) suffer from high iteration counts $n \propto a^{-2}$ as the continuum limit $a \rightarrow 0$ is approached, in addition to the naive volume factor due to an increasing number of lattice sites. Therefore, in Sec. III E we introduce other non-iterative (momentum) smearing methods that may be more suitable for small lattice spacings and that also allow for the construction of non-Gaussian shapes. These will be used by us in the near future.

Realizing momenta that are much larger than the hadron masses of interest is of fundamental importance in several modern applications, e.g., in direct determinations of (quasi) distribution amplitudes [10–12], of (quasi) (generalized) parton distributions [6–9] and moments of transverse momentum distributions [2–5]. The new method allowed us to extract nucleon masses (employing very moderate computational resources) up to momenta $p^2 \approx 7.9 \text{ GeV}^2$. This clearly makes the above observables amenable to lattice simulations in a realistic setting. For three-point functions, these methods potentially even allow for virtualities $Q^2 = 4p^2 \approx 30 \text{ GeV}^2$, switching a source momentum of $-p$ into a sink momentum of $+p$. The computation of these quantities with the new smearing is, depending on the observable, either in progress or planned.

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APPENDIX: BOOSTED (MOMENTUM) SMEARING

We describe how we introduce an anisotropy into (momentum) Wuppertal smearing to introduce a Lorentz contraction of the smearing function [32, 33] along the momentum direction proportional to $1/\gamma$, where in the continuum

$$\gamma = E(\mathbf{p})/m = \sqrt{1 + \frac{\mathbf{p}^2}{m^2}} \geq 1. \quad (49)$$

Note that m above denotes the hadron mass and the hadron momentum $\mathbf{p}$ differs from the momentum smearing parameter $\mathbf{k}$. We remark that as we are using equal Euclidean time interpolators there is no compelling reason why such a boost should be applied.

In order to “boost” the smearing function we need to replace

$$\nabla + i\mathbf{k} \mapsto \left(\frac{\nabla_{\parallel}}{\gamma} + \nabla_{\perp} \right) + i \left(\frac{k_{\parallel}}{\gamma} + k_{\perp} \right) \quad (50)$$

within Eq. (23), where $\nabla_{\parallel} = \mathbf{e}(\mathbf{e} \cdot \nabla)$, $\nabla_{\perp} = \nabla - \nabla_{\parallel}$ and

$$\mathbf{e} = \frac{\mathbf{p}}{|\mathbf{p}|} = \frac{\mathbf{k}}{|\mathbf{k}|}. \quad (51)$$

For the Laplacian this means

$$\Delta \mapsto \frac{\nabla_{\parallel}^2}{\gamma^2} + \nabla_{\perp}^2 = \left(\frac{1}{\gamma^2} - 1 \right) \nabla_{\parallel}^2 + \Delta, \quad (52)$$

where

$$\nabla_{\perp}^2 = \sum_{i,j} e_i e_j \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j}. \quad (53)$$

One can easily derive a corresponding iterative smearing: Generalizing Eqs. (9)–(14) and (24), we obtain

$$(\Phi_{\mathbf{k}}^* g)_{\mathbf{x}} = \frac{1}{N} \left\{ g_{\mathbf{x}} + \varepsilon' \left[\sum_{j=1}^{d-1} N_{ij} U_{\mathbf{x},j} e^{ik_j} g_{\mathbf{x}+\mathbf{j}} + \left(1 - \frac{1}{\gamma} \right) \sum_{i \neq j=1}^d \frac{e_i c_i}{2} (U_{\mathbf{x},j} U_{\mathbf{x}+\mathbf{j},-i} + U_{\mathbf{x},-i} U_{\mathbf{x}-i,j}) e^{ik_{-j-i}} g_{\mathbf{x}+\mathbf{j}-i} \right] \right\}, \quad (54)$$

where

$$N_j = N_j(\gamma, \mathbf{c}) = 1 + \left(\frac{1}{\gamma^2} - 1 \right) c_j \sum_{i=1}^d c_i. \quad (55)$$

The (arbitrary) normalization factor

$$N = N(\varepsilon', \gamma) = 1 + 2\varepsilon' \left(d - 1 + \frac{1}{\gamma} \right) \quad (56)$$

follows in the free case and is kept to avoid numerical overflow for high iteration counts. Like Eq. (24) this is most easily implemented, replacing the (APE smeared) links $U_{\mathbf{x},j}$ by $U_{\mathbf{x},j} e^{ik_j}$ and then iterating Eq. (54), using these modified transporters, instead of multiplying in the additional phase factors.

We remark that if the momentum is chosen parallel to a lattice axis, the second sum within Eq. (54) vanishes. In this case the only differences with respect to

the momentum Wuppertal smearing defined in Eq. (24) are that shifts in this direction carry a suppression factor $1/\gamma^2$, the normalization factor N and $\varepsilon' \neq \varepsilon$.

The free field solution for n iterations of Eq. (54) is a Gaussian with a phase factor $e^{ik \cdot \mathbf{x}}$ and variance,

$$\sigma^2 = 2na^2 \frac{\varepsilon'}{N(\varepsilon', \gamma)}, \quad (57)$$

in the directions perpendicular to the momentum, see Eq. (15). Parallel to the momentum the width is reduced by a factor $1/\gamma$, as it should be. The perpendicular variance above can be kept fixed, equating ε'/N with $\varepsilon/(1+2d\varepsilon)$. The resulting relation between the $\gamma = 1$ smearing parameter ε and the boosted parameter ε' reads:

$$\varepsilon' = \frac{\varepsilon}{1 + 2\varepsilon(1 - 1/\gamma^2)} \leq \varepsilon. \quad (58)$$

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P05：平凡化映射威尔逊流与 HMC

主题归属	规范理论、格点与算符基础
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页码范围：P05-p001 – P05-p029

Trivializing maps, the Wilson flow and the HMC algorithm

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Abstract

In lattice gauge theory, there exist field transformations that map the theory to the trivial one, where the basic field variables are completely decoupled from one another. Such maps can be constructed systematically by integrating certain flow equations in field space. The construction is worked out in some detail and it is proposed to combine the Wilson flow (which generates approximately trivializing maps for the Wilson gauge action) with the HMC simulation algorithm in order to improve the efficiency of lattice QCD simulations.

1. Introduction

The Nicolai map transforms interacting supersymmetric theories to non-interacting ones [1]. Supersymmetry is considered to be essential for the existence of these field transformations in view of the fact that their Jacobian is exactly cancelled by the fermion partition function.

In lattice gauge theory, a natural question to ask is whether there are field transformations that map the theory to its strong-coupling limit. In particular, if there are no matter fields, one is looking for substitutions

$$U = \mathcal{F}(V) \tag{1.1}$$

of the gauge field U in the functional integral whose Jacobian cancels the gauge-field action. Similarly to the Nicolai map, this kind of transformation maps the theory to a solvable one, but supersymmetry is not required and the Jacobian plays a different rôle.

$$U \xrightarrow{\mathcal{F}^{-1}} V \xrightarrow{\text{HMC}} V' \xrightarrow{\mathcal{F}} U'$$

Fig. 1. The proposed simulation algorithm for lattice QCD updates the gauge field U in three steps, following the arrows in this diagram, where the Hamilton function used in the HMC step has the standard kinetic term and includes the Jacobian of the field transformation $\mathcal{F}$ (cf. subsect. 2.4).

On a finite lattice, and if the gauge group is compact and connected, the existence of such trivializing maps is guaranteed by a general theorem on volume forms on compact manifolds (see ref. [2], Theorem 1.26, for example). One may be inclined to assume that these transformations are too complicated to be of any use. However, as explained later in this paper, it is possible to build up trivializing maps by integrating flows in field space, whose generators satisfy certain partial differential equations. The latter are quite tractable and can, to some extent, be solved analytically in the pure gauge theory. An application of trivializing maps, which can then be envisaged, is the acceleration of lattice QCD simulations.

The fact that the efficiency of the available simulation algorithms is unpredictable has always been a weakness of numerical lattice QCD. Already a while ago, empirical studies of the $SU(3)$ gauge theory by Del Debbio, Panagopoulos and Vicari [3] showed that the autocorrelation times of observables related to the topological charge of the gauge field tend to be large and appear to grow exponentially with the inverse of the lattice spacing. Moreover, Schaefer, Sommer and Virotta [4] recently found that the situation is, in this respect, essentially unchanged when the sea quarks are included in the simulations.

The rapid slowdown of the simulations at small lattice spacings may conceivably be overcome by combining approximately trivializing maps with the HMC simulation algorithm [5] (see fig. 1). Since the transformation moves the theory closer to the strong-coupling limit, where the HMC algorithm is known to be highly efficient, the autocorrelation times are, in general, expected to be reduced in this way. Evidently, for the combined algorithm to work out in practice, approximately trivializing maps must be found which are fairly simple and programmable. One of the goals of the present paper is thus to provide a solution to this problem.

2. Field transformations

Most concepts developed in this paper are expected to be widely applicable, but as explained above, the case of immediate interest is lattice QCD. In the following, the gauge group is therefore taken to be $SU(3)$. Since the quarks will play a spectator rôle, they will be added to the theory only in sect. 6, where the proposed simulation algorithm for lattice QCD is discussed.

2.1 Field space

The lattice theory is set up on a finite hypercubic lattice Λ with periodic boundary conditions. For notational convenience, the lattice spacing is set to unity. As usual, the gauge field variables $U(x, \mu) \in SU(3)$ are assumed to reside on the links (x, μ) of the lattice (where $x \in \Lambda$ and $\mu = 0, \dots, 3$). The expectation value of any observable $\mathcal{O}(U)$ is then given by the functional integral

$$\langle \mathcal{O} \rangle = \frac{1}{\mathcal{Z}} \int \mathcal{D}[U] \mathcal{O}(U) e^{-S(U)} \quad (2.1)$$

over the space of gauge fields. In this expression, $S(U)$ denotes the gauge-field action, $\mathcal{Z}$ the partition function and $\mathcal{D}[U]$ the product of the normalized $SU(3)$ -invariant integration measures of the link variables $U(x, \mu)$.

From a purely mathematical point of view, the space of lattice gauge fields is a power of $SU(3)$ and therefore a compact connected manifold. Field transformations are invertible maps of this manifold onto itself. Such transformations will always be required to be differentiable in both directions and orientation-preserving (here and below, “differentiable” means “infinitely often continuously differentiable”).

2.2 Right-invariant differential operators

Since the link variables $U(x, \mu)$ take values in a Lie group, it is natural to express differentiations with respect to them through a basis $\partial_{x,\mu}^a$, $a = 1, \dots, 8$, of differential operators that are invariant under the right-action of the group. The action of these operators on a differentiable function $f(U)$ of the gauge field is given by

$$\partial_{x,\mu}^a f(U) = \left. \frac{d}{dt} f(U_t) \right|_{t=0}, \quad U_t(y, \nu) = \begin{cases} e^{tT^a} U(x, \mu) & \text{if } (y, \nu) = (x, \mu), \\ U(y, \nu) & \text{otherwise,} \end{cases} \quad (2.2)$$

where T^a are the $SU(3)$ generators (see appendix A). In particular, $\partial_{x,\mu}^a$ transforms according to the adjoint representation under the left-action of $SU(3)$.

The operators $\partial_{x,\mu}^a$ go along with a basis of 1-forms on the field manifold,

$$\theta_{x,\mu}^a = -2 \operatorname{tr} \{ dU(x, \mu) U(x, \mu)^{-1} T^a \}, \quad (2.3)$$

such that

$$df(U) = \sum_{x,\mu} \theta_{x,\mu}^a \partial_{x,\mu}^a f(U) \quad (2.4)$$

for all functions $f(U)$.

2.3 Jacobian matrix

For any given gauge field $V(y, \nu)$, the transformation (1.1) produces another field $U(x, \mu) = [\mathcal{F}(V)](x, \mu)$. When considering such transformations, one needs to distinguish differentiations with respect to V from those with respect to U . The associated 1-forms must also be distinguished. In the following, all symbols carrying a hat represent quantities and operations referring to V .

The Jacobian matrix

$$[\mathcal{F}_*(V)](x, \mu; y, \nu)^{ab} = -2 \operatorname{tr} \{ \hat{\partial}_{y,\nu}^b U(x, \mu) U(x, \mu)^{-1} T^a \} \quad (2.5)$$

can be considered to be the kernel of a linear operator acting on link fields with values in $\mathfrak{su}(3)$. In particular,

$$\theta_{x,\mu}^a = \sum_{y,\nu} [\mathcal{F}_*(V)](x, \mu; y, \nu)^{ab} \hat{\theta}_{y,\nu}^b. \quad (2.6)$$

Since the functional integration measure $D[U]$ is proportional to the maximal product of these 1-forms, it follows that

$$D[U] = D[V] \det \mathcal{F}_*(V). \quad (2.7)$$

The Jacobian of the map (1.1) is thus $\det \mathcal{F}_*(V)$.

If the transformation satisfies

$$S(\mathcal{F}(V)) - \ln \det \mathcal{F}_*(V) = \text{constant}, \quad (2.8)$$

the substitution $U \rightarrow V$ of the integration variables in the functional integral maps the theory to the trivial one where the link variables are completely decoupled from

one another. The expectation values (2.1) are then given by

$$\langle \mathcal{O} \rangle = \int \mathcal{D}[V] \mathcal{O}(\mathcal{F}(V)). \quad (2.9)$$

Such trivializing maps thus contain the entire dynamics of the theory.

Although the remark is likely to remain an academic one, an intriguing observation is that the integral (2.9) can be simulated simply by generating uniformly distributed random gauge fields. Subsequent field configurations are uncorrelated in this case and there are therefore no autocorrelations in the data series for the observables $\mathcal{O}$.

2.4 Transformation behaviour of the HMC algorithm

The HMC algorithm [5] operates on the phase space associated to the field manifold. In particular, the transition $V \rightarrow V'$ in fig. 1 requires the equations of motion derived from the Hamilton function

$$\hat{H}(\hat{\pi}, V) = \frac{1}{2}(\hat{\pi}, \hat{\pi}) + S(\mathcal{F}(V)) - \ln \det \mathcal{F}_*(V) \quad (2.10)$$

to be integrated, where $\hat{\pi}(x, \mu) \in \mathfrak{su}(3)$ is the canonical momentum of $V(x, \mu)$.

Although the complete update algorithm for the field U described by fig. 1 looks different, it is in fact equivalent to the HMC algorithm with a non-standard Hamilton function. The equivalence can be established by noting that the transformation from V to U preserves the symplectic 2-form

$$\hat{\Omega} = \sum_{x, \mu} d\{\hat{\pi}^a(x, \mu) \hat{\theta}_{x, \mu}^a\}, \quad (2.11)$$

i.e. $\Omega = \hat{\Omega}$, if the momenta of the fields are transformed according to

$$\hat{\pi}^b(y, \nu) = \sum_{x, \mu} [\mathcal{F}_*(V)](x, \mu; y, \nu)^{ab} \hat{\pi}^a(x, \mu). \quad (2.12)$$

The evolution of the transformed fields is then governed by the Hamilton function

$$H(\pi, U) = \frac{1}{2}(\pi, K(U)\pi) + S(U) - \ln \det \mathcal{F}_*(V), \quad (2.13)$$

where

$$K(U) = \mathcal{F}_*(V)^T \mathcal{F}_*(V), \quad V = \mathcal{F}^{-1}(U). \quad (2.14)$$

Note that the Jacobian in eq. (2.13) cancels when the momenta are integrated over in the functional integral. The algorithm outlined in fig. 1 thus amounts to applying the HMC algorithm with a modified Hamilton function of the kind considered long ago by Duane et al. [6].

3. Transformations generated by flow equations

Flows in field space build up field transformations from infinitesimal transformations. The latter are generally easier to work with than integral transformations, because they refer to the current field only. Moreover, the differentiability and invertibility of the generated transformations is automatically guaranteed.

3.1 Flows in field space

An infinitesimal field transformation,

$$U \rightarrow U + \epsilon Z(U)U + \mathcal{O}(\epsilon^2), \quad (3.1)$$

is generated by a link field $[Z(U)](x, \mu)$ with values in $\mathfrak{su}(3)$. The continuous composition of such transformations amounts to integrating a flow equation

$$\dot{U}_t = Z_t(U_t)U_t \quad (3.2)$$

with respect to a fictitious time t . A simple choice for the generator of the flow is

$$[Z_t(U)]^a(x, \mu) = \partial_{x,\mu}^a \mathcal{W}_0(U), \quad (3.3)$$

$$\mathcal{W}_0(U) = \sum_{x, \mu \neq \nu} \text{tr}\{U(x, \mu, \nu)\}, \quad (3.4)$$

where $U(x, \mu, \nu)$ denotes the plaquette loop in the (μ, ν) -directions at the point x . In the following, this flow will be referred to as the “Wilson flow”.

If $Z_t(U)$ is a differentiable function of t and U , the flow equation (3.2) has a unique solution U_t for any specified initial value $U_0 = V$ and all $t \in (-\infty, \infty)$. Moreover, the solution is differentiable with respect to t and V (for a proof of these statements, see ref. [7], for example). It should be emphasized that the existence of the solution for all times is non-trivial and can only be guaranteed, without further assumptions, because the field manifold is compact.

3.2 Integrated transformations

At fixed t , the field U_t is a well-defined function of the initial field V . Through the integration of the flow equation, one thus obtains a differentiable transformation, $V \rightarrow U_t = \mathcal{F}_t(V)$, of the field space. The transformation is invertible and its inverse is differentiable, because the flow equation can be integrated backwards from t to 0. Moreover, since the Jacobian $\det \mathcal{F}_{t,*}(V)$ is equal to unity at $t = 0$ and does not pass through zero at any time, the transformation is also orientation-preserving and thus fulfils all requirements for an acceptable map of field space.

There is a useful compact expression for the Jacobian which is obtained starting from the equations

$$\frac{d}{dt} \ln \det \mathcal{F}_{t,*}(V) = \text{Tr} \{ \dot{\mathcal{F}}_{t,*}(V) \mathcal{F}_{t,*}(V)^{-1} \}, \quad (3.5)$$

$$\begin{aligned} [\dot{\mathcal{F}}_{t,*}(V)](x, \mu; y, \nu)^{ab} = & -2 \text{tr} \{ \partial_{y,\nu}^b [Z_t(U_t)](x, \mu) U_t(x, \mu) \} U_t(x, \mu)^{-1} T^a \\ & - \partial_{y,\nu}^b U_t(x, \mu) U_t(x, \mu)^{-1} [Z_t(U_t)](x, \mu) T^a. \end{aligned} \quad (3.6)$$

Noting

$$\partial_{y,\nu}^b = \sum_{x,\mu} [\mathcal{F}_*(V)](x, \mu; y, \nu)^{ab} \partial_{x,\mu}^a, \quad (3.7)$$

a few lines of algebra then lead to the formula

$$\ln \det \mathcal{F}_{t,*}(V) = \int_0^t ds \sum_{x,\mu} \{ \partial_{x,\mu}^a [Z_s(U)]^a(x, \mu) \}_{U=U_s}. \quad (3.8)$$

In the case of the Wilson flow, for example, the contribution of the Jacobian to the action of the field V ,

$$\ln \det \mathcal{F}_{t,*}(V) = -\frac{4b}{3} \int_0^t ds \mathcal{W}_0(U_s), \quad (3.9)$$

is proportional to the integral of the Wilson plaquette action along the flow.

4. Trivializing maps

Somewhat surprisingly, trivializing maps can, to some extent, be constructed explicitly in the pure gauge theory. The construction is explained in this section, assuming that the gauge action $S(U)$ is a sum of Wilson loops (plaquettes, rectangles, etc.).

4.1 Trivializing flows

If the generator $Z_t(U)$ of the flow (3.2) is such that

$$\int_0^t ds \sum_{x,\mu} \{ \partial_{x,\mu}^a [Z_s(U)]^a(x, \mu) \}_{U=U_s} = tS(U_t) + C_t, \quad (4.1)$$

where C_t may depend on t but not on the fields, the associated integrated transformations satisfy

$$S(\mathcal{F}_t(V)) - \ln \det \mathcal{F}_{t,*}(V) = (1-t)S(\mathcal{F}_t(V)) - C_t. \quad (4.2)$$

In particular, the transformation at $t=1$ is then a trivializing map.

Equation (4.1) is a rather implicit condition on the generator of the flow. However, when differentiated with respect to t , it assumes a more tractable form,

$$\sum_{x,\mu} \{ \partial_{x,\mu}^a [Z_t(U)]^a(x, \mu) - t \partial_{x,\mu}^a S(U) [Z_t(U)]^a(x, \mu) \} = S(U) + \dot{C}_t, \quad (4.3)$$

which involves the generator at time t only. Note that the differential condition (4.3) and the flow equation (3.2) imply eq. (4.1), i.e. it suffices to find a generator $Z_t(U)$ that satisfies eq. (4.3).

4.2 Existence of trivializing flows

Equation (4.3) is an inhomogeneous linear partial differential equation for the generator $Z_t(U)$. Since it is a scalar equation, one expects that there are many solutions. In the following, the solution will be obtained in the form

$$[Z_t(U)]^a(x, \mu) = -\partial_{x,\mu}^a \tilde{S}_t(U), \quad (4.4)$$

where the action $\tilde{S}_t(U)$ is to be determined.

When inserted in eq. (4.3), the ansatz (4.4) leads to the Laplace equation

$$\mathfrak{L}_t \bar{S}_t = S + \dot{C}_t, \quad (4.5)$$

$$\mathfrak{L}_t = \sum_{x,\mu} \{ -\partial_{x,\mu}^a \partial_{x,\mu}^a + t (\partial_{x,\mu}^a S) \partial_{x,\mu}^a \} \quad (4.6)$$

(for simplicity, the argument U is now often omitted). The operator $\mathfrak{L}_t$ is elliptic and symmetric with respect to the scalar product

$$(\phi, \psi) = \int \mathcal{D}[U] e^{-tS(U)} \phi(U)^* \psi(U). \quad (4.7)$$

$\mathfrak{L}_t$ has therefore a complete set of differentiable eigenfunctions and a purely discrete spectrum with no accumulation points (see ref. [8], sect. 1.6, for example). Moreover, since

$$(\phi, \mathfrak{L}_t \phi) = \sum_{x,\mu} (\partial_{x,\mu}^a \phi, \partial_{x,\mu}^a \phi) \geq 0, \quad (4.8)$$

the function $\phi(U) = 1$ is the only zero mode of $\mathfrak{L}_t$ and all other eigenfunctions have eigenvalues separated from the origin by a strictly positive spectral gap.

Now if one chooses C_t to be such that

$$\dot{C}_t = -(1, S)/(1, 1), \quad (4.9)$$

the zero-mode component is removed from the right-hand side of eq. (4.5) and

$$\bar{S}_t = \mathfrak{L}_t^{-1} (S + \dot{C}_t). \quad (4.10)$$

is then a well-defined expression that solves the equation. The differentiability of $\bar{S}_t$ with respect to t and U essentially follows from the ellipticity of $\mathfrak{L}_t$ (appendix E). A constructive proof of the existence of trivializing flows has thus been given.

4.3 Expansion in powers of t

The solution (4.10) is well defined but still quite implicit since it involves the inverse of an operator acting on functions of the gauge field. In this and the next subsection, the solution is worked out analytically in powers of t .

When the series

$$\tilde{S}_t = \sum_{k=0}^{\infty} t^k \tilde{S}^{(k)} \quad (4.11)$$

is inserted in eq. (4.5), the matching of the terms of a given order in t leads to the recursion

$$\mathfrak{L}_0 \tilde{S}^{(0)} = S + \dot{C}^{(0)}, \quad (4.12)$$

$$\mathfrak{L}_0 \tilde{S}^{(k)} = - \sum_{x,\mu} \partial_{x,\mu}^x S \partial_{x,\mu}^x \tilde{S}^{(k-1)} + \dot{C}^{(k)}, \quad k = 1, 2, \dots, \quad (4.13)$$

for the actions $\tilde{S}^{(k)}$. The Laplacian $\mathfrak{L}_0$ coincides with the colour-electric part of the Hamilton operator in lattice gauge theory in $4 + 1$ dimensions. In particular, its eigenfunctions are products of $SU(3)$ representation functions of the link variables. Sums of Wilson loops and products of Wilson loops, for example, are eigenfunctions of $\mathfrak{L}_0$ or can easily (algebraically) be decomposed into eigenfunctions.

The solution of the recursion,

$$\tilde{S}^{(0)} = \mathfrak{L}_0^{-1} S, \quad (4.14)$$

$$\tilde{S}^{(k)} = - \mathfrak{L}_0^{-1} \sum_{x,\mu} \partial_{x,\mu}^x S \partial_{x,\mu}^x \tilde{S}^{(k-1)}, \quad k = 1, 2, \dots, \quad (4.15)$$

is thus obtained in the form of sums of Wilson loops and products of Wilson loops. Note that, as already mentioned in subsect. 4.2, the constant function is the only zero mode of $\mathfrak{L}_0$. Since the smallest non-zero eigenvalue of $\mathfrak{L}_0$ is $\frac{1}{3}$, the right-hand sides of eqs. (4.14), (4.15) are therefore unambiguously determined up to an irrelevant additive constant.

4.4 Calculation of $\tilde{S}^{(0)}$ and $\tilde{S}^{(1)}$ in the Wilson theory

For illustration, the first two terms of the series (4.11) are now worked out explicitly for the plaquette action [9]

$$S_w(U) = -\frac{1}{6} \beta \mathcal{W}_0(U) \quad (4.16)$$

(where $\beta = 6/g_0^2$ denotes the inverse gauge coupling). A short calculation, using the completeness relation (A.5), shows that the leading term is

$$\tilde{S}^{(0)} = -\frac{1}{32} \beta \mathcal{W}_0 = \frac{3}{16} S_w. \quad (4.17)$$

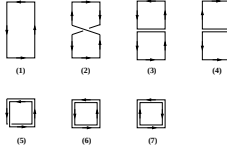


Fig. 2. Classes of loops and pairs of loops contributing to $\tilde{S}^{(1)}$ in the Wilson theory. The loops 5 – 7 reside on a single plaquette of the lattice. All other loops occupy two plaquettes which can lie in a plane or be at right angles in three dimensions.

To this order and up to a rescaling of the time parameter t , the trivializing flow in the Wilson theory thus coincides with the Wilson flow.

The expression to be worked out at the next order is

$$\tilde{S}^{(1)} = -\frac{1}{192} \beta^2 \Omega_0^{-1} \sum_{x,\mu} \partial_{x,\mu}^n \mathcal{W}_0 \partial_{x,\mu}^n \mathcal{W}_0. \quad (4.18)$$

The Wilson loops and products of Wilson loops that can occur at this point derive from the contractions of two plaquette loops with a common link. Altogether there are then seven classes C_i , $i = 1, \dots, 7$, of loops and pairs of loops to consider (see fig. 2). By summing the traces of the associated Wilson loops, each class C_i defines an action

$$\mathcal{W}_i = \sum_{C \in C_i} \text{tr}\{U(C)\} \quad \text{if } i = 1, 2, 5, \quad (4.19)$$

$$\mathcal{W}_i = \sum_{\{C, C'\} \in C_i} \text{tr}\{U(C)\} \text{tr}\{U(C')\} \quad \text{if } i = 3, 4, 6, 7, \quad (4.20)$$

where $U(C)$ denotes the ordered product of the link variables along the loop C . The sums in these equations extend over all possible positions of the loops on the lattice. Loops with opposite orientation are considered to be different and are both included in the sums.

Using the identity (A.5) again, some algebra now yields

$$\sum_{x,\mu} \partial_{x,\mu}^a \mathcal{W}_0 \partial_{x,\mu}^a \mathcal{W}_0 = \mathcal{W}_1 - \mathcal{W}_2 - \frac{4}{3}\mathcal{W}_3 + \frac{4}{3}\mathcal{W}_4 - 2\mathcal{W}_5 + \frac{2}{3}\mathcal{W}_6 - \frac{4}{3}\mathcal{W}_7 \quad (4.21)$$

up to an additive constant. Furthermore,

$$\mathfrak{L}_0 \mathcal{W}_1 = 8\mathcal{W}_1, \quad (4.22)$$

$$\mathfrak{L}_0 \mathcal{W}_2 = \frac{32}{3}\mathcal{W}_2 + \mathcal{W}_4, \quad (4.23)$$

$$\mathfrak{L}_0 \mathcal{W}_3 = 11\mathcal{W}_3 - \mathcal{W}_1, \quad (4.24)$$

$$\mathfrak{L}_0 \mathcal{W}_4 = \frac{16}{3}\mathcal{W}_4 + \mathcal{W}_2, \quad (4.25)$$

$$\mathfrak{L}_0 \mathcal{W}_5 = \frac{8}{3}\mathcal{W}_5 + 4\mathcal{W}_6, \quad (4.26)$$

$$\mathfrak{L}_0 \mathcal{W}_6 = \frac{8}{3}\mathcal{W}_6 + 4\mathcal{W}_5, \quad (4.27)$$

$$\mathfrak{L}_0 \mathcal{W}_7 = 12\mathcal{W}_7 + \text{constant}. \quad (4.28)$$

In the subspace of these functions, the operator $\mathfrak{L}_0$ can be easily inverted and the result

$$\begin{aligned} \tilde{S}^{(1)} = \frac{4}{192} g^2 \left\{ -\frac{4}{33}\mathcal{W}_1 + \frac{12}{119}\mathcal{W}_2 + \frac{4}{33}\mathcal{W}_3 - \frac{6}{119}\mathcal{W}_4 + \frac{3}{10}\mathcal{W}_5 \right. \\ \left. - \frac{4}{9}\mathcal{W}_6 + \frac{4}{9}\mathcal{W}_7 \right\} \end{aligned} \quad (4.29)$$

is thus obtained. Note that the smallness of the numerical coefficients in this formula is balanced, to some extent, by the number of loops per lattice point in the classes $\mathcal{C}_i$ (which are equal to 120, 12 and 6 for $i = 1, \dots, 4$, $i = 5, 6$ and $i = 7$ respectively).

4.5 Miscellaneous remarks

(a) *Higher orders.* The actions $\tilde{S}^{(k)}$, $k \geq 2$, can be computed algebraically following the steps taken in the previous subsection. Since all loops generated by contracting a plaquette loop with the loops at order $k-1$ must be considered, the work required for the calculation is however rapidly growing with k .

(b) *Locality and convergence.* The series (4.11) is an expansion in local terms whose footprint on the lattice increases proportionally to the order k . At the values of ℓ , where the expansion converges, the action $\tilde{S}_\ell$ is then guaranteed to be local as well.

The norm estimates in appendix E imply a lower bound on the convergence radius of the series, but this bound is rather poor and vanishes in the infinite-volume limit. It seems nevertheless plausible that the series has a non-zero convergence radius in this limit, because the inverse of the operator $\mathfrak{L}_t$ in eq. (4.10) is likely to remain bounded in a complex neighbourhood of $t = 0$. An analysis that takes the locality properties of $\mathfrak{L}_t$ into account will however be required to show this.

(c) *Truncation of the expansion.* If all terms in the series (4.11) of order $k \geq n$ are dropped, one obtains an approximately trivializing flow that satisfies eq. (4.2) up to an additive correction of order t^{n+1} . An at least partial cancellation of the action is achieved in this case as long as t is sufficiently small for the correction to be strongly suppressed.

(d) *Smoothing property.* The Wilson flow satisfies

$$\frac{d}{dt} S_w(U_t) = -\frac{d}{16} \sum_{x,\mu} \left\{ \partial_{x,\mu}^2 S_w(U) \partial_{x,\mu}^2 S_w(U) \right\}_{U=U_t} \leq 0 \quad (4.30)$$

and therefore lowers the Wilson action as t increases. To leading order in t , the trivializing flow constructed in this section for the Wilson theory thus has a smoothing effect on the gauge field. On the other hand, if the flow is followed in the reverse direction, the gauge field tends to become rougher.

(e) *Topological charge sectors.* In lattice QCD, the topological (instanton) sectors are not a property of the field manifold alone, but are expected to emerge dynamically when the continuum limit is approached. The fact that trivializing maps completely “straighten out” the sectors is therefore not in conflict with the topological properties of the field space.

(f) *Renormalization group.* By composing the trivializing map $U = \mathcal{F}_1(V)$ in the Wilson theory with its inverse at another value of the gauge coupling, one obtains a group of transformations whose only effect on the action is a shift of the coupling. The locality properties of these transformations are not transparent, however, and could be quite different from the ones of a Wilsonian “block spin” transformation.

5. Numerical integration of the Wilson flow

The discussion in sect. 1 now suggests to combine the HMC algorithm with the field transformations generated by the trivializing flow constructed in the previous sec-

tion. In particular, if the Wilson gauge action is used, the transformations generated by the Wilson flow may lead to an algorithm with improved sampling efficiency.

5.1 Forward integration

There is a wide range of numerical integration methods that can in principle be used to integrate the Wilson flow (see ref. [10], for example). The Euler scheme discussed in the following performs the integration in time steps of size ϵ and updates the link variables one after another according to

$$U(x, \mu) \rightarrow U'(x, \mu) = e^{\epsilon[Z(U)](x, \mu)} U(x, \mu), \quad (5.1)$$

where

$$[Z(U)](x, \mu) = T^a \partial_{x, \mu}^a \mathcal{W}_0(U). \quad (5.2)$$

Starting from the gauge field U_t at time t , the field at time $t + \epsilon$ is thus obtained by running through all links (x, μ) on the lattice and updating the link variable residing there. Note that the ordering of the links matters, since the old value of $U(x, \mu)$ is replaced by the new one before going to the next link.

The generator of the flow is explicitly given by

$$\begin{aligned} [Z(U)](x, \mu) = & - \sum_{\nu \neq \mu} \mathcal{P} \{ U(x, \mu) U(x + \hat{\mu}, \nu) U(x + \hat{\nu}, \mu)^{-1} U(x, \nu)^{-1} \\ & + U(x, \mu) U(x + \hat{\mu} - \hat{\nu}, \nu)^{-1} U(x - \hat{\nu}, \mu)^{-1} U(x - \hat{\nu}, \nu) \}, \end{aligned} \quad (5.3)$$

where $\hat{\mu}$ denotes the unit vector in direction μ and

$$\mathcal{P}\{M\} = \frac{1}{2}(M - M^\dagger) - \frac{1}{8}\text{tr}(M - M^\dagger) \quad (5.4)$$

projects any 3×3 matrix M to $\mathfrak{su}(3)$. The Euler integration of the Wilson flow thus amounts to applying a number of “stout smearing” steps [11], except that the link variables are here updated one by one rather than all at once.

5.2 Backward integration

The application of n Euler sweeps maps the initial field $V = U_0$ to the field $U = U_{ne}$ at time $t = ne$. If t is held fixed and n is taken to infinity, this map converges to the transformation obtained by integrating the Wilson flow exactly. However, the HMC algorithm may potentially be combined with the map defined by the Euler integrator

at fixed n and ϵ . For this proposition to be a viable option, the transformation must be invertible, i.e. one must be able to trace back the Euler integration by inverting the link update steps one by one in the reverse order.

The question is thus whether eq. (5.1) has a unique solution $U(x, \mu)$ given $U'(x, \mu)$ (and keeping all other link variables fixed). As explained in appendix D, the answer is affirmative, for arbitrary values of the field variables, if

$$|\epsilon| < \frac{1}{8}. \quad (5.5)$$

Moreover, in this range of ϵ , the solution $U(x, \mu) = e^{-\epsilon X_*} U'(x, \mu)$ can be obtained through the fixed-point iteration

$$X_0 = 0, \quad (5.6)$$

$$X_{n+1} = \left\{ [Z(U)](x, \mu) \right\}_{U(x, \mu) = e^{-\epsilon X_n} U'(x, \mu)}, \quad n = 0, 1, 2, \dots, \quad (5.7)$$

which converges to X_* at an exponential rate.

In appendix D it is also shown that the Jacobian of the transformation (5.1) is strictly positive in the range (5.5). The field transformations obtained through the Euler integration of the Wilson flow are therefore orientation-preserving diffeomorphisms of the field manifold, as required for acceptable maps of field space.

5.3 Jacobian matrix of the Euler integrator

The Euler step (5.1) amounts to applying the field transformation

$$[\mathcal{E}_{x, \mu}(U)](y, \nu) = \begin{cases} e^{\epsilon [Z(U)](x, \mu)} U(x, \mu) & \text{if } (y, \nu) = (x, \mu), \\ U(y, \nu) & \text{otherwise,} \end{cases} \quad (5.8)$$

to the current gauge field. An Euler sweep is then the composition product of these transformations over all links (x, μ) . It is straightforward to show that the Jacobian matrix of a composed transformation is the product of the Jacobian matrices of the factors. The Jacobian matrix of the Euler integrator is therefore an ordered product of the Jacobian matrices

$$[\mathcal{E}_{x, \mu, *}(U)](y, \nu; z, \rho)^{ac} = -2 \text{tr} \left\{ \partial_{\epsilon, \rho}^c [\mathcal{E}_{x, \mu}(U)](y, \nu) [\mathcal{E}_{x, \mu}(U)](y, \nu)^{-1} T^a \right\} \quad (5.9)$$

of the one-link transformations (5.8).

The matrix (5.9) can be expressed through the derivative

$$[Z_*(U)](y, \nu; z, \rho)^{bc} = \partial_{\epsilon, \rho}^c [Z(U)]^b(y, \nu) \quad (5.10)$$

of the generator of the Wilson flow. Explicitly, one finds that

$$\begin{aligned} [\mathcal{E}_{x,\mu,*}(U)](y, \nu; z, \rho)^{ac} &= \delta^{ac} \delta_{yz} \delta_{\nu\rho} + \delta_{xy} \delta_{\mu\nu} \left\{ (e^{\text{Ad } X} - 1)^{ac} \delta_{yz} \delta_{\nu\rho} \right. \\ &\quad \left. + \epsilon J(-X)^{ab\gamma} [Z_s(U)](y, \nu; z, \rho)^{bc} \right\}_{X=\epsilon[Z(U)](x,\mu)}, \end{aligned} \quad (5.11)$$

where use was made of the SU(3) formulae listed in appendix A and B.

5.4 Jacobian of the integrated transformations

The Euler integrator generates a sequence of fields

$$V = U_0 \rightarrow U_e \rightarrow U_{2e} \rightarrow \dots \rightarrow U_{ne} = U \quad (5.12)$$

by sweeping through the lattice n times and updating the link variables one by one in a specified order. In the following, the intermediate field configurations obtained starting from U_{ke} and updating the link variables on all links that come before (x, μ) will be denoted by $U_{ke, [x, \mu]}$. In particular, $U_{ke, [x, \mu]} = U_{ke}$ if (x, μ) is the first link and

$$U_{ke, [y, \nu]} = \mathcal{E}_{x, \mu}(U_{ke, [x, \mu]}) \quad (5.13)$$

if (y, ν) follows (x, μ) in the chosen link order.

Since the transformation $V \rightarrow U = \mathcal{F}_{ne}(V)$ is a composition product of one-link update steps, its Jacobian

$$\det \mathcal{F}_{ne,*}(V) = \prod_{k=0}^{n-1} \prod_{x, \mu} \det \mathcal{E}_{x, \mu,*}(U_{ke, [x, \mu]}) \quad (5.14)$$

factorizes into the product of the Jacobians of the steps. The latter coincide with the determinants of certain 8×8 matrices given explicitly in appendix C.

6. Proposed simulation algorithm for lattice QCD

With respect to the QCD simulation algorithms used to date, the combination of the transformations obtained through the Euler integration of the Wilson flow and the

HMC algorithm is expected to sample the topological sectors more quickly and to be generally more efficient. The use of the Wilson flow is suggested if the gauge action coincides with the Wilson plaquette action. For other actions, the appropriate flow can be easily constructed following the lines of sect. 4.

6.1 Choice of the parameters

The parameters of the Euler integrator are the integration step size ϵ and the number n of Euler sweeps that are applied. One also needs to choose a definite ordering of the links of the lattice.

The step size ϵ should be positive and not larger than, say, $1/16$ so that the invertibility of the Euler integrator is guaranteed within a safe margin. Some tuning of the integration time $n\epsilon$ will certainly be required in order to maximize the efficiency of the algorithm. Note that the unit of time differs from the one used in subsect. 4.4, i.e. setting $t = 1$ there corresponds to an integration time $n\epsilon = \beta/32$.

The ordering of the links can be chosen arbitrarily. One may, for example, first visit the links $(x, 0)$ on all even points x , then the links $(x, 0)$ on all odd points, then the links $(x, 1)$ on all even points, and so on. This ordering is well suited for parallel processing, since the link variables in a given direction on the even (odd) sites are decoupled from one another and can therefore be updated in parallel.

6.2 Force calculation

At the beginning of an update cycle, the current gauge field U is transformed to the field $V = \mathcal{F}_{ns}^{-1}(U)$ by applying n backward Euler sweeps to U . The force that drives the molecular-dynamics evolution of V is then given by

$$F(z, \rho)^c = \hat{\partial}_{z, \rho}^c \{S(\mathcal{F}_{ns}(V)) - \ln \det \mathcal{F}_{ns, \star}(V)\}, \quad (6.1)$$

where the action $S(U)$ now includes the usual sea-quark pseudo-fermion actions (for simplicity, the dependence on the pseudo-fermion fields is suppressed).

Each time the force is to be calculated, the current field V must be transformed to U again by applying n forward Euler sweeps (see fig. 3). The fields $U_0, U_1, \dots, U_{ns}$ generated in this process should be stored in memory so that they will be available when the force is propagated from U to V . Note that the intermediate fields

$$U_{k\star, [x, \mu]}(y, \nu) = \begin{cases} U_{k\star}(y, \nu) & \text{if } (y, \nu) \geq (x, \mu), \\ U_{k\star+\epsilon}(y, \nu) & \text{otherwise,} \end{cases} \quad (6.2)$$

are then also available.

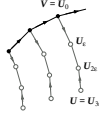


Fig. 3. The proposed algorithm evolves the field V using the standard HMC algorithm (thick line). The force that drives the molecular-dynamics evolution is obtained by forward integration of the Wilson flow and subsequent backward propagation of the derivatives of the action and the Jacobian of the flow ($n = 3$ in this figure).

The factorization (5.14) of the Jacobian implies

$$F(z, \rho)^c = \partial_{z, \rho}^c S(U_{ns}) - \sum_{k=0}^{n-1} \sum_{x, \mu} \partial_{z, \rho}^c \ln \det \mathcal{E}_{x, \mu, s}(U_{kx, [x, \mu]}). \quad (6.3)$$

Moreover,

$$\partial_{z, \rho}^c S(U_{ns}) = \sum_{y, \nu} \partial_{y, \nu}^c S(U) \Big|_{U=U_{ns}} \mathcal{F}_{ns, s}(y, \nu; z, \rho)^{ac}, \quad (6.4)$$

and there is a similar formula for the other terms in eq. (6.3) involving the Jacobian matrices of the transformations $V \rightarrow U_{kx, [x, \mu]}$. All these matrices, as well as the one in eq. (6.4), are products of the Jacobian matrices of the one-link transformations (5.8). The force can therefore be computed recursively as follows:

1. Set $U = U_{ns}$ and $F(z, \rho)^c = \partial_{z, \rho}^c S(U)$.
2. For k from $n-1$ to 0, run through all links (x, μ) in reverse order, set $U = U_{kx, [x, \mu]}$ and update the force according to

$$F(z, \rho)^c \rightarrow \sum_{y, \nu} F(y, \nu)^a [\mathcal{E}_{x, \mu, s}(U)](y, \nu; z, \rho)^{ac} - \partial_{z, \rho}^c \ln \det \mathcal{E}_{x, \mu, s}(U). \quad (6.5)$$

Note that the field U backtracks the forward integration of the Wilson flow in the course of the recursion. Since the Jacobian matrix $\mathcal{E}_{x, \mu, s}(U)$ differs from unity only

on the links sharing a plaquette with (x, μ) , the total computational effort required for the propagation (6.5) of the force is expected to be similar to the one required for n applications of a nearest-neighbour gauge-covariant difference operator to the force field.

6.3 Domain-decomposed algorithm

Field transformations can also be combined with the DD-HMC algorithm [12]. Only the so-called active link variables are transformed in this case, but the transformations may depend on the inactive field components.

In the pure gauge theory, it is then again possible to construct trivializing flows. They operate on the active link variables and contract the gauge action in each domain to a constant (i.e. an expression depending on the inactive link variables only). These flows are not the same as the ones constructed in sect. 4, but can be obtained by solving the differential equations derived there. In particular, the trivializing flow for the Wilson action is, to leading order, a slightly modified Wilson flow, where the plaquettes containing ν active links are given the weight $4/\nu$.

7. Concluding remarks

Trivializing maps and flows in field space have here been discussed with a particular application in mind. The underlying concepts are fairly general, however, and may have other uses in rigorous constructive work, numerical perturbation theory or in connection with renormalizable smoothing techniques, for example.

Whether the proposed combination of the Wilson flow and the HMC algorithm does indeed sample the topological sectors in lattice QCD more efficiently than the simulation algorithms used so far remains to be determined. As the lattice spacing is taken to smaller and smaller values, the quark fields may eventually have to be included in the flow and perhaps also the next-to-leading order correction discussed in sect. 4. Trivializing maps in the presence of matter fields is, in any case, a subject that deserves to be studied in its own right.

In the course of this work, I profited from many discussions with Filippo Palombi and Stefan Schaefer of various questions related to the slow topology-switching in current lattice QCD simulations. I also wish to thank Stefan Schaefer, Rainer Sommer and Francesco Viotto for sharing some of their simulation results before publication.

Appendix A. SU(3) notation

A.1 Group generators

The Lie algebra $\mathfrak{su}(3)$ of $SU(3)$ may be identified with the space of all anti-hermitian traceless 3×3 matrices. With respect to a basis T^a , $a = 1, \dots, 8$, of such matrices, the elements $X \in \mathfrak{su}(3)$ are given by

$$X = X^a T^a, \quad (\text{A.1})$$

where $(X^1, \dots, X^8) \in \mathbb{R}^8$ (repeated group indices are automatically summed over).

The generators T^a are assumed to satisfy the normalization condition

$$\text{tr}\{T^a T^b\} = -\frac{1}{2}\delta^{ab}. \quad (\text{A.2})$$

The structure of the Lie algebra is then encoded in the commutators

$$[T^a, T^b] = f^{abc} T^c, \quad (\text{A.3})$$

while the completeness of the generators implies

$$\{T^a, T^b\} = -\frac{1}{3}\delta^{ab} + id^{abc} T^c, \quad (\text{A.4})$$

$$T_{\alpha\beta}^a T_{\gamma\delta}^a = -\frac{1}{2} \left\{ \delta_{\alpha\delta} \delta_{\beta\gamma} - \frac{1}{3} \delta_{\alpha\beta} \delta_{\gamma\delta} \right\}. \quad (\text{A.5})$$

It follows from these equations that the structure constants f^{abc} and the tensor d^{abc} are both real. Moreover, f^{abc} is totally anti-symmetric in the indices and d^{abc} totally symmetric and traceless.

A.2 Adjoint representation

The representation space of the adjoint representation of $\mathfrak{su}(3)$ is the Lie algebra itself, i.e. the elements X of $\mathfrak{su}(3)$ are represented by linear transformations

$$\text{Ad } X : \mathfrak{su}(3) \mapsto \mathfrak{su}(3), \quad (\text{A.6})$$

$$\text{Ad } X \cdot Y = [X, Y] \quad \text{for all } Y \in \mathfrak{su}(3). \quad (\text{A.7})$$

The action of $\text{Ad } X$ on the group generators is given by

$$\text{Ad } X \cdot T^b = T^a (\text{Ad } X)^{ab}, \quad (\text{A.8})$$

where

$$(\text{Ad } X)^{ab} = -f^{abc} X^c \quad (\text{A.9})$$

is a real antisymmetric 8×8 matrix.

A.3 Matrix norms

The natural scalar product in $\mathfrak{su}(3)$ is

$$(X, Y) = X^a Y^a = -2 \text{tr}\{XY\}. \quad (\text{A.10})$$

In particular, $\|X\| = (X, X)^{1/2}$ is a possible definition of the norm of $X \in \mathfrak{su}(3)$.

Another useful matrix norm derives from the square norm

$$\|v\|_2 = \{|v_1|^2 + |v_2|^2 + |v_3|^2\}^{1/2} \quad (\text{A.11})$$

of complex colour vectors v . If A is any complex 3×3 matrix, one defines

$$\|A\|_2 = \max_{\|v\|_2=1} \|Av\|_2. \quad (\text{A.12})$$

This norm satisfies

$$\|A + B\|_2 \leq \|A\|_2 + \|B\|_2, \quad \|AB\|_2 \leq \|A\|_2 \|B\|_2, \quad (\text{A.13})$$

for all matrices A, B . Moreover, if A is hermitian or antihermitian, $\|A\|_2$ is equal to the maximum of the absolute values of its eigenvalues.

Appendix B. Properties of the $\text{SU}(3)$ exponential function

B.1 Lipschitz bound

For any $X, Y \in \mathfrak{su}(3)$, the relation

$$\|e^X - e^Y\|_2 = \|1 - e^{-X} e^Y\|_2 \quad (\text{B.1})$$

follows from the fact that e^X is unitary. Using the identity

$$1 - e^{-X} e^Y = \int_0^1 ds e^{-sX} (X - Y) e^{sY} \quad (\text{B.2})$$

and the subadditivity (A.13) of the norm, the Lipschitz bound

$$\|e^X - e^Y\|_2 \leq \|X - Y\|_2 \quad (\text{B.3})$$

is then obtained.

B.2 Differential of the exponential map

Let X be an element of the Lie algebra $\mathfrak{su}(3)$. A linear mapping $J(X) : \mathfrak{su}(3) \rightarrow \mathfrak{su}(3)$ is then defined by

$$J(X) \cdot Y = e^{-X} \left. \frac{d}{dt} e^{X+tY} \right|_{t=0} \quad \text{for all } Y \in \mathfrak{su}(3). \quad (\text{B.4})$$

$J(X)$ is referred to as the differential of the exponential map. Scaling the exponents by a parameter s , as above, one obtains the representation

$$J(X) \cdot Y = \int_0^1 ds e^{-sX} Y e^{sX} \quad (\text{B.5})$$

and thus the expansion

$$J(X) = 1 + \sum_{k=1}^{\infty} \frac{(-1)^k}{(k+1)!} (\text{Ad } X)^k \quad (\text{B.6})$$

which is absolutely convergent for any $X \in \mathfrak{su}(3)$.

The action of $J(X)$ on the group generators T^a is given by

$$J(X) \cdot T^b = T^a J(X)^{ab}, \quad (\text{B.7})$$

where $J(X)^{ab}$ is a real 8×8 matrix. Note that

$$J(X)^T = J(-X) = e^{\text{Ad } X} J(X), \quad J(X) \text{Ad } X = 1 - e^{-\text{Ad } X}. \quad (\text{B.8})$$

Moreover, eq. (B.5) implies

$$\|J(X) \cdot Y\|_2 \leq \|Y\|_2 \quad (\text{B.9})$$

for all $X, Y \in \mathfrak{su}(3)$.

Appendix C. Jacobian of the Euler step

The Jacobian matrix (5.11) of the Euler step is equal to unity except for some non-zero elements along the row $(y, \nu) = (x, \mu)$. Its determinant therefore coincides with the determinant of the $(x, \mu; x, \mu)$ -element

$$A^{ac} = \left\{ \left(e^{Ad X} \right)^{ac} + e J(-X)^{ab} [Z_*(U)](x, \mu; x, \mu)^{bc} \right\}_{X=e[Z(U)](x, \mu)} \quad (C.1)$$

of the matrix.

The derivative $[Z_*(U)](x, \mu; x, \mu)^{bc}$ can be worked out explicitly in terms of the plaquette sum

$$M = \sum_{\nu \neq \mu} \left\{ U(x, \mu) U(x + \hat{\mu}, \nu) U(x + \hat{\nu}, \mu)^{-1} U(x, \nu)^{-1} + \right. \\ \left. U(x, \mu) U(x + \hat{\mu} - \hat{\nu}, \nu)^{-1} U(x - \hat{\nu}, \mu)^{-1} U(x - \hat{\nu}, \nu) \right\}. \quad (C.2)$$

A few lines of algebra then lead to the expression

$$A^{ac} = B^{ac} + \frac{1}{2} e C^{ab} \left\{ i d^{bcd} \text{tr} \{ T^d (M + M^\dagger) \} - \frac{1}{3} \delta^{bc} \text{tr} \{ M + M^\dagger \} \right\} \quad (C.3)$$

where

$$B = \frac{1}{2} (e^{Ad X} + 1), \quad (C.4)$$

$$C = J(-X), \quad X = -e \mathcal{P} \{ M \}. \quad (C.5)$$

In particular,

$$\det A = 1 - \frac{1}{3} e \text{tr} \{ M + M^\dagger \} + O(e^2), \quad (C.6)$$

as expected from eq. (3.9).

Appendix D. Inversion of the Euler step

D.1 Basic norm bounds

For any complex 3×3 matrix M , the inequality

$$\|\mathcal{P}\{M\}\|_2 \leq \frac{4}{3}\|M\|_2 \quad (\text{D.1})$$

can be established in a few lines. One first observes that

$$\frac{1}{2}(M - M^\dagger) = ADA^{-1}, \quad A \in \text{SU}(3), \quad (\text{D.2})$$

where D is a diagonal matrix with diagonal elements $\lambda_1, \lambda_2, \lambda_3$. Setting

$$\bar{\lambda} = \frac{4}{3} \sum_{k=1}^3 \lambda_k, \quad (\text{D.3})$$

the estimates

$$\begin{aligned} \|\mathcal{P}\{M\}\|_2 &= \max_k |\lambda_k - \bar{\lambda}| \\ &\leq \frac{4}{3} \max_k |\lambda_k| = \frac{2}{3} \|M - M^\dagger\|_2 \leq \frac{4}{3} \|M\|_2 \end{aligned} \quad (\text{D.4})$$

then show that the inequality (D.1) holds for all matrices M .

An immediate consequence of the bound (D.1) and the Lipschitz bound (B.3) is that

$$\|\mathcal{P}\{A(e^X - e^Y)B\}\|_2 \leq \frac{4}{3}\|X - Y\|_2 \quad (\text{D.5})$$

for all $A, B \in \text{SU}(3)$ and $X, Y \in \mathfrak{su}(3)$.

D.2 Solution of eq. (5.1)

For a given link (x, μ) and any fixed $U'(x, \mu) \in \text{SU}(3)$, the function

$$f(X) = \{|Z(U)|\}(x, \mu) \Big|_{U(x, \mu) := e^{-\epsilon X} U'(x, \mu)} \quad (\text{D.6})$$

maps $X \in \mathfrak{su}(3)$ back to $\mathfrak{su}(3)$. Recalling eq. (5.3), the inequality (D.5) immediately implies that

$$\|f(X) - f(Y)\|_2 \leq k\|X - Y\|_2, \quad k = 8|\epsilon|. \quad (\text{D.7})$$

The function $f(X)$ is therefore a strict contraction if the integration step size ϵ is in the range (5.5) (which is assumed to be the case from now on).

It is not difficult to prove that strict contractions in a complete metric space have a unique fixed point (see ref. [13], Theorem V.18, for example). In the present case, the fixed point X_* can be computed by noting that the sequence $X_0 = 0, X_1, X_2, \dots$ generated through the recursion $X_{n+1} = f(X_n)$ satisfies

$$\|X_n - X_*\|_2 \leq k \|X_{n-1} - X_*\|_2 \quad (\text{D.8})$$

and therefore rapidly converges to X_* . The matrix

$$U(x, \mu) = e^{-\epsilon X_*} U'(x, \mu) \quad (\text{D.9})$$

then provides a solution of eq. (5.1). Moreover, there is no other solution, because the fixed point X_* of $f(X)$ is unique.

D.3 Positivity of the Jacobian

In order to prove that the determinant of the Jacobian matrix (C.1) is positive at all ϵ in the range (5.5), it suffices to show that the matrix has no zero mode, i.e. that

$$A^{ac} Y^c = 0 \quad (\text{D.10})$$

implies $Y = 0$. To this end, eq. (D.10) is first written in the form

$$Y^a = -\epsilon J(X)^{ab} W^b, \quad (\text{D.11})$$

where

$$X = \epsilon [Z(U)](x, \mu), \quad W^b = [Z_*(U)](x, \mu; x, \mu)^{bc} Y^c. \quad (\text{D.12})$$

Recalling eq. (5.10), the formula

$$W = \lim_{t \rightarrow 0} \frac{1}{t} \left\{ [Z(U)](x, \mu)|_{U(x, \mu) \rightarrow e^{tY} U(x, \mu)} - [Z(U)](x, \mu) \right\} \quad (\text{D.13})$$

may then be derived from which one infers that

$$\|W\|_2 \leq 8 \|Y\|_2. \quad (\text{D.14})$$

The inequality (D.5) has here been used again and also the fact that the norm is a continuous map from $\mathfrak{su}(3)$ to $\mathbb{R}$. The combination of eq. (D.11) and the bounds (B.9) and (D.14) now implies $\|Y\|_2 \leq k \|Y\|_2$ and thus $Y = 0$.

Appendix E. Differentiability of $\bar{S}_t$

The action $\bar{S}_t$ solves an elliptic partial differential equation with smooth coefficients and is therefore guaranteed (by elliptic regularity) to be a differentiable function of the gauge field U . In this appendix, the simultaneous differentiability in the time t is established by expanding $\bar{S}_t$ in powers of $t - t_0$ around any fixed time t_0 . Using Sobolev norms, the expansion can be shown to converge if t is sufficiently close to t_0 . The pointwise uniform convergence of the series and its derivatives, and therefore the differentiability of $\bar{S}_t$, then follows from Sobolev's lemma.

E.1 Sobolev spaces on the field manifold

The definition of the Sobolev spaces on a compact manifold is quite involved and will not be reviewed here. An introduction to the subject and the proof of all statements made in this subsection is given in the first three sections of ref. [8], for example.

Let C^∞ be the space of differentiable functions on the field manifold and H_k the associated Sobolev space of order $k \in \mathbb{Z}$. The latter is the completion of C^∞ with respect to a certain norm $\|\cdot\|_k$. A characteristic feature of these norms is that the bounds

$$\|\mathfrak{D}\phi\|_k \leq c_k \|\phi\|_{k+p} \quad (\text{E.1})$$

hold for all $\phi \in C^\infty$ and any differential operator $\mathfrak{D}$ of order p with smooth coefficients, where the constants c_k depend on $\mathfrak{D}$ but not on ϕ . Such differential operators thus extend to bounded linear operators from H_{k+p} to H_k . Moreover, $\|\phi\|_k \leq \|\phi\|_l$ if $k < l$ and therefore $H_l \subset H_k$.

A fairly concrete description of the Sobolev space H_k can be given when k is even and non-negative. For any $t \in \mathbb{R}$ and $j = 0, 1, 2, \dots$, another norm $\|\phi\|_{t,j}$ of $\phi \in C^\infty$ may be defined through

$$\|\phi\|_{t,j} = \|\phi\| + \|\mathfrak{L}_t^j \phi\|, \quad (\text{E.2})$$

where $\|\cdot\|$ is the norm associated to the scalar product (4.7). The fact that $\mathfrak{L}_t$ is a second-order elliptic differential operator then implies

$$\|\phi\|_{2j} \leq a_{t,j} \|\phi\|_{t,j}, \quad \|\phi\|_{t,j} \leq b_{t,j} \|\phi\|_{2j} \quad (\text{E.3})$$

for some constants $a_{t,j}, b_{t,j}$. In particular, H_{2j} is the completion of C^∞ with respect to the norm $\|\cdot\|_{t,j}$.

E.2 Properties of the inverse of $\mathfrak{L}_t$

Let $\mathfrak{P}_t$ be the orthogonal projector to the zero mode of $\mathfrak{L}_t$ in the Hilbert space with scalar product (4.7). Its action on any function $\phi \in C^\infty$ is given by

$$\mathfrak{P}_t \phi = (1, \phi)/(1, 1). \quad (\text{E.4})$$

Note that $\mathfrak{P}_t$ projects ϕ to a constant, but the constant depends on the definition of the scalar product $(\cdot, \cdot)$ and therefore on t . The action of the inverse $\mathfrak{G}_t$ of $\mathfrak{L}_t$ on ϕ is then determined by the equations

$$\mathfrak{L}_t \mathfrak{G}_t \phi = (1 - \mathfrak{P}_t) \phi, \quad \mathfrak{P}_t \mathfrak{G}_t \phi = 0. \quad (\text{E.5})$$

As already mentioned in subsect. 4.2, the ellipticity of $\mathfrak{L}_t$ and the absence of further zero modes imply that these equations have, for any fixed t , a unique differentiable solution $\mathfrak{G}_t \phi$. Evidently, $\tilde{S}_t = \mathfrak{G}_t S$.

Since $\mathfrak{L}_t$ has a spectral gap in the subspace orthogonal to the zero mode, $\mathfrak{G}_t$ is a bounded operator with respect to the norm $\|\cdot\|$. As a consequence, there exists a constant g_t such that

$$\|\mathfrak{G}_t \phi\|_{t,j} \leq g_t \|\phi\|_{t,j-1} \quad (\text{E.6})$$

for all $j = 1, 2, \dots$ and all $\phi \in C^\infty$. Moreover, recalling the inequalities (E.3), one concludes that

$$\|\mathfrak{G}_t \phi\|_{2j} \leq g_{t,j} \|\phi\|_{2j-2} \quad (\text{E.7})$$

for some other constants $g_{t,j}$.

E.3 Expansion of $\tilde{S}_t$

For any fixed time t_0 , the operator $\mathfrak{L}_t$ may be decomposed according to

$$\mathfrak{L}_t = \mathfrak{L}_{t_0} + (t - t_0) \mathfrak{L}', \quad \mathfrak{L}' = \sum_{x,\mu} (\partial_{x,\mu}^2 S) \partial_{x,\mu}. \quad (\text{E.8})$$

Each term in the power series

$$\psi_t = \sum_{n=0}^{\infty} (t_0 - t)^n (\mathfrak{G}_{t_0} \mathfrak{L}')^n \mathfrak{G}_{t_0} S \quad (\text{E.9})$$

is then a well-defined differentiable function of both t and U . Moreover, since

$$\|\mathfrak{G}_{t_0}\mathfrak{L}'\phi\|_{2j} \leq g_{t_0,j}\|\mathfrak{L}'\phi\|_{2j-2} \leq d_j g_{t_0,j}\|\phi\|_{2j-1} \leq d_j g_{t_0,j}\|\phi\|_{2j}, \quad (\text{E.10})$$

all $j = 1, 2, \dots$ and some constants d_j , it is clear that the series and all its derivatives with respect to t converge uniformly in the norm $\|\cdot\|_{2j}$ if t is sufficiently close to t_0 . Sobolev's lemma (statement (c) in lemma 1.3.5 of ref. [8]) then implies that the series converges pointwise and uniformly, together with all its derivatives in t and its derivatives in U up to some order proportional to j .

If t is in a neighbourhood of t_0 , where the convergence of the series and its derivatives up to order $l \geq 2$ is guaranteed, the action of the operator $\mathfrak{L}_t$ and the summation may be interchanged and one finds that

$$\mathfrak{L}_t \psi_t = S - \sum_{n=0}^{\infty} (t_0 - t)^n \mathfrak{P}_{t_0} (\mathfrak{L}'\mathfrak{G}_{t_0})^n S = (1 - \mathfrak{P}_t) S, \quad (\text{E.11})$$

the second equality being implied by the identities $\mathfrak{P}_t \mathfrak{L}_t \psi_t = 0$ and $\mathfrak{P}_t \mathfrak{P}_{t_0} = \mathfrak{P}_{t_0}$. As a consequence,

$$(1 - \mathfrak{P}_t) \psi_t = \mathfrak{G}_t S = \tilde{S}_t, \quad (\text{E.12})$$

which shows that $\tilde{S}_t$ is, in the specified neighbourhood of t_0 , l times continuously differentiable with respect to t and U . Since t_0 and l can be chosen arbitrarily, the simultaneous differentiability of $\tilde{S}_t$ is thus guaranteed at all times t and to all orders.

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Infinite N phase transitions in continuum Wilson loop operators.

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ABSTRACT: We define smoothed Wilson loop operators on a four dimensional lattice and check numerically that they have a finite and nontrivial continuum limit. The continuum operators maintain their character as unitary matrices and undergo a phase transition at infinite N reflected by the eigenvalue distribution closing a gap in its spectrum when the defining smooth loop is dilated from a small size to a large one. If this large N phase transition belongs to a solvable universality class one might be able to calculate analytically the string tension in terms of the perturbative Λ -parameter. This would be achieved by matching instanton results for small loops to the relevant large- N -universal function which, in turn, would be matched for large loops to an effective string theory. Similarities between our findings and known analytical results in two dimensional space-time indicate that the phase transitions we found only affect the eigenvalue distribution, but the traces of finite powers of the Wilson loop operators stay smooth under scaling.

KEYWORDS: $1/N$ Expansion, Lattice Gauge Field Theories.

Contents

1. Introduction.	1
2. Overview of lattice study.	4
2.1 Smearred lattice Wilson loops.	5
2.2 Lattice large N transitions survive in the continuum limit.	6
2.3 Smearing removes ultra-violet divergences.	8
3. Numerical details of lattice study.	9
3.1 The determination of $f_c(l)$.	10
3.1.1 Estimating the gap at infinite N .	10
3.1.2 Extrapolating to zero gap.	13
3.1.3 The continuum limit of $F_c(L, b)$.	14
3.2 A study of both sides of the transition under scaling.	15
3.2.1 Estimating the critical size from the small loop side.	17
3.2.2 Estimating the critical size from the large loop side.	17
3.3 General features of $f_c(l)$.	19
4. The continuum view.	21
4.1 Smearred continuum Wilson loops and hints of a fifth dimension.	21
4.2 Instantons close the gap.	22
4.3 A guess for the large N universality class.	23
5. Summary and Discussion.	26
5.1 Connections to other work.	26
5.2 Main new ideas in this work.	27
5.3 Plans for the future.	28

1. Introduction.

Most of the concrete results of this paper were obtained by methods of lattice field theory. These results make up a first component of a larger, rather ambitious and quite speculative program which is defined in the continuum, independently of any regularization. The outline of the program makes up the bulk of this section. The intention is to put the presentation of the lattice results that follows in later sections in perspective. Throughout the paper, we shall get back to continuum language to emphasize what structures are, in our view, continuum properties.

Pure four dimensional $SU(N)$ gauge theories are strongly interacting at large distances and weakly interacting at short scales. In $SU(3)$ the cross-over between these two regimes is relatively narrow and there are no good analytic approximation methods that apply to it; nor do we have a continuum way to calculate at strong coupling. The cross-over narrows as the number of colors, N , increases.

Here, in a direct continuation of the line of thought of some old work [1, 2], we report on an improved line of attack on this problem and take several steps in its direction. The old idea was that instantons indicate that in the limit of infinite N , at a particular sharply defined scale, there is a phase transition, and weak and strong coupling physics can be connected at that point. This idea has not been successfully implemented because too many parts were either wrong or missing. Much has been learned in the interim period and we would like to try again. Let us now briefly overview the new ingredients of our approach:

While the guess that the cross-over regime collapses to a point and becomes a phase transition at infinite N is old, we understand now that the scale at which the transition occurs is not a universal quantity that applies to all observables simultaneously; nor does it necessarily affect any local observables: it is not seen in the free energy density for example. The most interesting large N transitions do not have to be “bulk” phase transitions.

Recent lattice work on large N has produced such “bulk” transitions at global critical scales in the planar limit of QCD on a Euclidean torus, but the associated transitions take the system out of the zero temperature, infinite volume regime and depend on the global topology of space-time. This is not going to directly explain how a few particle high energy process hadronizes, for example.

Recent lattice work has also indicated that specific non-local observables undergo large N phase transitions as they are dilated [3, 4] and this happens while the Euclidean space-time volume stays infinite in all directions. The critical scales are observable dependent; one can imagine associating with these observables an entire family of dilated images and somewhere along the dilation axis the large N transition takes place.

If there were a string equivalent to planar QCD, one would guess that the dilation axis is somehow related to a fifth coordinate as has been often hypothesized [5, 6]. A meaningful higher dimensional formulation presupposes a local definition of the theory in the extended space; we shall have something to add to this speculation at the end of the paper, but it will be vague.

Another new ingredient we add to the old program is the modern understanding of universal features of certain large N transitions in matrix models. Although, on the face of it, there is no concept of locality in the index space associated with the symmetry group, the models admit a ’t Hooft double line expansion for two-index representations classified by the genus of two dimensional surfaces. A concept of two-dimensional locality applies to the physics on these surfaces and the associated Liouville type models represent universality classes of finite, but varying degrees of stability associated with the large N transitions [7]. Most of these models are exactly solvable.

We hope that the transitions we shall exhibit belong to some such universality class. We have a guess for the relevant universality class and it has an exactly solvable matrix

model representative. This means that at finite but large enough N , when the transition dissolves into a cross-over, there is a description of well chosen observables undergoing the cross-over in terms of universal scaling functions with a dependence on a limited number of non-universal parameters that typically come in as boundary conditions on the equations determining the universal, known scaling functions.

It is through these parameters that the cross-over regime connects to the weak and strong coupling regimes on either side. The four dimensional dilation parameter, in the vicinity of its critical value plays the role of a coupling that turns critical in the relevant version of matrix model description of the large N transition. By the mechanism of a (double) scaling limit the dilation parameter gets connected to a scale in the two dimensional physics that codifies the universal features of the large N transition. This connection will involve a nontrivial power and an associated dimensionful parameter.

The weak and strong coupling regimes are governed by their own universal behavior, field theoretical on the weak side, and (we guess) effective string like on the strong coupling side. Thus, the free, nonuniversal, parameters needed by each universal description, field theoretic, large N and string-like are matched to each other as the operator is dilated, and several universality mechanisms coexist in the same system, controlling separate scale ranges.

If the above scenario holds at $N = \infty$, we can apply it to pure $SU(3)$ gauge theory, and, hopefully provide a calculation of the string tension in terms of Λ_{QCD} which is accurate to order $\frac{1}{N} \approx 10$ percent. For a physicist, this might be of more value even than a mathematical proof of confinement. We are far from realizing this goal.

The first hurdle to overcome is to find good observables; we claim that the right question is not whether or not there is a large N phase transition in general, but rather, can one find good observables which undergo such a transition and can be controlled (by different theories) at very short and at very long distance scales respectively. Local observables are unlikely to undergo the transition. Therefore the cherished analyticity properties in four dimensional Euclidean momentum space stay protected, maintaining the basic unitarity and covariance properties of the field theoretical description of the physics.

In this paper we define a class of smeared Wilson loop observables on the lattice with the intent that they have finite continuum limits and obey the following properties:

- It is correct to think about the parallel transport round the loop as being realized by an $SU(N)$ matrix. This statement has to hold after the renormalization of the theory and of the operator have been carried out.
- When we scale the loop, a phase transition takes place in its eigenvalue distribution if $N = \infty$.
- The smeared loop operator is calculable in perturbation theory when it is scaled down.
- Its shape dependence is well described by an effective string theory when it is scaled up.

- It is possible to see by continuum non-perturbative methods (e. g. instantons) that the phase transition is approached when the loop is dilated up from being very small, so one can hope to calculate the matching at the weak coupling end of the cross-over.

At the large scale end one also needs a way to calculate a matching, and this is presumed to work by the large scale end of the cross-over having some string description (coming from the relevant universal matrix model) which can be joined smoothly to the right effective four dimensional effective string theory. That effective string theory could be, for example, similar to [8]. This we have not yet established, and is left for future work. Nevertheless, on the basis of the results we do have, we believe to have found non-local observables which are “good” enough.

In its essence the calculation we have outlined is an attempt to extend the logic of effective field theory [9] calculations to the strong-weak cross-over in pure gauge QCD. Also, the program is in the same spirit as the old CDG approach to instantons [10] where they tried to use instanton gas methods to set the parameters in an effective bag-model description of hadrons.

If our scenario works for pure glue, it would be possible to re-introduce fermions at a later stage, at least as probes with no back reaction and see whether one can construct hadronic observables containing fermionic fields that also undergo large N transitions. In particular it would be interesting to find a similar calculational program for spontaneous chiral symmetry breaking. One could check this numerically using exactly chiral fermions [11].

In the next section we introduce our observables on the lattice and present our numerical results in support of the claim that our observables have a continuum limit and undergo large N transitions under dilation. In subsequent sections we discuss a continuum view of the lattice construction and our guess for the large N universality class. The presentation becomes by necessity progressively more speculative. Ultimately, we hope to reduce the role of lattice simulations to a strictly qualitative one, providing support for the assumptions about the various universality classes that control the different regimes our calculation needs to connect. The calculation in itself would be entirely analytical. It goes without saying that any analytical step would need to be checked against numerical work, as this is uncharted territory.

2. Overview of lattice study.

We use an ordinary single plaquette Wilson action defined on a hypercubic lattice. We exploit twisted reduction [12] which allows a decent interval of lattice spacings to be realized on a small lattice of size $V = 7^4$. For twisting to work best, we need N to be given by 4 times a prime number. We work mostly at $N = 44$ with checks for finite N effects at $N = 28$. Our simulation method employs a combination of heat-bath and over-relaxation updates and equilibration is achieved in reasonable lengths of computer time. In most of our analysis we can ignore all statistical errors. As with any lattice work with few precedents to rely on, a substantial part of the effort is to identify useful ranges of the

many parameters that enter numerical work, so that the relevant physics questions can be addressed. Throughout, we use b for the lattice gauge coupling which is the inverse bare 't Hooft coupling; relatively to the conventional lattice coupling β , b already has the right power of N extracted:

$$b = \frac{\beta}{2N^2} = \frac{1}{g_{\text{YM}}^2 N} \quad (2.1)$$

For clarity purposes, we ignore a technicality in our presentation: For a twisted simulation b is taken as the negative of its value in the untwisted case; we shall quote the conventional, positive value of b .

2.1 Smeared lattice Wilson loops.

We consider only rectangular loops restricted to one plane of our hypercubic lattice and most of our work is done with square loops. However, a continuum approach would be concerned with the class of observables that contain smooth loops of arbitrary shape. Specifically, explicit calculations in the continuum would best be applied to circular loops contained in one flat plane in R^4 , as they have maximal symmetry.

In order to eliminate the perimeter and corner divergences [13] we construct the Wilson loop operators associated with our curves not from the $SU(N)$ link matrices $U_\mu(x)$ generated by the probability distribution determined by the single plaquette Wilson action, but, rather, from smeared versions of those. We employ APE smearing [14], defined recursively from $n = 0$, where the smeared matrix is equal to $U_\mu(x)$. Let $\Sigma_{U_\mu^{(n)}(x,f)}$ denote the ‘‘staple’’ associated with the link $U_\mu^{(n)}(x;f)$ in terms of the entire set of $U_\mu^{(n)}(y;f)$ matrices. One step in the recursion takes one from a set $U_\mu^{(n)}(x;f)$ to a set $U_\mu^{(n+1)}(x;f)$:

$$X_\mu^{(n+1)}(x;f) = (1 - |f|)U_\mu^{(n)}(x;f) + \frac{f}{6} \Sigma_{U_\mu^{(n)}(x,f)} \quad (2.2)$$

$$U_\mu^{(n+1)}(x;f) = X_\mu^{(n+1)}(x;f) \frac{1}{\sqrt{[X_\mu^{(n+1)}(x;f)]^\dagger X_\mu^{(n+1)}(x;f)}}$$

In the simulation, one never encounters a situation where the unitary projection in the above equations stalls because $X^{(n)}$ is singular. In other words, smearing is well defined with probability one. For a twisted simulation f is taken as the negative of its value in the untwisted case; we shall quote the conventional, positive value of f .

$U_\mu^{(n)}(x;f)$ transforms under gauge transformations the same way as $U_\mu(x)$ does. Our smeared Wilson loop operators, $\tilde{W}[L_1, l_2; f; n]$ are defined as ordered products round the $L_1 \times L_2$ rectangle restricted to a plane. L_a are integers and give the size of the loop in units of the lattice spacing. When the traversed link starts at site $x = (x_1, x_2, x_3, x_4)$ $x_\mu \in Z$ and connects to the neighboring site in the positive direction μ , $x + \mu$, the link matrix is $U^{(n)}(x;f)$, while when this oriented link is traversed in the opposite direction, the link matrix is $U^{(n)}(x;f)$. $\tilde{W}$ depends on the place the loop was opened, but its eigenvalues do not. The set of eigenvalues is gauge invariant under the fundamental gauge transformation operating on $U_\mu(x)$.

We established by numerical means, to be described in greater detail later, that as L_1, L_2, f, n are varied, at specific lattice couplings, the spectrum of $\tilde{W}[L_1, L_2; f; n]$ opens

a gap for very small and/or very smeared loops. This gap closes for very large and/or very lightly smeared loops.

2.2 Lattice large N transitions survive in the continuum limit.

Even restricting to just square loops, our lattice operators depend on many parameters: there is an overall scale, the side of the square, L ; there is a gauge coupling b which fixes the lattice spacing in physical units, and therefore also controls the scale; there are two parameters, f and n that control the APE smearing and are responsible of the taming of the perimeter and corner divergences in the traces of the operator in all irreducible $SU(N)$ representations simultaneously. Our construction implicitly adjusts the infinite number of undetermined finite parts associated with the renormalization of the trace in each representation so that the entire collection can be viewed as coming from one (fluctuating) $SU(N)$ matrix. This ensures that backtracking segments of the loop make no contribution to the operator.

Obviously, there is a substantial amount of arbitrariness in the choices we made when defining the regularized Wilson loop operators. If the construction is kept more general, this arbitrariness would reflect itself in an invariance under a large set of transformations in the continuum, perhaps under the form of a generalized renormalization group equation.

Our first objective is to find out how to adjust the parameter dependence in $\hat{W}$ such that $N = \infty$ transition points identified on the lattice, where the gap just opens, survive in the continuum limit in which the lattice coupling b is taken to infinity together with $L_{1,2}$ in such a way that the physical lengths $l_a = L_a a(b)$ are kept fixed. $a(b)$ is the lattice spacing at coupling b , and we express it in terms of the physical finite temperature deconfining transition t_c , by

$$a(b) = \frac{T_c(b)}{t_c} \quad (2.3)$$

where $\frac{1}{a(b)}$ is the lattice size of the compact direction on an $S^1 \times R^3$ lattice corresponding to the finite temperature transition critical coupling b .

We set the number of smearing steps n to be proportional to the perimeter square (we restricted the loop sizes to even $L_1 + L_2$), $n = \frac{(L_1 + L_2)^2}{4}$. The physical sizes of the loop are l_a and $L_a = \frac{l_a}{a(b)}$. We have set $n = \frac{(L_1(b) + L_2(b))^2}{4}$ because in physical terms the product fn is a length squared. This can be seen from the simple perturbative argument to be presented in the next subsection, or, more intuitively, by observing that smearing is a random walk that fattens the loop and the thickness grows as the square root of the number of smearing steps. Our choice for n makes f a dimensionless parameter in the physical sense; on the lattice f is actually bounded to an interval of order one.

We analyzed four sequences of $\hat{W}[L_1, L_2; f; n = \frac{(L_1 + L_2)^2}{4}]$ keeping the physical size fixed at four values, one for each sequence. We set $L_1 = L_2 = L$. By various means, to be described later, we varied f and found the critical value of f at which each member in each sequence just opens a gap in its spectrum at eigenvalue -1 . (So long as CP invariance is maintained, for any loop, the eigenvalue probability is extremal at ± 1 .) The sets of critical values for each sequence were then extrapolated to the $b \rightarrow \infty$ limit. In each sequence the physical size of the loop is kept fixed by definition. This gave four continuum critical

values of f for the four different physical loop sizes we selected. Throughout, the values of f used were within reasonable ranges, well within the maximally allowed lattice range of $0 < f < 0.75$ [15]. That it is actually possible to arrange that only such reasonable values of f are needed in the vicinity of the large N transitions is our most basic result and a nontrivial feature of the dynamics of planar pure Yang Mills theory. The collection of the continuum limits of the critical values of f at the transition points associated with each sequence identified by its physical scale defines a critical line, $f_c(l)$. For any finite b there were finite lattice spacing effects, but they were sufficiently small to keep all the lattice approximations to f in a good range.

Thus, we conclude that we have found a continuum critical line $f_c(l)$ in planar QCD where our continuum Wilson loop operators have a well defined spectrum and that spectrum is critical in the sense that its eigenvalue density, $\hat{\rho}(\theta)$ is nonzero for any $\theta \neq \pi$ but $\hat{\rho}(\pi) = 0$. Here, we denote the eigenvalues of the Wilson loop operator as $e^{i\theta}$, where θ is an angle on a circle.

It is not necessary to keep n quadratic in L_α for $M_\alpha \gg 1$; the loops that have a critical f are of sizes l for which Al is of order unity. For much larger loops, fn should be allowed to become proportional to a physical scale of order $\frac{1}{\Lambda}$ and not increase further as the physical loop size goes to infinity. All that is important is that for loop sizes of order $\frac{1}{\Lambda}$, n be adjusted as above, and that for small loops we let fn become small too. For small loops we wish to enter a perturbative regime, and therefore we should not eliminate the ultraviolet fluctuations at short distances, where they dominate.

Once the existence of the $f_c(l)$ line has been established in the continuum limit, with $\frac{df_c}{dl} > 0$, we can be reasonably sure that dilating a $\hat{W}[L, L; f; n]$ so that at lattice coupling b it passes through a point $L = \frac{1}{a(b)}$, $f = f_c(l) + \mathcal{O}(a^2(b))$, $n = L^2$ the spectrum of this $\hat{W}$ will go through the transition. We still need to make sure that once the gap is closed, the density at -1 starts building up continuously from zero; this would mean that the transition is continuous and likely to possess a universal character. If the transition were discontinuous, like, for example, the finite temperature transition in Polyakov loop operators, it is unlikely that a scaling regime exists on either side of the transition and a single scaling regime connecting the two sides (of the type we are hoping for) is ruled out. This is obviously the case in the finite temperature case because the transition there is not induced by taking $N \rightarrow \infty$, but exists for all N , and starts being first order at $N = 3$.

We addressed this point by carrying out additional simulations along another line in the physical (f, l) plane, which is not a constant l but, rather, has l varying by a substantial amount. We picked a convenient interval of lengths l and a line in the plane (f, l) given by $f = f_0 - f_1 l^2$ with constant $f_1 > 0$ chosen so that f varies between f_1 and f_2 with $f_1 \approx 0.1$ and $f_2 \approx 0.2$. This line intersects $f_c(l)$ at an angle not too close to zero, thus enhancing the variability in the spectrum as one takes $\hat{W}$ along the line between the two extremal lengths. There is no particular physical meaning attached to this second line, except that along it the loop is dilated and in addition, the amount of smearing decreased; both effects help closing the spectral gap that is present when the loop is small.

To go along the continuum line we fixed the lattice loop-size L and varied b . We kept $n = L^2$ also fixed, but varied f according to the formula $f = f(b) = f_0 - f_1(La(b))^2$.

Following $\hat{W}$ along this line we established that indeed the spectrum of $\hat{W}$ opens a gap as the band of lattice curves converging to $f_c(l)$ is crossed.

We avoided working close to the transition and estimated independently on each side where the corresponding phase would end, using various extrapolation methods: On the side where there is a gap, we studied the decrease of the gap to estimate when it will close. On the gap-less side, we estimated when $\tilde{\rho}(\pi)$ would vanish.

We repeated the procedure for three increasing values of L . In each case one covers similar (overlapping, but not identical) ranges of physical size l . In the common portions of the range, the larger the lattice value of L is, the smaller the lattice effects ought to be. We found general consistency with a scaling violation that goes as the lattice spacing squared.

The separate extrapolations in each phase produced independent estimates for end points of each phase which came out reasonably close to each other. This is consistent with a continuous large N transition at which the eigenvalue distribution of $\hat{W}$ opens a gap. The determinations of the critical points from each side were ordered in a way that was inconsistent with a discontinuous transition in which $\tilde{\rho}(\pi)$ decreases to a non-zero value and then drops to zero, jumping into a phase with a finite gap around π . Thus, we conclude that the transition is continuous and one can hope for some universal description in the vicinity of the transition point. Actually, it is hard to imagine what kind of dynamics could have produced a first order large N transition in the spectrum.

2.3 Smearing removes ultra-violet divergences.

The effect of smearing is easy to understand in perturbation theory where one supposes that each individual step in the smearing iteration can be linearized. Writing $U_\mu^{(n)}(x; f) = \exp(iA_\mu^{(n)}(x; f))$, and expanding in A_μ one finds [15], in lattice Fourier space:

$$A_\mu^{(n+1)}(q; f) = \sum_\nu h_{\mu\nu}(q) A_\nu^{(n)}(q; f) \quad (2.4)$$

with

$$h_{\mu\nu}(q) = f(q) \left(\delta_{\mu\nu} - \frac{\tilde{q}_\mu \tilde{q}_\nu}{q^2} \right) + \frac{\tilde{q}_\mu \tilde{q}_\nu}{q^2} \quad (2.5)$$

where $\tilde{q}_\mu = 2 \sin(\frac{q_\mu}{2})$ and

$$f(q) = 1 - \frac{f}{6} \tilde{q}^2 \quad (2.6)$$

The iteration is solved by replacing $f(q)$ by $f^n(q)$, where, for small enough f ,

$$f^n(q) \sim e^{-\frac{f^2}{6} \tilde{q}^2} \quad (2.7)$$

The condition $0 < f(q) < 1$ for all momenta, requires $0 < f < 0.75$. To get some feel for the numbers, a 4×4 loop, at a value of b that makes it critical, will have $n = 16$ and $f \sim 0.15$, which makes $\frac{f^2}{6} \sim 0.4$, which is an amount of fattening over a distance as large as the loop itself. This is why much larger loops should not be smeared with an fn factor

that keeps on growing as the perimeter squared; rather, for a square loop of side L , for example, the following choice would be appropriate:

$$f = \frac{\tilde{f}}{1 + M^2 L^2} \quad (2.8)$$

Here, M is a physical mass m expressed in lattice units, $M = ma(b)$. m is a typical hadronic scale chosen so that at the large N transition we found, ML is less than 0.01, say.

To reassure ourselves that such an extension of the cutoff dependence of f indeed would connect smoothly to large loops which show confinement with the known value of the string tension, we roughly estimated the string tension from our smeared loops, at the large size end of the ranges we studied. We found that the string tension comes out close to that of bare lattice Wilson loops, and this shows that our smeared Wilson loop operators are acceptable pure $SU(N)$ gauge theory string candidates.

In summary, the effect of smearing on the propagator of $A_\mu^{(n)}(x; f)$ is to eliminate the ultraviolet singularity that gives rise to perimeter and corner divergences in traces of bare Wilson loop operators. This works when the product fn varies as a physical length squared when the continuum limit is approached. For small loops we choose this physical length to be proportional to the loop itself. For large loops the physical length stabilizes at some hadronic scale. In this way, our operators become sensitive to the short distance dynamics when the scale is very small, but, when confinement becomes the dominating effect, for very large loops, the string tension of the smeared Wilson loops is the true, universal, string tension of the theory, unaffected by the smearing. The leading term in the effective string theory, describing the shape dependence of the Wilson loops when the loop is very large, is expected to be universal and the string tension enters only in a trivial manner, as a dimensional unit.

We stress that we are not executing a renormalization group transformation; our Wilson loop operators are defined for infinitely thin loops in space-time and for all loop sizes; rather, we are applying a regularization whose objective is to make the Wilson loop operators finite in the continuum. If we dealt with each irreducible representation separately, the elimination of perimeter and corner divergences would leave behind an infinite number of arbitrary finite parts. This freedom has been implicitly exploited by imposing the requirement of unitarity of the Wilson loop operators, which ensures what Polyakov has termed the “zigzag” symmetry [5]. Preserving the unitary character of the Wilson loop operator in turn makes the concept of an eigenvalue distribution well defined in the continuum and sets the stage for possible large N phase transitions. The zigzag symmetry also influences the form of the lattice loop equations at infinite N , which are obeyed by traces of bare Wilson loops in the fundamental representation; the effective string theory for large loops needs not obey the zigzag symmetry, but this symmetry could provide a useful constraint on its couplings.

3. Numerical details of lattice study.

In the previous section we described what our strategy was to address the question of existence of a continuous limit for the operators and the large N transition point. Also, we

made an effort to convince ourselves that the large N transitions indeed are continuous. We now proceed to describe the exact details of our simulations and the numbers that came out.

3.1 The determination of $f_c(l)$.

3.1.1 Estimating the gap at infinite N .

At infinite N , small $\tilde{W}$ loops will have a spectral gap on the lattice. We need a method to extract that gap from numerical data at finite N . As we dilate the loop, and/or reduce the amount of smearing, the gap shrinks. We need a way to extrapolate to the point where the gap vanishes. We wish to keep away from measurements for too small gaps, and enter that regime only by extrapolation.

Getting the infinite N value of the gap is difficult because subleading effects are large at the gap's edge. It is reasonable to assume that the universal behavior at the gap's edge is the same as at the edge of a gap in simple hermitian matrix models. Under this assumption, we have separate predictions for the behavior of the eigenvalue density and for the extremal eigenvalue. One can check numerically how well the universal forms work and we find them to work quite well. To fit onto the universal behavior we need to adjust a parameter that corresponds to the gap size at infinite N . This provides the determination of the gap sizes we use subsequently.

It is only on the side of the transition where the eigenvalue density has a gap that there is a regime where large N corrections are dominated by universal functions; therefore, we are able to get good estimates for the eigenvalue gap size at infinite N from data obtained at one single, but large N . For good measure, we checked that the predicted rational power dependence on N holds in the regimes in which we did our analysis. A comparison with some simulations at $N = 28$ convinced us that the power dependence on N is already at its known universal value at $N = 28, 44$.

We studied the eigenvalue distribution of $\tilde{W}[L, L; f; n = L^2; N]$ on the lattice for several values of L and b at a fixed large N . Let $e^{i\theta_k}$, $k = 1, \dots, N$ be the N eigenvalues of a fixed $L \times L$ loop on the lattice, $-\pi \leq \theta_1 < \theta_2 < \dots < \theta_{N-1} < \theta_N < \pi$. The sum of all angles is zero mod 2π . On a lattice of V sites there are $6V$ different $L \times L$ Wilson loops with a fixed orientation; we computed the eigenvalues of all such loops. In this manner, we obtain a histogram of the distribution $p_f(\theta; U)$ of all eigenvalues in a fixed gauge field background U . We also collected data for a histogram of the distribution $p_m(\theta_N; U)$ for the largest eigenvalue in a fixed gauge field background. In addition, we record the mean, variance, skewness and kurtosis pertaining to the largest eigenvalue:

$$\begin{aligned}\mu_N(U) &= \langle \theta_N \rangle_U; \\ \sigma_N^2(U) &= \langle \theta_N^2 \rangle_U - \langle \theta_N \rangle_U^2; \\ S_N(U) &= \frac{\langle \theta_N^3 \rangle_U - 3\langle \theta_N^2 \rangle_U \langle \theta_N \rangle_U + 2\langle \theta_N \rangle_U^3}{\sigma_N^3(U)}; \\ K_N(U) &= \frac{\langle \theta_N^4 \rangle_U - 4\langle \theta_N^3 \rangle_U \langle \theta_N \rangle_U + 6\langle \theta_N^2 \rangle_U \langle \theta_N \rangle_U^2 - 3\langle \theta_N \rangle_U^4}{\sigma_N^4(U)} - 3.\end{aligned}\quad (3.1)$$

If the eigenvalue distribution of $\hat{W}$ has a sizable gap, it is reasonable to ignore the influence of the two edges on each other. Then, the universal behavior at the gap's edge ought to be the same as that of the edge of the gap in simple Gaussian hermitian matrix model. Under this assumption, we compared our data with the behavior of the extremal eigenvalue [16] and the eigenvalue density close to the edge [17].

The universal distribution of the extremal eigenvalue, $p(s)$, is given by [16]

$$p(s) = \left[\int_s^\infty q^2(x) dx \right] e^{-\int_s^\infty (x-s)q^2(x) dx} \quad (3.2)$$

with q being the solution to the Painlevé II equation,

$$\frac{d^2 q}{ds^2} = sq + 2q^3; \quad (3.3)$$

satisfying the asymptotic condition

$$q(s) \sim Ai(s) \quad \text{as } s \rightarrow \infty. \quad (3.4)$$

The mean, variance, skewness and kurtosis of the above distribution are -1.7710868074 , 0.8131947928 , 0.2240842036 and 0.0934480876 . In order to match the lattice data for the distribution of θ_N to the above universal form, we define s according to

$$\theta_N = \frac{g}{\alpha} + \pi(1 - g) \quad (3.5)$$

and set

$$\alpha = \sqrt{\frac{0.8131947928}{\sigma_N^2}}, \quad g = 1 - \frac{1}{\pi} \left[\mu_N + \frac{1.7710868074}{\alpha} \right] \quad (3.6)$$

α is a non-universal scale factor and g is half the arc length of the gap on in the eigenvalue distribution on the unit circle, centered at -1 . The quantities, σ_N^2 and μ_N are the averages over the gauge field ensemble of the previously defined quantities, $\sigma_N^2(U)$ and $\mu_N(U)$. From the lattice data for $\sigma_N^2(U)$ and $\mu_N(U)$, we obtain an estimate for α and g with errors extracted by the jackknife method. Using these estimates, we compared the scaled lattice data with $p(s)$ for fixed f , b , N and V . One such comparison, at $f = 0.15$, $b = 0.361653$, $L = 4$, $N = 44$ and $V = 7^4$ with a gap of $0.1229(4)$, is shown in Figure 1 and there is good agreement. In spite of the fact that data is needed over the entire range of s to get a good estimate of the skewness and kurtosis, we find agreement between lattice data and the universal estimates to within 5% for the skewness and to within 20% for the kurtosis for a wide range of gap values. The estimate of the skewness and kurtosis do not depend on the extracted values of α and g .

The main result of the above numerical analysis is the half gap size g , for fixed f , L , b , $g(f, L, b)$.

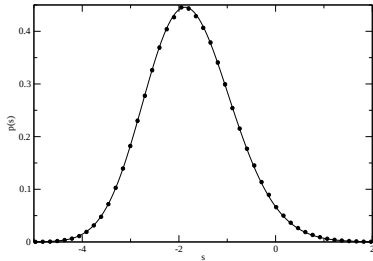


Figure 1: A comparison of the distribution of scaled extremal eigenvalue obtained on the lattice (circles) at $f = 0.15$, $b = 0.361653$, $L = 4$, $N = 44$ and $V = 7^4$ with that of the universal distribution (solid curve).

There is another way to obtain an estimate for g . A recent result provides the asymptotic corrections to the eigenvalue density of Gaussian hermitian matrices [17]. Using that formula, we fitted the eigenvalue density in terms of the scaled variable in (3.5) to

$$p_f(s) = c \left\{ [Ai'(s)]^2 - s[Ai(s)]^2 \right\} + d \left\{ 3s^2[Ai(s)]^2 - 2s[Ai'(s)]^2 - 3Ai(s)Ai'(s) \right\} \quad (3.7)$$

The variable c is fixed by the overall normalization and $\frac{d}{c} = -\frac{1}{20N^{1/2}}$ in the asymptotic formula [17] where N is the size of the matrix. The parameter d could be non-universal; also c could in some sense be non-universal, since it depends on the total number of eigenvalues, and it is not clear whether the Gaussian hermitian matrix model matches onto our model with the same N in both cases. Therefore, we fitted the lattice distribution to the above asymptotic formula with c and d as independent variables in addition to the two variables appearing in (3.5), producing a second determination of $g(f, L, b)$.

Figure 2 shows the comparison of the full eigenvalues distribution with the universal distribution near the edge. The universal distribution captures the tail for the same reason that the distribution of the extremal eigenvalue is in agreement with $p(s)$. But we also see good agreement with the first two oscillations near the edge and this includes more than the extremal eigenvalue. In general we found the determination of the gap to be consistent

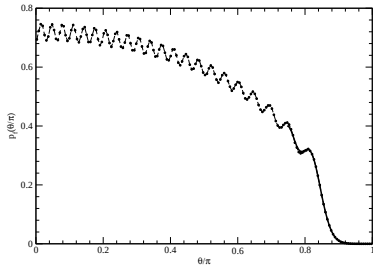


Figure 2: A comparison of the distribution of eigenvalue distribution obtained on the lattice (circles with dashed lines) at $f = 0.15$, $b = 0.361653$, $L = 4$, $N = 44$ and $V = 7^4$ with that of the universal distribution near the edge (solid curve).

with the one obtained using only the extremal eigenvalue. For an error estimate, we only used the extremal eigenvalue method.

3.1.2 Extrapolating to zero gap.

The next step is to extract the critical value of f at a fixed b and L at infinite N . This is done by assuming a fit of the form

$$g(f, L, b) = A(L, b) \sqrt{f - F_c(L, b)} \quad (3.8)$$

The square root dependence on external parameters that enter the probability law in an analytic manner is generic in matrix models and well supported by the data.

A plot of $g^2(f, L, b)$ as function of f at $b = 0.361653$, $N = 44$ is shown in figure 3. $F_c(L, b)$ is extracted using a range of f such that the gap, $g(f, L, b)$, is not too close to 0 nor is it too far from 0. If we go to too small gaps we fear unaccounted for rounding effects; if the gap is too large, there is little reason to use the universal square root behavior we postulated.

Figure 3 also shows the fit along with the estimate for $F_c(L, b)$.

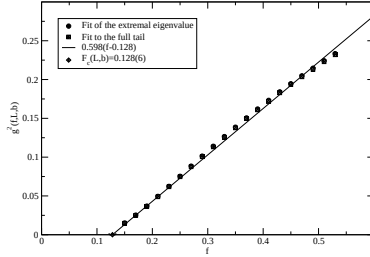


Figure 3: Plot of $g^2(f, L, b)$ as a function of f at $b = 0.361653$, $N = 44$ and $V = 7^4$. A fit to the data along with the extrapolated value for $F_c(L, b)$ is also shown.

3.1.3 The continuum limit of $F_c(L, b)$.

We now proceed to take the continuum limit of $F_c(L, b)$. We used two loop tadpole improved perturbation theory to keep the physical loop fixed as we varied b . Using [4], we have a numerical formula for the critical deconfinement temperature in lattice units, in terms of tadpole improved perturbation theory, with $e(b)$ being the average value of the plaquette for untwisted single plaquette gauge action at coupling b .

$$\frac{1}{T_c(b)} = 0.26 \left[\frac{11}{48\pi^2 b_f} \right]^{\frac{31}{12}} e^{\frac{24\pi^2 b_f}{11}}; \quad b_f = bc(b) \quad (3.9)$$

This will be used to determine the lattice spacing in physical units.

We set $lt_c = LT_c(b)$ and accumulate data at several values of the physical loop size: $lt_c = 0.42, 0.48, 0.58, 0.64$. At each of these physical sizes, we went through a sequence of three to four lattice spacings $a(b)$, and, for each b , we obtained a value for $F_c(L = \frac{1}{a(b)}, b)$, using the methods explained above.

For each lt_c , the $F_c(L = \frac{1}{a(b)}, b)$ values were then extrapolated to the continuum assuming

$$F_c(L = \frac{1}{a(b)}, b) = f_c(lt_c) + O(T_c^2(b)). \quad (3.10)$$

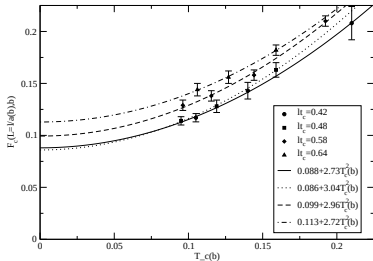


Figure 4: Plot of $F_c(L = \frac{1}{a(b)}, b)$ as a function of $T_c(b)$ for four different values of lt_c . Also shown are fits to the data producing continuum values for $f_c(lt_c)$.

where we recall that $T_c(b) = a(b)t_c$. (We shall use the slightly inconsistent notations $f_c(l)$ and $f_c(lt_c)$ for the same quantity, where the argument is either dimensional or made dimensionless by the appropriate power of the physical deconfining temperature t_c .)

The extrapolated results for $f_c(lt_c)$ are shown in figure 4.

Our final results are plotted in the (f, l) plane. The critical line $f_c(l) \equiv f_c(lt_c)$ divides this plane into two as shown in figure 5. In the upper part Wilson loops have a gap in their eigenvalue distribution, whereas in the lower part there is no gap. The band bounded from below by the critical zero-lattice-spacing line indicates the range of finite-lattice-spacing effects.

3.2 A study of both sides of the transition under scaling.

Finite lattice size effects are evident in figure 5. It is impractical to just scale the loop because the curve $f_c(l)$ varies when l is changed in the appropriate range by less than the lattice corrections. To see the effect of scaling we need to go on a line in the (f, l) plane that is not horizontal, which means we are going to vary the smearing as a function of the physical loop size. As usual, we trade l for the dimensionless variable lt_c .

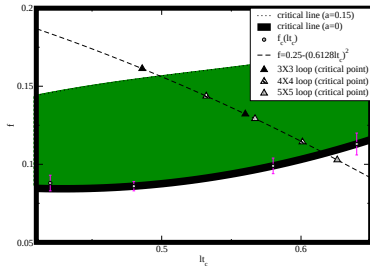


Figure 5: The critical line in the (f, lt_c) plane. Also shown is the crossing line used to investigate the phase transition induced by scaling the Wilson loops.

We chose the line (we shall refer to this line as the “crossing line”)

$$f = 0.25 - (0.6128lt_c)^2 \quad (3.11)$$

shown in figure 5 as it cuts across the entire finite band of lattice corrections bounded by the critical line $f_c(lt_c)$.

Consider a fixed loop of size L on the lattice for various lattice couplings b such that $f(b)$ is given by (3.11) with $lt_c = LT_c(b)$ and $T_c(b)$ given by (3.9). We wish to study the nature of the transition in the eigenvalue distribution along this line.

We already know that the distribution should have no gap for $l > l_c$ whereas it should have a gap for $l < l_c$. In addition to getting an estimate for $l_{c,c}$, we can also investigate the universality class of this transition by keeping L fixed and going along the crossing line given by (3.11). We set $L = 3, 4, 5$ and worked in the following ranges of b : for $L=3$, in $0.345 < b < 0.365$, for $L=4$, in $0.346 < b < 0.368$ and for $L=5$, in $0.351 < b < 0.369$. In each range, as b decreases the physical loop is scaled up. In this way one gets overlaps at the same physical loop size, once represented by a smaller value of L and a smaller value of b , and other times, by correspondingly larger values of these parameters. Because of scaling violations one will not get identical eigenvalue distributions; we need to see that, as b and

L increase, these corrections decrease and that a finite limiting eigenvalue distribution is approached. This is the eigenvalue distribution associated with the continuum loop of size l .

In Figure 5 we display, for each value of $L = 3, 4, 5$, the two critical lengths one obtains when extrapolating from each phase independently as triangles on the crossing line. As L is increased, we see that the distance between the two members of each pair decreases as L increases. Also, the middle points of the segments connecting the points in each pair appear to converge as L increases. This indicates that in the continuum limit there will be one single transition point on this line, a loop size at which a continuous transition in the large N spectrum of $\hat{W}$ occurs.

3.2.1 Estimating the critical size from the small loop side.

We again obtained estimates for the infinite N gap, using the extremal eigenvalue distribution ($p(s)$) and the eigenvalue density ($p_f(s)$) near the edge. The estimates for the gap came out consistent with each other and the results are plotted in figure 6. The critical size of the loop is extracted by again assuming a square root singularity written in the form

$$g^2(l_c) = A \ln \frac{l}{l_c} \quad (3.12)$$

For each value of $L = 3, 4, 5$ we got an estimate for l_c which is shown in figure 6. One sees that the lines $L = 4, 5$ are closer to each other than the lines $L = 3, 4$; the ratio of the approximate distances between the two pairs of lines is in rough agreement with a hypothesis of order $a^2(b)$ corrections.

3.2.2 Estimating the critical size from the large loop side.

For operators with no gap we need a method to extract the infinite N value of the eigenvalue density at angle π , $\hat{\rho}(\pi)$. Shrinking the loop and/or smearing it more makes $\hat{\rho}(\pi)$ decrease. We also need a method to extrapolate to the point where $\hat{\rho}(\pi)$ vanishes, without getting too close to it.

The estimation of the infinite N value of $\hat{\rho}(\pi)$ also suffers from large subleading effects, of order $\frac{1}{N}$ typically. There is a rapid fluctuation in the eigenvalue density reflective of the repulsion between eigenvalues. The rapid fluctuation is eliminated by Fourier transforming the numerically obtained binned eigenvalue distribution with respect to the angle for wavelengths kept large relative to $\frac{1}{N}$. This is equivalent to evaluating the traces of $\hat{W}^k$ for consecutive values of k , all kept much smaller than N . Keeping only those Fourier modes, we get our estimates for $\hat{\rho}(\pi)$; checks against a simulation at lower N 's shows that we are determining $\hat{\rho}(\pi)$ to reasonable accuracy.

The extrapolation of $\hat{\rho}(\pi)$ to zero as the loop size is decreased and/or its amount of smearing is increased is done in a similar way as when we deal with the gap. We find that the leading behavior is again square-root like, but this holds in a small range which can be extended by adding more powers of $(l - l_c)$. The determination of l_c by this method is seen to be quite robust.

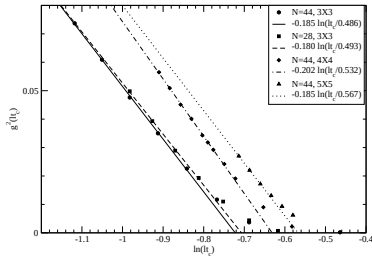


Figure 6: The square of the gap as a function of $\ln(l_c)$ along the crossing line.

In two dimensions the large N eigenvalue distribution for loops of arbitrary size is known in the continuum [18]. As we shall elaborate later, it is plausible to assume that the behavior close to the large N transition is universal, and holds also for our $\hat{\rho}(\theta)$ in four dimensions. Therefore, we postulate that $\hat{\rho}(\pi)$ vanishes as a square root at the critical value of any parameter that enters the underlying distribution controlling the smeared Wilson loop operators in an analytic manner. This is why we took $\hat{\rho}(\pi)$ to vanish as $\sqrt{l - l_c}$ as one approaches the appropriate critical size, l_c , from above.

As indicated above, for l large enough with respect to l_c we extract the nonzero value of $\hat{\rho}(\pi)$ by first representing the distribution for all angles as:

$$\hat{\rho}(\theta) = \sum_{k=0}^N f_k \cos(k\theta) \quad (3.13)$$

Next, we truncate the Fourier series and use the truncated series to provide the value of $\hat{\rho}(\pi)$.

$$\hat{\rho}(\pi) = \sum_{k=0}^{k_m} (-1)^k f_k \quad (3.14)$$

The value of k_m is chosen to be slightly less than $N/2$; this eliminates oscillations of frequency N . We checked that our choice for k_m was reasonable by looking for stability as

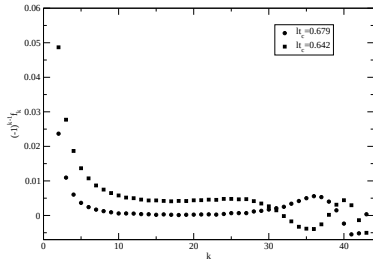


Figure 7: The fourier coefficients f_k as a function of k for two different loops in the gapless region. The measurements were obtained using an $L = 5$ loop at $N = 44$ at two different lattice spacings.

we vary k_m .

We found that for low enough k 's f_k oscillates in sign from $k = 1$ onwards. As Figure 7 shows, for k up to about $\frac{N}{2}$ the coefficients are smooth in k , and the sign oscillation is obeyed. We shall come back to this property later.

The values of $\hat{\rho}(\pi)$ we obtain are fitted to a square root in the critical region and the results are shown in figure 8. The estimates for $l_c l_c$ from $\hat{\rho}(\pi)$ tend to be higher than the ones coming from fitting the gap. This difference decreases as the spacing goes to zero.

The critical values for $l_c l_c$ obtained from the gap and $\hat{\rho}(\pi)$ are shown in figure 5. One can see that the difference between the two estimates shrinks and one can also see that the average of the two estimates approach a finite limit as the lattice spacing decreases.

The size and behavior of the scaling violations we see are consistent with usual expectations, reflecting contributions of higher dimensional continuum operators to the action and the operators. We leave for the future a more detailed investigation of these scaling violations, but note that the orders of magnitude are typical.

3.3 General features of $f_c(l)$.

In summary, using all of our data, a rough estimate for the critical size on our crossing line in the (f, l) plane, is $l_c l_c \approx 0.6$. $l_c \approx \sqrt{\sigma}$, where σ is the string tension and we used

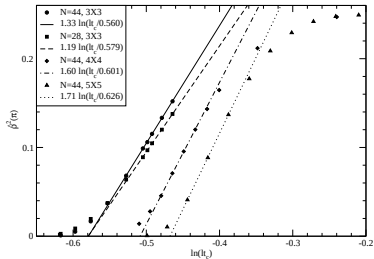


Figure 8: The square of the $\hat{\beta}(\pi)$ as a function of $\ln(l_c)$ along the crossing line.

the approximate relation $0.6\sigma = t_c$. In QCD units, l_c is about half a fermi, in general accordance with the accumulating body of numerical data which sets the transition from perturbative to string-like behavior at rather short scales, of less than one fermi. Clearly, varying the crossing line, the loop shape and/or the details of the continuum construction of the Wilson loop operator will produce slightly different values of l_c . However, the small variation over scales in the critical line $f_c(l)$ we find indicates that the transitions will occur in a restricted range of scales. Numerically, it seems that $f_c(l)$ asymptotes to a positive constant as $l \rightarrow 0$. This excludes choices of Wilson loop definitions with $l_c t_c \ll 1$. However, if it is somehow determined by further simulations on very fine lattices that $f_c(l) \rightarrow 0$ as $l \rightarrow 0$, such choices must be recognized as possible; in that case it is unlikely that an effective string description will hold for l too close to l_c . Even if it were possible, we would not choose such a definition of regulated continuum Wilson loop operators.

We have seen the scaling violations in the $F_c(L, b)$ approximations to $f_c(l)$ in a range of physical scales close to the large N transition point. It is impossible to decrease the physical size of the loops much more, because that would require too large values of b for the large N twisted reduction trick to work on a 7^4 lattice. Similarly, it is impossible to get to much larger loops because working at too small values of b takes the lattice system through a first order phase transition to a strong coupling phase which has no relevance to

continuum physics.

We found that $f_c(l)$ flattened out for small loops. It is possible that $f(l)$ does decrease to zero as l goes to zero, but if this at all happens, one would expect the effect to be logarithmic (i.e. $f(l) \sim \frac{1}{\log(l)}$), since $f(l)$ ought to behave roughly as the gauge coupling constant at scale l , which enters the perimeter divergence smearing is intended to eliminate. When l increases, $f_c(l)$ starts increasing more rapidly and eventually, $f_c(l)$ would violate its lattice bound. It is a strain to arrange for too large loops to have a spectral gap, because for large loops confinement is reflected by an almost uniform eigenvalue density.

Note however that confinement is not necessary in order to have our large N transitions; even if the eigenvalue density does not approach a uniform distribution for very large loops it remains possible that large loops have no gap in their spectrum. Thus, a similar analysis might apply also to models that do not confine.

4. The continuum view.

We would like to formulate everything we have been doing in continuum language because the basic phenomenon is, we claim, not a lattice artifact. So, we need to understand in continuum language what smearing means, what the smeared observables are, and how one could calculate them using perturbative and non-perturbative continuum methods. Next, we would need at least to consider how the large N transitions would fit into a continuum description and whether they are at all acceptable in a unitary field theory.

4.1 Smeared continuum Wilson loops and hints of a fifth dimension.

Our first task is to understand what smearing means in the continuum. The answer is that the quantity f_n plays the role of the fifth “time”, τ , in the Langevin equation when the noise has been set to zero:

$$\frac{\partial A_\mu}{\partial \tau} = D_\mu F_{\mu\nu} \quad (4.1)$$

This can be seen by comparing equations (2.4-2.7) to (4.1) for small f_n and small τ .

If we add a Zwanziger term [19] to the right hand side (it has no effect on our Wilson loops or on any other gauge invariant observable) and move it to the left hand side, the equation of motion gets a five dimensional form, with $x_5 \equiv \tau$:

$$F_{5\mu} = D_\mu F_{\mu\nu} \quad (4.2)$$

When a noise term is added to the right hand side one gets the Parisi-Wu stochastic quantization of gauge theories [20]. This setup has been shown to correspond to a topological, gauge invariant five dimensional theory [21], in which the fifth direction plays a special role (for example its units are length square as opposed to the units of the other four coordinates). There is Euclidean rotational invariance only in four dimensions. The bulk theory is trivial in the sense of a topological field theory, but the boundary has nontrivial dynamics. This setup is reminiscent of the concept of holography. The noise term is an essential ingredient in the construction, and it is an open and perhaps interesting question whether our work can be made to have a more precise relation to the one of [21].

In our paper we only looked at loops at one common fixed τ . This τ determined the thickness of the loop which we took as uniform. But, it makes perfect sense to also consider loop operators where τ varies round the loop, describing a string of varying thickness. Moreover, one would have to consider such loops if one looks for a loop equation that is local all around the loop. Indeed, the ordinary four dimensional equation of motion enters the τ evolution and therefore it is possible that at the expense of considering loops of every possible thickness simultaneously one can restore the locality of the Migdal-Makeenko equation that we lost when we simply fattened the loops staying in four dimensions.

In turn the Migdal-Makeenko equations are perhaps the right starting point to get at the fundamental (infinitely thin) large N QCD string, which would live now in five warped dimensions and might provide a description of four dimensional physics that is equivalent to that provided by four dimensional fat strings. This speculation suggests that our simple objective to get well defined continuum Wilson loops as unitary matrices in four dimensions did not lead us towards and extra dimension by sheer coincidence.

4.2 Instantons close the gap.

For a small loop the parallel transporter round it ought to be close to the identity matrix and have no eigenvalues around -1 . At infinite N , one expects a total suppression of eigenvalue far from 1, and a large gap. In matrix models, for finite but large N , this suppression is typically exponentially small in N .

Consider a circle of radius R in a plane and an instanton with center placed somewhere on the circumference of the circle. We can easily calculate the eigenvalues of the Wilson loop operator for the circle, opening it at the instanton center.

The instanton gauge field in a regular gauge is given by:

$$A_\mu = i \frac{x^2}{x^2 + \rho^2} U^\dagger \partial_\mu U, \quad (4.3)$$

where

$$U = \frac{x_4 + i \vec{x} \cdot \vec{\sigma}}{\sqrt{x^2}}. \quad (4.4)$$

Defining our circle as $(x_4 - R)^2 + x_3^2 = R^2$, with $x_{1,2} = 0$, we find:

$$\mathcal{P} e^{i \oint A_\mu dx_\mu} = e^{-i\pi(1 - \frac{\rho}{\sqrt{\rho^2 + 4R^2}})\sigma_3} \quad (4.5)$$

So long as the instanton is small enough and close enough to the periphery of the circle it would create some eigenvalue density at -1 . Large instantons are irrelevant for $AR \ll 1$, so one does not need to face the well known infrared problem of instanton calculus if the loop is small. This is a $e^{-N \cdot \text{const}}$ effect, likely to join smoothly onto the asymptotic behavior we would get from any matrix-model universal (double-) scaling function. This might be enough to select a unique solution to the differential equation of the matrix model and the latter may take us all the way to large R , where an effective string theory has an overlapping regime with the matrix model. Clearly, without some explicit calculations this is just a suggestion.

Note that the continuum version of smearing, as we defined it, will not affect the instanton since it sets to zero the right hand side of (4.1).

4.3 A guess for the large N universality class.

There are at least two obvious candidates for the universality class of the large N transitions in our smeared Wilson loops. If one just asks for the generic model dealing with a unitary matrix that opens a gap in its spectrum and which obeys a reality condition reflecting CP invariance, one will end up with a single unitary matrix model and the famous Gross-Witten [2] large N transition. The other immediate option is two dimensional QCD defined on a sphere of dimensionless area A which has the Douglas-Kazakov transition. The latter transition also occurs for two dimensional QCD on the infinite plane, but not for the partition function this time, but, rather for Wilson loop operators of area A ; the transition on the sphere in one case and in observables in the other are mathematically identical. We now argue that the second option is preferred, both theoretically and numerically.

It may be a bit confusing that we distinguish these two cases, as both are relevant to two dimensional QCD. The Gross-Witten transition is relevant to lattice QCD with a single plaquette action of the Wilson type. The single unitary matrix in the model is the parallel transporter round an elementary plaquette. As a function of the lattice coupling, this matrix opens a gap at some critical value as b is increased from zero and the trace of this matrix will be nonanalytic at the transition point. Now, consider the lattice model for which the analysis of Gross and Witten applies, but consider a loop larger than 1×1 . It is still true that its trace will be nonanalytic at the Gross-Witten transition point. But, the parallel transporter round the larger loop will open a spectral gap at a large value of b , and at that value the trace of the larger loop does not exhibit a nonanalytic behavior. In short, the nonanalytic behavior of the trace of the plaquette is a lattice effect which has no relevance to the continuum limit. Once we agree that we should avoid looking at lattice effects, it is obvious we should rather consider continuum two dimensional QCD as providing a representative of the large N universality class relevant to the transition we found for our smeared four dimensional Wilson loops $\tilde{W}$.

In continuum two dimensional planar QCD the spectrum of Wilson loop operators, for smooth simple loops, does not depend on the the shape of the curve but only on the enclosed area. As the loop is scaled, all that matters is that the area changes, and at infinite N it is known that its eigenvalue distribution undergoes a transition at a specific area (measured in units of the two dimensional gauge coupling constant) where it just opens a gap at -1. However, the traces of any finite power of the Wilson loop operator stay analytic as the area goes through the critical point of the spectrum. The expectation value of an n -wound Wilson loop in two dimensions still depends just on its fundamental area, which we make dimensionless using the 't Hooft coupling, and denote by A . The explicit formula at large N is known [23], given by the product of a polynomial and an exponential:

$$\frac{1}{N} \langle \text{Tr} W^n \rangle = \frac{1}{n} L_{n-1}^{(1)}(2An) e^{-An} \quad (4.6)$$

The dependence on scale hardly could be more analytic.

The Gross-Witten universality class and the one associated with continuum QCD are related but not necessarily identical. Still, both candidates produce a Painlevé II equation in the double scaling limit [24, 25].

Another point to keep in mind is that the string theory associated with the universal class of the Gross-Witten transition is quite degenerate and appears to differ significantly from any putative string theory dual of QCD [26]. On the other hand, there does exist a more tangible string description of two dimensional QCD due to Gross and Taylor [27] which also has sigma model representations [28]. The large N transition in Wilson loops of continuum two dimensional QCD on infinite space-time is of the same type as the Douglas-Kazakov transition in the partition function of planar two dimensional QCD defined on a sphere of area A [22].

There is another feature in favor of two dimensional QCD as the universality class and it has to do with instantons. When one takes the nonabelian instanton solution that we showed makes a non-zero contribution to the eigenvalue density at -1 for small loops in four dimensions and looks only at a two dimensional slice one gets two dimensional instantons that were argued to be related to the Douglas-Kazakov transition in [25].

All this adds up to putting the Gross-Witten transition in disfavor as a representative of the universality class of our transitions, and leaves us with the preferred option of two dimensional planar QCD. An explicit formula for the eigenvalue distribution of Wilson loops can be found in [18] and it is possible [29] to connect this formula to (4.6).

To see that we are not completely off in our guess we show a plot of several eigenvalue distributions we measure versus the exact infinite N continuum solution of Durhuus and Olesen [18]:

Figure 9 shows how the critical size is traversed and one sees directly the effect of the transition and that the Durhuus-Olesen distribution seems to work pretty well on both sides of the transition. An attempt to fit to the Gross-Witten distribution would fail, since on the gap-less side the Gross-Witten distribution would always cross the value $\frac{1}{2}$ in Figure 9 at $\theta = \pm \frac{\pi}{2}$.

Another point in favor of the universality class is evident from equation 4.6: The prefactor has oscillating signs for large areas. These are nothing but the oscillating signs we saw in the Fourier coefficients of Figure 7. These oscillating signs could be taken as an indication that an effective string representation of loops traversed multiple times requires some elementary two dimensional excitation of statistics different from that of bosons [30]. The simple effective string models one typically uses do not address loops traversed multiple numbers of times.

Because of the importance of the large N universality class to our program, we now turn to a more general description of what appears to be its fundamental representative family in terms of matrix models:

Imagine n $SU(N)$ matrices, independently and identically distributed according to a conjugation invariant measure $d\mu(U_i)$. The measure is peaked at $U_i = 1$ and is of the single trace type. As a function of each eigenvalue it decreases monotonically as one goes round the circle from 1 to -1 , where it has a minimum. The continuum Wilson loop operators, W , one defines, would behave in the vicinity of the large N transition point at which their eigenvalue distribution opens a gap at -1 , similarly to $W = \prod_{i=1}^n U_i$. The parameters of the measure and n can be adjusted to various critical behaviors in the eigenvalue distribution of W at infinite N , and likely produce a variety of double-scaling limits corresponding to

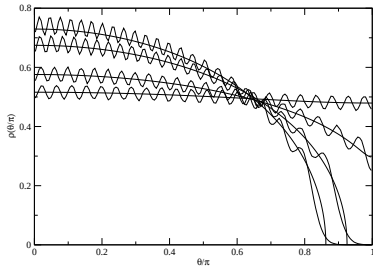


Figure 9: Fit of the distributions to the Durhms-Olesen distributions for four different sizes of Wilson loops, namely, $W_c = 0.740, 0.660, 0.560, 0.503$. The associated areas (k in the Durhms-Olesen notation) that describe the continuous curves are given by $k = 4.03, 2.30, 1.41, 1.15$ respectively.

large N fixed points of decreasing degrees of stability. The shape and scale of W map analytically into these parameters in the vicinity of the transition. Most plausibly, the most generic and stable transition is the relevant one.

This random matrix model is exactly soluble, as we now sketch, leaving details for the future:

One introduces n pairs of N -component Grassmann variables, $\bar{\psi}_i, \psi_i$ and proves by simple recursion that

$$\int \prod_i^n (d\bar{\psi}_i d\psi_i) e^{\sum_i \bar{\psi}_i \psi_i - \sum_i \bar{\psi}_i U_i \psi_{i+1}} = \det(z^n + W) \quad (4.7)$$

Above, we use the convention $\psi_{n+1} \equiv -\psi_1$.

Now, one can introduce auxiliary fields for the $SU(N)$ invariant bilinears $\bar{\psi}_i \psi_i$ and $\bar{\psi}_i \psi_{i+1}$. Averaging over the U_i variables can be done independently and the remaining integral can be dealt with at infinite N by ordinary saddle point methods.

Choosing an axial gauge on an infinite two dimensional lattice, it is easy to see that both on the lattice and in the continuum [18] the non-minimal Wilson loops of two dimensional

planar QCD are indeed described by the above universality class. Similarly, the Douglas-Kazakov critical point is also in this universality class, corresponding to the most stable critical point.

Intuitively, the universality class can be visualized as a nonabelian generalization of the following abelian situation: Let there be given a smooth loop C and let S be a surface of the topology of a disk bounded by C . On the surface S define a two dimensional $U(1)$ gauge field, with independently and identically distributed fluxes through small patches of surface that tile S . The phase of the parallel transport round C is then given by the sum of all these fluxes. The distribution of that sum will be generically governed by the central limit theorem. It is now obvious that we have a nonabelian generalization of this arrangement [31], of the type studied by Voiculescu [32].

5. Summary and Discussion.

We first give an overview of various ideas that have influenced us and to which we have not yet referred directly. Subsequently we proceed to a summary explaining what are the essential new points we have made in this paper. We finish with a short description of planned future work.

5.1 Connections to other work.

At the conceptual level, this paper has been influenced by the work of many others, in particular people in our own subfield of lattice field theory, even if this was not mentioned at any point in our presentation until now. While it is impossible to avoid omissions, we try here to draw attention to what stands out in our memory at this point.

That an extra dimension might be needed for a local string dual of the field theoretical description clearly is an idea with many sources. The relationship between this extra dimension and a space of eigenvalues has been emphasized by many, for example in papers by A. Jevicki [33]. The essential point is that for a hermitian or unitary matrix the eigenvalues are restricted to a one dimensional line and repel each other; as such they make up dynamically, at infinite N , a continuous one dimensional degree of freedom. In the context of matrix models this produces an analogue of the extra dimension we have seen in higher dimensional setups.

Using the lattice to make progress on the question of a possible string dual of non-abelian YM theory has preoccupied a sizable group of researchers. This new wave of lattice work on the large N limit has been started by M. Teper, and in various collaborations he has addressed many of the questions we address here [34]. In particular, the question of the cross-over has been studied in various contexts [35]. There is an interesting issue raised there suggesting a qualitative difference between three dimensional and four dimensional YM gauge theories with respect to observable dependent large N transitions. The immediate implication is that it would be interesting to repeat our work here in three Euclidean dimensions; it would be surprising to see three dimensions so different in this context from both two and four dimensions.

J. Kuti [36] and collaborators have studied in great detail the cross-over from field theoretical to string-like behavior and have shown how various string states reorganize themselves as one goes from short to long QCD strings. The philosophy of matching various field theories with known or suspected string-like excitations onto an effective string theory at long distances underlies much of this work. This work has not yet been extended to large values of N , but this is recognized as a possible valid avenue for further steps in this program.

We have been influenced significantly by papers by R. Brower [37]. On the one hand side he and collaborators have provided additional examples in which one can see the extra dimension being connected to standard scale decompositions of four dimensional processes of traditional interest. This interpretation is very natural when one thinks in terms of a warp factor in the context of the famous AdS/CFT correspondence in the conformal and related cases. It provides a most convincing resolution of the old puzzle of how a string theory can embody hard, field theoretic ultraviolet behavior.

It has also been stressed by Brower that one should be more ambitious than merely connecting onto an effective long distance string theory, and, that the extra dimension could provide the means to construct a complete stringy dual to YM, valid at all distances, albeit too difficult for actual calculation at short distances. It is with this in mind that although our strings are fat, we maintain the zigzag symmetry, which would be a potential clue about the infinitely thin string dual. Also, with this in mind, we insisted on finding observables that have a continuum field theoretical semiclassical expansion at small sizes, where the effects of short distance fluctuations reveal themselves.

The effective string description has been studied in various guises also by others [38], in addition to Polchinski and Strominger; the approaches focused both on open Dirichlet strings and on closed string wrapping round a large compact direction of space-time.

Another related topic is the issue of "Casimir scaling", in the context of the dependence of the string tension on the representation of the source and sink the $SU(N)$ string connects. It is known that "Casimir scaling" is exact when applied to charges in the fundamental and in the adjoint representation at $N = \infty$, so long as we take $N \rightarrow \infty$ first and let the minimal area spanned by the loop increase only after that [39]. We wish to mention here a recent paper [40] in the context of "Casimir scaling" at finite N where the Wilson loop eigenvalue distribution has been studied on the lattice for $SU(2)$.

Last but not least we wish to mention the old idea of preconfinement [41], in which one pictures a jet producing process in QCD as evolving by first creating (in the planar limit) finite energy clusters that already are color neutral and only subsequently hadronize. This preparation for hadronization is similar to the early signals of closing the gap in the eigenvalue spectrum, before the density becomes uniform up to corrections exponential in the minimal area spanned by the loop, where one has full confinement. As mentioned previously, one can have a large N transition without ever making it all the way to confinement.

5.2 Main new ideas in this work.

Our main change in vantage was a focus on the universal nature of large N transition which

we hope to turn into a calculation that can take us through the cross-over which connects the regime in which field theory is the most convenient description to the regime in which string theory is assumed to eventually become a convenient description.

We see the beginnings of such a calculational scheme and feel that were it successful, it would provide a concrete result on which to build in the search for a more elegant and fundamental dual representation of Yang Mills theory.

We found ourselves naturally led into something that looks like an extra dimension and one that is intimately related to scaling at that.

5.3 Plans for the future.

The first problem for the future is a more thorough investigation of the critical regime of the large N transitions and the numerical identification of the right universality class. In parallel an effort will be made to make the semiclassical calculation based on instantons at the short distance end concrete. The double scaling limits in the larger universality class would need to be studied in detail; assuming that our guess for this universality class is right, it should be possible to construct those double scaling limits exactly.

The connection onto a convenient effective string model at the strong coupling end is another direction of future activity; here, the main tool would be numerical at first, and we expect to draw from the experience of the many other workers on similar problems. Our main focus will be on keeping N large so the string coupling constant can be set to zero as is often done even for $N = 3$.

Also, we feel it might be useful to try to face the question whether an effective string theory for planar QCD is indeed not different from an effective string theory in a model which has nothing to do with non-abelian gauge theory. For example, on the face of it one would expect the zigzag symmetry to play no role at leading order in the effective string theory, but maybe this is not a correct assumption. An analogy is the chiral Lagrangian describing pions in QCD, which indeed could come from a fundamental theory that has no gauge fields. But, the complete decoupling at all scales of all the mesons, well known to occur at infinite N , is something the effective Lagrangian, by itself, does not incorporate in a structural way; rather, many of its free parameters need to be set to zero.

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P19: 格点算符非微扰重正化一般方法

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A GENERAL METHOD FOR NON-PERTURBATIVE RENORMALIZATION OF LATTICE OPERATORS

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Abstract

We propose a non-perturbative method for computing the renormalization constants of generic composite operators. This method is intended to reduce some systematic errors, which are present when one tries to obtain physical predictions from the matrix elements of lattice operators. We also present the results of a calculation of the renormalization constants of several two-fermion operators, obtained, with our method, by numerical simulation of QCD , on a $16^3 \times 32$ lattice, at $\beta = 6.0$. The results of this simulation are encouraging, and further applications to four-fermion operators and to the heavy quark effective theory are proposed.

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1 Introduction

Renormalization of lattice operators is a necessary ingredient to obtain many physical results from numerical simulations, such as meson decay constants, form factors, structure functions, mixing amplitudes, etc. In this paper we study the renormalization of composite operators in lattice *QCD*. We propose a method which avoids completely the use of lattice perturbation theory and allows a non-perturbative determination of the renormalization constants of any composite operator. For illustrative purposes, we consider some specific applications to matrix elements of two-fermion operators. The approach is particularly useful in those cases, such as the scalar or pseudoscalar densities, where it is not possible to use the chiral Ward identities to determine the renormalization constants non-perturbatively¹ [1]–[3]. Moreover, our proposal can be applied to many other cases, such as the renormalization of the four-fermion operators of the effective weak Hamiltonian, and to heavy-light currents in the Heavy Quark Effective Theory. Preliminary results have been presented in ref. [4]. A similar attempt, limited to the divergent part of the $\Delta I = 1/2$ penguin-operator, can be found in ref. [5].

Renormalized lattice operators must correspond, in the limit of infinite lattice cut-off, to finite chirally covariant operators, which obey the same renormalization conditions as those in the continuum. With just a few exceptions, lattice perturbation theory is used to evaluate the renormalization constants of lattice operators. The problem of mixing with lower dimensional operators requires however, a non-perturbative subtraction of all power divergences [1],[6]–[9]. Apart from this special case, the use of perturbation theory, in the computation of multiplicative constants and dimensionless mixing coefficients, is well justified, provided that the lattice spacing a is sufficiently small, i.e. $a^{-1} \gg \Lambda_{QCD}$. However, in those cases where the results from perturbation theory, obtained using the bare lattice coupling constant $\alpha_s^{LATT} = g_0^2/4\pi$ as an expansion parameter, can be checked using non-perturbative methods, there is generally a significant discrepancy. Several solutions to this problem have been proposed so far:

- In order to improve the convergence of the perturbative series, the au-

¹In the following we will denote the method to determine the renormalization constants, using the Ward identities, as the Ward identity method.

thors of ref. [10] have proposed a “tadpole improved” perturbation theory. Nevertheless, ignorance of higher order contributions still represents an important source of uncertainty in the extraction of physical results.

- In some limited cases, corresponding to finite operators, namely the vector and axial vector currents, and the ratio of pseudoscalar and scalar densities, a fully non-perturbative determination of the renormalization constants can be obtained with the use of chiral Ward identities [1]–[3].
- One may fix non-perturbative renormalization conditions directly on hadronic matrix elements. This procedure was used in ref. [7], to subtract the divergent part of the $\Delta I = 1/2$ operator. The price is that one has to sacrifice a physical prediction, for any subtraction imposed on hadronic states. Thus, when several renormalization conditions are necessary, one loses much predictive power [1, 6].

Our proposal is to impose renormalization conditions non-perturbatively, directly on quark and gluon Green functions, in a fixed gauge, with given off-shell external states, with large virtualities.

The method consists in mimicking what is usually done in perturbation theory. One fixes the renormalization conditions of a certain operator by imposing that suitable Green functions, computed between external off-shell quark and gluon states, in a fixed gauge, coincide with their tree level value. For example, if we consider the generic two-quark operator $O_T = \psi\Gamma\psi$, we may impose the condition

$$Z_\Gamma \langle p|O_T|p \rangle|_{p^2=-\mu^2} = \langle p|O_T|p \rangle_0, \quad (1)$$

where Γ is one of the Dirac matrices and $\langle p|O_T|p \rangle_0$ is the tree level matrix element. The extension to more complicated cases, including four-fermion operators and operator mixing, is straightforward. This procedure defines the same renormalized operators, i.e. the same Wilson coefficient functions, in all regularization schemes, provided they are expressed in terms of the same renormalized coupling constant. However, the coefficient functions now depend on the external states and on the gauge, which must be specified.

In principle this scheme completely solves the problem of large corrections in lattice perturbation theory, which are automatically included in the renormalization constants. Matching of lattice operators to the corresponding ones, renormalized in a continuum renormalization scheme (for definiteness, the $\overline{MS}$ scheme), then requires continuum perturbation theory only. When next-to-leading corrections are known, the error on the physical matrix elements is of $O((\alpha_s^{\overline{MS}})^2)$, in the continuum perturbative expansion, which is expected to have a better convergence. A continuum perturbative calculation of the matching conditions is a common step in all approaches, standard or “tadpole improved” perturbation theory and non-perturbative renormalization. Our proposal can be applied to any composite operator, unlike the Ward identity method, which is limited to a few, albeit very important, cases².

The feasibility of using Green functions computed between quark states, in a fixed gauge, was already discussed in ref. [3], in the context of non-perturbative renormalization, using Ward identities. Indeed, in the case of the axial current, it was shown that the signal was rather good and the result was in agreement with the corresponding hadronic gauge invariant determination.

As far as the renormalization conditions are concerned, our proposal is expected to work, whenever it is possible to fix the virtuality of the external states μ and to satisfy the condition $\Lambda_{QCD} \ll \mu \ll 1/a$, in order to keep under control both non-perturbative and discretization effects. We stress that this requirement is common to all methods. The results of our numerical investigation are encouraging: they suggest that there does exist a “window” in μ , where the method can be applied, at values of the lattice coupling constant used in current lattice simulations.

The plan for the remainder of the paper is as follows. In the next section the basic formulae necessary to define our non-perturbative method are presented, using as an example logarithmically divergent two-quark operators, such as the scalar and pseudoscalar densities. The discussion is generalised to all composite operators in section 3, and in the following section the implementation of the method in numerical simulations for two-quark operators

²It is still true that the statistical accuracy is in general slightly better with the Ward identities method, in those cases where it can be applied, see sec. 6.

is discussed. In section 5 we give some details about the perturbative evaluation of the renormalisation constants on lattices of finite volume. The values of the renormalisation constants obtained in a numerical simulation are presented in section 6, and are compared to the results from perturbation theory. The final section contains our conclusions, a summary of the numerical results and a discussion of possible further applications of the ideas presented in this paper.

2 Non-perturbative renormalization conditions

In this section, the non-perturbative renormalization scheme is presented in detail. We also discuss the conditions which must be satisfied for our method to be applicable.

To facilitate the discussion, it is convenient to classify composite operators into three main classes, according to their ultra-violet behaviour, as $a \rightarrow 0$:

1. **Finite operators:** these include the vector and axial vector currents. Another example is the ratio of the scalar and pseudoscalar renormalization constants, Z_S/Z_P . For these cases, the Ward identity method is also applicable.
2. **Logarithmically divergent operators:** this is a large family of operators, relevant to hadron phenomenology. It includes scalar and pseudoscalar densities, four-fermion $\Delta F = 2$ operators, some components of the energy-momentum tensor, etc.
3. **Power divergent operators:** these operators are present when mixing with lower dimensional operators is possible. This happens in regularizations which have an intrinsic mass scale, such as the lattice regularization. Examples are often encountered in phenomenological applications of lattice QCD , such as four-fermion operators relevant to $\Delta I = 1/2$ transitions or operators of order $1/m$ in the heavy quark effective theory (HQET) [11]–[15].

For simplicity, we first consider the two-fermion operators of class 2., and extend the discussion to other operators in this class, and also to those in

classes 1. and 3., in the next section. Thus, the formulae given below apply directly to the pseudoscalar and scalar densities. Throughout our discussion, we assume that discretization errors are negligible: in field theory language, this is equivalent to the statement that the renormalized Green functions do not differ appreciably from their values in the limit of an infinite ultra-violet cut-off. The discussion is presented in the limit of small quark masses and in Euclidean space-time. The discretisation of the quark action is assumed to be performed *à la* Wilson, characterized by explicit chiral symmetry breaking of the Lagrangian³. The extension to staggered fermions is straightforward.

Let us consider the forward amputated Green function $\Gamma_O(pa)$, of a two-fermion bare lattice operator $O(a)$, computed between off-shell quark states with four-momentum p , with $p^2 = \mu^2$, and in a fixed gauge, for example the Landau gauge. Without gauge-fixing all Green functions computed between quark and gluon external states are zero. This is also true when the gauge is not completely fixed, e.g. in the Coulomb gauge. We define the renormalized operator $O(\mu)$, by introducing the renormalization constant Z_O

$$O(\mu) = Z_O(\mu a, g(a))O(a). \quad (2)$$

Z_O is found, by imposing the renormalization condition

$$Z_O(\mu a, g(a))Z_\psi^{-1}(\mu a, g(a))\Gamma_O(pa)|_{p^2=\mu^2} = 1, \quad (3)$$

where Z_ψ is the field renormalization constant, to be defined below. This procedure defines a renormalized operator $O(\mu)$ which is independent of the regularization scheme [17]–[19]. It depends, however, on the external states and on the gauge. This does not affect the final results, which, combined with the continuum calculation of the renormalization conditions, at any given order of perturbation theory, will be gauge invariant and independent of the external states. Let us specify the different quantities entering eq. (3). Γ_O is defined in terms of the expectation value⁴ of the non-amputated Green function $G_O(pa)$, and of the quark propagator $S(pa)$

$$\Gamma_O(pa) = \frac{1}{12}\text{tr} \left(\Lambda_O(pa) \hat{P}_O \right), \quad (4)$$

³This includes the SW-Clover action [16].

⁴“Expectation value” means, as usual, that one averages the Green functions over the gauge field configurations, generated by Monte Carlo simulation, see sec. 4.

where

$$\Lambda_O(pa) = S(pa)^{-1} G_O(pa) S(pa)^{-1}. \quad (5)$$

$\hat{P}_O$ is a suitable projector on the tree-level operator: $\hat{P}_O = \hat{1}$ ($\hat{P}_O = \gamma_5$) for the scalar (pseudoscalar) density. The factor $1/12$ ensures the correct overall normalization of the trace (colour $\times$ spin = 12). Projectors are very convenient when defining Green functions, particularly in the non-perturbative case. They have been extensively used in refs. [18, 19]. Of course one can also use other definitions of Z_O .

$Z_\psi^{1/2}$ is the renormalization constant of the fermion field. It can be defined in different ways, some of which are equivalent perturbatively. Beyond perturbation theory, the most natural definition of Z_ψ is obtained from the amputated Green function of the conserved vector current V^C . Indeed, one knows that for V^C the renormalization constant is equal to one:

$$Z_{V^C}^{-1} = \Gamma_{V^C} \times Z_\psi^{-1} = \frac{1}{48} \text{tr} \left(\Lambda_{V^C}(pa) \gamma_\mu \right) \big|_{p^2=\mu^2} \times Z_\psi^{-1} = 1, \quad (6)$$

which implies

$$Z_\psi = \frac{1}{48} \text{tr} \left(\Lambda_{V^C}(pa) \gamma_\mu \right) \big|_{p^2=\mu^2}. \quad (7)$$

Equations (3)–(7) completely define our method. In the remainder of this section, we discuss some important aspects concerning its applicability.

In the above formulae, for simplicity, we have always considered only forward matrix elements. In general, one has the freedom to define the renormalization conditions at different external momenta p and p' ($p \neq p'$). The virtualities of the quark states must be much larger than Λ_{QCD} . The reason is that, in order to obtain the physical result, we have to combine the matrix element of the renormalized operator $O(\mu)$, with a Wilson coefficient function. The latter is computed in continuum perturbation theory, by expanding in $a_s^{\overline{MS}}$ at a scale of order μ . Thus, for the validity of this perturbative calculation, μ must be large. An important question in our program is whether it is possible to find, on the lattice, a scale μ which is sufficiently low, in order to have small $O(a)$ effects, and sufficiently large, in order to have small higher order corrections. The range of μ , where the above conditions are satisfied, depends on the value of g_0^2 of the numerical simulation.

We stress that, if such a window does not exist, in current lattice simulations, an accurate matching of lattice operators to continuum ones becomes impossible, not only in our approach, but also with any other method.

It might be expected that the condition $\mu \gg \Lambda_{QCD}$ ensures that perturbation theory is valid. When spontaneous symmetry breaking occurs however, as is the case in QCD , a large value of μ may not be enough, because of the presence of the Goldstone boson, the pion in our case. At low momentum transfers $q = p - p'$, Green functions can receive a non-perturbative contribution from the pion pole, which cannot be computed. The naïve expectation is that this effect vanishes as $1/q^2$, at large q^2 . However, with fermion fields, this contribution is proportional to $1/p^2 = 1/\mu^2$, even when $q^2 = 0$. The proof is given in the appendix. We conclude that a large value of μ solves the problem. At low values of μ^2 , however, for the pseudoscalar density and the axial current, we expect to find visible effects, see secs. 3 and 6.

A danger to the non-perturbative scheme may come from the presence of Gribov copies. We do not address this problem here; it has been investigated in refs. [20, 21]. The results of ref. [21] indicate that, at least for the quantities of interest in the present study, the “Gribov uncertainty”, in current lattice simulations, is at most of the same order as the statistical error.

One may imagine applying the renormalization conditions at very large values of β , on lattices which have small physical volumes. This might seem to overcome the problem of the existence of the window in μ . For physical applications however, it is necessary to evolve the renormalization constants to smaller values of β . Non-perturbative contributions, which are not detected at large β 's, may then become important. This demonstrates the necessity, for any method, of the existence of a window $\Lambda_{QCD} \ll \mu \ll 1/a$, where a is the lattice spacing used in numerical simulations of physical quantities. Only then are non-perturbative effects and lattice artefacts small simultaneously.

3 Applications of the method to generic composite operators

In the previous section, using the renormalization of the scalar and pseudoscalar densities as a prototype, the general strategy of the non-perturbative

scheme has been presented. In the present section the necessary modifications needed in other cases are briefly described. We will follow the classification of operators introduced in sec. 2.

3.1 Finite operators

In the case of vector and axial vector currents, we do not have the freedom of fixing the renormalization conditions in an arbitrary way: the renormalization conditions are acceptable only when they are compatible with the Ward identities. To be specific, one can apply eq. (3) to the local vector current in order to determine Z_{V^L}

$$Z_{V^L} = \frac{Z_\psi}{\Gamma_{V^L}}, \quad (8)$$

We now show that this definition of Z_{V^L} , which uses the general prescription given in the previous section, coincides with the one derived from the vector current Ward identity on quark states [1, 22]⁵

$$q_\mu \left[Z_{V^L} \Lambda_{V^L}^\mu(p + \frac{q}{2}, p - \frac{q}{2}) \right] = -i \left(S^{-1}(p + \frac{q}{2}) - S^{-1}(p - \frac{q}{2}) \right). \quad (9)$$

where $\Lambda_{V^L}^\mu(p + q/2, p - q/2)$ is the amputated vertex with momentum transfer q . By applying $\partial/\partial q_\mu$, at $q = 0$, on both sides of eq. (9), one gets

$$Z_{V^L} \left(\Lambda_{V^L}^\mu(p) + q_\mu \frac{\partial}{\partial q_\mu} \Lambda_{V^L}^\mu(p + \frac{q}{2}, p - \frac{q}{2})|_{q=0} \right) = -i \frac{\partial}{\partial p_\rho} S^{-1}(p). \quad (10)$$

The second term on the l.h.s. vanishes when $q \rightarrow 0$. By tracing eq. (10) with γ_ρ , we then recover the definition of Z_{V^L} of eq. (8).

One is tempted to follow the same path for the local axial vector current, i.e. to define

$$Z_{A^L} = \frac{Z_\psi}{\Gamma_{A^L}}. \quad (11)$$

However this definition of Z_{A^L} is not equivalent to the one obtained by the Ward identity

$$q_\mu \left[Z_{A^L} \Lambda_{A^L}^\mu(p + \frac{q}{2}, p - \frac{q}{2}) \right] = -i \left(\gamma_5 S^{-1}(p + \frac{q}{2}) + S^{-1}(p - \frac{q}{2}) \gamma_5 \right). \quad (12)$$

⁵For the remainder of this subsection we work in lattice units, setting $a = 1$.

To see this we proceed as for the vector current, applying $\partial/\partial q_\mu$ to both sides of eq. (12).

$$Z_{A^L} \left(\Lambda_{A^L_\mu}(p) + q_\mu \frac{\partial}{\partial q_\mu} \Lambda_{A^L_\mu}(p + \frac{q}{2}, p - \frac{q}{2}) \Big|_{q=0} \right) =$$

$$- \frac{i}{2} \left(\gamma_5 \frac{\partial}{\partial p_\mu} S^{-1}(p) - \frac{\partial}{\partial p_\mu} S^{-1}(p) \gamma_5 \right). \quad (13)$$

In the axial case, however, the second term on the l.h.s. does not vanish, as $q \rightarrow 0$ and, indeed, this term is necessary to saturate the Ward identity, in presence of a massless Goldstone boson. As demonstrated in the appendix however, this term becomes negligible for high values of p^2 . In this case, by tracing eq. (13), one recovers (11). In conclusion, we expect to find a value of Z_{A^L} close to those obtained from the Ward identity on hadron or on quark states [3], by using eq. (11), at large μ^2 .

Z_S/Z_P is another finite quantity, which is completely fixed by the Ward identities, even though the renormalization conditions on Z_S and Z_P are separately arbitrary. In other words one is free to decide the renormalization conditions of one of the two operators, the scalar density say, and the value of Z_P will automatically follow. The procedure of sec. 2, respects the Ward identities and thus guarantees that the correct value of Z_S/Z_P is obtained. This is clearly true only at large values of μ^2 , for the same reasons as for the axial vector current.

3.2 Logarithmically divergent operators

For this category of operator only the question of operator-mixing is left to be discussed. There are two kinds of mixing: the one which can occur also in the continuum and the one induced by the explicit lattice chiral symmetry breaking. We are only concerned here with the latter case, and hence restrict our discussion to operators which renormalize multiplicatively in the continuum. To illustrate our arguments, we consider the four-fermion operator

$$O^{\Delta S=2} = \left(\bar{s} \gamma^\mu (1 - \gamma_5) d \right) \left(\bar{s} \gamma_\mu (1 - \gamma_5) d \right), \quad (14)$$

which is relevant to the calculation of the $K^0-\bar{K}^0$ mixing amplitude. Because of the Wilson term, this operator mixes with operators of different chiralities [1, 6, 23]

$$O_{\text{cont}}^{\Delta S=2} = Z_O \left(O_{\text{latt}}^{\Delta S=2} + \sum_i Z_i O^i \right), \quad (15)$$

where the operators O^i are given by $O^1 = (sd)(\bar{s}d)$, $O^2 = (s\gamma_5 d)(\bar{s}\gamma_5 d)$, etc. [23]. The mixing coefficients Z_i are finite functions of $g_0^2(a)$. The over-all renormalization Z_O is necessary to eliminate the logarithmic divergence.

We now sketch the renormalization procedure for $O^{\Delta S=2}$. Let us start from the four-point amputated, Green function of $O_{\text{latt}}^{\Delta S=2}$, between quark states with momentum $p^2 = \mu^2$

$$\Lambda_{\alpha\beta\gamma\delta}^{ABCD}(pa), \quad (16)$$

where $\alpha, \beta, \dots$ and $A, B, \dots$ are spin and colour indices respectively. To find the mixing coefficients Z_i , one uses suitable projectors $\hat{P}$, as was done for two-quark operators⁶, for example

$$\text{tr}(\Lambda \hat{P}) = \Lambda_{\alpha\beta\gamma\delta}^{AABB} (\gamma_5)_{\beta\alpha} (\gamma_5)_{\delta\gamma}. \quad (17)$$

By projecting on all the possible colour-spin structures, corresponding to the operators O^i , one obtains a system of linear equations in the mixing coefficients Z_i . Using the Z_i , determined in this way, one can compute the Green function of the subtracted operator $O^* = O_{\text{latt}}^{\Delta S=2} + \sum_i Z_i O^i$. From the amputated vertex of O^* , one can finally remove the logarithmic divergence, by imposing the renormalization condition

$$Z_O(\mu, g(a)) Z_\psi^{-2}(\mu, g(a)) \Gamma_O^*(pa)|_{p^2=\mu^2} = 1, \quad (18)$$

where

$$\Gamma_O^* \sim \text{tr}(\Lambda^* \hat{P}_{L \times L}). \quad (19)$$

and $\hat{P}_{L \times L}$ is the suitable projector on the $\gamma_\mu(1-\gamma_5) \otimes \gamma^\mu(1-\gamma_5)$ Dirac tensor.

⁶ In order to simplify the presentation, we do not discuss here the complications arising from the different possible colour structures.

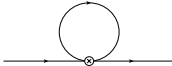


Figure 1: Diagram which mixes the four fermion operator in eq. (20) with the scalar and pseudoscalar densities. The mixing is induced by the chiral breaking term present in the Wilson and Clover actions.

3.3 Power divergent operators

Composite operators are divergent in $1/a$ when they can mix with lower dimensional operators. Examples are given by the operators relevant to deep inelastic scattering and by the penguin-operators appearing in the $\Delta S = 1/2$ Hamiltonian. Let us consider the $\Delta S = 1/2$ operator

$$O^{\Delta S=1/2} = (\bar{s}\gamma_\mu(1 - \gamma_5)d) \sum_i (\bar{q}_i\gamma^\mu(1 + \gamma_5)q_i). \quad (20)$$

$O_{\text{latt}}^{\Delta S=1/2}$ mixes with the scalar and pseudoscalar densities, even at zeroth order in α_s , through the diagram in fig. 1⁷

$$O_{\text{cont}}^{\Delta S=1/2} = Z_O O_{\text{latt}}^{\Delta S=1/2} + Z_1(\bar{s}d) + Z_2(\bar{s}\gamma_5 d) + \dots \quad (21)$$

To find the coefficients $Z_{1,2}$, one imposes that the two-point Green function of the operator on the r.h.s. of eq. (21), on off-shell quark states, is zero [5]. Once all the divergent parts have been removed, one can impose the renormalization conditions, necessary to remove the over-all logarithmic divergence, as discussed above.

⁷ $O_{\text{latt}}^{\Delta S=1/2}$ also mixes with the chromo-magnetic operator, e.g. $\bar{s}\sigma_{\mu\nu}G^{\mu\nu}d$, which is not being considered here.

4 Implementation of the method in numerical simulations

To implement the ideas of the previous section, let us consider the generic local two-fermion operator

$$O_{\Gamma}(x) = \bar{\psi}(x)\Gamma\psi(x) \quad (22)$$

where Γ is any Dirac matrix. The lattice, bare Green function of O_{Γ} , between off-shell quark states, can be obtained from

$$G_O(x, y) = \langle \psi(x)O_{\Gamma}(0)\bar{\psi}(y) \rangle = \frac{1}{N} \sum_{i=1}^N S_i(x|0)\Gamma S_i(0|y), \quad (23)$$

where $i = 1, N$ labels the gauge field configurations and $S_i(x|0)$ is the quark propagator on the single configuration i , obtained by inverting the discretized Dirac operator. We denote by $S_i(x|y)$ the inverse Dirac operator, which is not translationally invariant, in contrast to $S(x - y)$, which is the usual quark propagator. The gauge field configurations are generated with some standard, gauge-invariant algorithm; the gauge fields and the quark propagators are then rotated into the Landau gauge, which will be defined in sec. 6. From eq. (23), using $S_i(x|y) = \gamma_5 S_i(y|x)\gamma_5$, one gets

$$\begin{aligned} G_O(pa) &\equiv \int d^4x d^4y e^{-i(p \cdot x - p \cdot y)} G_O(x, y) \\ &= \frac{1}{N} \sum_{i=1}^N \left(\int d^4x S_i(x|0) e^{-i p \cdot x} \right) \Gamma \left(\int d^4y S_i(0|y) e^{i p \cdot y} \right) \\ &= \frac{1}{N} \sum_{i=1}^N S_i(p|0) \Gamma \gamma_5 \left(\int d^4y S_i^\dagger(y|0) e^{i p \cdot y} \right) \gamma_5 \\ &= \frac{1}{N} \sum_{i=1}^N S_i(p|0) \Gamma \left(\gamma_5 S_i^\dagger(p|0) \gamma_5 \right), \end{aligned} \quad (24)$$

where we have defined

$$S_i(p|y) = \int d^4x S_i(x|y) e^{-i(p \cdot x)}, \quad (25)$$

because, on a single configuration, the quark propagator is not translationally invariant. Translational invariance is recovered after averaging over the configurations. We can then write

$$S(pa) = \langle S(p|0) \rangle = \frac{1}{N} \sum_{i=1}^N S_i(p|0). \quad (26)$$

Eqs. (24) and (26) give $G_O(pa)$ and $S(pa)$ in terms of quark propagators, computed by inverting the Dirac operator in the presence of a background gauge field configuration, generated by numerical simulation. With these quantities at hand, one can apply the strategy and the formulae, introduced in sec. 2. We add some information, that can help the reader in practical applications. The quark wave-function renormalization constant Z_ψ has been computed using

$$Z_\psi = \Gamma_{V^L} \times Z_{V^L} = \frac{1}{48} \text{tr} \left(\Lambda_{V^L}(pa) \gamma_\mu \right) \big|_{p^2=\mu^2} \times Z_{V^L} \quad (27)$$

where $\Lambda_{V^L}(pa)$ is defined as in eq. (5); the trace is taken over colours and spins and Z_{V^L} is the renormalization constant of the local vector current $V_\mu^L(x) = \bar{\psi}(x) \gamma_\mu \psi(x)$. Equations (7) and (27) are two equivalent definitions of Z_ψ . We have used eq. (27) instead of eq. (7), because the former is more convenient when, as in our case, one has only computed quark propagators which stem from the origin. The three-point $(q - V - q)$ Green function is obtained by inserting the local vector current in the origin. Z_{V^L} can be obtained with very high accuracy from ratios of matrix elements of the conserved and local currents [2, 24]. From the amputated Green functions

$$\begin{aligned} \Gamma_1(pa) &= \frac{1}{12} \text{tr} \left(\Lambda_1(pa) \hat{1} \right), \\ \Gamma_\gamma(pa) &= \frac{1}{12} \text{tr} \left(\Lambda_\gamma(pa) \gamma_5 \right), \\ \Gamma_{\gamma_5 \gamma}(pa) &= \frac{1}{48} \text{tr} \left(\Lambda_{\gamma_5 \gamma}(pa) \gamma_5 \gamma_\mu \right), \end{aligned} \quad (28)$$

by imposing the renormalization conditions (3), at several values of $p^2 = \mu^2$, to be specified in sec. 6, we have determined $Z_{S,P,A}(\mu, g(a))$, as functions of μ^2 .

If we wish to compute $Z_{V\perp}$ using the procedure defined in sec. 3 we clearly cannot take Z_ψ from eq. (27). From the Ward identity we have

$$Z_\psi = -i \frac{1}{12} \text{tr} \left(\frac{\partial S(p)^{-1}}{\partial p} \right) \Big|_{p^2=\mu^2} \quad (29)$$

where $\partial S(p)^{-1}/\partial p = 1/4 \times \gamma_p \partial S(p)^{-1}/\partial p_p$. To avoid derivatives with respect to a discrete variable, we have used

$$Z'_\psi(\mu, g(a)) = \frac{\text{tr} \left(-i \sum_{\lambda=1,4} \gamma_\lambda \sin(p_\lambda a) S^{-1}(pa) \right)}{4 \sum_{\lambda=1,4} \sin^2 p_\lambda a} \Big|_{p^2=\mu^2}. \quad (30)$$

In one loop perturbation theory, $Z'_\psi = Z_\psi$, in the Landau gauge⁸. By using

$$\Gamma_{\gamma_p}(pa) = \frac{1}{48} \text{tr} \left(\Lambda_{\gamma_p}(pa) \gamma_p \right), \quad (31)$$

and imposing the renormalization conditions (3), we have determined $Z_{V\perp}$ at several μ^2 . From the general discussion of the previous section, one expects that $Z_{V\perp}$, defined through the present procedure, is independent of μ^2 , up to terms of $O(a)$, as has been effectively found, see sec. 6.

In actual numerical simulations, the renormalization procedure could be obscured by lattice artefacts, i.e. terms of $O(\mu a)$, which are present when one imposes the renormalization conditions at large values of μ^2 , required to avoid large non-perturbative or higher order effects. As shown in ref.[25] (we use here the same notation), in order to reduce discretization errors, due to the finite value of the lattice spacing, one can compute correlation functions using a nearest-neighbour improved fermion action [16]

$$S_F^I = S_F^W + a^4 \sum_{x,\mu\nu} \left[-i g_0 \frac{ar}{4} \bar{\psi}(x) \sigma_{\mu\nu} F_{\mu\nu}(x) \psi(x) \right], \quad (32)$$

where S_F^W is the usual Wilson action, r is the Wilson parameter and $F_{\mu\nu}$ is the lattice field strength tensor. Moreover, it is also necessary to use improved operators O_F^I defined as

$$O_F^I(x) = \bar{\psi}(x) \overline{R}_x \Gamma \overline{R}_x \psi(x), \quad (33)$$

⁸In general, they differ by a finite term of order α_s .

where the rotations of the fermion fields are defined as

$$\overline{R}_x = \left(1 - a\frac{r}{4}(\overline{D}_x - m_0)\right) \quad , \quad \overline{R}_y = \left(1 + a\frac{r}{4}(\overline{D}_y + m_0)\right). \quad (34)$$

By using the improved action and operators, one eliminates discretization errors of $O(a)$, and only those of $O(\alpha_s a)$ or $O(a^2)$, or higher, remain. In eq. (33), $\overline{R}_x$ and $\overline{R}_y$ can be replaced by

$$\overline{R}_x = 1 - a\frac{r}{2}\overline{D}_x \quad , \quad \overline{R}_y = 1 + a\frac{r}{2}\overline{D}_y, \quad (35)$$

in computations of the hadron spectrum and of on-shell operator matrix elements. As discussed in refs.[26, 27], the improvement procedure is equivalent to expressing all correlation functions in terms of “effective” quark propagators

$$S_i^{eff}(x|0) = \left(1 - a\frac{r}{2}\overline{D}_x\right)S_i^I(x|0)\left(1 + a\frac{r}{2}\overline{D}_{y=0}\right) \quad (36)$$

where $S_i^I(x|0)$ is the fermion propagator of the improved theory from the point x to the origin. In order to eliminate terms of $O(a)$ in the off-shell correlation functions, used in this study, a further step is necessary [3]: we have to shift the quark propagator by a local term, which in Fourier space appears as a constant, to be added to its diagonal components

$$S_{\alpha\beta}^{AB}(p\mu) = S_{\alpha\beta}^{eff,AB}(p\mu) + \frac{r}{2}\delta^{AB}\delta_{\alpha\beta}$$

In the following, the vector current V_μ^L , the axial vector current A_μ^L , the pseudoscalar density P and the scalar density S are improved operators, obtained by using eq. (36).

5 Lattice perturbation theory

In order to gain some insight into the non-perturbative results presented in the next section, we have performed the corresponding one-loop perturbative calculation both in the continuum and on the lattice in the Landau gauge. This calculation is an extension of the one presented in ref. [27].

The standard calculation of the one-loop corrections corresponds to a computation of the renormalization constants with a fixed ultra-violet cut-off on the loop momenta, $|p_i| \leq \pi/a$, neglecting terms of $O(a)$ in the final result. The renormalization constants obtained in this limit depend on the quark mass m_q and on the external momenta only through the logarithmically divergent terms, which are related to the anomalous dimension of the operator. Since, at one loop, these terms are universal, i.e. they are the same on the lattice and in any continuum regularization, the renormalization constants, necessary to relate lattice to continuum renormalized operators, are independent of the quark mass and external momenta (they depend however on the lattice spacing and on the continuum renormalization scale μ). We call this limit “Standard Perturbation Theory” (STP). The non-perturbative calculation however, is performed on a lattice of finite size, and at a fixed value of the lattice spacing and quark mass: thus we cannot make terms of $O(a)$ arbitrarily small. In particular, since we want to study the μ -dependence of the renormalization constants, it is important to know at which values of μ (and m_q), the perturbative results, obtained on a finite lattice and with a fixed lattice spacing, differ from those obtained in SPT. In order to follow closely the non-perturbative computation, the perturbative results have also been obtained, on a lattice of size $16^3 \times 32$. In this way, we keep all terms of $O(\alpha_s a)$ or $O(a^2)$, or higher, in the final result (we call this procedure “Discrete Perturbation Theory”, DPT). The loop integrals were computed numerically, both in SPT and in DPT.

The non-perturbative results have been obtained by tracing suitable amputated Green functions, computed between off-shell quark states, in the Landau gauge, cf. eq.(4). In perturbation theory, this procedure differs from the standard one, where one identifies the correction from the term proportional to the tree-level matrix element of the operator under consideration [18, 19]. The difference is due to finite terms which, in the standard procedure, are considered as part of the matrix element of the operator, whereas with the projectors, are considered as part of the one-loop corrections. At one-loop order, these terms are the same in the continuum and on the lattice (in the limit $a \rightarrow 0$) and cancel when one computes the difference between the continuum and the lattice renormalization constants. To be more specific, let us consider the diagrams in fig. 2, computed in Naïve Dimensional

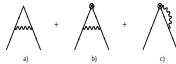


Figure 2: Vertex diagrams with the Clover action. The second and third diagrams refer to the contribution from the rotated part of the operator.

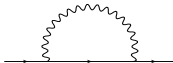


Figure 3: Self energy diagram.

Regularization (NDR), in the Feynman gauge

$$\Lambda_{\psi}(\frac{p}{\mu}) = \langle p | \bar{\psi} \gamma_{\mu} \psi | p \rangle = \left[1 + \frac{\alpha_s}{4\pi} C_F \left(\frac{p^2}{\mu^2} \right)^{-\epsilon} \left(\left(\frac{1}{\epsilon} + 1 \right) \gamma_{\mu} - 2 \frac{\not{p} \not{p}_{\mu}}{p^2} \right) \right], \quad (37)$$

where $C_F = (N^2 - 1)/2N$, $1/\bar{\epsilon} = 1/\epsilon - \ln 4\pi - \gamma_E$ and p is the external momentum. Usually, one takes the term proportional to $(1/\bar{\epsilon} + 1)$, i.e. the coefficient of γ_{μ} , as the one loop contribution to the bare operator. With the projector (see eq.(4)) the last term in eq. (37) also gives a contribution, so that the factor now becomes $(1/\bar{\epsilon} + 1/2)$. In table 1, we give the one-loop corrections obtained, with the projectors and in the 't Hooft and Veltman (HV) dimensional renormalization scheme [28], for the two-quark operators considered in this paper. The NDR regularization differs from the HV one by the definition of the anti-commuting properties of γ_5 . For details see for example [19]. In NDR the corrections to the axial current and pseudoscalar density coincide with those to the vector and scalar density respectively. By writing the gluon propagator as $G(k) = (-\delta_{\mu\nu}k^2 + (1 - \lambda)k_{\mu}k_{\nu})/k^4$, we denote as “Landau” the contribution to the one-loop corrections proportional to $(1 - \lambda)$.

With the $\overline{\text{MS}}$ procedure, after the subtraction of the pole, the irreducible

Operator	Feynman	Landau
γ_μ	$1/\bar{\epsilon} + 1/2$	$-1/\bar{\epsilon} - 1/2$
$\gamma_\mu \gamma_5$	$1/\bar{\epsilon} + 5/2$	$-1/\bar{\epsilon} - 1/2$
1	$4/\bar{\epsilon} + 6$	$-1/\bar{\epsilon} - 2$
γ_5	$4/\bar{\epsilon} + 10$	$-1/\bar{\epsilon} - 2$
Σ	$-1/\bar{\epsilon} - 1$	$1/\bar{\epsilon} + 1$

Table 1: The results of the calculation of the one-loop diagram of fig. 2 for different two-quark operators, in the HV scheme, are given. We also give the result of the calculation of the self-energy diagram of fig. 3. The mass of the quark has been taken to be zero, the inverse quark propagator is written as $i\not{p}\Sigma(p^2)$ and a factor $(\alpha_s/4\pi)C_F(p^2/\mu^2)^{-\epsilon}$ is omitted.

vertex of fig. 2 has the form⁹

$$\Gamma_O(p/\mu) = 1 + \frac{\alpha_s}{4\pi} C_F \left(-\gamma^{(O)} \ln(p^2/\mu^2) + C_O^{\overline{\text{MS}}} \right). \quad (38)$$

From table 1, one finds $C_{\gamma_\mu}^{\overline{\text{MS}}} = 1/2$, $C_1^{\overline{\text{MS}}} = 6$, etc., in the Feynman gauge and $C_{\gamma_\mu}^{\overline{\text{MS}}} = -1/2$, $C_5^{\overline{\text{MS}}} = -2$, etc., for the Landau (longitudinal) terms.

On the lattice, one gets a similar expression

$$\Gamma_O(pa) = 1 + \frac{\alpha_s}{4\pi} C_F \left(-\gamma^{(O)} \ln(p^2 a^2) + C_O^{\text{Lat}} \right). \quad (39)$$

The values of C_O^{Lat} , for the Feynman and Landau corrections are given in table 2. In table 2, we have denoted by “Rotated Feynman” the contribution coming from the rotated part of the two-fermion operators, see eqs. (33) and (34). The Landau contribution to the rotated part of the operators vanishes. This is required by the fact that the renormalization constants which relate lattice to continuum operators are gauge invariant.

Following refs. [27]–[29], we write

$$O(\mu) = \left(1 + \frac{\alpha_{\overline{\text{MS}}}}{4\pi} C_F \left(-\gamma^{(0)} \ln(\mu a)^2 + \Delta_O \right) \right) O(a), \quad (40)$$

⁹ $\Gamma_O(p/\mu)$ is obtained from $\Lambda_O(p/\mu)$ using a projector, as described in eq. (4).

Operator	Feynman	Rotated Feynman	Landau
γ_μ	6.62	-3.52	-4.29
$\gamma_\mu \gamma_5$	5.09	-11.68	-4.29
1	16.11	-12.00	-5.79
γ_5	19.18	4.31	-5.79
Σ	8.21	—	4.79

Table 2: The values of C_O^{Latt} for different two-quark operators and for the self-energy (C_Σ), with the Clover action. The mass of the quark has been taken to be zero.

where Δ_O can be computed from tables 1 and 2

$$\Delta_O = C_O^{\overline{\text{MS}}} - C_O^{\text{Latt}} + C_\Sigma^{\overline{\text{MS}}} - C_\Sigma^{\text{Latt}}. \quad (41)$$

In the difference between continuum and lattice one-loop corrections, all Landau pieces cancel, as expected. The results for $\Delta_{\gamma_\mu}, \Delta_{\gamma_\mu \gamma_5}, \dots$ all agree with those of ref. [27].

On a lattice, for a given quark mass and external momenta, there are terms of order a which can become important at large values of m_q and p . Since with our method we have to impose the renormalization conditions at values of the momenta which are large, it is important to monitor, at least in perturbation theory, the values of masses and momenta at which $O(a)$ effects become important. The Green function in DPT has a form similar to (39)

$$\Gamma_O(pa) = 1 + \frac{\alpha_s}{4\pi} C_F \left(-\gamma^{(O)} \ln(p^2 a^2) + C_O^{\text{Latt}}(pa, m_q a) \right), \quad (42)$$

where, however, the “constant” C_O^{Latt} now depends on the terms of $O(a)$. Thus, in the calculation of the one-loop corrections in DPT, all the cut-off dependent terms have been included. On a finite volume, the propagators of massless particles at zero momentum must be regulated. The results of Feynman diagrams depend on the regulator, however this dependence vanishes as the volume is increased. In the results presented below, we have set the zero-momentum terms in both the quark and gluon propagators to zero. We observe a small difference in the results from SPT and DPT, at low

values of μ^2 , which is due to this choice. At the value of the quark mass at which we have performed the simulation, mass corrections are negligible.

The coupling constant appearing in eqs. (39) and (42) is the bare lattice coupling. In ref. [10] it was argued that it is possible to optimize the convergence of the perturbative series by reorganizing the expansion in terms of a “boosted” coupling α_s^V , defined in terms of a physical quantity. In the comparison with the non-perturbative results below we follow the suggestion of ref. [10], and compute the perturbative corrections using an effective coupling

$$\alpha_s^V = \frac{1}{\langle \frac{1}{3} \text{Tr} U_{\text{plaq}} \rangle} \alpha_s^{LATT} \simeq 1.68 \alpha_s^{LATT}, \quad (43)$$

where $\alpha_s^{LATT} = g_0^2/4\pi$ and $\langle \frac{1}{3} \text{Tr} U_{\text{plaq}} \rangle$ is the expectation value of the plaquette. We call this expansion “Boosted Discrete Perturbation Theory” (BDPT). Analogously we denote the “Boosted Standard Perturbation Theory” by BSPT.

The results of the calculation of the renormalization constants in BDPT will be compared to the non-perturbative results in the next section.

6 Non-perturbative results

In this section, we give the main results of our numerical study of the non-perturbative renormalization procedure proposed in this paper and a comparison of these results with perturbation theory.

We have performed a simulation, by generating 36 independent gluon field configurations, on a $16^3 \times 32$ lattice, at $\beta = 6.0$. The quark propagators have been computed for a single value of the quark mass ($am_q \sim 0.07$), corresponding to a value of the hopping parameter $K = 0.1425$. All the Green functions have been computed in the lattice Landau gauge, which is obtained by minimizing the functional

$$\text{tr} \left[\sum_{\mu=1}^4 \left(U_{\mu}(x) + U_{\mu}^{\dagger}(x) \right) \right]. \quad (44)$$

Possible effects from Gribov copies or spurious solutions [30] have not been considered. As shown below, for those quantities which can be determined

also in a gauge invariant way, the non-perturbative results obtained on quark state in a fixed gauge are in good agreement with those obtained using the Ward identity method [3].

Vector current: In fig. 4, the renormalization constant of the local vector current Z_{V^L} , obtained using eqs. (30) and (31), is given. As expected, Z_{V^L} is independent of the scale, within statistical errors, up to large values of μ^2 , where distortions due to lattice discretization become important. This is a consequence of the equivalence, which can be established up to terms of $O(a_s a)$, of the method used here to determine Z_{V^L} , and of the Ward identity for the local vector current, see sec. 3. By using the points at $\mu^2 a^2 \sim 1$, corresponding to $\mu \sim 2$ GeV, we get $Z_{V^L}^{\text{NP}} = 0.84(1)$, to be compared with $Z_{V^L}^{\text{WI}} = 0.824(2)$ from the Ward identity method, and $Z_{V^L}^{\text{BPPT}} = 0.83$ ($0.86-0.87$ on a $16^3 \times 32$ lattice, at $\mu^2 a^2 \sim 1$) from BSPT. The three methods are in good agreement.

Scalar density: Z_S is an ideal quantity with which to check the validity of our method for a number of reasons: it cannot be determined using the Ward identity method, it is logarithmically divergent and gauge dependent, and it is not affected by the pole of the Goldstone boson (in contrast to the pseudoscalar density and the axial current), sec. 2.

In fig. 5, Z_S is shown as a function of the renormalization scale. We report separately the results, obtained by using Z_ψ , eq.(27), and Z'_ψ , eq.(30). Notice that the points, corresponding to different definitions of Z_ψ , are in very good agreement where $\mu^2 a^2$ is not too large, and discretization errors are small. This can also be seen by comparing the results in BSPT (dashed curve) to those in BDPT (continuous curve). We do not expect agreement with perturbation theory, either at low μ^2 , where higher order effects become very large, or at high μ^2 , where lattice distortions are important, as shown by the continuous curve. The numerical results follow the theoretical expectations and we find good agreement between the non-perturbative determination of Z_S and the predictions from boosted perturbation theory for $0.3 \leq \mu^2 a^2 \leq 1$ ($1.1 \text{ GeV} \leq \mu \leq 2 \text{ GeV}$). Notice the Z_S (Z_P) is defined here from the off-shell Green function and depends on the anomalous dimension of the scalar (pseudoscalar) density, and hence on $\log(\mu^2 a^2)$. Moreover, the result is gauge dependent, even in the continuum, as can be shown by an explicit calculation in one-loop perturbation theory.

Axial vector current: The local axial-vector current shares some of the features of the vector current. Its renormalization constant is finite at all orders in perturbation theory [31] and can be determined from Ward identities. However, the axial-vector current is coupled to the would-be Goldstone boson of QCD , which can give an important contribution at low μ^2 . Z_{A^L} , which like Z_{V^L} , should be independent of μ , is shown as a function of μ^2 in fig. 6.

We interpret the strong μ -dependence of Z_{A^L} , at low μ^2 , as the non-perturbative effect of the pseudoscalar state. Unfortunately, there is no clear sign of the existence of a plateau between the non-perturbative regime and the large μ region, where lattice artefacts become important¹⁰. Without the information coming from the Ward identity method [3], it would be difficult to determine Z_{A^L} confidently. Nevertheless, we do observe that, at $\mu^2 a^2 \sim 1$, just before lattice artefacts become large in DPT, the results are close to the value determined with the Ward identity, $Z_A^{\text{WI}} = 1.06(2)$ [3, 21], see fig. 6. As observed in ref. [3], perturbation theory gives values of Z_{A^L} smaller than one, while the Ward identity method gives values larger than one, for $\beta = 6.0$ -6.2. A value larger than one is also suggested by our non-perturbative results.

Pseudoscalar density: The pseudoscalar density shares the main features of the scalar density, with two important differences. Firstly it is coupled to the would-be Goldstone boson, and secondly the one-loop perturbative corrections, with the Clover action, are quite large. It is then not surprising that the non-perturbative value of the corresponding renormalization constant Z_P lie well below the perturbative result, as shown in fig. 7. Had we used standard perturbation theory in fig. 7, instead of boosted perturbation theory, the discrepancy would have been even worse. The discrepancy could have been anticipated from the results of ref. [3], where it was shown that the ratio Z_S/Z_P , determined by using the Ward identity method, was significantly larger than the result obtained in boosted perturbation theory. Combining this information with the agreement between the non-perturbative determination of Z_S and the result from BPT, one would conclude that the difference is due to Z_P . This may also be due to the fact that Z_P has larger one-loop finite corrections, $\sim 35\%$, and that perhaps the higher order terms, which are not accounted for by one loop boosted

¹⁰The plateau could eventually appear more clearly at larger values of β .

perturbation theory, are important.

We also present, in fig. 8, Z_S/Z_P as a function of μ^2 . The ratio Z_S/Z_P has a behaviour similar to Z_{A^*} . Some discretization effects appear to be smaller than those observed in Z_S and Z_P separately. The numerical results support, qualitatively, the theoretical interpretation. In the intermediate range of μ , this ratio stays almost constant. At small or large values of μ the results are unstable, due to non-perturbative or discretization effects respectively. At $\mu^2 a^2 \sim 1$, the value of this ratio obtained from our non-perturbative method is in reasonable agreement with that determined with the Ward identity method, $Z_S/Z_P = 1.64(5)$ [3]¹¹.

7 Conclusion

We have proposed a new non-perturbative method to renormalize lattice composite operators. It can be used in all cases and is particularly useful when the Ward identity method is not applicable. This method avoids the need to perform any calculations using lattice perturbation theory in the computation of physical quantities from lattice simulations. The success of our proposal is subject to the existence, at current values of β , of a window between the non-perturbative region at low momenta and the region of large momenta, where discretization errors become important. This limitation is however, common to all methods (with the exception of the Ward identity method, which can only be applied to a few cases, corresponding to finite operators). The window, necessary to implement the renormalization programme described here, seems to exist already at $\beta = 6.0$ and we expect that the range of momenta, useful for non-perturbative renormalization, will become larger as β increases. We are planning to apply the approach presented in this paper to the renormalization of the $\Delta S = 2$ four fermion operator (14) and of the heavy-light axial vector current in the static theory.

¹¹The value published in ref. [3], using half of the present statistics, was $Z_P/Z_S = 0.64(2)$, corresponding to $Z_S/Z_P = 1.56(5)$.

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Appendix

In this appendix we demonstrate that the non-perturbative contributions to forward two-quark matrix elements are suppressed as inverse powers of p^2 , where p is the external momentum. Let us consider the non-amputated, two-fermion Green function

$$F_O(p) = \int d^4x d^4y e^{-ip \cdot z} \langle \bar{q}(0) \Gamma q(x) O_{\Gamma}(y) \rangle,$$

where $O_{\Gamma}(y) = \bar{q}(y) \Gamma q(y)$ and Γ is one of the Dirac matrices. It is natural to worry about the infrared behaviour of $F_O(p)$, because it contains the insertion of a zero-momentum operator. On general grounds, we expect infrared divergences to appear in Green functions computed at exceptional momenta. In physical terms, this is due to the fact that the matrix elements of O_{Γ} , at zero momentum, carry information about the physical mass spectrum, which is inaccessible to perturbation theory. In particular, if the operator O_{Γ} is the pseudoscalar density (γ_5) and we work in the chiral limit, a contribution from a Goldstone pole, corresponding to the pion, should appear in $F_O(p)$. The presence of non-perturbative contributions to $F_O(p)$ is signaled by the appearance of infrared divergences in the perturbative expansion. This is what happens in general. For quark degrees of freedom in the limit of large p^2 however, the Operator Product Expansion (OPE) guarantees that $F_O(p)$ can be reliably computed at all orders in perturbation theory. The argument goes as follows. In the limit of large p^2 , the dominant contribution to $F_O(p)$ comes from regions of integration (in x and y) where the integrand is singular. This implies that, in this limit, we have to study the behaviour of

$$F_O(x, y) = \langle \bar{q}(0) \Gamma q(x) O_{\Gamma}(y) \rangle,$$

when $x \sim 0$ and $x \sim y$. In both cases of course, we can use the OPE in the following form¹²

$$q_a(x) O_{\Gamma}(y) \rightarrow A_{as}^{\Gamma}(x-y) \Gamma_{as} q_s(y) + \dots,$$

¹²From a technical point of view, OPE is a weak operator statement and, when O_{Γ} is inserted in a Green function containing other operators, a slightly modified form would be appropriate. It is easy to convince oneself, however, that the additional terms, required in the general case, are not relevant to the present discussion.

for $x \sim y$, and

$$\bar{q}(x)\Gamma q(0) \rightarrow B_{\Gamma}(x^2)\bar{q}(0)\Gamma q(0) + \dots,$$

for $x \sim 0$.

Since QCD is asymptotically free, the behaviour of $A_{n\beta}^{\Gamma}(x-y)$ and $B_{\Gamma}(x)$ can be computed in perturbation theory. The interaction gives rise to logarithmic corrections to the free field behaviour

$$A_{\Gamma}(x) = \not{C}_{\Gamma}(x^2) \equiv \not{\not{A}_{\Gamma} \ln^{\alpha_{\Gamma}}(\mu^2 x^2) + \dots}$$

$$B_{\Gamma}(x^2) = b_{\Gamma} \ln^{\beta_{\Gamma}}(\mu^2 x^2) + \dots$$

Using the above equations, in the limit $p^2 \rightarrow \infty$, we find

$$\int d^4x d^4y e^{-ipx} \left(\langle \bar{q}(0)\Gamma A_{\Gamma}(x-y)\Gamma q(y) \rangle + B_{\Gamma}(x^2) \langle \bar{q}(0)\Gamma q(0)O_{\Gamma}(y) \rangle \right) =$$

$$-ip_{\mu}C_{\Gamma}(p^2)\tilde{S}_{\Gamma}^{\mu}(p) + B_{\Gamma}(p^2)\tilde{\Delta}_{\Gamma}(0),$$

where

$$\tilde{S}_{\Gamma}^{\mu}(p) = \int d^4y e^{-ip \cdot y} \langle \bar{q}(0)\Gamma \gamma^{\mu} \Gamma q(y) \rangle \quad \tilde{\Delta}_{\Gamma}(0) = \int d^4y \langle \bar{q}(0)\Gamma q(0)O_{\Gamma}(y) \rangle.$$

The term $-ip_{\mu}\tilde{C}_{\Gamma}(p^2)\tilde{S}_{\Gamma}^{\mu}(p)$ is computable in perturbation theory. On dimensional ground, for $p^2 \rightarrow \infty$, it behaves as

$$-ip_{\mu}\tilde{C}_{\Gamma}(p^2)\tilde{S}_{\Gamma}^{\mu}(p) \rightarrow c_{\Gamma} \frac{\ln^{\alpha_{\Gamma}}(p^2/\mu^2)}{p^2}.$$

As for $\tilde{B}_{\Gamma}(p^2)$, we find

$$\tilde{B}_{\Gamma}(p^2) \rightarrow d_{\Gamma} \frac{\ln^{\beta_{\Gamma}}(p^2/\mu^2)}{p^4}.$$

$\tilde{B}_{\Gamma}(p^2)$ is, however, multiplied by $\tilde{\Delta}_{\Gamma}(0)$, which is a non-perturbative quantity. We have shown however, that, in the large p^2 limit, the perturbative contribution dominates by one power of p^2 over the non-perturbative one.

This is the reason why, even at an exceptional external momentum, $F_O(p)$ is infrared safe in perturbation theory, when p^2 is large. It is straightforward to trace the suppression of the infrared divergence, by analyzing the corresponding Feynman diagrams. For example, the infrared divergent part of the diagram in fig. 9-a is suppressed by one power of p^2 , due to the momentum flow through the lower gluon line.

It is interesting to see what would happen in a (fictitious) theory, with quarks represented by scalar fields. In this case the infrared divergence is not suppressed, as can be seen from the diagram in fig. 9-b. The general argument goes as follows

$$F(p^2) = \int d^4x d^4y e^{-ipx} \langle \bar{\phi}(0) \phi(x) J(y) \rangle,$$

with

$$J(y) = \bar{\phi}(y) \phi(y).$$

When $x \sim 0$ and $x \sim y$, the OPE gives

$$\phi(x) J(y) \rightarrow A \left((x-y)^2 \right) \phi(y) + \dots \quad \bar{\phi}(x) \phi(0) \rightarrow B(x^2) \bar{\phi}(0) \phi(0) + \dots,$$

with

$$A(x^2) = \frac{a \ln^a(\mu^2 x^2)}{x^2} + \dots \quad B(x^2) = b \ln^b(\mu^2 x^2) + \dots,$$

and we would get

$$F(p^2) \rightarrow \bar{A}(p^2) \bar{S}(p^2) + \bar{B}(p^2) \bar{\Delta}(0)$$

with

$$\bar{S}(p^2) = \int d^4y e^{-ipy} \langle \bar{\phi}(0) \phi(y) \rangle \quad \bar{\Delta}(0) = \int d^4y \langle \bar{\phi}(0) \phi(0) J(y) \rangle.$$

One now obtains

$$\bar{A}(p^2) \bar{S}(p^2) \rightarrow c \frac{\ln^\gamma(p^2/\mu^2)}{p^4}$$

and

$$\bar{B}(p^2) \rightarrow d \frac{\ln^4(p^2/\mu^2)}{p^4}.$$

The latter equations imply that non-perturbative and perturbative contributions have the same p^2 dependence. Thus, we expect the appearance of infrared divergences in $F(p^2)$.

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Figure 4: Z_{V^L} as a function of $\mu^2 a^2$. Since the lattice is not symmetric in space and time, the results obtained by using the time-component of the local current (V_0) and the space components, averaged in the three directions, (V_k) are shown separately. The dashed line is $Z_{V^L}^{\text{BSPT}}$ from BSPT, and the curve is from BDPT, on a volume of size $16^3 \times 32$, see sec. 5, see sec. 5. The straight (continuous) line is the result obtained using the Ward identities method [3, 21].

Figure 5: Z_S as a function of $\mu^2 a^2$. We give the renormalization constant obtained by using the two possible definitions of the quark wave function renormalization, denoted by Z_ψ , eq. (27), and Z'_ψ , eq. (30). The dashed curve is Z_S^{BSPT} , from boosted standard perturbation theory in the infinite volume limit, and the full curve is the result obtained in BDPT.

Figure 6: Z_{A^L} as a function of $\mu^2 a^2$. We give the renormalization constant, obtained by using the time component A_0 and the space components A_k , separately. The dashed line is Z_A^{BSPT} , from boosted standard perturbation theory, and the full curve is from BDPT . The straight (continuous) line is the result obtained using the Ward identities method [3, 21].

Figure 7: Z_P as a function of $\mu^2 a^2$. We give the renormalization constant obtained by using the two possible definitions of the quark wave function renormalization, denoted by Z_ψ , eq. (27), and Z'_ψ , eq. (30). The dashed curve is Z_P^{BSPT} , from boosted standard perturbation theory in the infinite volume limit, and the full curve is the result obtained in BDPT.

Figure 8: Z_S/Z_P as a function of $\mu^2 a^2$. The dashed line corresponds to BSPT, the full line comes from the determination of this ratio, using the Ward identity method [3].

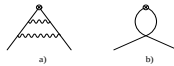


Figure 9: Diagrams with standard, a), and scalar quarks, b).

